

0001. A blocked pitot head with a clear static source causes the airspeed indicator to ;

- a) react like an altimeter.
- b) read like a vertical speed indicator.
- c) freeze at zero.
- d) operate normally.

Answer ;

a) react like an altimeter.

---

0002. A "Bourdon Tube" is used in ;

- a) vibration detectors.
- b) turbine temperature probes.
- c) smoke detectors.
- d) pressure sensors.

Answer ;

d) pressure sensors.

---

0003. A closed loop control system in which a small power input controls a much larger power output in a strictly proportionate manner is known as ;

- a) a servomechanism.
- b) a feedback control circuit.
- c) an amplifier.
- d) an autopilot.

Answer ;

a) a servomechanism.

---

0004. A control system consisting of four pendulous vanes is used in ;

- a) a directional gyro indicator.
- b) a strap down inertial system.
- c) an air driven artificial horizon.
- d) a gyromagnetic indicator.

Answer ;

c) an air driven artificial horizon.

---

0005. A D-ATIS is ;

- a) an ATIS message received by datalink.
- b) a short ATIS that only includes parameters that have changed from previous ATIS record.
- c) an ATIS message broadcasted on HF when out of the VHF radio range from the airport.
- d) a Diversion ATIS for the alternate airport.

Answer ;

a) an ATIS message received by datalink.

---

0006. A directional gyro consists of a ;

NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.

- a) 1 degrees-of-freedom vertical axis gyro.
- b) 2 degrees-of-freedom horizontal axis gyro.
- c) 1 degrees-of-freedom horizontal axis gyro.
- d) 2 degrees-of-freedom vertical axis gyro.

Answer ;

b) 2 degrees-of-freedom horizontal axis gyro.

---

0007. A directional gyro is ;

- 1. a gyroscope free around two axis
- 2. a gyroscope free around one axis
- 3. capable of self-orientation around an earth-tied direction
- 4. incapable of self-orientation around an earth-tied direction

The combination which regroups all of the correct statements is ;

NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.

- a) 1 , 4.
- b) 1 , 3.
- c) 2 , 4.
- d) 2 , 3.

Answer ;

a) 1 , 4.

---

**0008.** A directional gyro is corrected for an apparent drift due to the earth's rotation at latitude 30°S. During a flight at latitude 60°N , a drift rate of 15.5°/h to the right is observed. The apparent wander due to change of aircraft position is ;

- a) 2.5°/h to the right.
- b) 2.5°/h to the left.
- c) 5°/h to the right.
- d) 5°/h to the left.

Answer ;

**d) 5°/h to the left.**

---

**0009.** A dynamic pressure measurement circuit is constituted of the following pressure probes ;

- a) total pressure and standard pressure.
- b) total pressure and static pressure.
- c) total pressure only.
- d) static pressure only.

Answer ;

**b) total pressure and static pressure.**

---

**0010.** A flight control system is fail-operational if , in the event of a failure ;

- a) the approach only can be completed automatically.
- b) the landing is not completed automatically.
- c) the approach , flare and landing can be completed automatically.
- d) there is no significant out-of-trim condition or deviation of flight path or attitude but the landing is not completed automatically.

Answer ;

**c) the approach , flare and landing can be completed automatically.**

---

**0011.** A flight control system is fail-passive if , in the event of a failure ;

- a) the system operates as a fail-operational system.
- b) there is no significant out-of-trim condition or deviation of flight path or attitude but the landing is not completed automatically.
- c) there is no significant out-of-trim condition or deviation of flight path or attitude but the landing is completed automatically.
- d) the system operates as a fail-operational hybrid landing system.

Answer ;

**b) there is no significant out-of-trim condition or deviation of flight path or attitude but the landing is not completed automatically.**

---

**0012.** A flight control system which can , in the event of a failure , complete automatically the approach , flare and landing is called fail ;

- a) soft.
- b) operational.
- c) hard.
- d) passive.

Answer ;

**b) operational.**

---

**0013.** A float type fuel gauge ;

1. gives a mass information.
2. gives information independent of aircraft's manoeuvres and attitude changes.
3. gives information all the more accurate as the tank is full.
4. is typically a DC powered system.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 3 , 4.
- c) 4.
- d) 1.

Answer ;

**c) 4.**

---

- 0014.** A flux valve detects the horizontal component of the earth's magnetic field ;
1. the flux valve is made of a pair of soft iron bars.
  2. the information can be used by a "flux gate" compass or a directional gyro.
  3. the flux gate valve signal comes from the magnetic compass.
  4. the accuracy on the value of the magnetic field indication is less than 0.5°.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 1 , 2 , 4.
- c) 1 , 2.
- d) 1 , 4.

**Answer ;**

- a) 2 , 4.**
- 

- 0015.** A flux valve detects the horizontal component of the earth's magnetic field ;
1. the flux valve is made of a pair of soft iron bars
  2. the information can be used by a "flux gate" compass or a directional gyro
  3. the flux gate valve casing is dependent on the aircraft three inertial axis.
  4. the accuracy on the value of the magnetic field indication is less than 0.5°

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4.
- b) 2 , 4.
- c) 1 , 2 , 4.
- d) 1 , 2 , 3 , 4.

**Answer ;**

- b) 2 , 4.**
- 

- 0016.** A free gyro has its rotor axis horizontal and is aligned with the geographic meridian. If this stationary free gyro is situated at latitude 60°N , the apparent drift rate according to the earthbound observer is ;

- a) 7.5°/h to the right.
- b) 13°/h to the left.
- c) 13°/h to the right.
- d) 7.5°/h to the left.

**Answer ;**

- c) 13°/h to the right.**
- 

- 0017.** A free gyro has the axis of the spinning rotor horizontal and aligned with the geographic meridian. If this free gyro is situated at latitude 60°N , the apparent drift rate according to the earthbound observer is ;

- a) 13°/h to the right.
- b) 13°/h to the left.
- c) 7.5°/h to the left.
- d) 7.5°/h to the right.

**Answer ;**

- a) 13°/h to the right.**
- 

- 0018.** A Full Authority Digital Engine Control ( FADEC ) has the following functions ;
1. flow regulation ( fuel , decelerations and accelerations monitoring ).
  2. automatic starting sequence.
  3. transmissions of engine data to the pilot's instruments.
  4. thrust management and protection of operation limits.
  5. monitoring of the thrust reversers.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4 , 5.
- b) 2 , 4 , 5.
- c) 1 , 2 , 3 , 4 , 5.
- d) 1 , 3 , 5.

**Answer ;**

- c) 1 , 2 , 3 , 4 , 5.**
- 

- 0019.** A gravity erector system is used to correct the errors on ;

- a) a directional gyro.
- b) an artificial horizon.
- c) a turn indicator.
- d) a gyromagnetic compass.

**Answer ;**

- b) an artificial horizon.**
-

0020. A gravity type erector is used in a vertical gyro device to correct errors on ;

- a) a directional gyro unit.
- b) a gyromagnetic indicator.
- c) an artificial horizon.
- d) a turn indicator.

Answer ;

c) an artificial horizon.

---

0021. A ground proximity warning system ( GPWS ) , when mandatory installed on board an aircraft , must in all cases generate ;

- a) a visual alarm to which a sound alarm can be.
- b) a sound alarm or a visual alarm.
- c) a sound and visual alarm.
- d) at least one sound alarm to which a visual alarm can be added.

Answer ;

d) at least one sound alarm to which a visual alarm can be added.

---

0022. A gyromagnetic compass is a system which always consists of ;

- 1. a horizontal axis gyro.
- 2. a vertical axis gyro.
- 3. an earth's magnetic field detector.
- 4. an erection mechanism to maintain the gyro axis horizontal.
- 5. a torque motor to make the gyro precess in azimuth.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4 , 5.
- b) 2 , 3 , 5.
- c) 1 , 4.
- d) 2 , 3.

Answer ;

a) 1 , 3 , 4 , 5.

---

0023. A laser gyro can measure ;

- a) an acceleration and a rotation motion.
- b) an acceleration and a speed.
- c) an acceleration.
- d) a rotation motion.

Answer ;

d) a rotation motion.

---

0024. A leak in the pitot total pressure line of a non-pressurized aircraft to an airspeed indicator would cause it to ;

- a) under-read in a climb and over-read in a descent.
- b) over-read.
- c) over-read in a climb and under-read in a descent.
- d) under-read.

Answer ;

d) under-read.

---

0025. A magnetic tachometer consists of ;

- a) a single-phase generator connected to an asynchronous motor.
- b) a single-phase generator connected to a synchronous motor.
- c) a permanent magnet turning inside a non magnetic drag cup.
- d) a three phase generator connected to a synchronous motor.

Answer ;

c) a permanent magnet turning inside a non magnetic drag cup.

---

0026. A manifold pressure gauge of a piston engine measures ;

- a) absolute airpressure entering the carburettor.
- b) absolute pressure in intake system near the inlet valve.
- c) fuel pressure leaving the carburettor.
- d) vacuum in the carburettor.

Answer ;

b) absolute pressure in intake system near the inlet valve.

---

- 0027.** A MAYDAY datalink message can be sent to the ATC via the ;
- a) ACAS ( Aircraft Collision Avoidance System ) application.
  - b) ADS application.
  - c) AFN application.
  - d) CPDLC application.

**Answer ;**

**d) CPDLC application.**

---

- 0028.** A millivoltmeter measuring the electromotive force between the “hot junction” and the “cold junction” of a thermocouple can be directly graduated in temperature values provided that the temperature of the ;
- a) hot junction is maintained constant.
  - b) cold junction is maintained constant.
  - c) hot junction is maintained at 15 °C.
  - d) cold junction is maintained at 15 °C.

**Answer ;**

**b) cold junction is maintained constant.**

---

- 0029.** A pilot engages the control wheel steering ( CWS ) of a conventional autopilot and carries out a manoeuvre in roll. When the control wheel is released , the autopilot will ;
- a) roll wings level and maintain the heading obtained at that moment.
  - b) restore the flight attitude and the rate of turn selected on the autopilot control display unit.
  - c) maintain the track and the flight attitude obtained at that moment.
  - d) maintain the flight attitude obtained at that moment.

**Answer ;**

**d) maintain the flight attitude obtained at that moment.**

---

- 0030.** A pilot wishes to turn left on to a northerly heading with 10° bank at a latitude of 50° North. Using a direct reading compass , in order to achieve this he must stop the turn on an approximate heading of ;
- a) 015°
  - b) 330°
  - c) 030°
  - d) 355°

**Answer ;**

**c) 030°**

---

- 0031.** A pilot wishes to turn left on to a southerly heading with 20° bank at a latitude of 20° North. Using a direct reading compass , in order to achieve this he must stop the turn on an approximate heading of ;
- a) 160°
  - b) 170°
  - c) 190°
  - d) 200°

**Answer ;**

**a) 160°**

---

- 0032.** A pilot wishes to turn right on to a southerly heading with 20° bank at a latitude of 20° North. Using a direct reading compass , in order to achieve this he must stop the turn on an approximate heading of ;
- a) 200°
  - b) 150°
  - c) 180°
  - d) 170°

**Answer ;**

**a) 200°**

---

- 0033.** A pilot wishes to turn right through 90° on to North at rate 2 at latitude of 40 North using a direct reading compass. In order to achieve this the turn should be stopped on an indicated heading of approximately ;
- a) 010°
  - b) 330°
  - c) 360°
  - d) 030°

**Answer ;**

**b) 330°**

---

0034. A pitot tube covered by ice which blocks the ram air inlet will affect the following instrument(s) ;
- a) airspeed indicator , altimeter and vertical speed indicator.
  - b) vertical speed indicator only.
  - c) airspeed indicator only.
  - d) altimeter only.

Answer ;

c) airspeed indicator only.

---

0035. A radio altimeter can be defined as a ;
- a) self-contained on-board aid used to measure the true height of the aircraft.
  - b) ground radio aid used to measure the true height of the aircraft.
  - c) self-contained on-board aid used to measure the true altitude of the aircraft.
  - d) ground radio aid used to measure the true altitude of the aircraft.

Answer ;

a) self-contained on-board aid used to measure the true height of the aircraft.

---

0036. A radio altimeter uses ;
- a) two antennas ; both of them for simultaneous transmission and reception.
  - b) four antennas ; two for the transmission and two for the reception.
  - c) a single antenna for simultaneous transmission and reception.
  - d) two antennas ; one for the transmission and an other for the reception.

Answer ;

d) two antennas ; one for the transmission and an other for the reception.

---

0037. A radio altimeter uses the following frequency band ;
- a) HF ( High Frequency ).
  - b) VLF ( Very Low Frequency ).
  - c) SHF ( Super High Frequency ).
  - d) VHF ( Very High Frequency ).

Answer ;

c) SHF ( Super High Frequency ).

---

0038. A radio altimeter uses the following frequency band ;
- a) 4200 KHz to 4400 KHz.
  - b) 1600 KHz to 1660 KHz.
  - c) 1600 GHz to 1660 GHz.
  - d) 4200 MHz to 4400 MHz.

Answer ;

d) 4200 MHz to 4400 MHz.

---

0039. A rate gyro is used in a ;
1. directional gyro indicator.
  2. turn co-ordinator.
  3. artificial horizon.

The combination regrouping all the correct statements is ;

- a) 1.
- b) 1 , 2 , 3.
- c) 2.
- d) 1 , 2.

Answer ;

c) 2.

---

0040. A ring laser gyro can measure ;
- a) rotation about its sensitive axis.
  - b) accelerations in all direction.
  - c) accelerations about its sensitive axis.
  - d) rotation in all directions.

Answer ;

a) rotation about its sensitive axis.

---

0041. A ring laser gyro is ;
- a) a device which measures angular movements.
  - b) a device which measures the earth rate precession.
  - c) an optical accelerometers.
  - d) used for stabilising the INS platform.

Answer ;

- a) a device which measures angular movements.
- 

0042. A servo-assisted altimeter is more accurate than a simple altimeter because the small movements of ;
- a) the capsules are detected by a very sensitive electro-magnetic pick-off.
  - b) the capsules are not taken into account.
  - c) the capsules are inhibited.
  - d) the pointers are detected by a very sensitive electro-magnetic pick-off.

Answer ;

- a) the capsules are detected by a very sensitive electro-magnetic pick-off.
- 

0043. A slaved directional gyro derives it's directional signal from ;
- a) the air-data-computer.
  - b) the flux valve.
  - c) the flight director.
  - d) a direct reading magnetic compass.

Answer ;

- b) the flux valve.
- 

0044. A stability augmentation system ( SAS ) in a helicopter ;
- a) assists in static stability only.
  - b) assists in static and dynamic stability.
  - c) assists in dynamic stability only.
  - d) has full control of dynamic stability.

Answer ;

- a) assists in static stability only.
- 

0045. A stall warning system is based on a measure of ;
- a) groundspeed.
  - b) angle of attack.
  - c) airspeed.
  - d) attitude.

Answer ;

- b) angle of attack.
- 

0046. A Stand-by-horizon or emergency attitude indicator ;
- a) is automatically connected to the primary vertical gyro if the alternator fails.
  - b) contains its own separate gyro.
  - c) is fully independent of external energy resources in an emergency situation.
  - d) only works if there is a complete electrical failure.

Answer ;

- b) contains its own separate gyro.
- 

0047. A strapdown inertial system consists in ;
- a) a platform free of the aircraft chassis which includes gyroscopes and accelerometers.
  - b) accelerometers attached to the aircraft chassis and gyroscopes which are not.
  - c) gyroscopes attached to the aircraft chassis and accelerometers which are not.
  - d) a platform attached to the aircraft chassis and which includes gyroscopes and accelerometers.

Answer ;

- d) a platform attached to the aircraft chassis and which includes gyroscopes and accelerometers.
- 

0048. A synchroscope is used on aircraft to ;
- a) reduce the vibration of each engine.
  - b) achieve optimum control of on-board voltages.
  - c) set several engines to the same speed.
  - d) reduce the rpm of each engine.

Answer ;

- c) set several engines to the same speed.
-

0049. A TCAS 2 generates a resolution advisory ( RA ) when ;
- a) the intruder becomes “proximate traffic”.
  - b) a potential collision threat exists.
  - c) the intruder becomes “other traffic”.
  - d) a serious collision threat exists.

Answer ;

- d) a serious collision threat exists.
- 

0050. A TCAS 2 ( Traffic Collision Avoidance System ) provides ;
- a) a simple intruding airplane proximity warning.
  - b) the intruder relative position and possibly an indication of a collision avoidance manoeuvre within the vertical plane only.
  - c) the intruder relative position and possibly an indication of a collision avoidance manoeuvre within the horizontal plane only.
  - d) the intruder relative position and possibly an indication of a collision avoidance manoeuvre within both the vertical and horizontal planes.

Answer ;

- b) the intruder relative position and possibly an indication of a collision avoidance manoeuvre within the vertical plane only.
- 

0051. A TCAS Resolution Advisory ( RA ) voice message “CLIMB - CLIMB NOW” repeated twice is generated ;
- a) after a “DESCEND” RA when a reversal in the vertical manoeuvre sense is required.
  - b) each time a “CLIMB” RA is announced by a voice message.
  - c) when cleared of conflict , and below the cruise altitude.
  - d) after a “CLIMB” RA when the climb vertical rate is too weak.

Answer ;

- a) after a “DESCEND” RA when a reversal in the vertical manoeuvre sense is required.
- 

0052. A temperature sensor has a recovery factor of 0.95. The temperature measured is equal to ;
- a) 95% of the ram air temperature ( RAT ).
  - b) static air temperature ( SAT ) + 95% of the ram rise.
  - c) 95% of the static air temperature ( SAT ).
  - d) ram air temperature ( RAT ) + 95% of the ram rise.

Answer ;

- b) static air temperature ( SAT ) + 95% of the ram rise.
- 

0053. A thermocouple can be made of ;
- a) a three wire coil.
  - b) two metal conductors of the same nature fixed together at two points.
  - c) two metal conductors of different nature fixed together at a single point.
  - d) a single wire coil.

Answer ;

- c) two metal conductors of different nature fixed together at a single point.
- 

0054. A thermocouple consists of ;
- a) a junction of two dissimilar metals.
  - b) several layers of identical metals, but with different thickness.
  - c) two layers of identical metals , both having the same size.
  - d) two layers of identical metals , one being longer than the other.

Answer ;

- a) a junction of two dissimilar metals.
- 

0055. A thermocouple type temperature sensing is composed of ;
- a) two identical metals joined together at one end ( called hot junction or measure junction ).
  - b) two identical metals joined together at both ends ( called hot and cold junctions ).
  - c) a single-wire metal winding.
  - d) two dissimilar metals joined together at one end ( called hot junction or measure junction ).

Answer ;

- d) two dissimilar metals joined together at one end ( called hot junction or measure junction ).
-



0056. A thermocouple type thermometer consists of ;
- a) two metal conductors of the same type connected at two points.
  - b) a single-wire metal winding.
  - c) a Wheatstone bridge connected to a voltage indicator.
  - d) two metal conductors of different type connected at two points.

Answer ;

- d) two metal conductors of different type connected at two points.
- 

0057. A three-phase electrical tachometer consists of ;
- a) a three-phase generator , a synchronous motor and a magnetic tachometer.
  - b) three associated dynamos.
  - c) three speed probes and a phonic wheel.
  - d) a speed probe and a phonic wheel.

Answer ;

- a) a three-phase generator , a synchronous motor and a magnetic tachometer.
- 

0058. A three-phase electrical tachometer utilises a generator feeding ;
- a) a synchronous motor turning a drag cup.
  - b) directly a voltmeter.
  - c) directly a drag cup.
  - d) directly a galvanometer.

Answer ;

- a) a synchronous motor turning a drag cup.
- 

0059. A turn indicator is an instrument which indicates rate of turn. Rate of turn depends upon ;
- 1. bank angle.
  - 2. aeroplane speed.
  - 3. aeroplane weight.

The combination regrouping the correct statements is ;

- a) 2 and 3.
- b) 1 and 3.
- c) 1 and 2.
- d) 1 , 2 and 3.

Answer ;

- c) 1 and 2.
- 

0060. A vertical speed indicator measures the difference between ;
- a) the total pressure and the static pressure.
  - b) the total instantaneous pressure and the total pressure at a previous moment.
  - c) the dynamic pressure and the static pressure.
  - d) the instantaneous static pressure and the static pressure at a previous moment.

Answer ;

- d) the instantaneous static pressure and the static pressure at a previous moment.
- 

0061. A vibration indicator receives a signal from different sensors ( accelerometers ). It indicates the ;
- a) vibration period expressed in seconds.
  - b) vibration frequency expressed in Hz.
  - c) acceleration measured by the sensors , expressed in g.
  - d) vibration amplitude at a given frequency.

Answer ;

- d) vibration amplitude at a given frequency.
- 

0062. A  $V_{MO}$ - $M_{MO}$  warning device consists of an alarm connected to ;
- a) a barometric aneroid capsule and an airspeed sensor subjected to dynamic pressure.
  - b) a barometric aneroid capsule subjected to a static pressure and an airspeed sensor subjected to a dynamic pressure.
  - c) a barometric aneroid capsule subjected to a dynamic pressure and an airspeed sensor subjected to a static pressure.
  - d) a barometric aneroid capsule and an airspeed sensor subjected to a static pressure.

Answer ;

- b) a barometric aneroid capsule subjected to a static pressure and an airspeed sensor subjected to a dynamic pressure.
-

0063. About a magnetic compass ;

- a) errors of parallax are due to the oscillation of the compass rose.
- b) acceleration errors are due to the angle of dip.
- c) acceleration errors are due to Schuler oscillations.
- d) turning error is due to the vertical component of the earth's magnetic field.

Answer ;

- d) turning error is due to the vertical component of the earth's magnetic field.
- 

0064. According to AMC 25-11 concerning the electronic display systems , the amber/yellow colours are associated with the following alert ;

- a) advisory.
- b) warning.
- c) status.
- d) caution.

Answer ;

- d) caution.
- 

0065. According to AMC 25-11 concerning the electronic display systems , the colour associated with a caution-type alert is ;

- a) red.
- b) white.
- c) magenta.
- d) amber or yellow.

Answer ;

- d) amber or yellow.
- 

0066. According to AMC 25-11 concerning the electronic display systems , the colour associated with a warning-type alert is ;

- a) amber.
- b) red.
- c) magenta.
- d) yellow.

Answer ;

- b) red.
- 

0067. According to AMC 25-11 concerning the electronic display systems , the colour associated with an armed mode is ;

- a) magenta.
- b) white or cyan.
- c) yellow.
- d) green.

Answer ;

- b) white or cyan.
- 

0068. According to AMC 25-11 concerning the electronic display systems , the colour associated with an engaged mode is ;

- a) magenta.
- b) green.
- c) amber.
- d) white.

Answer ;

- b) green.
- 

0069. According to AMC 25-11 concerning the electronic display systems , the colours used are ;

- a) red for caution & abnormal sources , white for flight envelope limits.
- b) amber for the system limits , white for autopilot or flight director engaged modes.
- c) red for warnings , magenta for autopilot or flight director engaged modes.
- d) red for flight envelope and system limits , green for autopilot or flight director engaged modes.

Answer ;

- d) red for flight envelope and system limits , green for autopilot or flight director engaged modes.
- 

0070. According to AMC 25-11 concerning the electronic display systems , the green colour is associated with the following indication ;

- a) engaged mode.
- b) advisory.
- c) armed mode.
- d) status.

Answer ;

- a) engaged mode.
-

- 0071.** According to AMC 25-11 concerning the electronic display systems , the red colour is associated with the following alert ;
- a) warning.
  - b) status.
  - c) caution.
  - d) advisory.

**Answer ;**

**a) warning.**

---

- 0072.** According to AMC 25-11 concerning the electronic display systems , the white/cyan colours are associated with the following indication ;

- a) warning.
- b) armed mode.
- c) engaged mode.
- d) caution.

**Answer ;**

**b) armed mode.**

---

- 0073.** According to AMC 25-11 concerning the electronic display systems , when exceeding the limits of the flight envelope, the colour accepted to alert the flight crew is ;

- a) magenta.
- b) red.
- c) amber.
- d) yellow.

**Answer ;**

**b) red.**

---

- 0074.** According to CS 25 , a CVR ( Cockpit Voice Recorder ) must record ;

- 1. communications transmitted from or received in the aeroplane by radio.
- 2. communications of flight-crew members on the flight deck.
- 3. communications of flight-crew members on the flight deck using the aeroplane's interphone system.
- 4. audio signals identifying navigation or approach aids introduced into a headset or speaker.

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 1 , 2 , 3 , 4.
- c) 1 , 3.
- d) 2 , 4.

**Answer ;**

**b) 1 , 2 , 3 , 4.**

---

- 0075.** According to CS 25 , section 3 concerning the Electronic Display Systems , the white/cyan colours are associated with the following indication ;

- a) warning.
- b) engaged mode.
- c) caution.
- d) armed mode.

**Answer ;**

**d) armed mode.**

---

- 0076.** According to CS25 , the definition of the control wheel steering mode ( CWS ) is ;

“Where the pilot has the ability to make inputs to the ;

- a) flight director by movement of the normal control wheel”.
- b) automatic pilot by movement of the normal control wheel”.
- c) flight director by movement of the alternate control wheel”.
- d) automatic pilot by movement of the alternate control wheel”.

**Answer ;**

**b) automatic pilot by movement of the normal control wheel”.**

---

**0077.** According to the range of pressure to be measured ( low , medium , high ) , different types of pressure sensors are used. Classify the following sensors by order of increasing pressure for which they are suitable ;

1. bellows type.
  2. Bourdon tube type.
  3. aneroid capsule.
- a) 3 , 2 , 1.  
b) 2 , 1 , 3.  
c) 1 , 2 , 3.  
d) 3 , 1 , 2.

**Answer ;**

**d) 3 , 1 , 2.**

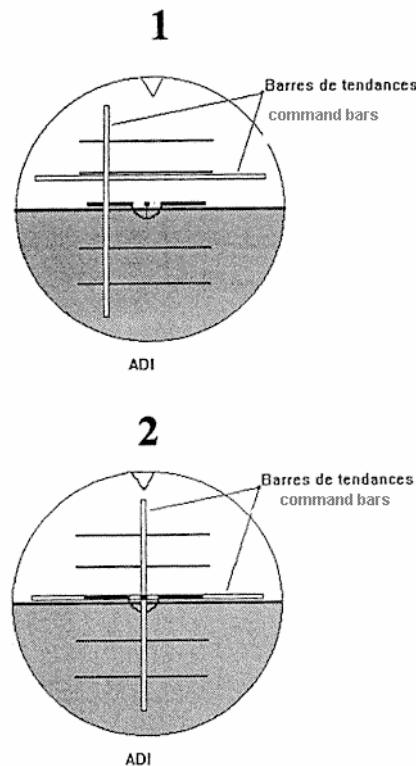
**0078.** After an aircraft has passed through a volcanic cloud which has blocked the total pressure probe inlet of the airspeed indicator , the pilot begins a stabilized descent and finds that the indicated airspeed ;

- a) decreases abruptly towards zero.  
b) increases abruptly towards  $V_{NE}$ .  
c) increases steadily.  
d) decreases steadily.

**Answer ;**

**d) decreases steadily.**

**0079.** ( For this question use annex 022-11232A )



**Annex 022-11232A**

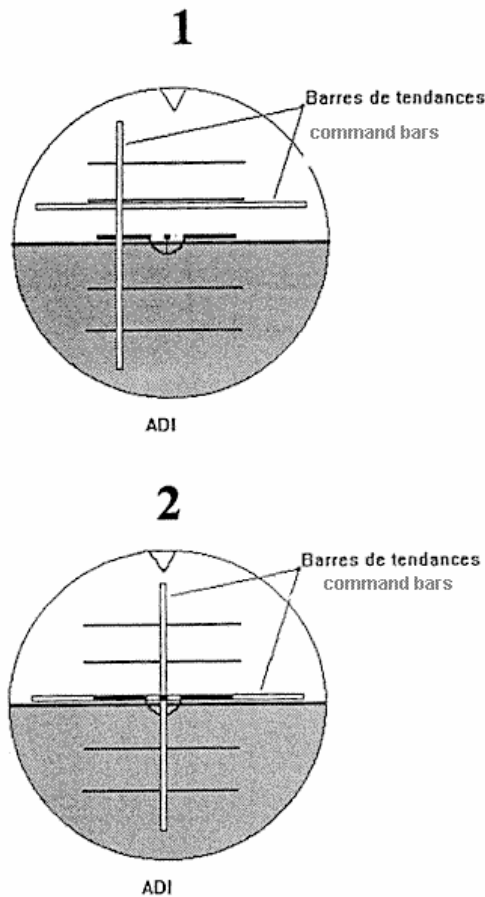
After having programmed your flight director , you see that the indications of your ADI ( Attitude Director indicator ) are as represented in diagram N°1 of the appended annex. On this instrument , the command bars indicate that you must ;

- a) increase the flight attitude and bank your aeroplane to the right until the command bars recentre on the symbolic aeroplane.  
b) increase the flight attitude and bank your airplane to the left until the command bars recentre on the symbolic aeroplane.  
c) decrease the flight attitude and bank your airplane to the right until the command bars recentre on the symbolic aeroplane.  
d) decrease the flight attitude and bank your airplane to the left until the command bars recentre on the symbolic aeroplane.

**Answer ;**

**b) increase the flight attitude and bank your airplane to the left until the command bars recentre on the symbolic aeroplane.**

0080. ( For this question use annex 022-10217A )



**Annex 022-10217A**

After having programmed your flight director , you see that the indications of your ADI ( Attitude Director Indicator ) are as represented in diagram N°1 of the appended annex. On this instrument , the command bars indicate that you must bank your airplane to the left and ;

- a) increase the flight attitude until the command bars recentre on the symbolic airplane.
- b) increase the flight attitude until the command bars recentre on the horizon.
- c) decrease the flight attitude until the command bars recentre on the symbolic airplane.
- d) decrease the flight attitude until the command bars recentre on the horizon.

Answer ;

- a) increase the flight attitude until the command bars recentre on the symbolic airplane.

0081. After the initial 90 degrees of a turn at constant pitch and bank , a classic artificial horizon indicates ;

- a) too much nose-up and too much bank.
- b) pitch and bank correct.
- c) too much nose-up and too little bank.
- d) too much nose-up and bank correct.

Answer ;

- c) too much nose-up and too little bank.

0082. Alarms are standardised and follow a code of colours. Those requiring action but not immediately , are signalled by the colour ;

- a) flashing red.
- b) red.
- c) green.
- d) amber.

Answer ;

- d) amber.

**0083.** Among the following functions of an autopilot , those related to the airplane guidance are ;

1. pitch attitude holding.
2. horizontal wing holding.
3. indicated airspeed or Mach number holding.
4. altitude holding.
5. VOR axis holding.
6. yaw damping.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 and 6.
- b) 3 , 4 and 5.
- c) 1 , 2 and 6.
- d) 1 , 3 , 4 and 5.

**Answer ;**

**b) 3 , 4 and 5.**

---

**0084.** Among the following functions of an autopilot , those related to the airplane stabilization are ;

1. pitch attitude holding.
2. IAS or Mach number holding.
3. horizontal wing holding.
4. VOR radial holding.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 1 , 3.
- c) 1 , 2 , 3 , 4.
- d) 1 , 2 , 3.

**Answer ;**

**b) 1 , 3.**

---

**0085.** Among the following functions of an autopilot , those related to the airplane stabilization are ;

1. pitch attitude holding.
2. horizontal wing holding.
3. displayed heading or inertial track holding.
4. indicated airspeed or Mach number holding.
5. yaw damping.
6. VOR axis holding.

The combination regrouping all the correct statements is ;

- a) 3 , 4 , 5 , 6.
- b) 1 , 2.
- c) 2 , 4 , 5.
- d) 1 , 2 , 5.

**Answer ;**

**b) 1 , 2.**

---

**0086.** Among the systematic errors of the “directional gyro” , the error due to the earth rotation make the north reference turn in the horizontal plane. At a mean latitude of 45°N , this reference turns by ;

- a) 10.5°/hour to the right.
- b) 15°/hour to the right.
- c) 7.5°/hour to the left.
- d) 7.5°/hour to the right.

**Answer ;**

**a) 10.5°/hour to the right.**

Drift due to Earth rate will make the gyro rotate clockwise ( to the right ) in the northern hemisphere at a rate of ;  
 $15 \times \sin ^{\circ} \text{lat} = 15 \times \sin 45 = 10.5^{\circ} / \text{hr}$

---

**0087.** An aeroplane is cruising at FL 60 with a TAS of 100 kt in standard atmosphere. In these conditions ;

1. the TAS is approximately 10% higher than the IAS.
2. the difference between the EAS and the CAS is negligible.
3. the speed displayed on the airspeed indicator is a CAS if the position error and instrument error are zero.

The combination regrouping all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 3.
- c) 1 , 2 , 3.
- d) 1 , 2.

**Answer ;**

**c) 1 , 2 , 3.**

---

0088. An aeroplane is cruising at FL 220. The auto-throttle maintains a constant CAS. If the OAT increases , the Mach number ;
- a) decreases.
  - b) decreases if OAT is lower than standard temperature , increases in the opposite case.
  - c) remains constant.
  - d) increases.

Answer ;

c) remains constant.

0089. An aeroplane is flying at FL300 with a TAS of 470 kt in standard conditions. The Mach number is ;

- a) 0.83
- b) 0.80
- c) 0.82
- d) 0.53

Öncelikle FL300'de ISA'ya göre olması gereken sıcaklık bulunur ;  $(2 \times 30) - 15 = -45^{\circ}\text{C}$ . İkinci adım olarak DR kompitüründe MACH NO INDEX oku  $-45$ 'e hizalanır. Dış skalada yer alan 47 değerin ( 470 kt TAS ) karşısında , iç skalada yaklaşık olarak 0.80 Mach değeri okunur.

Answer ;

b) 0.80

0090. An aeroplane is in steady climb. The auto-throttle maintains a constant calibrated airspeed. If the total temperature remains constant , the Mach number ;

- a) increases.
- b) decreases if the static temperature is lower than the standard temperature.
- c) decreases.
- d) remains constant.

Answer ;

a) increases.

0091. An aeroplane is in steady climb. The auto-throttle maintains a constant Mach number. If the total temperature remains constant , the calibrated airspeed ;

- a) remains constant.
- b) decreases if the static temperature is lower than the standard temperature , increases if higher.
- c) increases.
- d) decreases.

Answer ;

d) decreases.

0092. An aeroplane is in steady cruise at flight level 270. The auto-throttle maintains a constant calibrated airspeed. If the static air temperature decreases , the Mach number ;

- a) decreases.
- b) remains constant.
- c) increases.
- d) increases if the outside temperature is higher than the standard temperature , decreases if lower.

Answer ;

b) remains constant.

0093. An aeroplane is in steady cruise at flight level 270. The auto-throttle maintains a constant calibrated airspeed. If the static air temperature increases , the Mach number ;

- a) decreases.
- b) increases.
- c) remains constant.
- d) decreases if the outside temperature is higher than the standard temperature , increases if lower.

Answer ;

c) remains constant.

0094. An aeroplane is in steady cruise at flight level 290. The auto-throttle maintains a constant Mach number. If the total temperature decreases , the calibrated airspeed ;

- a) decreases.
- b) decreases if the outside temperature is lower than the standard temperature , increases if higher.
- c) remains constant.
- d) increases.

Answer ;

c) remains constant.

**0095.** An aeroplane is in steady descent. The auto-throttle maintains a constant calibrated airspeed. If the total temperature remains constant, the Mach number ;

- a) decreases.
- b) increases.
- c) increases if the static temperature is lower than the standard temperature, decreases if higher.
- d) remains constant.

**Answer ;**

**a) decreases.**

---

**0096.** An air data computer ;

- 1. supplies the ground speed and the drift ( angle ).
- 2. determines the total temperature and the true altitude.
- 3. receives the static pressure and the total pressure.
- 4. supplies the true airspeed to the inertial unit.
- 5. determines the Mach number, the outside ( static ) air temperature.

The combination regrouping all the correct statements is ;

- a) 2, 3 and 4.
- b) 3, 4 and 5.
- c) 2 and 5.
- d) 1 and 2.

**Answer ;**

**b) 3, 4 and 5.**

---

**0097.** An airborne instrument, equipped with a gyro with 1 degree of freedom and a horizontal spin axis is a ;

NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.

- a) turn indicator.
- b) fluxgate compass.
- c) directional gyro.
- d) gyromagnetic compass.

**Answer ;**

**a) turn indicator.**

---

**0098.** An airborne instrument, equipped with a gyro with 2 degrees of freedom and a horizontal spin axis is ;

NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.

- a) a directional gyro.
- b) an artificial horizon.
- c) a turn indicator.
- d) a fluxgate compass.

**Answer ;**

**a) a directional gyro.**

---

**0099.** An aircraft flies steadily on a heading 270°. The flight director is engaged in the heading select mode ( HDG SEL ), heading 270° selected. If a new heading 360° is selected, the vertical trend bar ;

- a) deviates to the right and will be centred as soon as you roll the aircraft to the bank angle calculated by the flight director.
- b) deviates to its right stop as long as the aeroplane is more than 10° off the new selected heading.
- c) deviates to the right and remains in that position until the aircraft has reached heading 360°.
- d) disappears, the new heading selection has deactivated the HDG mode.

**Answer ;**

**a) deviates to the right and will be centred as soon as you roll the aircraft to the bank angle calculated by the flight director.**

---

**0100.** An aircraft is descending from FL 390 to ground level at maximum speed. The limits in speed will be ;

- a) the  $V_{MO}$  only.
- b) initially the  $V_{MO}$ , then the  $M_{MO}$  below a certain flight level.
- c) the  $M_{MO}$  only.
- d) initially the  $M_{MO}$ , then the  $V_{MO}$  below a certain flight level.

**Answer ;**

**d) initially the  $M_{MO}$ , then the  $V_{MO}$  below a certain flight level.**

---



- 0101.** An aircraft is equipped with an autopilot and an auto-throttle. When the altitude hold mode ( ALT HOLD ) is active ;
- a) the indicated airspeed ( IAS ) is maintained constant by the autopilot by means of elevator.
  - b) the indicated airspeed ( IAS ) is maintained constant by the auto-throttle system.
  - c) the true airspeed ( TAS ) is maintained constant by the auto-throttle system.
  - d) the true airspeed ( TAS ) is maintained constant by the autopilot by means of elevator.

Answer ;

- b) the indicated airspeed ( IAS ) is maintained constant by the auto-throttle system.
- 

- 0102.** An aircraft is equipped with one altimeter that is compensated for position error and another one altimeter that is not. Assuming all other factors are equal , during a straight symmetrical flight ;

- a) the greater the speed , the lower the error between the two altimeters.
- b) the greater the speed , the greater the error between the two altimeters.
- c) the lower the speed , the greater the error between the two altimeters.
- d) the error between the two altimeters does not depend on the speed.

Answer ;

- b) the greater the speed , the greater the error between the two altimeters.
- 

- 0103.** An aircraft is flying at flight level FL180 and Mach number 0.36. Its onboard thermometer reads TAT =  $-5^{\circ}\text{C}$ . Considering that the probe recovery coefficient is 0.84 , the present weather conditions compared with the standard atmosphere are ;

- a) Standard +  $20^{\circ}\text{C}$ .
- b) Standard.
- c) Standard +  $30^{\circ}\text{C}$ .
- d) Standard +  $10^{\circ}\text{C}$ .

Answer ;

- d) Standard +  $10^{\circ}\text{C}$ .
- 

- 0104.** An aircraft is in a steady left turn , with too much left rudder ( skidding turn ). The yaw damper system ;

- a) acts only in case of slipping turns ( turn with too much bank ).
- b) acts on the left rudder pedal to provide a balanced turn.
- c) acts on the right rudder pedal to provide a balanced turn.
- d) is not designed to provide a balanced turn.

Answer ;

- d) is not designed to provide a balanced turn.
- 

- 0105.** An aircraft is in a steady right turn , with not enough right rudder ( slipping turn ). The yaw damper system ;

- a) acts on the left rudder pedal to provide a balanced turn.
- b) acts only in case of skidding turns ( turn with not enough bank ).
- c) acts on the right rudder pedal to provide a balanced turn.
- d) is not designed to provide a balanced turn.

Answer ;

- d) is not designed to provide a balanced turn.
- 

- 0106.** An aircraft is turning right , with too much bank ( slipping turn ). The yaw damper system ;

- a) is not designated to provide a balanced turn.
- b) acts only in case of skidding turns ( turn with not enough bank ).
- c) acts on the right rudder pedal to provide a balanced turn.
- d) acts on the left rudder pedal to provide a balanced turn.

Answer ;

- a) is not designated to provide a balanced turn.
- 

- 0107.** An aircraft takes-off on a runway with an alignment of  $045^{\circ}$  , the compass is made for the northern hemisphere. During rolling take-off , the compass indicates ;

- a) a value above  $045^{\circ}$  in the northern hemisphere.
- b) a value above  $045^{\circ}$  in the southern hemisphere.
- c) a value below  $045^{\circ}$ .
- d)  $045^{\circ}$ .

Answer ;

- c) a value below  $045^{\circ}$ .
-

- 0108.** An airplane is cruising at FL 190. The auto-throttle maintains a constant CAS. If the OAT decreases , the Mach number ;
- a) decreases if OAT is lower than standard temperature , increases in the opposite case.
  - b) increases.
  - c) remains constant.
  - d) decreases.

Answer ;

**c)** remains constant.

---

- 0109.** An airplane is cruising at FL 220. The auto-throttle maintains a constant CAS. If the OAT decreases , the Mach number ;
- a) decreases.
  - b) remains constant.
  - c) increases.
  - d) decreases if OAT is lower than standard temperature , increases in the opposite case.

Answer ;

**b)** remains constant.

---

- 0110.** An airplane is flying at FL140 with a CAS of 260 kt in standard conditions. The Mach number is ;
- a) 0.43
  - b) 0.41
  - c) 0.53
  - d) 0.51

Answer ;

**d)** 0.51

---

- 0111.** An airplane is in steady cruise at flight level 290. The auto-throttle maintains a constant Mach number. If the total temperature decreases , the calibrated airspeed ;
- a) decreases.
  - b) remains constant.
  - c) decreases if the outside temperature is lower than the standard temperature , increases if higher.
  - d) increases.

Answer ;

**b)** remains constant.

---

- 0112.** An airplane is in steady cruise at flight level 290. The auto-throttle maintains a constant Mach number. If the total temperature increases , the calibrated airspeed ;
- a) increases.
  - b) increases if the static temperature is higher than the standard temperature , decreases if lower.
  - c) remains constant.
  - d) decreases.

Answer ;

**c)** remains constant.

---

- 0113.** An airspeed indicator displays ;
- a) TAS.
  - b) CAS.
  - c) IAS.
  - d) EAS.

Answer ;

**c)** IAS.

---

- 0114.** An airspeed indicator includes a capsule , inside this capsule is ;
- a) a very low residual pressure and outside is static pressure.
  - b) static pressure and outside is dynamic pressure.
  - c) total pressure and outside is static pressure.
  - d) dynamic pressure and outside is static pressure.

Answer ;

**c)** total pressure and outside is static pressure.

---

- 0115.** An altimeter contains one or more aneroid capsules. Inside these capsule is ;
- a) static pressure and outside a very low residual pressure.
  - b) static pressure and outside is dynamic pressure.
  - c) a very low residual pressure outside is static pressure.
  - d) dynamic pressure and outside is static pressure.

Answer ;

**c)** a very low residual pressure outside is static pressure.

---

- 0116.** An “altitude warning system” must at least warn the crew ;
1. when approaching the pre-selected altitude.
  2. when the airplane is approaching the ground too fast.
  3. in case of a given deviation above or below the pre-selected altitude ( at least by an aural warning ).
  4. in case of excessive vertical speed.
  5. when approaching the ground with the gear retracted.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4.
- b) 2 , 4 , 5.
- c) 1 , 2 , 3 , 4 , 5.
- d) 1 , 3.

**Answer ;**

- d) 1 , 3.**
- 

- 0117.** An aneroid capsule ;
1. measures differential pressure.
  2. measures absolute pressure.
  3. is used for low pressure measurement.
  4. is used for very high pressure measurement.

The combination regrouping all the correct statements is ;

- a) 2 , 3.
- b) 1 , 4.
- c) 1 , 3.
- d) 2 , 4.

**Answer ;**

- a) 2 , 3.**
- 

- 0118.** An angle of attack sensor may consist of ;
1. an inertial system computing the difference between flight path and flight attitude.
  2. a conical slotted probe which positions itself to determine the angle of attack.
  3. a vane detector which positions the rotor of a synchro.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 2 , 3.
- c) 1 , 2.
- d) 1 , 3.

**Answer ;**

- b) 2 , 3.**
- 

- 0119.** An automatic ILS approach can be flown only ;
- a) within a range of levels of turbulence and no limit for crosswinds.
  - b) within a range of crosswinds and levels of turbulence.
  - c) without limits for crosswind or turbulence.
  - d) within a range of crosswinds and no limit for turbulence.

**Answer ;**

- b) within a range of crosswinds and levels of turbulence.**
- 

- 0120.** An automatic landing system necessitating that the landing be continued manually in the case of a system failure during an automatic approach is called “FAIL ....” ;

- a) “OPERATIONAL”
- b) “PASSIVE”
- c) “SAFE”
- d) “REDUNDANT”

**Answer ;**

- b) “PASSIVE”**
- 

- 0121.** An autopilot capable of holding at least altitude and heading mode is compulsory ;

- a) on multi-pilot airplanes.
- b) for IFR or night flights with only one pilot.
- c) on airplanes over 5.7 t.
- d) for VFR and IFR flights with only one pilot.

**Answer ;**

- b) for IFR or night flights with only one pilot.**
-

0122. An autopilot system ;

- a) must provide at least aircraft guidance functions.
- b) may provide automatic take off functions.
- c) must provide at least aircraft stabilisation functions.
- d) must provide automatic take off functions.

Answer ;

- c) must provide at least aircraft stabilisation functions.
- 

0123. An EFIS includes the following components ;

- a) FMS , symbol generator , display unit(s).
- b) ADC , inertial navigation computer , display unit(s).
- c) symbol generator , display unit(s) , control panel.
- d) display unit(s) only.

Answer ;

- c) symbol generator , display unit(s) , control panel.
- 

0124. An inertial navigation system ;

- a) can only operate when interfacing with the radionavigation equipments.
- b) can only operate when communicating with ground installations.
- c) can operate as stand alone equipment without any interface with other navigation equipments.
- d) can only operate when interfacing with the GPS equipment.

Answer ;

- c) can operate as stand alone equipment without any interface with other navigation equipments.
- 

0125. An inertial Navigation System ( INS ) is ;

- a) a radio navigation system.
- b) a self contained system which operates without signals from the ground.
- c) a system which operates on the Doppler principle.
- d) a hyperbolic navigation system.

Answer ;

- b) a self contained system which operates without signals from the ground.
- 

0126. An inertial reference and navigation system is a “strapdown” system when ;

- a) gyros and accelerometers are mounted on a stabilised platform in the aircraft.
- b) gyros and accelerometers need satellite information input to obtain a vertical reference.
- c) the gyroscopes and accelerometers are part of the unit’s fixture to the aircraft structure.
- d) only the gyros , and not the accelerometers , are part of the unit’s fixture to the aircraft structure.

Answer ;

- c) the gyroscopes and accelerometers are part of the unit’s fixture to the aircraft structure.
- 

0127. An outer loop Automatic Flight Control System ( AFCS ) , is a system which ;

- a) contains a parallel actuator which provides control through 100% of the control range and moves the control inputs only.
- b) controls the aircraft about the lateral axis only ( PITCH ).
- c) contains a parallel actuator which provides control through 100% of the control range and moves both the control inputs and the cockpit control stick.
- d) controls the aircraft about the longitudinal axis only ( ROLL ).

Answer ;

- c) contains a parallel actuator which provides control through 100% of the control range and moves both the control inputs and the cockpit control stick.
- 

0128. Any vibration displayed on an engine vibration monitoring system for a turbojet ;

- a) is presented without any amplification or filtering.
- b) varies inversely as the square of the engine speed.
- c) indicates rotor imbalance.
- d) is directly proportional to engine speed.

Answer ;

- c) indicates rotor imbalance.
-

**0129.** As a result of the failure of the central air data computer ( CADC ) , the inertial navigation system ( INS ) will no longer provide information about ;

- a) the ground speed.
- b) the time ( TIME ) at the next waypoint ( WPT ).
- c) the wind direction and speed.
- d) the drift.

**Answer ;**

**c) the wind direction and speed.**

---

**0130.** Assuming that the CAS remains constant , if the total pressure probe is blocked , the IAS ;

- a) remains constant during all the phases of the flight.
- b) remains constant during level flight , increases during a climb and decreases during a descent.
- c) increases during level flight , remains constant during a climb and a descent.
- d) remains constant during level flight , decreases during a climb and increases during a descent.

**Answer ;**

**b) remains constant during level flight , increases during a climb and decreases during a descent.**

---

**0131.** Assuming the flight level and Mach number remain constant , when the OAT decreases ;

- a) IAS decreases and TAS increases.
- b) IAS increases and TAS decreases.
- c) IAS and TAS increase.
- d) IAS remains constant and TAS decreases.

**Answer ;**

**d) IAS remains constant and TAS decreases.**

---

**0132.** Assuming the flight level and Mach number remain constant , when the OAT increases ;

- a) IAS increases and TAS decreases.
- b) IAS decreases and TAS increases.
- c) IAS remains constant and TAS increases.
- d) IAS and TAS decrease.

**Answer ;**

**c) IAS remains constant and TAS increases.**

---

**0133.** At a given altitude , the hysteresis error of an altimeter varies substantially with the ;

- a) aircraft attitude.
- b) time passed at this altitude.
- c) static temperature.
- d) mach number of the aircraft.

**Answer ;**

**b) time passed at this altitude.**

---

**0134.** At a low bank angle , the measurement of rate-of-turn actually consists in measuring the ;

- a) roll rate of the aircraft.
- b) yaw rate of the aircraft.
- c) pitch rate of the aircraft.
- d) bank of the aircraft.

**Answer ;**

**b) yaw rate of the aircraft.**

---

**0135.** At flight level and Mach number constant , if OAT decreases the CAS ;

- a) remains constant.
- b) decreases.
- c) decreases if OAT is lower than standard temperature , increases in the opposite case.
- d) increases.

**Answer ;**

**a) remains constant.**

---

**0136.** At flight level and Mach number constant , if OAT increases the CAS ;

- a) decreases if OAT is lower than standard temperature , increases in the opposite case.
- b) remains constant.
- c) increases.
- d) decreases.

**Answer ;**

**b) remains constant.**

---

0137. Automatic Flight Control System ( AFCS ) parallel actuator position is ;
- a) passed to the pilot via control stick position.
  - b) displayed in the cockpit as a function of input and output signals.
  - c) displayed to the pilot by movement on the ADI only.
  - d) not displayed in the cockpit due to short duration of operation.

Answer ;

- a) passed to the pilot via control stick position.
- 

0138. Automatic Flight Control System ( AFCS ) series actuator is ;
- a) passed to the pilot via control stick position.
  - b) displayed in the cockpit as a function of input and output signals.
  - c) displayed to the pilot by movement on the ADI/EADI.
  - d) not displayed in the cockpit due to short duration of operation.

Answer ;

- d) not displayed in the cockpit due to short duration of operation.
- 

0139. Below the tropopause in standard conditions , when descending at a constant Mach number ;
- a) the difference between surrounding condition and ISA must be known to deduce the TAS variation.
  - b) TAS decreases.
  - c) TAS remains constant.
  - d) TAS increases.

Answer ;

- d) TAS increases.
- 

0140. Below the tropopause in standard conditions , when climbing at a constant Mach number ;
- a) the difference between surrounding condition and ISA must be known to deduce the TAS variation.
  - b) TAS remains constant.
  - c) TAS increases.
  - d) TAS decreases.

Answer ;

- d) TAS decreases.
- 

0141. Below the tropopause , when descending at constant CAS ;
- a) Mach number increases and the velocity of sound decreases.
  - b) Mach number decreases and the velocity of sound increases.
  - c) Mach number and the velocity of sound decrease.
  - d) Mach number and the velocity of sound increase.

Answer ;

- b) Mach number decreases and the velocity of sound increases.
- 

0142. Below the tropopause , with no temperature inversion , when climbing at constant Mach number ;
- a) the difference between surrounding conditions and ISA must be known to deduce the TAS variation.
  - b) TAS remains constant.
  - c) TAS decreases.
  - d) TAS increases.

Answer ;

- c) TAS decreases.
- 

0143. Below the tropopause , with no temperature inversion , when descending at constant Mach number ;
- a) the difference between surrounding conditions and ISA must be known to deduce the TAS variation.
  - b) TAS decreases.
  - c) TAS increases.
  - d) TAS remains constant.

Answer ;

- c) TAS increases.
- 

0144. Calibrated Air Speed ( CAS ) is ;
- a) Equivalent Air Speed ( EAS ) corrected for density error.
  - b) Equivalent Air Speed ( EAS ) corrected for compressibility and density errors.
  - c) Indicated Air Speed ( IAS ) corrected for compressibility error.
  - d) Indicated Air Speed ( IAS ) corrected for position and instrument errors.

Answer ;

- d) Indicated Air Speed ( IAS ) corrected for position and instrument errors.
-

**0145.** Calibrated Air Speed ( CAS ) is obtained from Indicated Air Speed ( IAS ) by correcting for the ;

- a) position and density errors.
- b) density error.
- c) position and instrument errors.
- d) instrument error.

**Answer ;**

**c) position and instrument errors.**

---

**0146.** Calibrated Air Speed ( CAS ) is obtained from Indicated Air Speed ( IAS ) by correcting for the following errors ;

- 1. position.
- 2. compressibility.
- 3. instrument.
- 4. density.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4.
- b) 3.
- c) 1 , 3 , 4.
- d) 1 , 3.

**Answer ;**

**d) 1 , 3.**

---

**0147.** CAS can be obtained from the following data ;

- a) TAS and density altitude.
- b) EAS and density altitude.
- c) EAS and pressure altitude.
- d) TAS and pressure altitude.

**Answer ;**

**c) EAS and pressure altitude.**

---

**0148.** CAS is equal to ;

- a) EAS corrected for compressibility and density errors.
- b) EAS corrected for density error.
- c) IAS corrected for compressibility error.
- d) IAS corrected for position and instrument errors.

**Answer ;**

**d) IAS corrected for position and instrument errors.**

---

**0149.** CAS is obtained from IAS by correcting for the ;

- a) instrument error.
- b) density error.
- c) position and density errors.
- d) position and instrument errors.

**Answer ;**

**d) position and instrument errors.**

---

**0150.** Command bars of the flight director may be present on the ;

- 1. HSI
- 2. CDU
- 3. ADI

The combination that regroups all of the correct statements is ;

- a) 3.
- b) 1 , 3.
- c) 1.
- d) 1 , 2 , 3.

**Answer ;**

**a) 3.**

---

**0151.** Command bars of the flight director may be present on the ;

1. HIS
2. EICAS
3. CDU
4. ADI

The combination containing all of the correct statements is ;

- a) 1 and 3 only.
- b) 1 only.
- c) 4 only.
- d) 1 and 4 only.

**Answer ;**

**c) 4 only.**

---

**0152.** Compared with a conventional gyro , a laser gyro ;

- a) consumes a lot of power.
- b) is influenced by temperature.
- c) has a fairly long starting cycle.
- d) has a longer life cycle.

**Answer ;**

**d) has a longer life cycle.**

---

**0153.** Compared with a stabilised platform inertial system , a strapdown inertial system ;

1. can aligned while the aircraft is moving.
2. has a quicker alignment phase.
3. is more reliable in time.

The combination that regroupes all the correct statements is ;

- a) 2 , 3.
- b) 1 , 2.
- c) 1 , 2 , 3.
- d) 3.

**Answer ;**

**a) 2 , 3.**

---

**0154.** Compared with a stabilised platform inertial system , a strapdown inertial system ;

1. has a longer alignment phase in time
2. has a shorter alignment phase in time.
3. is more reliable in time.
4. is less reliable in time.

The combination that regroupes all of the correct statements is ;

- a) 1 , 3.
- b) 2 , 3.
- c) 1 , 4.
- d) 2 , 4.

**Answer ;**

**b) 2 , 3.**

---

**0155.** Comparing the media used to transmit datalink communications , the appropriate classification from the slowest to the fastest data transmission rate is ;

- a) HF datalink , VHF datalink , SATCOM.
- b) SATCOM , VHF datalink , HF datalink.
- c) SATCOM , HF datalink , VHF datalink.
- d) HF datalink , SATCOM , VHF datalink.

**Answer ;**

**a) HF datalink , VHF datalink , SATCOM.**

---



- 0156.** Comparing the radio navigation system and the inertial navigation system ;
1. the radio position is accurate when in DME range.
  2. the radio position may be obtained whatever the position on the earth.
  3. the inertial position may be obtained whatever the position on the earth.

The combination regrouping all the correct statements is ;

- a) 2.
- b) 3.
- c) 1 , 3.
- d) 1 , 2.

**Answer ;**

**c) 1 , 3.**

---

- 0157.** Components of the FMS are ;
1. CDU ( Control and Display Unit ).
  2. Database.
  3. FMC ( Flight Management Computer ).
  4. Electronic check-lists.
  5. GPWS mode controller.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 3 , 4.
- c) 1 , 3.
- d) 1 , 2 , 4 , 5.

**Answer ;**

**a) 1 , 2 , 3.**

---

- 0158.** Concerning a fail-operational flight control system , in the event of a failure ;
1. the system will operate as a fail-passive system.
  2. the landing is not completed automatically.
  3. the landing is completed automatically.

The combination that regroups all of the correct statements is ;

- a) 3.
- b) 2.
- c) 1 , 2.
- d) 1 , 3.

**Answer ;**

**d) 1 , 3.**

---

- 0159.** Concerning a fail-passive flight control system , in the event of a failure ;
1. there may be a significant deviation of flight path or attitude.
  2. there is no significant deviation of flight path or attitude.
  3. the landing is not completed automatically.
  4. the landing is completed automatically.

The combination that regroups all of the correct statements is ;

- a) 1 , 4.
- b) 2 , 3.
- c) 1 , 3.
- d) 1 , 4.

**Answer ;**

**b) 2 , 3.**

---

- 0160.** Concerning a fail-passive flight control system , in the event of a failure ;
1. there may be a significant deviation of flight path or attitude.
  2. there is no significant deviation of flight path or attitude.
  3. there is no significant out-of-trim condition.
  4. there may be a significant out-of-trim condition.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 2 , 3.
- c) 2 , 4.
- d) 1 , 4.

**Answer ;**

**b) 2 , 3.**

---

0161. Concerning a fuel gauge system , a pilot is more interested by the ;
- a) mass of the fuel than the volume of the fuel.
  - b) volume of the fuel than the permittivity of the fuel.
  - c) volume of the fuel than the mass of the fuel.
  - d) volume of the fuel than the density of the fuel.

Answer ;

- a) mass of the fuel than the volume of the fuel.
- 

0162. Concerning the airspeed indicator , IAS is ;
- a) the indicated airspeed corrected for instrument error only.
  - b) the indicated reading on an instrument presumed to be perfect.
  - c) the indicated airspeed corrected for instrument and position errors.
  - d) the indicated reading on the instrument.

Answer ;

- d) the indicated reading on the instrument.
- 

0163. Concerning the command bars of a flight director ;
- a) autopilot must be first disengaged to remove them.
  - b) it is possible to remove them by switching the flight director OFF.
  - c) autopilot must be first engaged before removing them.
  - d) it is not possible to remove them on some types of flight director systems.

Answer ;

- b) it is possible to remove them by switching the flight director OFF.
- 

0164. Concerning the direct reading magnetic compass , the turning error ;
- a) decreases with the magnetic longitude.
  - b) does not depend on the magnetic latitude.
  - c) increases with the magnetic latitude.
  - d) decreases with the magnetic latitude.

Answer ;

- c) increases with the magnetic latitude.
- 

0165. Concerning the directional gyro , the apparent drift rate due to the earth's rotation is a function of ;
- a) magnetic longitude.
  - b) latitude.
  - c) longitude.
  - d) latitude and longitude.

Answer ;

- b) latitude.
- 

0166. Concerning the flexible take-off mode , the temperature selected in the FMS is ;
- a) higher than the ambient airfield temperature , in order to achieve a reduced power setting.
  - b) lower than the ambient airfield temperature , in order to achieve a reduced power setting.
  - c) higher than the ambient airfield temperature , in order to achieve a increased power setting.
  - d) lower than the ambient airfield temperature , in order to achieve a increased power setting.

Answer ;

- a) higher than the ambient airfield temperature , in order to achieve a reduced power setting.
- 

0167. Concerning the flight warning system ( FWS ) , advisory messages may be ;
- a) any colour except green.
  - b) any colour except red , and preferably not amber.
  - c) any colour except amber.
  - d) red.

Answer ;

- b) any colour except red , and preferably not amber.
- 

0168. Concerning the flight warning system ( FWS ) , caution messages ;
- a) are the next highest priority alert messages after warning messages.
  - b) indicate that immediate recognition and corrective or compensatory action by the crew is required.
  - c) are inhibited by advisory messages.
  - d) are the highest priority alert messages.

Answer ;

- a) are the next highest priority alert messages after warning messages.
-

0169. Concerning the flight warning system ( FWS ) , if aural signals are provided , the signal for ;
- a) a caution should always take priority over that for a warning.
  - b) a warning should always take priority over that for a caution.
  - c) an advisory should always take priority over that for a caution.
  - d) an advisory should always take priority over that for a warning.

Answer ;

- b) a warning should always take priority over that for a caution.
- 

0170. Concerning the flight warning system ( FWS ) , warning messages ;
- a) are inhibited by caution messages.
  - b) are next highest priority alert messages after warning messages.
  - c) indicate that subsequent crew action may be required.
  - d) are highest priority alert messages.

Answer ;

- d) are highest priority alert messages.
- 

0171. Concerning the FMS ( Flight Management System ) , entering a cost index of zero ;
- a) modifies only cruising airspeed.
  - b) does not influence fuel consumption.
  - c) results in maximum trip fuel.
  - d) results in minimum trip fuel.

Answer ;

- d) results in minimum trip fuel.
- 

0172. Concerning the FMS ( Flight Management System ) , entering a cost index of zero results in ;
- a) minimum airspeed.
  - b) maximum cruising airspeed.
  - c) minimum range airspeed.
  - d) maximum range airspeed.

Answer ;

- d) maximum range airspeed.
- 

0173. Concerning the FMS ( Flight Management System ) , entering a high cost index results in ;
- a) minimum trip fuel.
  - b) high airspeed and high fuel trip.
  - c) maximum range airspeed.
  - d) minimum airspeed.

Answer ;

- b) high airspeed and high fuel trip.
- 

0174. Concerning the FMS ( Flight Management System ) , the cost index is determined by dividing ;
- a) aircraft cruise speed by fuel cost.
  - b) fuel cost by aircraft cruise speed.
  - c) fuel cost by aircraft operating cost.
  - d) aircraft operating cost by fuel cost.

Answer ;

- d) aircraft operating cost by fuel cost.
- 

0175. Concerning the FMS , the parameters used to work out the vertical flight profile are ;
1. zero fuel weight.
  2. cost index.
  3. fuel quantity.
  4. oxygen quantity available for flight crew.
  5. minimum safe enroute altitude.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 3 , 4 , 5.
- c) 1 , 2 , 3 , 5.
- d) 1 , 2 , 5.

Answer ;

- a) 1 , 2 , 3.
-

0176. Concerning the pitot and static system , the static pressure error ;
- a) is caused by disturbed airflow around the static ports.
  - b) affects the alternate static port only.
  - c) is a direct effect of heating of the static ports.
  - d) is a direct effect of a blockage of the static port.

Answer ;

a) is caused by disturbed airflow around the static ports.

---

0177. Concerning the pitot and static system , the static pressure error varies according to ;
- 1. altimeter setting.
  - 2. speed.
  - 3. angle of attack.

The combination that regroups all of the correct statements is ;

- a) 1 and 3.
- b) 1 and 2.
- c) 1 , 2 and 3.
- d) 2 and 3.

Answer ;

d) 2 and 3.

---

0178. Concerning the radio altimeter , the maximum range for indication is typically from touchdown to ;
- a) greater than 10000 ft.
  - b) 500 ft.
  - c) 10000 ft.
  - d) 2500 ft.

Answer ;

d) 2500 ft.

---

0179. Concerning the TCAS ( Traffic Collision Avoidance System ) ;
- a) In one of the system modes , the warning: "TOO LOW TERRAIN" is generated.
  - b) Resolution Advisory ( RA ) must not be followed without obtaining clearance from ATC.
  - c) No protection is available against aircraft not equipped with a serviceable SSR transponder.
  - d) In one of the system modes , the warning "PULL UP" is generated.

Answer ;

c) No protection is available against aircraft not equipped with a serviceable SSR transponder.

---

0180. Concerning the TCAS 2 ;
- 1. Neither advisory nor traffic display is provided for aircraft that do not have an operating transponder.
  - 2. TCAS 2 operation is independent of ground-based air traffic control.
  - 3. TCAS 2 has feeds from the radioaltimeter.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 2.
- c) 1 , 3.
- d) 1 , 2 , 3.

Answer ;

d) 1 , 2 , 3.

---

0181. Concerning the TCAS 2 ;
- a) No protection available against aircraft not equipped with a serviceable SSR transponder.
  - b) In one of the system modes , the warning "TOO LOW TERRAIN" is generated.
  - c) In one of the system modes , the warning "PULL UP" is generated.
  - d) No protection is available against aircraft not equipped with a serviceable FMS.

Answer ;

a) No protection available against aircraft not equipped with a serviceable SSR transponder.

---

0182. Concerning the TCAS 2 , when receiving a resolution advisory ( RA ) , crew members should ;
- a) disregard this RA if they have a mode S transponder.
  - b) immediately engage autopilot which will be temporarily controlled by the TCAS.
  - c) try to establish visual contact with the intruder before taking any evasive action.
  - d) immediately initiate the required manoeuvre.

Answer ;

d) immediately initiate the required manoeuvre.

---

- 0183.** Considering a flight director of the “command bars” type ;
1. the vertical bar is always associated with the roll channel.
  2. the vertical bar may be associated with the pitch channel.
  3. the horizontal bar may be associated with the roll channel.
  4. the horizontal bar is always associated with the pitch channel.

The combination regrouping all the correct statements is ;

- a) 1 , 4.
- b) 1 , 2 , 3 , 4.
- c) 1 , 3 , 4.
- d) 1 , 2 , 4.

**Answer ;**

- a) 1 , 4.**
- 

- 0184.** Considering a flight director of the “command bars” type ;
- a) the horizontal bar may be associated with the roll channel.
  - b) the vertical bar is associated with the roll channel.
  - c) the horizontal bar is associated with the roll channel.
  - d) the vertical bar is associated with the pitch channel.

**Answer ;**

- b) the vertical bar is associated with the roll channel.**
- 

- 0185.** Considering a flight director of the “command bars” type ;
- a) the vertical bar is associated with the pitch channel.
  - b) the horizontal bar is associated with the pitch channel.
  - c) the vertical bar may be associated with the pitch channel.
  - d) the horizontal bar is associated with the roll channel.

**Answer ;**

- b) the horizontal bar is associated with the pitch channel.**
- 

- 0186.** Considering a modern thrust computer , during a steady climb ;
- a) N1 and N2 remain constant.
  - b) N1 is automatically adjusted as aircraft climbs and N2 remains constant.
  - c) N2 is automatically adjusted as aircraft climbs and N1 remains constant.
  - d) N1 is automatically adjusted as aircraft climbs.

**Answer ;**

- d) N1 is automatically adjusted as aircraft climbs.**
- 

- 0187.** Considering a stabilised platform inertial system ;
1. the rate gyros and the accelerometers are mounted on the same platform.
  2. the rate gyros and the accelerometers are mounted on two separated platforms.
  3. the principle of operation requires at least 2 rate gyros.
  4. the principle of operation requires at least 2 accelerometers.

The combination that regroups all of the correct statements is ;

- a) 1 , 4.
- b) 2 , 4.
- c) 1 , 3.
- d) 2 , 3.

**Answer ;**

- a) 1 , 4.**
- 

- 0188.** Considering a stabilised platform inertial system , this platform ;
1. can be servo-controlled in azimuth.
  2. is kept levelled during alignment phase only.
  3. is always kept levelled.

The combination that regroups all of the correct statements is ;

- a) 2.
- b) 1 , 2.
- c) 3.
- d) 1 , 3.

**Answer ;**

- d) 1 , 3.**
-

- 0189.** Considering a strapdown inertial system , the IRU ( Inertial Reference Unit ) measures ;
- a) accelerations and angular rates.
  - b) linear accelerations only.
  - c) angular accelerations only.
  - d) angular rates only.

**Answer ;**

- a) accelerations and angular rates.
- 

- 0190.** Considering a strapdown inertial system , the operating principle requires the use of at least ;
- a) 2 laser gyros and 2 accelerometers.
  - b) 2 laser gyros and 3 accelerometers.
  - c) 3 laser gyros and 2 accelerometers.
  - d) 3 laser gyros and 3 accelerometers.

**Answer ;**

- d) 3 laser gyros and 3 accelerometers.
- 

- 0191.** Considering an air driven artificial horizon , when an airplane accelerates during take-off run , the result is ;
- a) a right or left wing down indication depending on the runway direction.
  - b) a false nose-up indication.
  - c) a false nose-down indication.
  - d) a correct and constant pitch indication.

**Answer ;**

- b) a false nose-up indication.
- 

- 0192.** Considering an air driven artificial horizon , when an airplane decelerates during the landing , the result is ;
- a) a false nose-up indication.
  - b) a false nose-down indication.
  - c) a right or left wing down indication depending on the runway direction.
  - d) a correct and constant pitch indication.

**Answer ;**

- b) a false nose-down indication.
- 

- 0193.** Considering an airspeed indicator , a second stripped needle , if installed , indicates ;
- a) maximum operating speed (  $V_{MO}$  ).
  - b) never-exceed speed (  $V_{NE}$  ).
  - c) never-exceed speed (  $V_{NE}$  ) or maximum operating speed (  $V_{MO}$  ) , depending on which is the higher.
  - d) never-exceed speed (  $V_{NE}$  ) or maximum operating speed (  $V_{MO}$  ) , depending on which is the lowest.

**Answer ;**

- a) maximum operating speed (  $V_{MO}$  ).
- 

- 0194.** Considering the relationship between CAS and EAS ;
- a) EAS may be lower or greater than CAS , depending on pressure altitude.
  - b) EAS is always greater than or equal to CAS.
  - c) EAS is always lower than or equal to CAS.
  - d) EAS may be lower or greater than CAS , depending on density altitude.

**Answer ;**

- c) EAS is always lower than or equal to CAS.
- 

- 0195.** Different pressure sensors are used according to the intensity of the pressure measured ( low , medium or high ). Classify the following sensors by order of increasing pressure for which they are suitable ;

- 1. bellows type.
  - 2. Bourdon tube type.
  - 3. aneroid capsule type.
- a) 1 , 2 , 3.
  - b) 3 , 2 , 1.
  - c) 3 , 1 , 2.
  - d) 2 , 1 , 3.

**Answer ;**

- c) 3 , 1 , 2.
-

0196. Direct reading magnetic compass errors are ;
- a) due to Schuler oscillations.
  - b) parallax errors due to compass rose oscillations.
  - c) due to north change , depending on the bank angle and magnetic heading.
  - d) due to lateral gusts which occur when the aircraft is heading eastward or westward.

Answer ;

- c) due to north change , depending on the bank angle and magnetic heading.
- 

0197. Due to its conception , the altimeter measures a ;

- a) a temperature altitude.
- b) a true altitude.
- c) a density altitude.
- d) a pressure altitude.

Answer ;

- d) a pressure altitude.
- 

0198. Due to the rotation of the earth , the apparent drift of a horizontal free gyroscope at a latitude of 30°S is ;

- a) 11° per hour to the right.
- b) 15° per hour to the right.
- c) 2° per hour to the left.
- d) 7.5° per hour to the left.

Answer ;

- d) 7.5° per hour to the left.
- 

0199. Due to the rotation of the earth , the apparent drift of a horizontal free gyroscope at a latitude of 30°N is ;

- a) 2° per hour to the right.
- b) 11° per hour to the left.
- c) 7.5° per hour to the right.
- d) 15° per hour to the left.

Answer ;

- c) 7.5° per hour to the right.
- 

0200. Due to the rotation of the earth , the apparent drift of a horizontal free gyroscope at a latitude of 45°N is ;

- a) 7° per hour to the left.
- b) 11° per hour to the right.
- c) 2° per hour to the right.
- d) 15° per hour to the left.

Answer ;

- b) 11° per hour to the right.
- 

0201. During a category III automatic approach , the position signals in the vertical plane under 200 ft are based on ;

- a) an altimeter set to the QNH.
- b) an altimeter set to the QFE.
- c) an altimeter set to 1013 hPa.
- d) a radio altimeter.

Answer ;

- d) a radio altimeter.
- 

0202. During a climb at a constant calibrated airspeed ( CAS ) below the tropopause in ISA conditions ;

- a) the Mach number decreases and the speed of sound increases.
- b) the Mach number and the speed of sound decrease.
- c) the Mach number and the speed of sound increase.
- d) the Mach number increases and the speed of sound decreases.

Answer ;

- d) the Mach number increases and the speed of sound decreases.
- 

0203. During a climb at a constant calibrated airspeed ( CAS ) below the tropopause in standard conditions ;

- a) TAS decreases and Mach number increases.
- b) TAS and Mach number decrease.
- c) TAS increases and Mach number decreases.
- d) TAS and Mach number increase.

Answer ;

- d) TAS and Mach number increase.
-

0204. During a climb at a constant CAS below the tropopause in standard atmosphere ;
- a) TAS and Mach number decrease.
  - b) TAS and Mach number increase.
  - c) TAS increases and Mach number decreases.
  - d) TAS decreases and Mach number increases.

Answer ;

b) TAS and Mach number increase.

---

0205. During a climb at a constant CAS below the tropopause in standard atmosphere ;
- a) the Mach number decreases and the speed of sound increases.
  - b) the Mach number increases and the speed of sound decreases.
  - c) the Mach number and the speed of sound increase.
  - d) the Mach number and the speed of sound decrease.

Answer ;

b) the Mach number increases and the speed of sound decreases.

---

0206. During a climb at a constant IAS below the tropopause in ISA conditions ;
- a) the Mach number and the true airspeed decrease.
  - b) the Mach number increases and the true airspeed decreases.
  - c) the Mach number decreases and the true airspeed increases.
  - d) the Mach number and the true airspeed increase.

Answer ;

d) the Mach number and the true airspeed increase.

---

0207. During a climb at a constant Mach number below the tropopause in ISA conditions ;
- a) CAS and TAS increase.
  - b) CAS increases and TAS decreases.
  - c) CAS and TAS decrease.
  - d) CAS decreases and TAS increases.

Answer ;

c) CAS and TAS decrease.

---

0208. During a climb at a constant Mach number below the tropopause in ISA conditions ;
- a) IAS and TAS increase.
  - b) IAS increases and TAS decreases.
  - c) IAS and TAS decrease.
  - d) IAS decreases and TAS increases.

Answer ;

c) IAS and TAS decrease.

---

0209. During a climb , if the total pressure head is rapidly clogged up by ice , the sensed total pressure remains constant and ;
- a) the static pressure decreases , implying a decreasing IAS.
  - b) the static pressure increases , implying a decreasing IAS.
  - c) the static pressure decreases , implying an increasing IAS.
  - d) the IAS remains also constant.

Answer ;

c) the static pressure decreases , implying an increasing IAS.

---

0210. During a climb , the total pressure probe of the airspeed indicator becomes blocked ; if the pilot maintains a constant indicated airspeed , the true airspeed ;

- a) decreases by 1% per 600 ft.
- b) decreases until reaching the stall speed.
- c) increases by 1% per 600 ft.
- d) increases until reaching VMO.

Answer ;

b) decreases until reaching the stall speed.

---

0211. During a climb with the autopilot engaged in the V/S mode ( holding of vertical speed ) , the auto-throttle ;
- a) is not available.
  - b) can be operated in the N1 / EPR mode ( holding of N1 or EPR ).
  - c) can be operated in the SPEED ( holding of IAS ) or N1 ( holding of N1 ) modes.
  - d) can be operated in the SPEED mode ( holding of IAS ).

Answer ;

d) can be operated in the SPEED mode ( holding of IAS ).

---



**0212.** During a deceleration phase at constant attitude , the control system of the artificial horizon results in the horizon bar indicating a ;

- a) constant attitude.
- b) nose up attitude.
- c) nose down attitude.
- d) nose up followed by a nose down attitude.

**Answer ;**

- c) nose down attitude.**
- 

**0213.** During a descent at a constant calibrated airspeed ( CAS ) below the tropopause in ISA conditions ;

- a) Mach number and TAS decrease.
- b) Mach number increases and TAS decreases.
- c) Mach number decreases and TAS increases.
- d) Mach number and TAS increase.

**Answer ;**

- a) Mach number and TAS decrease.**
- 

**0214.** During a descent at a constant CAS below the tropopause in standard atmosphere ;

- a) Mach number decreases and TAS increases.
- b) Mach number and TAS increase.
- c) Mach number and TAS decrease.
- d) Mach number increases and TAS decreases.

**Answer ;**

- c) Mach number and TAS decrease.**
- 

**0215.** During a descent at a constant IAS below the tropopause in ISA conditions ;

- a) Mach number and TAS increase.
- b) Mach number and TAS decrease.
- c) Mach number decreases and TAS increases.
- d) Mach number increases and TAS decreases.

**Answer ;**

- b) Mach number and TAS decrease.**
- 

**0216.** During a descent at a constant Mach number below the tropopause in ISA conditions ;

- a) IAS decreases and TAS increases.
- b) IAS and TAS decrease.
- c) IAS increases and TAS decreases.
- d) IAS and TAS increase.

**Answer ;**

- d) IAS and TAS increase.**
- 

**0217.** During a descent at a constant Mach number below the tropopause in ISA conditions ;

- a) CAS and TAS increase.
- b) CAS and TAS decrease.
- c) CAS decreases and TAS increases.
- d) CAS increases and TAS decreases.

**Answer ;**

- a) CAS and TAS increase.**
- 

**0218.** During a final approach , the flight director is engaged in the G/S mode ( holding of ILS Glide Slope ). If the horizontal command bar is deviating upward , it means that ;

- a) the pitch attitude must be increased.
- b) the pitch attitude must be reduced.
- c) the aircraft is above the glide slope.
- d) the aircraft is below the glide slope.

**Answer ;**

- a) the pitch attitude must be increased.**
-

**0219.** During a final approach , the flight director is engaged in the LOC mode ( holding of localizer axis ). If the vertical command bar is deviating to the left , it means that the aircraft ;

- a) must be rolled to the right.
- b) is right of the LOC axis.
- c) must be rolled to the left.
- d) is left of the LOC axis.

**Answer ;**

**c) must be rolled to the left.**

---

**0220.** During a final approach , the flight director system is engaged in the G/S mode ( holding of ILS Glide Slope ). The position of the horizontal command bar indicates ;

- 1. the position of the aircraft relative to the ILS Glide Slope.
- 2. the correction on the pitch to be applied to join and follow the ILS Glide Slope.
- 3. the pitch attitude of the aircraft.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 1 , 2.
- c) 1.
- d) 2.

**Answer ;**

**d) 2.**

---

**0221.** During a final approach , if the flight director system is engaged in the G/S mode ( holding of ILS Glide Slope ) , the position of the horizontal command bar indicates ;

- a) the instantaneous deviation between the aircraft position and the ILS Glide Slope.
- b) the correction on the pitch to be applied to join and follow the ILS Glide Slope.
- c) the pitch attitude of the aircraft.
- d) the position of the aircraft relative to the ILS Glide Slope.

**Answer ;**

**b) the correction on the pitch to be applied to join and follow the ILS Glide Slope.**

---

**0222.** During a final approach , the flight director system is engaged in the LOC mode ( holding of localizer axis ). The position of the vertical command bar indicates ;

- 1. the position of the aircraft relative to the localizer axis.
- 2. the roll attitude of the aircraft.
- 3. the correction on the bank to be applied to join and follow the Localizer axis.

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 1.
- c) 3.
- d) 1 , 3.

**Answer ;**

**c) 3.**

---

**0223.** During a final approach , the flight director system is engaged in the LOC mode ( holding of Localizer axis ). The position of the vertical command bar indicates ;

- a) the roll attitude of the aircraft.
- b) the correction on the bank to be applied to join and follow the Localizer axis.
- c) the position of the aircraft relative to the Localizer axis.
- d) the instantaneous deviation between the aircraft position and the Localizer axis.

**Answer ;**

**b) the correction on the bank to be applied to join and follow the Localizer axis.**

---

**0224.** During a manual ILS final approach , the auto-throttle ;

- a) is not available.
- b) can be operated in the SPEED mode ( holding of IAS ).
- c) can be operated in the N1 ( EPR ) mode ( holding of N1 or EPR ).
- d) can not be engaged because autopilot is not engaged.

**Answer ;**

**b) can be operated in the SPEED mode ( holding of IAS ).**

---

**0225.** During an acceleration phase at constant attitude , the control system of the artificial horizon results in the horizontal bar indicating a ;

- a) constant attitude.
- b) nose-down followed by a nose-up attitude.
- c) nose-up attitude.
- d) nose-down attitude.

**Answer ;**

**c) nose-up attitude.**

---

**0226.** During an autocoupled ILS approach followed by an automatic landing , the guidance signals in the vertical plane under 200 ft are computed according to the ;

- a) radio altitude.
- b) barometric altitude with the altimeter set to the QNH.
- c) barometric altitude with the altimeter set to the 1013.25.
- d) barometric altitude with the altimeter set to the QFE.

**Answer ;**

**a) radio altitude.**

---

**0227.** During an automatic ILS final approach , the auto-throttle ;

- a) can be operated in the N1 / EPR mode ( holding of N1 or EPR ).
- b) can be operated in the SPEED ( holding of IAS ) or N1 (holding of N1 ) modes.
- c) is not available.
- d) can be operated in the SPEED mode ( holding of IAS ).

**Answer ;**

**d) can be operated in the SPEED mode ( holding of IAS ).**

---

**0228.** During an automatic landing , between 50 ft AGL and touch down , the autopilot maintains ;

- a) a vertical speed according to the GPS height.
- b) a vertical speed according to the radio altimeter height.
- c) a constant flight path angle with reference to the ground.
- d) a constant vertical speed.

**Answer ;**

**b) a vertical speed according to the radio altimeter height.**

---

**0229.** During deceleration following a landing in a northerly direction , a magnetic compass made for the southern hemisphere indicates ;

- a) no apparent turn.
- b) an apparent turn to the east.
- c) no apparent turn only on southern latitudes.
- d) an apparent turn to the west.

**Answer ;**

**a) no apparent turn.**

---

**0230.** During deceleration following a landing in a southerly direction , a magnetic compass made for the northern hemisphere indicates ;

- a) an apparent turn to the west.
- b) no apparent turn only on northern latitudes.
- c) no apparent turn.
- d) an apparent turn to the east.

**Answer ;**

**c) no apparent turn.**

---

**0231.** During deceleration following a landing in a westerly direction , a magnetic compass made for the northern hemisphere indicates ;

- a) an apparent turn to the south.
- b) no apparent turn.
- c) an apparent turn to the north.
- d) no apparent turn only on northern latitudes.

**Answer ;**

**a) an apparent turn to the south.**

---

0232. During deceleration following a landing in an easterly direction , a magnetic compass made for the northern hemisphere indicates ;

- a) an apparent turn to the north.
- b) an apparent turn to the south.
- c) no apparent turn.
- d) no apparent turn only on northern latitudes.

Answer ;

**b)** an apparent turn to the south.

---

0233. During deceleration following a landing in an easterly direction , a magnetic compass made for the southern hemisphere indicates ;

- a) an apparent turn to the north.
- b) no apparent turn only on southern latitudes.
- c) no apparent turn.
- d) an apparent turn to the south.

Answer ;

**a)** an apparent turn to the north.

---

0234. During deceleration following a landing in a westerly direction , a magnetic compass made for the southern hemisphere indicates ;

- a) no apparent turn.
- b) no apparent turn only on southern latitudes.
- c) an apparent turn to the south.
- d) an apparent turn to the north.

Answer ;

**d)** an apparent turn to the north.

---

0235. During descent , the total pressure probe of the airspeed indicator becomes blocked. In this case ;

- 1. IAS becomes greater than CAS.
- 2. IAS becomes lower than CAS.
- 3. maintaining IAS constant ,  $V_{MO}$  may be exceeded.
- 4. maintaining IAS constant , aircraft may stall.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 2 , 4.
- c) 2 , 3.
- d) 1 , 4.

Answer ;

**c)** 2 , 3.

---

0236. During large control inputs from an automatic flight control system ( AFCS ) , the control stick in the cockpit is moved to inform the pilot of the action. This is ;

- a) achieved by series actuator.
- b) achieved by a parallel actuator.
- c) a false statement ; the information is displayed to the pilot via the ADI , HSI and AFCS controller.
- d) achieved by the flight director.

Answer ;

**b)** achieved by a parallel actuator.

---

0237. During the approach , a crew reads on the radio altimeter the value of 650 ft. This is an indication of the true ;

- a) altitude of the aircraft.
- b) height of the aircraft with regard to the runway.
- c) height of the aircraft with regard to the ground at any time.
- d) height of the lowest wheels with regard to the ground at any time.

Answer ;

**d)** height of the lowest wheels with regard to the ground at any time.

---

0238. During the approach , the radio altimeter indicates 950 ft. This is ;

- a) the height of the pilot eyes with regard to the ground.
- b) the relative height of the aircraft above airport level ( AAL ).
- c) the height of the lowest wheels with regard to the ground.
- d) the relative height of the aircraft with regard to the runway.

Answer ;

**c)** the height of the lowest wheels with regard to the ground.

---

0239. During the ground run take-off phase , the auto-throttles allow to ;

- a) maintain V2 under 1.500 ft.
- b) hold and maintain the Mach number.
- c) hold N1.
- d) hold IAS.

Answer ;

c) hold N1.

---

0240. EAS can be obtained from the following data ;

- a) TAS and pressure altitude.
- b) IAS and density altitude.
- c) TAS and density altitude.
- d) CAS and pressure altitude.

Answer ;

d) CAS and pressure altitude.

---

0241. EAS is equal to ;

- a) CAS corrected for compressibility error.
- b) TAS corrected for compressibility and density errors.
- c) CAS corrected for density error.
- d) TAS corrected for compressibility error.

Answer ;

a) CAS corrected for compressibility error.

---

0242. EAS is equal to ;

- a) IAS corrected for position , instrument and compressibility errors.
- b) TAS corrected for compressibility error.
- c) IAS corrected for compressibility error.
- d) CAS corrected for density error.

Answer ;

a) IAS corrected for position , instrument and compressibility errors.

---

0243. Electrical requirements for an alumel/chromel indicating system are ;

- a) 28VDC for sensor plus power for gauge lighting.
- b) 3 phase AC for sensor plus 26VAC for gauge lighting.
- c) power for gauge lighting only.
- d) 3 phase AC for sensor plus 28VDC for gauging.

Answer ;

c) power for gauge lighting only.

---

0244. Equivalent Air Speed ( EAS ) is ;

- a) True Air Speed ( TAS ) corrected for compressibility error.
- b) Indicated Air Speed ( IAS ) corrected for compressibility error.
- c) Calibrated Air Speed ( CAS ) corrected for density error.
- d) Indicated Air Speed ( IAS ) corrected for position , instrument and compressibility errors.

Answer ;

d) Indicated Air Speed ( IAS ) corrected for position , instrument and compressibility errors.

---

0245. Equivalent Air Speed ( EAS ) is ;

- a) True Air Speed ( TAS ) corrected for compressibility and density errors.
- b) Calibrated Air Speed ( CAS ) corrected for density error.
- c) True Air Speed ( TAS ) corrected for compressibility error.
- d) Calibrated Air Speed ( CAS ) corrected for compressibility error.

Answer ;

d) Calibrated Air Speed ( CAS ) corrected for compressibility error.

---

0246. Equivalent Air Speed ( EAS ) is obtained from Calibrated Air Speed ( CAS ) by correcting for ;

- a) instrument error.
- b) position error.
- c) density error.
- d) compressibility error.

Answer ;

d) compressibility error.

---

0247. Equivalent Air Speed ( EAS ) is obtained from Calibrated Air Speed ( CAS ) by correcting for the following errors ;
1. position.
  2. compressibility.
  3. instrument.
  4. density.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 1 , 2 , 3 , 4.
- c) 4.
- d) 2.

Answer ;

d) 2.

---

0248. Equivalent Air Speed ( EAS ) is obtained from Indicated Air Speed ( IAS ) by correcting for the following errors ;
1. instrument.
  2. position.
  3. density.
  4. compressibility.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2 , 4.
- c) 1 , 2 , 3 , 4.
- d) 1 , 2.

Answer ;

b) 1 , 2 , 4.

---

0249. Except for airplanes under 5.7 t airworthiness certificate of which is subsequent to 31 march 1998 , a flight data recording system must be able to store the recorded data for a minimum of the last ;

- a) 10 hours.
- b) 30 minutes.
- c) 60 minutes.
- d) 25 hours.

Answer ;

d) 25 hours.

---

0250. Flight Director Information supplied by an FD computer is presented in the form of command bars on the following instrument ;

- a) ADI Attitude Director Indicator.
- b) RMI Radio Magnetic Indicator.
- c) HSI Horizontal Situation Indicator.
- d) BDHI Bearing Distance Heading Indicator.

Answer ;

a) ADI Attitude Director Indicator.

---

0251. Flying manually during a final approach , the flight director system is engaged in the G/S mode ( holding of ILS Glide Slope ). If the aircraft is above the ILS Glide Slope , the horizontal command bar ;

- a) cannot be centred.
- b) will be centred only when establish on the ILS Glide Slope.
- c) may be centred if the pilot is correcting to come back on to the ILS Glide Slope.
- d) is automatically centred since the G/S mode is engaged.

Answer ;

c) may be centred if the pilot is correcting to come back on to the ILS Glide Slope.

---

0252. Flying manually during a final approach , the flight director system is engaged in the G/S mode ( holding of ILS Glide Slope ). If the aircraft is below the ILS Glide Slope , the horizontal command bar ;

- a) is automatically centred since the G/S mode is engaged.
- b) deviates downward , whatever the attitude of the aircraft is.
- c) deviates upward , whatever the attitude of the aircraft is.
- d) may be centred if the pilot is correcting to come back on the ILS Glide Slope.

Answer ;

d) may be centred if the pilot is correcting to come back on the ILS Glide Slope.

---

**0253.** Flying manually during a final approach , the flight director system is engaged in the LOC mode ( holding of localizer axis ). If the aircraft is left of the Localizer axis , the vertical command bar ;

- a) is automatically centred since the LOC mode is engaged.
- b) deviates to the right , whatever the attitude of the aircraft is.
- c) deviates to the left , whatever the attitude of the aircraft is.
- d) may be centred if the pilot is correcting to come back on the Localizer axis.

**Answer ;**

- d)** may be centred if the pilot is correcting to come back on the Localizer axis.
- 

**0254.** Flying manually during a final approach , the flight director system is engaged in the LOC mode ( holding of localizer axis ). If the aircraft is right of the Localizer axis , the vertical command bar ;

- a) cannot be centred.
- b) will be centred only when establish on the Localizer axis.
- c) may be centred if the pilot is correcting to come back on the Localizer axis.
- d) is automatically centred since the LOC mode is engaged.

**Answer ;**

- c)** may be centred if the pilot is correcting to come back on the Localizer axis.
- 

**0255.** For a directional gyro , the system which detects the local vertical supplies ;

- a) a levelling erection torque motor.
- b) a torque motor on the sensitive axis.
- c) a nozzle integral with the outer gimbal ring.
- d) two torque motors arranged horizontally.

**Answer ;**

- a)** a levelling erection torque motor.
- 

**0256.** For an aircraft EFIS equipped ;

- 1. the ND displays flight director command bars.
- 2. the PFD displays the altimeter setting.
- 3. the ND displays the attitude of aircraft.
- 4. the flight mode annunciator is part of the ND.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 1 , 4.
- c) 2.
- d) 1 , 3.

**Answer ;**

- c)** 2.
- 

**0257.** For an aircraft flying a true track of 360° between the 5°S and 5°N parallels , the precession error of the directional gyro due to apparent drift is equal to ;

- a) 15°/hour.
- b) approximately 0°/hour.
- c) -5°/hour.
- d) +5°/hour.

**Answer ;**

- b)** approximately 0°/hour.
- 

**0258.** For an FMS designed with the lateral navigation ( LNAV ) capability coupled to the autopilot , the FMS lateral command output is ;

- a) a roll angle or a heading target.
- b) a lateral acceleration.
- c) a calibrated airspeed ( CAS ) and a cross track distance ( XTK ).
- d) a longitudinal acceleration and a roll rate.

**Answer ;**

- a)** a roll angle or a heading target.
-

**0259.** For an FMS designed with the vertical navigation ( VNAV ) capability coupled to the autopilot , the FMS vertical command output can be ;

1. an angle of attack.
2. a flight path angle.
3. a speed target.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 2.
- c) 1 , 2.
- d) 2 , 3.

**Answer ;**

**d) 2 , 3.**

---

**0260.** For compatibility between the different components of a flight warning system , the priority from the highest to the lowest is ;

- a) Stall , Windshear , GPWS , TCAS.
- b) Stall , TCAS , GPWS , Winshear.
- c) TCAS , Stall , Windshear , GPWS.
- d) GPWS , Stall , TCAS , Windshear.

**Answer ;**

**a) Stall , Windshear , GPWS , TCAS.**

---

**0261.** For most FMS the Fuel prediction function , which computes the remaining fuel along the flight plan , takes into account the following situations ;

1. the additional drag resulting in a flight carried out with the landing gear extended.
2. the current wind computed or the resulting ground speed.
3. the additional drag resulting in a flight carried out with the flaps stucked , partly extended.
4. the additional drag resulting in a missing fuselage or wing element in compliance with the CDL.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 2 , 4.
- c) 3.
- d) 2.

**Answer ;**

**d) 2.**

---

**0262.** For TAS calculations , the ADC uses the following parameters ;

1. SAT.
2. TAT.
3. static pressure.
4. total pressure.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4.
- b) 2 , 3 , 4.
- c) 1 , 2 , 3 , 4.
- d) 3 , 4.

**Answer ;**

**b) 2 , 3 , 4.**

---

**0263.** For the same TAS , when pressure altitude decreases below the tropopause ;

- a) the difference between surrounding conditions and ISA must be known to deduce the Mach number variation.
- b) Mach number remains constant.
- c) Mach number increases.
- d) Mach number decreases.

**Answer ;**

**d) Mach number decreases.**

---

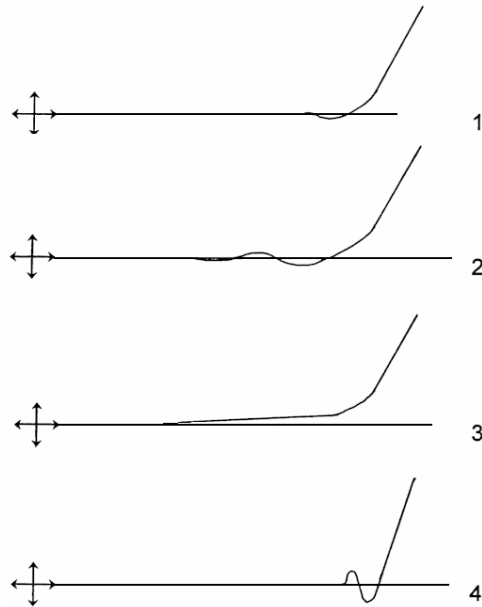


0264. For the same TAS , when pressure altitude increases below the tropopause ;
- Mach number increases.
  - Mach number decreases.
  - Mach number remains constant.
  - the difference between surrounding conditions and ISA must be known to deduce the Mach number variation.

Answer ;

a) Mach number increases.

0265. ( For this question use annex 022-10179A )



**Annex 022-10179A**

Four scenarios of VOR axis interception are represented in the appended annex. The one corresponding to the optimal interception path calculated by a flight director is number ;

- 4.
- 2.
- 3.
- 1.

Answer ;

b) 2.

0266. Gas turbine engine rotational speed ( RPM ) is usually sensed using either ;
- a 28VDC tachometer generator or a DC phonic wheel.
  - a single phase AC tachometer generator or an AC phonic wheel system.
  - a 28VDC tachometer generator or an AC phonic wheel.
  - a 3 phase AC tachometer generator or an AC phonic wheel system.

Answer ;

d) a 3 phase AC tachometer generator or an AC phonic wheel system.

0267. Given ;

Pt = total pressure  
Ps = static pressure  
Pd = dynamic pressure

- $Pd = Pt + Ps$
- $Pt = Pd + Ps$
- $Pd = Pt / Ps$
- $Ps = Pt + Pd$

Answer ;

b)  $Pt = Pd + Ps$

0268. Given:

M is the Mach number  
Ts is the static temperature  
Tt is the total temperature

- a)  $T_s = T_t (0.2 \cdot M^2)$
- b)  $T_s = T_t (1 + 0.2 \cdot M^2)$
- c)  $T_s = T_t / (1 + 0.2 \cdot M^2)$
- d)  $T_s = T_t / (0.2 \cdot M^2)$

Answer ;

c)  $T_s = T_t / (1 + 0.2 \cdot M^2)$

---

0269. Given ;

Pt = total pressure  
Ps = static pressure  
Pso = static pressure at sea level

Calibrated airspeed ( CAS ) is a function of ;

- a)  $P_t / P_s$
- b)  $P_t - P_{so}$
- c)  $P_t - P_s$
- d)  $(P_t - P_{so}) / P_s$

Answer ;

c)  $P_t - P_s$

---

0270. Given ;

Zp = pressure altitude  
Zd = density altitude

CAS can be obtained from the following data ;

- a) EAS and Zd.
- b) TAS and Zd.
- c) TAS and Zp.
- d) EAS and Zp.

Answer ;

d) EAS and Zp.

---

0271. Given ;

Pt = total pressure  
Ps = static pressure

Dynamic pressure is ;

- a)  $(P_t - P_s) / P_s$
- b)  $P_t / P_s$
- c)  $P_t - P_s$
- d)  $(P_t - P_s) / P_t$

Answer ;

c)  $P_t - P_s$

---

0272. Given ;

Pt = total pressure  
Ps = static pressure  
Pso = static pressure at sea level

Dynamic pressure is ;

- a)  $(P_t - P_{so}) / P_{so}$
- b)  $(P_t - P_s) / P_s$
- c)  $P_t - P_s$
- d)  $P_t - P_{so}$

Answer ;

c)  $P_t - P_s$

---

0273. Given ;

Zp = pressure altitude

Zd = density altitude

TAS can be obtained from the following data ;

- a) EAS and Zp.
- b) EAS and Zd.
- c) CAS and Zp.
- d) CAS and Zd.

Answer ;

**b) EAS and Zd.**

0274. Given:

Pt : total pressure

Ps : static pressure

Pd : dynamic pressure

The airspeed indicator is fed by ;

- a) Pt – Pd
- b) Ps – Pt
- c) Pd
- d) Pd – Ps

Answer ;

**c) Pd**

0275. Given:

Pt : total pressure

Ps : static pressure

Pd : dynamic pressure

The airspeed indicator measures ;

- a) Pt – Ps
- b) Pt – Pd
- c) Pd – Ps
- d) Ps – Pt

Answer ;

**a) Pt – Ps**

0276. Given ;

Pt : total pressure

Ps : static pressure

Pd : dynamic pressure

The altimeter is fed by ;

- a) Ps – Pt.
- b) Pt – Pd.
- c) Pd – Ps.
- d) Pd.

Answer ;

**b) Pt – Pd.**

0277. Given ;

Mach number M = 0.70

measured impact temperature = –48°C

the recovery factor ( Kr ) of the temperature probe = 0.85

The OAT is ;

- a) –51°C
- b) –65°C
- c) –45°C
- d) –68°C

Answer ;

**b) –65°C**

Çözüm ;

OAT ( SAT ) yukarıdaki iki formülle de hesaplanabilir. İlk formülün kullanılabilmesi için DR kompitüründen TAS'ın bulunması gereklidir. Bunun için DR kompitüründe “Mach No Index” oku –48 değerine hizalanır. İç skalada yer alan 70 değerinin ( 0.70 M ) karşısında , dış skalada TAS değeri 412 kt olarak okunur.

Değerleri ilk formüle yerleştirdiğimizde ;

$$\text{SAT} = -48 - \left( \frac{412}{100} \right)^2 = -48 - ( 412^2 : 100^2 ) = -48 - ( 412 : 100 ) = -48 - ( 169744 : 10000 ) = -48 - ( 16.9744 ) = 64.9744 \approx -65^\circ\text{C} \text{ olarak bulunur.}$$

0278. Given ;

E = electromotive force ( emf )  
T = hot junction temperature  
K = constant

The relationship that applies to a thermocouple is ;

- a)  $E = K \times T$
- b)  $E = K \times T^2$
- c)  $E = K \times \text{SQRT}(T)$
- d)  $E = K + T$

Answer ;

a)  $E = K \times T$

---

0279. Given ;

E = electromotive force ( emf )  
Tc = cold junction temperature  
Th = hot junction temperature  
K = constant

The relationship that applies to a thermocouple is ;

- a)  $E = K \times T_c$
- b)  $E = K \times T_h$
- c)  $E = K \times T_h^2$
- d)  $E = K \times T_c^2$

Answer ;

b)  $E = K \times T_h$

---

0280. Given ;

Ts the static temperature ( SAT )  
Tt the total temperature ( TAT )  
Kr the recovery coefficient  
M the Mach number

The total temperature can be expressed approximately by the formula ;

- a)  $T_t = T_s ( 1 + 0.2 M^2 )$
- b)  $T_t = T_s ( 1 - 0.2 M^2 )$
- c)  $T_t = T_s ( 1 + 0.2 K_r.M^2 )$
- d)  $T_t = T_s / ( 1 + 0.2 K_r.M^2 )$

Answer ;

c)  $T_t = T_s ( 1 + 0.2 K_r.M^2 )$

---

0281. Given ;

MH = magnetic heading of the aircraft  
Omega = yawing rate of the aircraft

The yaw damper computer sends a motion order to the rudder if ;

- a) the derivative of MH according to time is equal to zero.
- b) the derivative of Omega according to time is not equal to zero.
- c) the derivative of Omega according to time is equal to zero.
- d) the derivative of MH according to time is not equal to zero.

Answer ;

b) the derivative of Omega according to time is not equal to zero.

---

0282. Given the following parameters , in a capacitance fuel gauge , the correct formula is ;

A = area of plates  
D = distance between plates  
E = dielectric permittivity

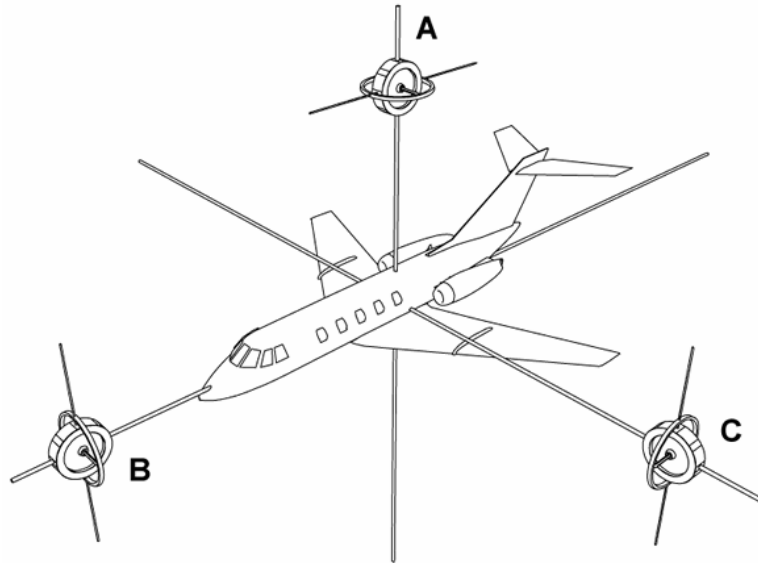
- a) Capacitance =  $A \times D / E$
- b) Capacitance =  $E \times D / A$
- c) Capacitance =  $E \times A / D$
- d) Capacitance =  $E \times D \times A$

Answer ;

c) Capacitance =  $E \times A / D$

---

0283. ( For this question refer to Image I )



**Image I**

Graphic at reference shows three gyro assemblies ; A , B and C. Among these gyros ;  
one is a roll gyro ( noted 1 )  
one is a pitch gyro ( noted 2 )  
one is a yaw gyro ( noted 3 )

The correct matching of gyros and assemblies is ;

- a) 1B , 2C , 3A.
- b) 1A , 2B , 3C.
- c) 1B , 2A , 3C.
- d) 1C , 2B , 3A.

Answer ;

- a) 1B , 2C , 3A.

---

0284. “Heading hold” is normally ;

- a) not regarded as a basic function.
- b) a function of a stability augmentation system.
- c) regarded as a basic function of a stability augmentation system.
- d) regarded as a basic function of automatic stabilisation equipment.

Answer ;

- d) regarded as a basic function of automatic stabilisation equipment.

---

0285. Heading information from the gyromagnetic compass flux gate is transmitted to the ;

- a) error detector.
- b) amplifier.
- c) erecting system.
- d) heading indicator.

Answer ;

- a) error detector.

---

0286. Heading information given by a gyro platform , is given by a gyro with ;

NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.

- a) 2 degrees-of-freedom in the vertical axis.
- b) 1 degree-of-freedom in the vertical axis.
- c) 1 degree-of-freedom in the horizontal axis.
- d) 2 degrees-of-freedom in the horizontal axis.

Answer ;

- d) 2 degrees-of-freedom in the horizontal axis.
-

**0287.** If a manifold pressure gauge consistently registers atmospheric pressure , the cause is probably ;

- a) leak in pressure gauge line.
- b) too high float level.
- c) fuel of too low volatility.
- d) ice in induction system.

**Answer ;**

- a) leak in pressure gauge line.
- 

**0288.** If a TCAS Resolution Advisory ( RA ) requires an immediate descent while at the same time ATC requests an immediate climb , the crew should ;

- a) consider both requests and remain at the same level.
- b) comply with the ATC request.
- c) start a climb first and then follow the TCAS RA.
- d) follow the TCAS RA and inform the ATC.

**Answer ;**

- d) follow the TCAS RA and inform the ATC.
- 

**0289.** If an aircraft is flying ( with flaps and landing gear retracted ) in proximity to terrain and its GPWS ( Ground Proximity Warning System ) get activated , because it is detecting that the aeroplane has an excessive rate of descent , the system provides the following aural warning signals ;

- a) “SINK RATE , SINK RATE” followed by “WHOOOP WHOOP PULL UP” ( twice )
- b) “TOO LOW , TERRAIN” ( twice ) followed by “TOO LOW GEAR” ( twice )
- c) “TERRAIN , TERRAIN” followed by “WHOOOP WHOOP PULL UP” ( twice )
- d) “DON’T SINK , DON’T SINK”

**Answer ;**

- a) “SINK RATE , SINK RATE” followed by “WHOOOP WHOOP PULL UP” ( twice )
- 

**0290.** If an aircraft maintaining a constant CAS and flight level is flying from a cold air mass into warmer air ;

- a) TAS decreases.
- b) Mach number decreases.
- c) TAS increases.
- d) Mach number increases.

**Answer ;**

- c) TAS increases.
- 

**0291.** If an aircraft maintaining a constant CAS and flight level is flying from a warm air mass into colder air ;

- a) Mach number decreases.
- b) TAS increases.
- c) TAS decreases.
- d) Mach number increases.

**Answer ;**

- c) TAS decreases.
- 

**0292.** If an EPR is set at a constant barometric pressure , with an increasing OAT , the thrust ;

- a) decreases.
- b) remains constant.
- c) varies according to the characteristics of the engine.
- d) increases.

**Answer ;**

- b) remains constant.
- 

**0293.** If crew awareness is required and subsequent crew action may be required , the flight warning system ( FWS ) generates ;

- a) an advisory message.
- b) a call message.
- c) a caution message.
- d) a warning message.

**Answer ;**

- c) a caution message.
-

**0294.** If, during a descent ; the pneumatic altimeter reading is constant , the vertical speed indicator shows zero , the IAS is increasing , the most likely explanation is that ;

- a) the antenna of the radio altimeter is completely iced up.
- b) the static ports are completely blocked.
- c) the total pressure head is completely blocked.
- d) there is a leakage in the static pressure line.

**Answer ;**

- b)** the static ports are completely blocked.
- 

**0295.** If immediate crew awareness is required and subsequent crew action will be required , the flight warning system ( FWS ) generates ;

- a) a warning message.
- b) an urgency message.
- c) an advisory message.
- d) a caution message.

**Answer ;**

- a)** a warning message.
- 

**0296.** If immediate recognition and corrective or compensatory action by the crew is required , the flight warning system ( FWS ) generates ;

- a) a warning message.
- b) a caution message.
- c) an advisory message.
- d) a distress message.

**Answer ;**

- a)** a warning message.
- 

**0297.** If , in the event of a failure , the approach , flare and landing can be completed by the remaining part of the automatic system , such an automatic landing system is considered as ;

- a) fail-hard.
- b) fail-passive.
- c) fail-operational.
- d) fail-soft.

**Answer ;**

- c)** fail-operational.
- 

**0298.** If , in the event of a failure , there is no significant out-of-trim condition or deviation of flight path or attitude but the landing is not completed automatically , such an automatic landing system is considered as ;

- a) fail-passive.
- b) fail-safe.
- c) fail-operational.
- d) fail-redundant.

**Answer ;**

- a)** fail-passive.
- 

**0299.** If OAT decreases when at a constant Mach number ;

- a) TAS increases.
- b) TAS decreases only if the flight level remains constant.
- c) TAS remains constant only if the flight level remains constant.
- d) TAS decreases.

**Answer ;**

- d)** TAS decreases.
- 

**0300.** If OAT decreases when at a constant TAS ;

- a) Mach number remains constant.
- b) Mach number decreases.
- c) the difference between surrounding conditions and ISA must be known to deduce the Mach number variation.
- d) Mach number increases.

**Answer ;**

- d)** Mach number increases.
-

0301. If OAT decreases whilst maintaining a constant CAS and flight level ;

- a) TAS increases.
- b) TAS remains constant.
- c) Mach number increases.
- d) Mach number remains constant.

Answer ;

d) Mach number remains constant.

---

0302. If OAT increases when at a constant Mach number ;

- a) TAS increases.
- b) TAS decreases.
- c) TAS decreases only if the flight level remains constant.
- d) TAS remains constant only if the flight level remains constant.

Answer ;

a) TAS increases.

---

0303. If OAT increases when at a constant TAS ;

- a) the difference between surrounding conditions and ISA must be known to deduce the Mach number variation.
- b) Mach number remains constant.
- c) Mach number increases.
- d) Mach number decreases.

Answer ;

d) Mach number decreases.

---

0304. If OAT increases whilst maintaining a constant CAS and flight level ;

- a) TAS decreases.
- b) Mach number remains constant.
- c) Mach number decreases.
- d) TAS remains constant.

Answer ;

b) Mach number remains constant.

---

0305. If the acceleration of an aircraft is zero , its velocity ;

- a) is constant.
- b) will decrease.
- c) will increase.
- d) is always zero.

Answer ;

a) is constant.

---

0306. If the computed aircraft position becomes less accurate , the Enhanced GPWS ( EGPWS) function(s) affected is (are) ;

- 1. the excessive rate of descent.
- 2. the terrain display on the navigation display.
- 3. the flight into terrain when not in landing conditions.
- 4. the excessive downward deviation from an ILS.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 2.
- c) 1 , 2 , 3.
- d) 3.

Answer ;

b) 2.

---



0307. If the computed aircraft position becomes less accurate , the Enhanced GPWS ( EGPWS ) function(s) affected is (are) ;
1. the “FIVE HUNDRED” voice call out.
  2. the excessive rate of descent.
  3. the terrain display on the navigation display.
  4. the negative climb rate or altitude loss after takeoff.

The combination that regroups all of the correct statements is ;

- a) 3.
- b) 3 , 4.
- c) 1 , 2.
- d) 4.

Answer ;

a) 3.

- 
0308. If the GPWS ( Ground Proximity Warning System ) activates , and alerts the pilot with an aural warning “DON’T SINK” ( twice times ) , it is because ;

- a) the aircraft experiences an unexpected proximity to terrain , without landing-flap selected.
- b) at too low altitude , the aircraft has an excessive rate of descent.
- c) during take-off or missed approach manoeuvre , the aircraft has started to loose altitude.
- d) the aircraft experiences an unexpected proximity to the terrain , with landing gear retracted.

Answer ;

c) during take-off or missed approach manoeuvre , the aircraft has started to loose altitude.

- 
0309. If the navigation function of an INS is inoperative and the control rotary switch is set to ATT , the output data of the INS are ;

- a) attitude and heading.
- b) attitude and ground speed.
- c) attitude , TAS and heading.
- d) ground speed and heading.

Answer ;

a) attitude and heading.

- 
0310. If the navigation function of an INS is inoperative and the control rotary switch is set to ATT , the output data of the INS is (are) ;

- a) position and attitude.
- b) attitude and ground speed.
- c) attitude and heading.
- d) position only.

Answer ;

c) attitude and heading.

- 
0311. If the navigation function of an INS is inoperative , the back up mode if existing , used to operate the INS is ;

- a) the ATT mode which supplies attitude and heading data.
- b) the ATT mode which allows to maintain pitch attitude only.
- c) the GS mode which supplies ground speed and heading data.
- d) the OFF mode which turns off the navigation mode but recovers the heading mode.

Answer ;

a) the ATT mode which supplies attitude and heading data.

- 
0312. If the outside temperature at 35 000 feet is  $-40^{\circ}\text{C}$  , the local speed of sound is ;

- a) 595 kt.
- b) 307 kt.
- c) 686 kt.
- d) 247 kt.

Answer ;

a) 595 kt.

Çözüm ;

DR kompitüründe , Mach No. Index okunu  $-40^{\circ}\text{C}$  konumuna getirin. Bu kurulumu göre LSS ( Local Speed of Sound ) , orta skalada Mach 1 ( 10 rakamı ) değerinin üzerinde yer alacaktır. 10 rakamının üzerinde yer alan değer 59.5’dir yani , bu durumda ses hızı 595 kt olur.

0313. If the pilot increases the reference pressure using the sub-scale setting knob , the altitude indicated by the altimeter ;
- a) decreases if QNH > 1013 hPa.
  - b) decreases.
  - c) increases.
  - d) decreases if QNH < 1013 hPa.

Answer ;

c) increases.

---

0314. If the pitot tube becomes blocked during a descent , the airspeed indicator ;
- a) indicates a constant speed.
  - b) over-reads.
  - c) under-reads or over-reads , depending on the air density.
  - d) under-reads.

Answer ;

d) under-reads.

---

0315. If the pitot tube ices up during a flight , the affected equipment(s) is(are) ;
- 1. the altimeter.
  - 2. the variometer.
  - 3. the airspeed indicator.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 1 , 2 , 3.
- c) 1.
- d) 3.

Answer ;

d) 3.

---

0316. If the position data ( lat , long ) is no longer computed an IRS , the affected system(s) is (are) ;
- a) the TCAS.
  - b) the ADC and the TCAS.
  - c) the FMS.
  - d) the FMS and the TCAS.

Answer ;

c) the FMS.

---

0317. If the static intakes are completely clogged up by ice during a climb , the VSI shows ;
- a) zero.
  - b) a constant rate of climb , even if the aircraft is levelling out.
  - c) a descent if the outside static pressure is less than pressure in the VSI gauge.
  - d) an increasing rate of climb if the ambient static pressure decreases.

Answer ;

a) zero.

---

0318. If the static ports are completely clogged up by ice during a climb , the vertical speed indicator shows ;
- a) zero.
  - b) an increasing rate of climb if the ambient static pressure decreases.
  - c) a constant rate of climb , even if the aircraft is levelling out.
  - d) a descent if the outside static pressure is less than the pressure in the instrument's gauge.

Answer ;

a) zero.

---

0319. If the static source to an airspeed indicator ( ASI ) becomes blocked during a descent the instrument will ;
- a) over-read.
  - b) continue to indicate the speed applicable to that at the time of the blockage.
  - c) under-read.
  - d) read zero.

Answer ;

a) over-read.

---

0320. If the static source to an altimeter becomes blocked during a climb , the instrument will ;
- a) gradually return to zero.
  - b) under-read by an amount equivalent to the reading at the time that the instrument became blocked.
  - c) continue to indicate the reading at which the blockage occurred.
  - d) over-read.

Answer ;

- c) continue to indicate the reading at which the blockage occurred.
- 

0321. If the static source of an altimeter becomes blocked during a descent the instrument will ;
- a) under-read.
  - b) continue to display the reading at which the blockage occurred.
  - c) indicate a height equivalent to the setting on the millibar subscale.
  - d) gradually indicate zero.

Answer ;

- b) continue to display the reading at which the blockage occurred.
- 

0322. If the tanks of your airplane only contain water , the capacitor gauges indicate ;
- a) a mass equal to zero.
  - b) the exact mass of water contained in the tanks.
  - c) a mass equal to the mass of a same volume of fuel.
  - d) a mass of water different from zero , but inaccurate.

Answer ;

- d) a mass of water different from zero , but inaccurate.
- 

0323. If the total pressure intake on the pitot tube is rapidly clogged up by ice during flight , what effect will it have on the airspeed indication during a climb ?

- a) As the total pressure in the pitot static system is trapped , the airspeed indicator will indicate a constant airspeed.
- b) The total pressure is trapped while the static pressure decreases , implying a decreasing indicated airspeed.
- c) The total pressure is trapped while the static pressure increases , implying a decreasing indicated airspeed.
- d) The total pressure is trapped while the static pressure decreases , implying an increasing indicated airspeed.

Answer ;

- d) The total pressure is trapped while the static pressure decreases , implying an increasing indicated airspeed.
- 

0324. If warning , caution , or advisory lights are installed in the cockpit , they must , unless otherwise approved by the Authority , be amber for ;

- a) other urgency messages.
- b) advisory lights.
- c) warning lights.
- d) caution lights.

Answer ;

- d) caution lights.
- 

0325. If warning , caution , or advisory lights are installed in the cockpit , they must , unless otherwise approved by the Authority , be green for ;

- a) caution lights.
- b) safe operation lights.
- c) advisory lights.
- d) warning lights.

Answer ;

- b) safe operation lights.
- 

0326. If warning , caution , or advisory lights are installed in the cockpit , they must , unless otherwise approved by the Authority , be red for ;

- a) advisory lights.
- b) warning lights.
- c) caution lights.
- d) other distress messages.

Answer ;

- b) warning lights.
-

0327. If , with the automatic flight control system ( AFCS ) engaged , the pilot presses the force trim release button to allow a manual input to the flight controls ;

- a) artificial feel is produced by the parallel actuators.
- b) the pilot must overcome the force of the spring unit to make the input.
- c) the series actuator will run to damp the pilot's input.
- d) the spring units will become ineffective.

Answer ;

- d) the spring units will become ineffective.
- 

0328. In a 3-phase synchronous motor type tachometer indicator ;

- 1. the transmitter is a direct current generator.
- 2. the voltage is proportional to the transmitter drive speed.
- 3. the frequency is proportional to the transmitter drive speed.
- 4. the speed indicating element is a galvanometer.
- 5. the speed indicating element is an asynchronous motor driving a magnetic tachometer.

The combination regrouping all the correct statements is ;

- a) 1 , 4.
- b) 3 , 5.
- c) 2 , 5.
- d) 1 , 2.

Answer ;

- b) 3 , 5.
- 

0329. In a datalink system , the downlink communications consist in transmitting data ;

- a) from the flight deck to the cabin.
- b) from the ground to the aircraft.
- c) from the aircraft to the ground.
- d) from the airline maintenance to the aircraft.

Answer ;

- c) from the aircraft to the ground.
- 

0330. In a datalink system , the uplink communications consist in transmitting data ;

- a) from the ground to the aircraft.
- b) from the flight deck to the cabin.
- c) from the aircraft to the airline maintenance.
- d) from the aircraft to the ground.

Answer ;

- a) from the ground to the aircraft.
- 

0331. In a directional gyro , gimballing errors are due to ;

- a) a banked attitude.
- b) an apparent weight and an apparent vertical.
- c) the aircraft's movement over the earth.
- d) the vertical component of the earth's magnetic field.

Answer ;

- a) a banked attitude.
- 

0332. In a FMS , MCDUs are used pre-flight to manually initialise with dispatch information the ;

- 1. Inertial Reference System ( when installed ).
- 2. Flight Director and Auto Flight Control System.
- 3. Air Data Computer.
- 4. Flight Management Computer.

The combination regrouping all of the correct statements is ;

- a) 2.
- b) 2 , 3 , 4.
- c) 1 , 3.
- d) 1 , 4.

Answer ;

- d) 1 , 4.
-

0333. In a gyromagnetic compass , the direction of the earth's magnetic field is given by the ;

- a) direct indicating compass.
- b) ADC.
- c) flux valve.
- d) directional gyro.

Answer ;

c) flux valve.

---

0334. In a gyromagnetic compass , the flux valve feeds the ;

- a) error detector.
- b) heading indicator.
- c) erecting system.
- d) amplifier.

Answer ;

a) error detector.

---

0335. In a gyromagnetic compass , the gyro axis ;

- 1. is maintained vertical.
- 2. is maintained horizontal.
- 3. is servo-controlled in azimuth.
- 4. is free in azimuth.

The combination that regroups all of the correct statements is ;

- a) 2 , 4.
- b) 1 , 4.
- c) 2 , 3.
- d) 1 , 3.

Answer ;

c) 2 , 3.

---

0336. In a gyromagnetic compass , the signal feeding the precession amplifier comes from ;

- a) directional gyro unit.
- b) error detector.
- c) directional gyro erection device.
- d) flux valve.

Answer ;

b) error detector.

---

0337. In a radio altimeter , the height measurement is based upon ;

- a) a pulse transmission , for which time between transmission and reception is measured on a circular scanning screen.
- b) a wave transmission , for which the frequency shift by DOPPLER effect after ground reflection is measured.
- c) a frequency modulation wave , for which the frequency variation between the transmitted wave and the received wave after ground reflection is measured.
- d) a triangular amplitude modulation wave , for which modulation phase shift between transmitted and received waves after ground reflection is measured.

Answer ;

c) a frequency modulation wave , for which the frequency variation between the transmitted wave and the received wave after ground reflection is measured.

---

0338. In a stabilised platform inertial system , the accelerations are measured in a trihedron which is ;

( NB : "aircraft trihedron" = pitch , roll and yaw axis )

- a) fixed in absolute space.
- b) merged with the aircraft's trihedron.
- c) free from the aircraft trihedron.
- d) merged with only two axis of the aircraft trihedron ; the roll axis and the pitch axis.

Answer ;

c) free from the aircraft trihedron.

---

0339. In a standard atmosphere and at the sea level , the calibrated airspeed ( CAS ) is ;

- a) higher than the true airspeed ( TAS ).
- b) equal to the true airspeed ( TAS ).
- c) lower than the true airspeed ( TAS ).
- d) independent of the true airspeed ( TAS ).

Answer ;

b) equal to the true airspeed ( TAS ).

---

0340. In a standard atmosphere and at the sea level , the equivalent airspeed ( EAS ) is ;

- a) independent of the true airspeed ( TAS ).
- b) lower than the true airspeed ( TAS ).
- c) higher than the true airspeed ( TAS ).
- d) equal to the true airspeed ( TAS ).

Answer ;

d) equal to the true airspeed ( TAS ).

---

0341. In a steep turn , the northerly turning error on a magnetic compass on the northern hemisphere is ;

- a) none on a 090° heading in a right turn.
- b) none on a 270° heading in a left turn.
- c) equal to 180° on a 090° heading in a right turn.
- d) equal to 180° on a 270° heading in a right turn.

Answer ;

c) equal to 180° on a 090° heading in a right turn.

---

0342. In a strapdown inertial system , the accelerations are measured in a trihedron which is fixed regarding to the ;

- a) aircraft's trihedron ( pitch , roll and yaw axis ).
- b) absolute space.
- c) earth's trihedron ( X , Y , Z ).
- d) earth's trihedron ( longitude , latitude ).

Answer ;

a) aircraft's trihedron ( pitch , roll and yaw axis ).

---

0343. In a transport airplane , an autopilot comprises , in addition to the mode display devices , the following fundamental elements ;

- 1. Airflow valve.
- 2. Sensors.
- 3. Comparators.
- 4. Computers.
- 5. Amplifiers.
- 6. Servo-actuators.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4 , 6.
- b) 1 , 2 , 6.
- c) 2 , 3 , 4 , 5.
- d) 2 , 3 , 4 , 5 , 6.

Answer ;

d) 2 , 3 , 4 , 5 , 6.

---

0344. In a turn at a constant angle of bank , the turn indicator reading is ;

- a) proportional to the aircraft weight.
- b) independent to the aircraft true airspeed.
- c) proportional to the aircraft true airspeed.
- d) inversely proportional to the aircraft true airspeed.

Answer ;

d) inversely proportional to the aircraft true airspeed.

---

0345. In a yaw damper system , sensing of a disturbance in yaw is usually by ;

- a) measuring the rudder angular movement.
- b) a rate gyro or an accelerometer.
- c) measuring the roll rate.
- d) measuring the ailerons angular movement.

Answer ;

b) a rate gyro or an accelerometer.

---

0346. In an Air Data Computer ( ADC ) , aeroplane altitude is calculated from ;

- a) the difference between absolute and dynamic pressure at the fuselage.
- b) measurement of absolute barometric pressure from a static source on the fuselage.
- c) measurement of elapsed time for a radio signal transmitted to the ground surface and back.
- d) measurement of outside air temperature ( OAT ).

Answer ;

b) measurement of absolute barometric pressure from a static source on the fuselage.

---

0347. In an auto-pilot slaved powered control circuit , the system which ensures synchronisation ;
- a) can itself , when it fails , prevent the automatic pilot from being engaged.
  - b) intervenes only when the automatic pilot has been engaged.
  - c) is inhibited when the automatic pilot is engaged.
  - d) prevents uncommanded surface deflection when the automatic pilot is disengaged.

Answer ;

- a) can itself , when it fails , prevent the automatic pilot from being engaged.
- 

0348. In an autopilot system ;
- a) the outer loops provide the stability functions and the inner loops provide the guidance functions.
  - b) the inner loops provide the stability functions and the outer loops provide the guidance functions.
  - c) the outer loops provide the stability functions only.
  - d) the inner loops provide the guidance functions only.

Answer ;

- b) the inner loops provide the stability functions and the outer loops provide the guidance functions.
- 

0349. In an autopilot system , the basic stabilisation modes are ;
- 1. altitude hold.
  - 2. pitch attitude hold.
  - 3. roll attitude hold.
  - 4. IAS hold

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 1.
- c) 2 , 3 , 4.
- d) 2 , 3.

Answer ;

- d) 2 , 3.
- 

0350. In an autopilot system , the flight path modes are ;
- 1. Pitch attitude hold.
  - 2. IAS and Mach number hold.
  - 3. Altitude hold.
  - 4. Glide slope intercept and track.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4.
- b) 1 , 2 , 3 , 4.
- c) 4.
- d) 3.

Answer ;

- a) 2 , 3 , 4.
- 

0351. In an autopilot system , the function consisting in controlling the movements around the center of gravity of the aircraft is provided by the ;

- a) inner loop systems.
- b) outer loop systems.
- c) synchronization system.
- d) Flight Management System ( FMS ).

Answer ;

- a) inner loop systems.
- 

0352. In an autopilot system , the functions consisting in controlling the path of the aircraft are the ;
- a) attitude functions.
  - b) stability functions.
  - c) guidance functions.
  - d) inner loop functions.

Answer ;

- c) guidance functions.
-

0353. In an engine vibration monitoring system for a turbojet any vibration produced by the engine is ;
- fed directly to the cockpit indicator without amplification or filtering.
  - directly proportional to engine speed.
  - amplified and filtered before being fed to the cockpit indicator.
  - inversely proportional to engine speed.

Answer ;

**c) amplified and filtered before being fed to the cockpit indicator.**

---

0354. In an inertial navigation system , integrating once the speed in gives ;
- a position.
  - an instantaneous acceleration.
  - a distance travelled.
  - an average acceleration.

Answer ;

**c) a distance travelled.**

---

0355. In an inertial navigation system , the integration process ;
- amounts to making a time division.
  - amounts to making a time multiplication.
  - enable to get accelerations from position.
  - enable to get position from accelerations.

The combination that regroupes all of the correct statements is ;

- 1 , 4.
- 2 , 4.
- 1 , 3.
- 2 , 3.

Answer ;

**b) 2 , 4.**

---

0356. In an inertial navigation system , the integration process makes a ;
- time multiplication.
  - time division.
  - distance multiplication.
  - distance division.

Answer ;

**a) time multiplication.**

---

0357. In an inertial navigation system , the principle used to obtain position is ;
- single integration of acceleration according to time.
  - double integration of acceleration according to time.
  - single integration of speed according to time.
  - double integration of speed according to time.

The combination that regroupes all of the correct statements is ;

- 1.
- 2 or 3.
- 3.
- 1 or 4.

Answer ;

**b) 2 or 3.**

---

0358. In an inertial navigation system , the principle used to obtain the change in speed is ;
- double integration of acceleration according to time.
  - double integration of position according to time.
  - single integration of acceleration according to time.
  - single integration of position according to time.

Answer ;

**c) single integration of acceleration according to time.**

---

0359. In an inertial navigation system , to know the distance travelled ;
- it is necessary to integrate once the speed in time , and to know the initial speed and the initial position.
  - it is necessary to integrate once the speed in time , and to know the initial speed only.
  - it is necessary to integrate once the speed in time , and to know the initial position only.
  - integrating once the speed in time is sufficient.

Answer ;

**d) integrating once the speed in time is sufficient.**

---



0360. In an inertial navigation system , to know the distance travelled , it is necessary to ;
1. integrate once the speed in time.
  2. to know the initial position.
  3. to know the initial speed.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1.
- c) 1 , 2.
- d) 1 , 3

Answer ;

- b) 1.
- 

0361. In an inertial reference system ( IRS ) , should the platform be displaced from the horizontal , it would oscillate with a period , called Schuler period , of about ;

- a) 84 seconds.
- b) 24 minutes.
- c) 64 seconds.
- d) 84 minutes.

Answer ;

- d) 84 minutes.
- 

0362. In an inertial reference system ( IRS ) , the alignment sequence consists in ;

1. search and alignment on the local vertical.
2. search for the true North.
3. definition of the latitude.
4. definition of the longitude.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2 , 4.
- c) 1 , 2 , 3 , 4.
- d) 3 , 4.

Answer ;

- a) 1 , 2 , 3.
- 

0363. In case of accidental closing of an aircraft's left static pressure port ( rain , birds ) , the altimeter ;

- a) keeps on providing reliable reading in all situations.
- b) overreads the altitude in case of a side-slip to the right and displays the correct information during symmetric flight.
- c) overreads the altitude in case of a side-slip to the left and displays the correct information during symmetric flight.
- d) underreads the altitude.

Answer ;

- c) overreads the altitude in case of a side-slip to the left and displays the correct information during symmetric flight.
- 

0364. In case of excessive descent rate , the GPWS generates the following aural warnings ;

- a) "DON'T SINK , DON'T SINK"
- b) "TERRAIN , TERRAIN" followed by "WHOOOP WHOOOP PULL UP" ( twice )
- c) "SINK RATE , SINK RATE" followed by "WHOOOP WHOOOP PULL UP" ( twice )
- d) "TOO LOW , TERRAIN" ( twice ) followed by "TOO LOW GEAR" ( twice )

Answer ;

- c) "SINK RATE , SINK RATE" followed by "WHOOOP WHOOOP PULL UP" ( twice )
- 

0365. In case of impending stall , the flight warning system ( FWS ) generates ;

- a) an advisory message.
- b) an urgency message.
- c) a warning message.
- d) a caution message.

Answer ;

- c) a warning message.
-

0366. In case of static blockage , the airspeed indicator ;
- a) under-reads in descent only.
  - b) over-reads in climb and under-reads in descent.
  - c) over-reads in climb only.
  - d) under-reads in climb and over-reads in descent.

Answer ;

d) under-reads in climb and over-reads in descent.

---

0367. In low altitude radio altimeters , the height measurement ( above the ground ) is based upon ;
- a) a wave transmission , for which the frequency shift by DOPPLER effect after ground reflection is measured.
  - b) a frequency modulation wave , for which the frequency variation between the transmitted wave and the received wave after ground reflection is measured.
  - c) a pulse transmission , for which time between transmission and reception is measured on a circular scanning screen.
  - d) a triangular amplitude modulation wave , for which modulation phase shift between transmitted and received waves after ground reflection is measured.

Answer ;

b) a frequency modulation wave , for which the frequency variation between the transmitted wave and the received wave after ground reflection is measured.

---

0368. In order to align an inertial reference system ( IRS ) , it is required to insert the local geographical coordinates. This enables the IRS to ;

- a) compare the longitude it finds with that entered by the operator.
- b) find true north.
- c) initialise the FMS flight plan.
- d) compare the latitude it finds with that entered by the operator.

Answer ;

d) compare the latitude it finds with that entered by the operator.

---

0369. In order to align an IRS , it is required to insert the local geographical coordinates. This enables the IRS to ;
- 1. compare the computed latitude with the one entered by the pilot.
  - 2. compare the computed longitude with the one entered by the pilot.
  - 3. know the longitude.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 3.
- c) 1 , 2.
- d) 2 , 3.

Answer ;

b) 1 , 3.

---

0370. In order to know in which mode the auto-throttles are engaged , the crew will check the ;

- a) ND ( Navigation Display ).
- b) PFD ( Primary Flight Display ).
- c) throttles position.
- d) TCC ( Thrust Control Computer ).

Answer ;

b) PFD ( Primary Flight Display ).

---

0371. In order to measure temperature the cylinder head temperature ( CHT ) gauge utilises a ;

- a) bourdon tube.
- b) ratiometer circuit.
- c) thermocouple consisting of two dissimilar metals.
- d) wheatstone bridge circuit.

Answer ;

c) thermocouple consisting of two dissimilar metals.

---

0372. In standard atmosphere at sea level , the EAS is ;

- a) lower than the TAS.
- b) higher than the TAS.
- c) equal to the TAS.
- d) independent of the TAS.

Answer ;

c) equal to the TAS.

---

0373. In standard atmosphere , when climbing at constant CAS ;
- a) TAS decreases and Mach number increases.
  - b) TAS and Mach number increase.
  - c) TAS increases and Mach number decreases.
  - d) TAS and Mach number decrease.

Answer ;

b) TAS and Mach number increase.

---

0374. In standard atmosphere , when descending at constant CAS ;
- a) TAS first increases and then remains constant above the tropopause.
  - b) TAS increases.
  - c) TAS remains constant.
  - d) TAS decreases.

Answer ;

d) TAS decreases.

---

0375. In the absence of position and instrument errors ;
- a) CAS = EAS.
  - b) IAS = CAS.
  - c) CAS = TAS.
  - d) IAS = EAS.

Answer ;

b) IAS = CAS.

---

0376. In the absence of position and instrument errors , CAS is equal to ;
- a) IAS and EAS.
  - b) IAS.
  - c) TAS.
  - d) EAS.

Answer ;

b) IAS.

---

0377. In the absence of position and instrumental errors , IAS is equal to ;
- a) KAS.
  - b) EAS.
  - c) CAS.
  - d) TAS.

Answer ;

c) CAS.

---

0378. In the building principle of a gyroscope , the best efficiency is obtained through the concentration of the mass ;
- a) on the periphery and with a low rotation speed.
  - b) on the periphery and with a high rotation speed.
  - c) close to the axis and with a low rotation speed.
  - d) close to the axis and with a high rotation speed.

Answer ;

b) on the periphery and with a high rotation speed.

---

0379. In the case of altitude loss during the initial climb after take off , GPWS generates an aural alert ;
- a) "TERRAIN AHEAD".
  - b) "TOO LOW FLAPS".
  - c) "DON'T SINK".
  - d) "TOO LOW GEAR".

Answer ;

c) "DON'T SINK".

---

0380. In the event of a conflict , the TCAS 2 ( Traffic Collision alert and Avoidance System ) presents warnings to the crew such as ;

- a) "Glide Slope".
- b) "Too low terrain".
- c) "Climb" or "Descent".
- d) "Turn left" or "Turn right".

Answer ;

c) "Climb" or "Descent".

---

0381. In the event of a failure , a fail-operational flight control system will operate as a ;

- a) fail-safe system.
- b) fail-passive system.
- c) fail-redundant system.
- d) fail-operational hybrid system.

Answer ;

b) fail-passive system.

---

0382. In the following formula  $EAS = CAS \times K$  , the compressibility factor K ;

- 1. may be greater than 1
- 2. is always lower or equal to 1
- 3. depends on Mach number only
- 4. depends on pressure altitude only

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 4.
- c) 1 , 3.
- d) 2 , 4.

Answer ;

a) 2 , 3.

---

0383. In the gyromagnetic compass , the heading information from the flux valve is sent to the ;

- a) amplifier.
- b) erector system.
- c) error detector.
- d) heading indicator.

Answer ;

c) error detector.

---

0384. In the northern hemisphere , during deceleration following a landing in a northerly direction , a direct reading magnetic compass indicates ;

- a) an apparent turn to the west.
- b) no apparent turn.
- c) an apparent turn to the south.
- d) an apparent turn to the east.

Answer ;

b) no apparent turn.

---

0385. In the northern hemisphere , during deceleration following a landing in a southerly direction , a direct reading magnetic compass indicates ;

- a) no apparent turn.
- b) an apparent turn to the east.
- c) an apparent turn to the west.
- d) an apparent turn to the north.

Answer ;

a) no apparent turn.

---

0386. In the northern hemisphere , during a take-off run in a easterly direction , a direct reading magnetic compass indicates ;

- a) no apparent turn.
- b) an apparent turn to the north.
- c) an apparent turn to the south.
- d) an apparent turn to approximately the heading 100°.

Answer ;

b) an apparent turn to the north.

---

0387. In the northern hemisphere , during a take-off run in a westerly direction , a direct reading magnetic compass indicates ;

- a) an apparent turn to approximately the heading 255°.
- b) an apparent turn to the north.
- c) no apparent turn.
- d) an apparent turn to the south.

Answer ;

b) an apparent turn to the north.

---

0388. Indication of Mach number is obtained from ;
- a) A kind of echo sound comparing velocity of sound with indicated speed.
  - b) Indicated speed ( IAS ) compared with true air speed ( TAS ) from the air data computer.
  - c) Indicated speed and altitude using a speed indicator equipped with an altimeter type aneroid.
  - d) An ordinary airspeed indicator scaled for Mach numbers instead of knots.

Answer ;

- c) Indicated speed and altitude using a speed indicator equipped with an altimeter type aneroid.
- 

0389. Lights indicating a hazard which may require immediate corrective action must be ;

- a) yellow.
- b) amber.
- c) magenta.
- d) red.

Answer ;

- d) red.
- 

0390. Lights indicating the possible need for future corrective action must be ;

- a) red.
- b) red and white striped.
- c) amber.
- d) magenta.

Answer ;

- c) amber.
- 

0391. Mach Trim is a device to compensate for ;

- a) the effects of fuel transfer between the main tanks and the tank located in the horizontal tail.
- b) backing of the aerodynamic centre at high Mach numbers by moving the elevator to nose up.
- c) weight reduction resulting from fuel consumption during the cruise.
- d) the effects of temperature variation during a climb or descent at constant Mach.

Answer ;

- b) backing of the aerodynamic centre at high Mach numbers by moving the elevator to nose up.
- 

0392. Machmeter readings are subject to ;

- a) density error.
- b) setting error.
- c) temperature error.
- d) position pressure error.

Answer ;

- d) position pressure error.
- 

0393. Magnetic compass errors are ;

- a) parallax errors due to compass rose oscillations.
- b) due to Schuler oscillations.
- c) due to the lateral gusts which occur when the airplane is heading eastward or westward.
- d) about north change , depending on the bank angle and magnetic heading.

Answer ;

- d) about north change , depending on the bank angle and magnetic heading.
- 

0394. Maintaining CAS and flight level constant , a fall in ambient temperature results in ;

- a) higher TAS because air density increases.
- b) lower TAS because air density increases.
- c) lower TAS because air density decreases.
- d) higher TAS because air density decreases.

Answer ;

- b) lower TAS because air density increases.
- 

0395. Mode "Localizer ARM" active on Flight Director means ;

- a) System is armed for localizer approach and coupling will occur upon capturing centre line.
- b) Coupling has occurred and system provides control data to capture the centreline.
- c) Localizer is armed and coupling will occur when flag warning disappears.
- d) Localizer ALARM , making localizer approach not authorized.

Answer ;

- a) System is armed for localizer approach and coupling will occur upon capturing centre line.
-

0396. Modern low altitude radio altimeters emit waves in the following frequency band ;

- a) HF ( High Frequency ).
- b) VLF ( Very Low Frequency ).
- c) UHF ( Ultra High Frequency ).
- d) SHF ( Super High Frequency ).

Answer ;

d) SHF ( Super High Frequency ).

---

0397. On a display of a TCAS 2 ( Traffic alert and Collision Avoidance System ) , the traffic being cause of a resolution advisory ( RA ) is ;

- a) a red full square.
- b) an amber solid circle.
- c) a white or cyan empty lozenge.
- d) a white or cyan solid lozenge.

Answer ;

a) a red full square.

---

0398. On a large aeroplane and according to the CS25 ; the regulatory margin between the stall and stall warning is ;

- a) 5 kt or 5% of the CAS whichever is the lower.
- b)  $V_{S0} + 5$  kt.
- c)  $V_{S0} + 5\%$ .
- d) 5 kt or 5% of the CAS whichever is the greater.

Answer ;

d) 5 kt or 5% of the CAS whichever is the greater.

---

0399. On a large aeroplane and according to the CS25 ; the stall warning system must provide an alarm with sufficient margin to prevent inadvertent stalling ;

- a) with the flaps down and gear up only.
- b) with the flaps and landing gear in any normal position.
- c) with the flaps fully extended and gear down only.
- d) with the flaps not retracted only , whatever the position of the landing gear.

Answer ;

b) with the flaps and landing gear in any normal position.

---

0400. On a large aeroplane and according to the CS25 ; when the speed is reduced , the stall warning must begin ;

- a) at the stall speed  $\pm 5\%$ .
- b) at a speed exceeding the stall speed by not less than 5 kt or 5% CAS.
- c) at a speed exceeding the stall speed by not less than 10 kt or 10% CAS.
- d) exactly at the stall speed.

Answer ;

b) at a speed exceeding the stall speed by not less than 5 kt or 5% CAS.

---

0401. On a large aeroplane , the computer of a stall warning system receives information about the ;

- 1. angle of attack.
- 2. engine RPM.
- 3. configuration.
- 4. pitch and bank attitude.
- 5. sideslip.

The combination that regroups all of the correct statements is ;

- a) 1 , 5.
- b) 1 , 3.
- c) 1 , 2 , 3 , 4.
- d) 1 , 2 , 3 , 4 , 5.

Answer ;

b) 1 , 3.

---

**0402.** On a large aeroplane , the input data of the stall warning system are ;

1. angle of attack.
2. weight.
3. bank angle.
4. configuration ( slats / flaps ).

The combination that regroups all of the correct statements is ;

- a) 1 , 2.
- b) 2 , 3 , 4.
- c) 1 , 2 , 4.
- d) 1 , 4.

**Answer ;**

- d) 1 , 4.**
- 

**0403.** On a modern aircraft , the flight director modes are displayed on the ;

- a) upper strip of the ND ( Navigation Display ).
- b) upper strip of the PFD ( Primary Flight Display ).
- c) control panel of the flight director only.
- d) upper strip of the ECAM ( Electronic Centralized A/C Management ).

**Answer ;**

- b) upper strip of the PFD ( Primary Flight Display ).**
- 

**0404.** On a modern transport category airplane , the engagement of the automatic pilot is checked on the display of ;

- a) the PFD ( Primary Flight Display ).
- b) the ECAM ( Electronic Centralized Aircraft Monitoring ) left screen.
- c) the ND ( Navigation Display ) of the pilot in command.
- d) the ND ( Navigation Display ).

**Answer ;**

- a) the PFD ( Primary Flight Display ).**
- 

**0405.** On a TCAS 2 ( Traffic Collision Avoidance System ) , a corrective “resolution advisory” ( RA ) is a “resolution advisory” ;

- a) asking the pilot to modify the heading of his aircraft.
- b) which does not require any action from the pilot but on the contrary asks him not to modify his current vertical speed rate.
- c) asking the pilot to modify the speed of his aircraft.
- d) asking the pilot to modify effectively the vertical speed of his aircraft.

**Answer ;**

- d) asking the pilot to modify effectively the vertical speed of his aircraft.**
- 

**0406.** On a TCAS 2 ( Traffic Collision Avoidance System ) , a preventive resolution advisory ;

- a) advises the pilot to keep the vertical speed within given limits.
- b) asks the pilot to modify the heading of his aircraft.
- c) asks the pilot to modify effectively the vertical speed of his aircraft.
- d) asks the pilot to modify the speed of his aircraft.

**Answer ;**

- a) advises the pilot to keep the vertical speed within given limits.**
- 

**0407.** On a TCAS 2 ( Traffic Collision Avoidance System ) , the preventive “resolution advisory” ( RA ) is a “resolution advisory” ;

- a) that advises the pilot to avoid certain deviations from the current vertical rate but does not require any change to be made to that rate.
- b) asking the pilot to modify the heading of his aircraft.
- c) asking the pilot to modify effectively the vertical speed of his aircraft.
- d) asking the pilot to modify the speed of his aircraft.

**Answer ;**

- a) that advises the pilot to avoid certain deviations from the current vertical rate but does not require any change to be made to that rate.**
- 

**0408.** On the airspeed indicator of a twin-engine aeroplane , the blue radial line corresponds to the ;

- a) minimum ground control speed.
- b) single-engine holding speed.
- c) best single-engine rate of climb.
- d) minimum air control speed.

**Answer ;**

- c) best single-engine rate of climb.**
-

- 0409.** On the display of a TCAS 2 ( Traffic alert and Collision Avoidance System ) , a proximate traffic is represented by ;
- a) an amber solid circle.
  - b) a white or cyan solid lozenge.
  - c) a red full square.
  - d) a white or cyan empty lozenge.

**Answer ;**

- b) a white or cyan solid lozenge.**
- 

- 0410.** On the display of a TCAS 2 ( Traffic alert and Collision Avoidance System ) , a resolution advisory ( RA) is represented by ;
- a) a red full square.
  - b) an amber solid circle.
  - c) a white or cyan empty lozenge.
  - d) a white or cyan solid lozenge.

**Answer ;**

- a) a red full square.**
- 

- 0411.** On the display of a TCAS 2 ( Traffic alert and Collision Avoidance System ) , a traffic advisory ( TA ) is represented by ;
- a) an amber solid circle.
  - b) a red full square.
  - c) a white or cyan empty lozenge.
  - d) a white or cyan solid lozenge.

**Answer ;**

- a) an amber solid circle.**
- 

- 0412.** On the display of a TCAS 2 ( Traffic alert and Collision Avoidance System ) , the traffic being cause of a resolution advisory ( RA ) is represented by ;
- a) a white or cyan empty lozenge.
  - b) a white or cyan solid lozenge.
  - c) a red full square.
  - d) an amber solid circle.

**Answer ;**

- c) a red full square.**
- 

- 0413.** On the ground , during a left turn , the turn indicator indicates ;
- a) needle in the middle , ball to the right.
  - b) needle to the left , ball to the left.
  - c) needle to the left , ball to the right.
  - d) needle in the middle , ball to the left.

**Answer ;**

- c) needle to the left , ball to the right.**
- 

- 0414.** On the ground , during a right turn , the turn indicator indicates ;
- a) needle to the right , ball to the right.
  - b) needle in the middle , ball to the right.
  - c) needle to the right , ball to the left.
  - d) needle in the middle , ball to the left.

**Answer ;**

- c) needle to the right , ball to the left.**
- 

- 0415.** On the INS control panel , the rotary knob can be selected to OFF , NAV or ATT positions. The correct statement is ;
- a) ATT is the normal system setting.
  - b) NAV is the normal system setting , the OFF position is the back up position in case of failure of the navigation function.
  - c) ATT is the normal system setting , the NAV position inhibits the attitude data.
  - d) NAV is the normal system setting , the ATT position is the back up position in case of failure of the navigation function.

**Answer ;**

- d) NAV is the normal system setting , the ATT position is the back up position in case of failure of the navigation function.**
-



- 0416.** On the navigation display ( ND ) of an EFIS equipped aircraft , the colours used are ;
- a) magenta or white for the active waypoint ( TO waypoint ) , green for light precipitation.
  - b) amber or white for the active waypoint ( TO waypoint ) , yellow for high precipitation.
  - c) white or blue for the active waypoint ( TO waypoint ) , red for medium precipitation.
  - d) magenta or red for the active waypoint ( TO waypoint ) , amber for medium precipitation.

**Answer ;**

- a)** magenta or white for the active waypoint ( TO waypoint ) , green for light precipitation.
- 

- 0417.** One of the disadvantages of an electronic tachometer is the ;
- a) influence of temperature on the indication.
  - b) generation of spurious signals at the commutator.
  - c) high influence of line resistance on the indication.
  - d) necessity of a power supply.

**Answer ;**

- d)** necessity of a power supply.
- 

**0418.** One of the errors inherent in a ring laser gyroscope occurs at low input rotation rates tending towards zero when a phenomenon known as “lock-in” is experienced. What is the name of the technique , effected by means of a piezo-electric motor , that is used ;

- a) dither.
- b) zero drop.
- c) beam lock.
- d) cavity rotation.

**Answer ;**

- a)** dither.
- 

- 0419.** Parallax error is ;
- a) due to temperature effect.
  - b) due to the effect of aircraft accelerations.
  - c) due to pressure effect.
  - d) a reading error.

**Answer ;**

- d)** a reading error.
- 

- 0420.** Should a defect arise in the stability augmentation system ( SAS ) during flight ;
- a) the pilot will be advised by an amber warning.
  - b) an audio alarm will attract the pilots attention to check the AFCS control panel.
  - c) the control stick will move to a pre-set patum causing the pilot to revert to manual control.
  - d) the pilot is not informed unless the second channel fails also.

**Answer ;**

- a)** the pilot will be advised by an amber warning.
- 

**0421.** Some of the FMS have a navigation mode called Dead Reckoning mode ( DR ) , computing airspeed , heading , wind data , ground speed and time. This mode is ;

- a) a back up navigation mode to compute a FMS position when the other navigation sensors are no longer operating.
- b) an operating mode used to intercept radials To or From a flight plan waypoint.
- c) the normal navigation mode for FMS which do not use inertial navigation systems INS to compute the aircraft position.
- d) a navigation mode used to monitor the FMS position.

**Answer ;**

- a)** a back up navigation mode to compute a FMS position when the other navigation sensors are no longer operating.
- 

- 0422.** Sound propagates through the air at a speed which only depends on ;
- a) pressure.
  - b) temperature and the pressure.
  - c) density.
  - d) temperature.

**Answer ;**

- d)** temperature.
-

0423. Speed of sound is proportional to ;
- a) the square of the TAS.
  - b) the square root of the absolute temperature.
  - c) the square root of the TAS.
  - d) the square of the absolute temperature.

Answer ;

**b) the square root of the absolute temperature.**

---

0424. TAS can be obtained from the following data ;

- a) EAS and pressure altitude.
- b) CAS and density altitude.
- c) EAS and density altitude.
- d) CAS and pressure altitude.

Answer ;

**c) EAS and density altitude.**

---

0425. TAS is equal to ;

- a) CAS corrected for density error only.
- b) EAS corrected for compressibility and density errors.
- c) CAS corrected for compressibility and density errors.
- d) EAS corrected for compressibility error only.

Answer ;

**c) CAS corrected for compressibility and density errors.**

---

0426. TAS is equal to ;

- a) CAS corrected for compressibility error.
- b) CAS corrected for density error.
- c) EAS corrected for density error.
- d) EAS corrected for compressibility error.

Answer ;

**c) EAS corrected for density error.**

---

0427. TAS is equal to ;

- a) IAS corrected for compressibility and density errors only.
- b) IAS corrected for instrument , position , compressibility and density errors.
- c) CAS corrected for instrument , compressibility and density errors.
- d) CAS corrected for instrument , position , compressibility and density errors.

Answer ;

**b) IAS corrected for instrument , position , compressibility and density errors.**

---

0428. TAS is obtained from EAS by correcting for ;

- a) compressibility error.
- b) density error.
- c) instrument error.
- d) position and instrument errors.

Answer ;

**b) density error.**

---

0429. TCAS 2 ( Traffic Collision Avoidance System ) uses for its operation ;

- a) the replies from the transponders of other aircrafts.
- b) the echoes of collision avoidance radar system especially installed on board.
- c) the echoes from the ground air traffic control radar system.
- d) both the replies from the transponders of other aircraft and the ground-based radar echoes.

Answer ;

**a) the replies from the transponders of other aircrafts.**

---

0430. The ACARS allows air/ground datalink communications for ;

- a) Airline Operational and Public Communications ( AOC & PC ).
- b) Airline Operational and Air Traffic Communications ( AOC & ATC ).
- c) Air Traffic Control Communications only ( ATC ).
- d) Airline Operational Communications only ( AOC ).

Answer ;

**b) Airline Operational and Air Traffic Communications ( AOC & ATC ).**

---

0431. The accelerometers of a strap-down Inertial Reference System are in line with ;
- a) the local meridian and parallel.
  - b) the geographical directions.
  - c) the aircraft axes.
  - d) the local vertical and the local meridian.

Answer ;

c) the aircraft axes.

---

0432. The accuracy of the altitude computed by a stand alone inertial system ;
- a) is bounded.
  - b) decreases proportionally with flight time.
  - c) decreases exponentially with flight time.
  - d) is poor at the beginning of the flight.

Answer ;

c) decreases exponentially with flight time.

---

0433. The ADC uses the following parameters as input data ;
- a) total pressure , static pressure , TAT , EGT.
  - b) static pressure , total pressure , TAT.
  - c) TAS , baro altitude , TAT.
  - d) Mach number , bro altitude , CAS.

Answer ;

b) static pressure , total pressure , TAT.

---

0434. The ADC uses the following parameters as input data ;
- a) static pressure , total pressure , total air temperature.
  - b) TAS , CAS , altitude , total air temperature.
  - c) TAS , altitude , vertical speed , SAT.
  - d) static pressure and total pressure only.

Answer ;

a) static pressure , total pressure , total air temperature.

---

0435. The ADS is a FANS application performed ;
- a) manually , the pilot has to define the set of data to downlink and the type of report contract.
  - b) semi automatically , the pilot has to define the set of data to downlink but cannot define the type of report contract.
  - c) semi automatically , the pilot cannot define the set of data to downlink but can define the type of report contract.
  - d) automatically , without any crew action to define the set of data to downlink or the type of report contract.

Answer ;

d) automatically , without any crew action to define the set of data to downlink or the type of report contract.

---

0436. The ADS is a FANS application which consists in ;
- a) broadcasting aircraft position and intents to the other aircraft in the vicinity.
  - b) establishing automatic air/ground connection to verify if the datalink communication can be performed.
  - c) connecting the aircraft to the appropriate ATC center.
  - d) sending automatically aircraft surveillance data to the air traffic controller.

Answer ;

d) sending automatically aircraft surveillance data to the air traffic controller.

---

0437. The advantages of a DC generator tachometer are ;
1. easy transmission of the information.
  2. independence of the information relative to the airborne electrical power supply.
  3. freedom from any spurious current due to the commutator.

The combination regrouping all the correct statements is ;

- a) 2 , 3.
- b) 1 , 3.
- c) 1 , 2.
- d) 1 , 2 , 3.

Answer ;

c) 1 , 2.

---

**0438.** The advantages of a float type fuel gauge are(is) ;

1. easy construction.
2. independence of indications with regard to aircraft attitude.
3. independence of indications with regard to the accelerations.
4. independence of indications with regard to temperature variations.

The combination that regroups all of the correct statements is ;

- a) 1 , 4.
- b) 1 , 2 , 3 , 4.
- c) 1.
- d) 2 , 3 , 4.

**Answer ;**

**c) 1.**

---

**0439.** The advantages of an “electric” fuel ( float ) gauge are ;

1. easy construction.
2. independence of indications with regard to airplane attitude.
3. independence of indications with regard to the accelerations.
4. independence of indications with regard to temperature variations.
5. independence of indications with regard to vibrations.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 , 4 , 5.
- b) 2 , 3 , 4.
- c) 2 , 3 , 4 , 5.
- d) 1.

**Answer ;**

**d) 1.**

---

**0440.** The advantages of an electrical induction tachometer are ;

1. the display is not sensitive to line resistance.
2. the measurement is independent of aircraft power supply.
3. the measurement is independent of temperature variations.
4. the option to use without restriction several indicators connected in parallel to a single transmitter.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4.
- b) 1 , 2 , 3 , 4.
- c) 1 , 3 , 4.
- d) 1 , 2 , 4.

**Answer ;**

**d) 1 , 2 , 4.**

---

**0441.** The advantages of single-phase AC generator tachometer are ;

1. the suppression of spurious signals due to a DC generator commutator.
2. the importance of line resistance on the information value.
3. the independence of the information in relation to the airborne electrical power supply.
4. the ease of transmission of the information.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4.
- b) 2 , 4.
- c) 1 , 3.
- d) 1 , 2 , 3 , 4.

**Answer ;**

**c) 1 , 3.**

---

**0442.** The aircraft radio equipment which emits on a frequency of 4400 MHz is the ;

- a) weather radar.
- b) primary radar.
- c) high altitude radio altimeter.
- d) radio altimeter.

**Answer ;**

**d) radio altimeter.**

---

0443. The airplane outside air temperature “probe” measures the ;
- a) “total” air temperature minus kinetic heating effects in order to obtain the static temperature.
  - b) “total” air temperature minus compressibility effects in order to obtain the static temperature.
  - c) “static” air temperature minus kinetic heating effects in order to obtain the total temperature.
  - d) “static” air temperature minus compressibility effects in order to obtain the total temperature.

Answer ;

- a) “total” air temperature minus kinetic heating effects in order to obtain the static temperature.
- 

0444. The “airspeed hold” mode can be engaged and maintained during ;

- a) turns only.
- b) climbs , descents , turns and power changes.
- c) climbs , descents and power changes only.
- d) climbs and descents only.

Answer ;

- b) climbs , descents , turns and power changes.
- 

0445. The airspeed indicator of a twin-engine aircraft comprises different sectors and colour marks. The blue line corresponds to the ;

- a) minimum control speed , or  $V_{MC}$ .
- b) maximum speed in operations , or  $V_{MO}$ .
- c) speed not to be exceeded , or  $V_{NE}$ .
- d) optimum climbing speed with one engine inoperative , or  $V_Y$ .

Answer ;

- d) optimum climbing speed with one engine inoperative , or  $V_Y$ .
- 

0446. The alignment of a gyro stabilized inertial platform consists in positionning the platform relative to ;

- a) the vertical axis and true north.
- b) the pitch and roll axis.
- c) the pitch axis only.
- d) the roll axis only.

Answer ;

- a) the vertical axis and true north.
- 

0447. The alignment of a strapdown inertial system consists in ;

- a) measuring the earth rotation and local gravitation to position the reference trihedron.
- b) positioning the accelerometers.
- c) positioning the gyroscopes and accelerometers relative to the fuselage axis.
- d) positioning the platform relative to the local vertical and true north.

Answer ;

- a) measuring the earth rotation and local gravitation to position the reference trihedron.
- 

0448. The alignment of an inertial system can be successfully performed ;

- a) in all phases of flight outside areas of turbulence.
- b) when the aircraft is taxiing.
- c) in all phases of flight.
- d) when the aircraft is stationary.

Answer ;

- d) when the aircraft is stationary.
- 

0449. The alignment phase of a gyro stabilized platform consists in ;

- a) levelling the platform and determining its orientation.
- b) aligning the platform axis with the aircraft pitch axis only.
- c) aligning the platform axis with the aircraft axis ( pitch , roll , yaw ).
- d) aligning the platform axis with the aircraft roll axis only.

Answer ;

- a) levelling the platform and determining its orientation.
-

0450. The alignment sequence of an IRS consists of ;
1. searching for the local vertical.
  2. searching for the true north.
  3. searching for the latitude.
  4. searching for the longitude.
  5. comparison between the computed longitude and the one entered by the pilot.
  6. comparison between the computed latitude and the one entered by the pilot.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4 , 5 , 6.
- b) 1 , 2 , 3 , 6.
- c) 1 , 2 , 4 , 5.
- d) 3 , 4.

Answer ;

- b) 1 , 2 , 3 , 6.
- 

0451. The alignment time of a strapdown inertial system takes longer time when the aircraft is ;

- a) close to the equator.
- b) at a high longitude.
- c) at a location where the magnetic variation is greater than 15 degrees.
- d) at a high latitude.

Answer ;

- d) at a high latitude.
- 

0452. The alternate static source is used ;

- a) when the drain holes freeze.
- b) when the static ports become blocked.
- c) to compensate the hysteresis of the aneroid capsule.
- d) to compensate the static pressure error.

Answer ;

- b) when the static ports become blocked.
- 

0453. The alternate static source of a light non-pressurized aeroplane is located in the flight deck , when used ;

- a) it has no influence on airspeed indicator reading.
- b) the airspeed indicator indicates a consistent decreasing speed.
- c) the airspeed indicator tends to over-read.
- d) the airspeed indicator tends to under-read.

Answer ;

- c) the airspeed indicator tends to over-read.
- 

0454. The alternate static source of a light non-pressurized aeroplane is located in the flight deck , when used , the altimeter ;

- a) is blocked.
- b) tends to over-read.
- c) indicates zero.
- d) tends to under-read.

Answer ;

- b) tends to over-read.
- 

0455. The alternate static source of a light non-pressurized aircraft is located in the flight deck , as the alternate static source is opened , the vertical airspeed indicator may ;

- a) indicate a slight momentary descent.
- b) be blocked.
- c) indicate a high rate of descent.
- d) indicate a momentary climb.

Answer ;

- d) indicate a momentary climb.
- 

0456. The alternate static source of a light non-pressurized aircraft is located in the flight deck , when used , the static pressure sensed is likely to be ;

- a) lower than ambient pressure if QNH is greater than 1013 hPa.
- b) lower than ambient pressure due to aerodynamic suction.
- c) higher than ambient pressure due to aerodynamic suction.
- d) higher than ambient pressure if QNH is greater than 1013 hPa.

Answer ;

- b) lower than ambient pressure due to aerodynamic suction.
-

**0457.** The alternate static source of non-pressurized aircraft is located in the cabin. As the alternate static source is opened , the vertical airspeed indicator may ;

- a) show a slight momentary descent.
- b) show a momentary climb.
- c) be blocked.
- d) show a high rate of descent.

**Answer ;**

- b) show a momentary climb.**
- 

**0458.** The altimeter consists of one or several aneroid capsules located in a sealed casing. The pressures in the aneroid capsule (i) and casing (ii) are respectively ;

- a) (i) total pressure (ii) static pressure.
- b) (i) vacuum ( or a very low pressure ) (ii) static pressure.
- c) (i) static pressure at time t (ii) static pressure at time t - dt.
- d) (i) static pressure (ii) total pressure.

**Answer ;**

- b) (i) vacuum ( or a very low pressure ) (ii) static pressure.**
- 

**0459.** The altimeter indicates true altitude ;

- a) when the temperature on the ground is +15°C with a lapse rate of 2°C per 1000 feet , and correct QFE is set.
- b) when the temperature on the ground is +15°C with a lapse rate of 2°C per 1000 feet , and correct QNH is set.
- c) in ISA conditions only.
- d) when pressure at mean sea level is 10135,25 hPa , with ground temperature of +15°C and density equal to 1,225 kg/m<sup>3</sup>.

**Answer ;**

- c) in ISA conditions only.**
- 

**0460.** The altimeter is subject to static pressure error. This error results from ;

- a) frictions inside the instrument.
- b) imperfect elasticity of the aneroid capsules.
- c) incorrect pressure sensing caused by disturbed airflow around the static ports.
- d) cabin pressure slightly lower than outside air pressure due to airflow over the fuselage.

**Answer ;**

- c) incorrect pressure sensing caused by disturbed airflow around the static ports.**
- 

**0461.** The altimeter is subject to static pressure error. This error varies according to ;

- a) angle of attack and OAT.
- b) TAS and altimeter setting.
- c) TAS and OAT.
- d) TAS and angle of attack.

**Answer ;**

- d) TAS and angle of attack.**
- 

**0462.** The altimeter is subject to the position error , this error varies substantially with the ;

- a) Mach number.
- b) deformation of the aneroid capsule.
- c) flight time at high altitude.
- d) OAT.

**Answer ;**

- a) Mach number.**
- 

**0463.** The altimeter is supplied with ;

- a) static pressure.
- b) total pressure.
- c) dynamic pressure.
- d) differential pressure.

**Answer ;**

- a) static pressure.**
-

**0464.** The altimeter of your aircraft indicates 10000 ft with a subscale-setting of 1013.25 mb. OAT is +5°C. The pressure altitude of the aircraft is ;

- a) 9600 ft.
- b) 697 hPa.
- c) 10400 ft.
- d) 10000 ft.

**Answer ;**

**d) 10000 ft.**

**Açıklama ;**

Altimetreye standart basınç değeri olan 1013.25 set edildiğinde bu bize basınç irtifasını ( pressure altitude ) verir. Dolayısıyla 10000 feet olarak altimetrede okunan değer basınç irtifası değeridir. Burada sıcaklığın bir önemi yoktur.

**0465.** The altimeter of your aircraft indicates 11000 ft with a subscale-setting of 1013.25 mb. QNH is 1023 hPa. OAT is +3°C. The pressure altitude of the aircraft is ;

- a) 670 hPa.
- b) 11740 ft.
- c) 10260 ft.
- d) 11000 ft.

**Answer ;**

**d) 11000 ft.**

**Açıklama ;**

Altimetreye standart basınç değeri olan 1013.25 set edildiğinde bu bize basınç irtifasını ( pressure altitude ) verir. Dolayısıyla 11000 feet olarak altimetrede okunan değer basınç irtifası değeridir. Burada sıcaklığın veya QNH'in bir önemi yoktur.

**0466.** The altimeter of your aircraft indicates 12000 ft with a subscale-setting of 1013.25 mb. QNH is 999 hPa. The pressure altitude of the aircraft is ;

- a) 12420 ft.
- b) 12000 ft.
- c) 11580 ft.
- d) 644 hPa.

**Answer ;**

**b) 12000 ft.**

**Açıklama ;**

Altimetreye standart basınç değeri olan 1013.25 set edildiğinde bu bize basınç irtifasını ( pressure altitude ) verir. Dolayısıyla 12000 feet olarak altimetrede okunan değer basınç irtifası değeridir. Burada QNH'in bir önemi yoktur.

**0467.** The altimeter of your aircraft indicates 15000 ft with a subscale-setting of 1013.25 mb. OAT is -21°C. The pressure altitude of the aircraft is ;

- a) 572 hPa.
- b) 14640 ft.
- c) 15000 ft.
- d) 15360 ft.

**Answer ;**

**c) 15000 ft.**

**Açıklama ;**

Altimetreye standart basınç değeri olan 1013.25 set edildiğinde bu bize basınç irtifasını ( pressure altitude ) verir. Dolayısıyla 15000 feet olarak altimetrede okunan değer basınç irtifası değeridir. Burada sıcaklığın bir önemi yoktur.

**0468.** The altimeter of your aircraft indicates 16000 ft with a subscale-setting of 1013.25 mb. QNH is 993 hPa. OAT is -3°C. The pressure altitude of the aircraft is ;

- a) 14300 ft
- b) 17700 ft
- c) 549 hPa.
- d) 16000 ft.

**Answer ;**

**d) 16000 ft.**

**Açıklama ;**

Altimetreye standart basınç değeri olan 1013.25 set edildiğinde bu bize basınç irtifasını ( pressure altitude ) verir. Dolayısıyla 16000 feet olarak altimetrede okunan değer basınç irtifası değeridir. Burada sıcaklığın veya QNH'in bir önemi yoktur.



0469. The altitude alerting system ;

- a) alerts the flight crew upon approaching a pre-selected altitude.
- b) generates a signal once the aircraft is steady at the pre-selected altitude.
- c) alerts the flight crew in case of ground proximity.
- d) alerts the flight crew in case of an autopilot disengagement.

Answer ;

- a) alerts the flight crew upon approaching a pre-selected altitude.
- 

0470. The angle of attack transmitter placed laterally on the forward part of the fuselage supplies an electrical signal which can indicate the angular position of ;

- 1. a specific slaved pitot probe.
- 2. a vane detector.
- 3. a conical slotted probe.

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 1 , 3.
- c) 2 , 3.
- d) 1 , 2 , 3.

Answer ;

- c) 2 , 3.
- 

0471. The apparent wander of a directional gyro is  $15^\circ/h$  ;

- a) at the latitude  $45^\circ$ .
- b) at the North pole.
- c) at the equator.
- d) at the latitude  $30^\circ$ .

Answer ;

- b) at the North pole.
- 

0472. The artificial horizon uses a gyroscope with ;

( Note : the degree(s) of freedom of a gyro does not take into account its rotor spin axis. )

- a) two degrees of freedom , and its rotor spin axis is continuously maintained to local vertical by an automatic erecting system.
- b) one degree of freedom , and its rotor spin axis is continuously maintained to local vertical by an automatic erecting system.
- c) one degree of freedom , and its rotor spin axis is continuously maintained in the horizontal plane by an automatic erecting system.
- d) two degrees of freedom , and its rotor spin axis is continuously maintained in the horizontal plane by an automatic erecting system.

Answer ;

- a) two degrees of freedom , and its rotor spin axis is continuously maintained to local vertical by an automatic erecting system.
-

0473. ( For this question use annex 022-9771A )

**Table d'Atmosphère Standard Internationale**

| ALTITUDE<br>Pieds | TEMPERATURE   |        | PRESSION    |       |          |          | RAPPORT<br>de<br>PRESSION<br>$\delta = P/P_0$ | IDENSITE<br>RELATIVE<br>$\delta = \rho/\rho_0$ | $\sqrt{\text{DENSITE}}$ | VITESSE<br>du<br>SON (a)<br>kt | ALTITUDE<br>Metres |
|-------------------|---------------|--------|-------------|-------|----------|----------|-----------------------------------------------|------------------------------------------------|-------------------------|--------------------------------|--------------------|
|                   | °C            | °F     | hPa         | P S I | In<br>Hg | mm<br>Hg |                                               |                                                |                         |                                |                    |
| <b>45.000</b>     | <b>-56,5</b>  | - 69,7 | <b>147</b>  | 2,14  | 4,36     | 110,7    | 0,1415                                        | 0,1936                                         | 0,440                   | 574                            | <b>13.716</b>      |
| <b>44.000</b>     | <b>-56,5</b>  | - 69,7 | <b>155</b>  | 2,24  | 4,57     | 116,0    | 0,1527                                        | 0,2031                                         | 0,451                   | 574                            | <b>13.411</b>      |
| <b>43.000</b>     | <b>-56,5</b>  | - 69,7 | <b>162</b>  | 2,35  | 4,79     | 121,7    | 0,1602                                        | 0,2131                                         | 0,462                   | 574                            | <b>13.106</b>      |
| <b>42.000</b>     | <b>-56,5</b>  | - 69,7 | <b>170</b>  | 2,47  | 5,03     | 127,8    | 0,1681                                        | 0,2236                                         | 0,473                   | 574                            | <b>12.802</b>      |
| <b>41.000</b>     | <b>-56,5</b>  | - 69,7 | <b>179</b>  | 2,19  | 5,28     | 134,1    | 0,1764                                        | 0,2346                                         | 0,484                   | 574                            | <b>12.497</b>      |
| <b>40.000</b>     | <b>-56,5</b>  | - 69,7 | <b>188</b>  | 2,72  | 5,54     | 140,7    | 0,1851                                        | 0,2462                                         | 0,496                   | 574                            | <b>12.192</b>      |
| <b>39.000</b>     | <b>-56,5</b>  | - 69,7 | <b>197</b>  | 2,81  | 5,81     | 147,6    | 0,1942                                        | 0,2583                                         | 0,508                   | 574                            | <b>11.887</b>      |
| <b>38.000</b>     | <b>-56,5</b>  | - 69,7 | <b>206</b>  | 2,99  | 6,10     | 154,9    | 0,2018                                        | 0,2710                                         | 0,521                   | 574                            | <b>11.582</b>      |
| <b>37.000</b>     | <b>-56,5</b>  | - 69,7 | <b>217</b>  | 3,14  | 6,40     | 162,6    | 0,2138                                        | 0,2843                                         | 0,533                   | 574                            | <b>11.278</b>      |
| <b>36.000</b>     | <b>-56,3</b>  | - 69,4 | <b>227</b>  | 3,30  | 6,71     | 170,4    | 0,2243                                        | 0,2981                                         | 0,546                   | 574                            | <b>10.973</b>      |
| <b>35.000</b>     | <b>-54,3</b>  | - 65,8 | <b>238</b>  | 3,46  | 7,04     | 178,8    | 0,2353                                        | 0,3099                                         | 0,557                   | 576                            | <b>10.668</b>      |
| <b>34.000</b>     | <b>-52,4</b>  | - 62,3 | <b>250</b>  | 3,63  | 7,38     | 187,5    | 0,2467                                        | 0,3220                                         | 0,567                   | 579                            | <b>10.363</b>      |
| <b>33.000</b>     | <b>-50,4</b>  | - 58,7 | <b>262</b>  | 3,80  | 7,74     | 196,6    | 0,2586                                        | 0,3345                                         | 0,578                   | 582                            | <b>10.058</b>      |
| <b>32.000</b>     | <b>-48,4</b>  | - 55,1 | <b>274</b>  | 3,98  | 8,11     | 206,0    | 0,2709                                        | 0,3473                                         | 0,589                   | 584                            | <b>9.754</b>       |
| <b>31.000</b>     | <b>-46,4</b>  | - 51,6 | <b>287</b>  | 4,17  | 8,49     | 215,6    | 0,2837                                        | 0,3605                                         | 0,600                   | 587                            | <b>9.449</b>       |
| <b>30.000</b>     | <b>-44,4</b>  | - 48,0 | <b>301</b>  | 4,36  | 8,89     | 225,8    | 0,2970                                        | 0,3741                                         | 0,611                   | 589                            | <b>9.144</b>       |
| <b>29.000</b>     | <b>-42,5</b>  | - 44,4 | <b>315</b>  | 4,57  | 9,30     | 236,2    | 0,3107                                        | 0,3881                                         | 0,623                   | 591                            | <b>8.839</b>       |
| <b>28.000</b>     | <b>-40,5</b>  | - 40,9 | <b>329</b>  | 4,78  | 9,73     | 247,1    | 0,3250                                        | 0,4025                                         | 0,634                   | 594                            | <b>8.534</b>       |
| <b>27.000</b>     | <b>-38,5</b>  | - 37,3 | <b>344</b>  | 4,99  | 10,17    | 258,3    | 0,3398                                        | 0,4173                                         | 0,646                   | 597                            | <b>8.230</b>       |
| <b>26.000</b>     | <b>-36,5</b>  | - 33,7 | <b>360</b>  | 5,22  | 10,63    | 270,0    | 0,3552                                        | 0,4325                                         | 0,658                   | 599                            | <b>7.925</b>       |
| <b>25.000</b>     | <b>-34,5</b>  | - 30,2 | <b>376</b>  | 5,45  | 11,10    | 281,9    | 0,3711                                        | 0,4481                                         | 0,669                   | 602                            | <b>7.620</b>       |
| <b>24.000</b>     | <b>-32,5</b>  | - 26,8 | <b>393</b>  | 5,70  | 11,60    | 294,6    | 0,3876                                        | 0,4642                                         | 0,681                   | 604                            | <b>7.315</b>       |
| <b>23.000</b>     | <b>-30,6</b>  | - 23,0 | <b>410</b>  | 5,95  | 12,11    | 307,6    | 0,4047                                        | 0,4806                                         | 0,693                   | 607                            | <b>7.010</b>       |
| <b>22.000</b>     | <b>-28,6</b>  | - 19,5 | <b>428</b>  | 6,21  | 12,64    | 321,1    | 0,4223                                        | 0,4976                                         | 0,705                   | 609                            | <b>6.706</b>       |
| <b>21.000</b>     | <b>-26,6</b>  | - 15,9 | <b>446</b>  | 6,47  | 13,18    | 334,8    | 0,4406                                        | 0,5150                                         | 0,718                   | 612                            | <b>6.401</b>       |
| <b>20.000</b>     | <b>-24,6</b>  | - 12,3 | <b>466</b>  | 6,75  | 13,75    | 349,3    | 0,4596                                        | 0,5328                                         | 0,730                   | 614                            | <b>6.096</b>       |
| <b>19.000</b>     | <b>-22,6</b>  | - 8,8  | <b>485</b>  | 7,04  | 14,34    | 364,2    | 0,4791                                        | 0,5511                                         | 0,742                   | 617                            | <b>5.791</b>       |
| <b>18.000</b>     | <b>-20,7</b>  | - 5,2  | <b>506</b>  | 7,34  | 14,94    | 379,5    | 0,4994                                        | 0,5699                                         | 0,755                   | 619                            | <b>5.486</b>       |
| <b>17.000</b>     | <b>-18,7</b>  | - 1,6  | <b>527</b>  | 7,65  | 15,57    | 395,5    | 0,5203                                        | 0,5892                                         | 0,768                   | 622                            | <b>5.182</b>       |
| <b>16.000</b>     | <b>-16,7</b>  | + 1,9  | <b>549</b>  | 7,97  | 16,22    | 412,0    | 0,5420                                        | 0,6089                                         | 0,780                   | 624                            | <b>4.877</b>       |
| <b>15.000</b>     | <b>-14,7</b>  | + 5,5  | <b>572</b>  | 8,29  | 16,89    | 429,0    | 0,5644                                        | 0,6292                                         | 0,793                   | 626                            | <b>4.572</b>       |
| <b>14.000</b>     | <b>-12,7</b>  | + 9,1  | <b>595</b>  | 8,63  | 17,58    | 446,1    | 0,5875                                        | 0,6500                                         | 0,806                   | 629                            | <b>4.267</b>       |
| <b>13.000</b>     | <b>-10,7</b>  | + 12,6 | <b>619</b>  | 8,99  | 18,29    | 464,6    | 0,6113                                        | 0,6713                                         | 0,819                   | 631                            | <b>3.962</b>       |
| <b>12.000</b>     | <b>-8,8</b>   | + 16,2 | <b>644</b>  | 9,35  | 19,03    | 483,4    | 0,6360                                        | 0,6932                                         | 0,833                   | 634                            | <b>3.658</b>       |
| <b>11.000</b>     | <b>-6,8</b>   | + 19,8 | <b>670</b>  | 9,72  | 19,79    | 502,7    | 0,6614                                        | 0,7155                                         | 0,846                   | 636                            | <b>3.353</b>       |
| <b>10.000</b>     | <b>-4,8</b>   | + 23,3 | <b>697</b>  | 10,11 | 20,58    | 522,7    | 0,6877                                        | 0,7385                                         | 0,859                   | 638                            | <b>3.048</b>       |
| <b>9.000</b>      | <b>-2,8</b>   | + 26,9 | <b>724</b>  | 10,50 | 21,39    | 543,3    | 0,7148                                        | 0,7619                                         | 0,873                   | 641                            | <b>2.743</b>       |
| <b>8.000</b>      | <b>-0,8</b>   | + 30,5 | <b>753</b>  | 10,92 | 22,23    | 564,6    | 0,7428                                        | 0,7860                                         | 0,887                   | 643                            | <b>2.438</b>       |
| <b>7.000</b>      | <b>+ 1,1</b>  | + 34,0 | <b>782</b>  | 11,34 | 23,09    | 586,5    | 0,7716                                        | 0,8106                                         | 0,900                   | 645                            | <b>2.134</b>       |
| <b>6.000</b>      | <b>+ 3,1</b>  | + 37,6 | <b>812</b>  | 11,78 | 23,98    | 609,1    | 0,8014                                        | 0,8358                                         | 0,914                   | 648                            | <b>1.829</b>       |
| <b>5.000</b>      | <b>+ 5,1</b>  | + 41,2 | <b>843</b>  | 12,23 | 24,90    | 632,5    | 0,8321                                        | 0,8616                                         | 0,928                   | 650                            | <b>1.524</b>       |
| <b>4.000</b>      | <b>+ 7,1</b>  | + 44,7 | <b>875</b>  | 12,69 | 25,84    | 656,3    | 0,8637                                        | 0,8881                                         | 0,942                   | 652                            | <b>1.219</b>       |
| <b>3.000</b>      | <b>+ 9,1</b>  | + 48,3 | <b>908</b>  | 13,17 | 26,82    | 681,2    | 0,8962                                        | 0,9151                                         | 0,957                   | 655                            | <b>914</b>         |
| <b>2.000</b>      | <b>+ 11,0</b> | + 51,9 | <b>942</b>  | 13,66 | 27,82    | 706,6    | 0,9298                                        | 0,9427                                         | 0,971                   | 657                            | <b>610</b>         |
| <b>1.000</b>      | <b>+ 13,0</b> | + 55,4 | <b>977</b>  | 14,17 | 28,86    | 733,0    | 0,9644                                        | 0,9710                                         | 0,985                   | 659                            | <b>305</b>         |
| <b>0</b>          | <b>+ 15,0</b> | + 59,0 | <b>1013</b> | 14,70 | 29,92    | 760,0    | 1,0000                                        | 1,0000                                         | 1,000                   | 661                            | <b>0</b>           |
| <b>1.000</b>      | <b>+ 17,0</b> | + 62,5 | <b>1050</b> | 15,23 | 31,02    | 787,9    | 1,0366                                        | 1,0295                                         | 1,015                   | 664                            | <b>-305</b>        |

**Annex 022-9771A**

The atmospheric pressure at FL 70 in a “standard +10” atmosphere is ;

- a) 942.13 hPa.
- b) 644.41 hPa.
- c) 781.85 hPa.
- d) 1 013.25 hPa.

Answer ;

**c) 781.85 hPa.**

0474. The attitude data computed by an IRS can be used by the ;

- a) stall warning system.
- b) GPWS.
- c) TCAS.
- d) auto pilot system.

Answer ;

**d) auto pilot system.**

0475. The aural alert associated with the mode 1 ( excessive descent rate ) of the GPWS is ;

- a) "SINKRATE".
- b) "TERRAIN".
- c) "DON'T SINK".
- d) "TOO LOW TERRAIN".

Answer ;

- a) "SINKRATE".
- 

0476. The aural alert associated with the mode 2 ( excessive terrain closure rate ) of the GPWS is ;

- a) "TERRAIN".
- b) "SINKRATE".
- c) "CLIMB".
- d) "DON'T SINK".

Answer ;

- a) "TERRAIN".
- 

0477. The aural alert associated with the mode 3 ( altitude loss after take-off or go-around ) of the GPWS is ;

- a) "SINKRATE".
- b) "TERRAIN".
- c) "TOO LOW TERRAIN".
- d) "DON'T SINK".

Answer ;

- d) "DON'T SINK".
- 

0478. The aural alert associated with the mode 5 ( excessive deviation below Glideslope ) of the GPWS is ;

- a) "TOO LOW GLIDESLOPE"
- b) "MONITOR GLIDESLOPE".
- c) "CAUTION GLIDESLOPE"
- d) "GLIDESLOPE".

Answer ;

- d) "GLIDESLOPE".
- 

0479. The aural alert(s) associated with the mode 1 ( excessive descent rate ) of the GPWS is (are) ;

- 1. "TERRAIN".
- 2. "DON'T SINK".
- 3. "SINKRATE".
- 4. "PULL UP".

The combination that regroups all of the correct statements is ;

- a) 3 , 4.
- b) 2 , 4.
- c) 1.
- d) 2.

Answer ;

- a) 3 , 4.
- 

0480. The aural alert(s) associated with the mode 3 ( altitude loss after take-off or go-around ) of the GPWS is (are) ;

- 1. "DON'T SINK".
- 2. "SINKRATE".
- 3. "TERRAIN".
- 4. "PULL UP".

The combination that regroups all of the correct statements is ;

- a) 2.
- b) 2 , 4.
- c) 1.
- d) 3 , 4.

Answer ;

- c) 1.
-

**0481.** The aural alert(s) associated with the mode 4 of the GPWS ( unsafe terrain clearance while not in the landing configuration ) is (are) ;

1. "TOO LOW GEAR".
2. "TOO LOW TERRAIN".
3. "TOO LOW FLAPS".
4. "PULL UP".

The combination that regroups all of the correct statements is ;

- a) 2.
- b) 1 , 2 , 3.
- c) 2 , 4.
- d) 1 , 2.

**Answer ;**

**b) 1 , 2 , 3.**

---

**0482.** The aural alert(s) associated with the mode 4 of the GPWS ( unsafe terrain clearance while not in the landing configuration ) is (are) ;

- a) "TOO LOW FLAPS" , "TOO LOW GEAR" , "GLIDE SLOPE".
- b) "TOO LOW" , "CHECK FLAPS" , "CHECK GEAR".
- c) "TOO LOW , TERRAIN" only.
- d) "TOO LOW GEAR" , "TOO LOW FLAPS" , "TOO LOW TERRAIN".

**Answer ;**

**d) "TOO LOW GEAR" , "TOO LOW FLAPS" , "TOO LOW TERRAIN".**

---

**0483.** The aural alert(s) associated with the mode 5 ( excessive deviation below Glideslope ) of the GPWS is (are) ;

1. "TERRAIN".
2. "TOO LOW GLIDESLOPE".
3. "GLIDESLOPE".
4. "PULL UP".

The combination that regroups all of the correct statements is ;

- a) 1.
- b) 2.
- c) 2 , 4.
- d) 3.

**Answer ;**

**d) 3.**

---

**0484.** The automatic pitch trim ;

1. ensures the aeroplane is properly trimmed when the autopilot is engaged.
2. permits the elevator to always be in neutral position with respect to horizontal stabiliser.
3. ensures the aeroplane is properly trimmed when the autopilot is disengaged.

The combination regrouping all the correct statements is ;

- a) 2 , 3.
- b) 1 , 2.
- c) 1 , 2 , 3.
- d) 1 , 3.

**Answer ;**

**a) 2 , 3.**

---

**0485.** The automatic power control system ( auto-throttle ) of a transport airplane has the following mode(s) ;

1. capture and holding of speeds.
2. capture and holding of Mach number.
3. capture and holding of flight angle of attack.
4. capture and holding of N1 or EPR ( Engine Power Ratio ).
5. capture and holding of flight paths.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 2 , 3 , 5.
- c) 1 , 4 , 5.
- d) 2 , 4.

**Answer ;**

**a) 1 , 2 , 4.**

---

- 0486.** The automatic synchronisation function of an autopilot control system ;
1. operates only when the autopilot is engaged.
  2. prevents the aircraft's control system from jerking when disengaging the autopilot.
  3. prevents the aircraft's control system from jerking when engaging the autopilot.
- The combination that regroups all of the correct statements is ;
- a) 3.
  - b) 1 , 2.
  - c) 1 , 2 , 3.
  - d) 1 , 3.

**Answer ;**

**a) 3.**

---

- 0487.** The automatic synchronisation function of an autopilot control system ;
1. operates only when the autopilot is engaged.
  2. prevents the aircraft's control system from jerking when disengaging the autopilot.
  3. enables the cancellation of the rudder control signals.
  4. prevents the aircraft's control system from jerking when engaging the autopilot.
- The combination regrouping all the correct statements is ;
- a) 1 , 2 , 4.
  - b) 1 , 4.
  - c) 1 , 2.
  - d) 4.

**Answer ;**

**d) 4.**

---

- 0488.** The automatic trim is a component of the autopilot pitch channel. Its function is to ;
- a) set the attitude to an instantaneous value before engaging the autopilot.
  - b) reset the attitude , after engaging ( the autopilot ).
  - c) automatically disengage the autopilot in the case of an excessive pitch up.
  - d) transfer a stabilized aeroplane to the pilot during autopilot disengagement.

**Answer ;**

**d) transfer a stabilized aeroplane to the pilot during autopilot disengagement.**

---

- 0489.** The auto-throttle ;
1. enable to catch and to maintain the N1 RPM.
  2. enable to catch and to maintain the N2 RPM.
  3. enable to catch and to maintain an airplane indicated airspeed ( IAS ).
  4. is always engaged automatically at the same time as the autopilot.
- The combination regrouping all the correct statements is ;
- a) 1 and 4.
  - b) 1 , 3 and 4.
  - c) 2 and 3.
  - d) 1 and 3.

**Answer ;**

**d) 1 and 3.**

---

- 0490.** The auto-throttle ;
1. can capture and maintain the N1 RPM.
  2. can capture and maintain the N2 RPM.
  3. can capture and maintain an IAS.
  4. is always engaged automatically at the same time as the autopilot.
- The combination that regroups all of the correct statements is ;
- a) 1 , 3.
  - b) 1 , 4.
  - c) 2 , 3.
  - d) 1 , 3 , 4.

**Answer ;**

**a) 1 , 3.**

---

- 0491.** The autothrottle system ;
- a) can be engaged in the TAS mode - holding of constant TAS - during a final approach.
  - b) can be engaged in the GS mode - holding of constant Ground Speed - during a final approach.
  - c) is automatically disconnected when the aircraft is on the ground.
  - d) can be used for take-off.

**Answer ;**

- d)** can be used for take-off.
- 

- 0492.** The auto-throttle system of a transport aeroplane has the following mode(s) ;

- 1. capture and holding of IAS.
- 2. capture and holding of Mach number.
- 3. capture and holding of flight angle of attack.
- 4. capture and holding of N1 or EPR.
- 5. capture and holding of flight paths.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 4 , 5.
- c) 2 , 4.
- d) 1 , 2 , 3 , 5.

**Answer ;**

- a)** 1 , 2 , 4.
- 

- 0493.** The auto-throttles enables to hold ;

- 1. a true airspeed.
- 2. a Mach number.
- 3. an indicated airspeed.
- 4. a N1 thrust.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 2 , 3.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3 , 4.

**Answer ;**

- c)** 2 , 3 , 4.
- 

- 0494.** The basic modes of an autopilot consist in ;

- a) controlling the movement of the centre of gravity of the aircraft.
- b) controlling the path of the aircraft vertically only.
- c) controlling the path of the aircraft horizontally or vertically.
- d) stabilizing the aircraft around its centre of gravity.

**Answer ;**

- d)** stabilizing the aircraft around its centre of gravity.
- 

- 0495.** The basic on-board datalink communication system is typically composed by the following sub systems ;

- 1. Communication Management Unit ( CMU ).
- 2. Multi-purpose Control and Display Unit ( MCDU ).
- 3. Communication unit ( VHF , HF , Satcom ).

The combination which regroups all of the correct statements is ;

- a) 1 , 3.
- b) 1 , 2.
- c) 1 , 2 , 3.
- d) 2 , 3.

**Answer ;**

- c)** 1 , 2 , 3.
- 

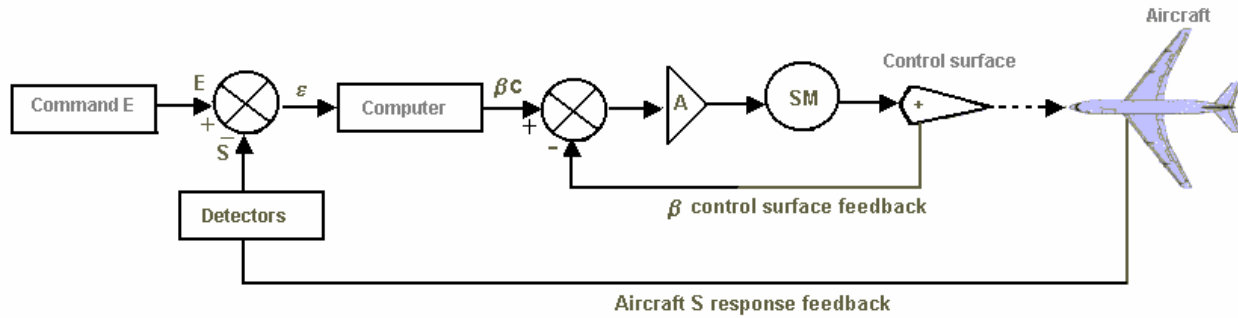
- 0496.** The basic principle of a capacitance fuel gauge system is that the ;

- a) capacity of a capacitor depends on the nature of the dielectric in which it is immersed.
- b) capacity of a capacitor depends only on the density of the liquid on which it is immersed.
- c) internal resistance of a capacity depends on the nature of the dielectric in which it is immersed.
- d) electromotive force of a capacity depends on the nature of the dielectric in which it is immersed.

**Answer ;**

- a)** capacity of a capacitor depends on the nature of the dielectric in which it is immersed.
-

0497. ( For this question use annex 022-3880A )



### Annex 022-3880A

The block diagram of an auto-pilot is shown in the annex. For each control channel ( pitch , roll and yaw ) the piloting law is the relationship between the deflection of the control surface commanded by the computer (  $\beta c$  ) and the ;

- offset EPSILON at the computer input.
- real deflection of the control surface ( BETA control surface feedback ).
- aircraft response S.
- pilot command E.

Answer ;

- a) offset EPSILON at the computer input.

0498. The “Bourdon tube” is used to measure ;

- quantity.
- pressure.
- a flow rate.
- temperature.

Answer ;

- b) pressure.

0499. The calculator combined with the stick shaker system of a modern transport airplane receives information about the ;

- angle of attack.
- engine RPM.
- configuration.
- pitch and bank attitude.
- sideslip.

The combination regrouping all the correct statements is ;

- 1 and 5.
- 1 , 2 , 3 and 4.
- 1 , 2 , 3 , 4 and 5.
- 1 and 3.

Answer ;

- d) 1 and 3.

0500. The calibrated airspeed ( CAS ) or Mach holding mode is carried out by ;

- the autopilot pitch channel in the climb mode at a constant calibrated airspeed ( CAS ) or Mach number.
- the auto-throttles in the climb mode at a constant calibrated airspeed ( CAS ) or Mach number.
- the autopilot pitch channel in the altitude or glide path holding mode.
- the auto-throttles in the altitude or glide path holding mode.

The combination regrouping all the correct statements is ;

- 2 and 4.
- 1 and 4.
- 2 and 3.
- 1 and 3.

Answer ;

- b) 1 and 4.

0501. The CAS is obtained from IAS by correcting for the following errors ;

1. position.
2. compressibility.
3. instrument.
4. density.

The combination that regroups all of the correct statements is ;

- a) 3.
- b) 1 , 3 , 4.
- c) 2 , 3 , 4.
- d) 1 , 3.

Answer ;

- d) 1 , 3.
- 

0502. The center of gravity of the compass rose of a direct reading magnetic compass lies below the pivot point in order to reduce the influence of the ;

- a) parallax error.
- b) position error.
- c) magnetic inclination.
- d) magnetic variation.

Answer ;

- c) magnetic inclination.
- 

0503. The characteristics of the directional gyro used in a gyro stabilised compass system are ;

NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.

- a) two degrees of freedom , whose horizontal axis corresponding to the reference direction is maintained in the horizontal plane by an automatic erecting system.
- b) one degree of freedom , whose vertical axis , aligned with the real vertical to the location is maintained in this direction by an automatic erecting system.
- c) two degrees of freedom , whose axis aligned with the vertical to the location is maintained in this direction by an erecting system.
- d) one degree of freedom , whose horizontal axis is maintained in the horizontal plane by an automatic erecting system.

Answer ;

- a) two degrees of freedom , whose horizontal axis corresponding to the reference direction is maintained in the horizontal plane by an automatic erecting system.
- 

0504. The characteristics of the earth which are being used during the alignment of an INS platform are ;

- a) earth rotation and gravity.
- b) longitude and gravity.
- c) earth magnetic field and earth rotation.
- d) earth rotation and longitude.

Answer ;

- a) earth rotation and gravity.
- 

0505. The characteristics of the gyroscope used in gyromagnetic compass system are ;

NB : the rotor spin axis is not counted for the number of degrees of freedom of the gyro.

- a) one degree of freedom , whose axis is maintained in the direction of the local vertical by an erecting system.
- b) one degree of freedom , whose horizontal axis is maintained in the horizontal plane by an erecting system.
- c) two degrees of freedom , whose horizontal axis corresponding to the reference direction is maintained in the horizontal plane by an erecting system.
- d) two degrees of freedom , whose axis is maintained in the direction of the local vertical by an erecting system.

Answer ;

- c) two degrees of freedom , whose horizontal axis corresponding to the reference direction is maintained in the horizontal plane by an erecting system.
- 

0506. The characteristics of the SATCOM transmissions , used for datalink communications are ;

- a) line of sight limitation , no variable quality of signals disturbed by ionospheric conditions.
- b) line of sight limitation , variable quality of signals depending on ionospheric conditions.
- c) no line of sight limitation , variable quality of signals depending on ionospheric conditions.
- d) no line of sight limitation , no variable quality of signals disturbed by ionospheric conditions.

Answer ;

- d) no line of sight limitation , no variable quality of signals disturbed by ionospheric conditions.
-



0507. The command bars of a flight director ;
- a) are always displayed during take-off.
  - b) are displayed only if the autopilot is engaged.
  - c) may be displayed when flying manually.
  - d) are always displayed when the autopilot is engaged.

Answer ;

c) may be displayed when flying manually.

---

0508. The command bars of a flight director ;
- a) are displayed only when flying manually.
  - b) are always displayed during take-off.
  - c) are displayed only when the autopilot is engaged.
  - d) may be displayed when flying manually or with the autopilot engaged.

Answer ;

d) may be displayed when flying manually or with the autopilot engaged.

---

0509. The command bars of a flight director are generally represented on an ;
- a) RMI ( Radio Magnetic Indicator ).
  - b) ADI ( Attitude Director Indicator ).
  - c) HSI ( Horizontal Situation Indicator ).
  - d) ILS ( Instrument Landing System ).

Answer ;

b) ADI ( Attitude Director Indicator ).

---

0510. The compass heading can be derived from the magnetic heading by reference to a ;
- a) map showing the isogonic lines.
  - b) magnetic variation correction card.
  - c) compass deviation card.
  - d) map showing the isoclinic lines.

Answer ;

c) compass deviation card.

---

0511. The commands sent out by the yaw damper computer ;
- a) are added to or subtracted from the rudder deflection orders sent out by the pilot or the autopilot.
  - b) are inhibited when the pilot acts on the rudder pedals.
  - c) inhibit the rudder deflection orders sent out by the pilot or the autopilot.
  - d) are inhibited when the autopilot is engaged.

Answer ;

a) are added to or subtracted from the rudder deflection orders sent out by the pilot or the autopilot.

---

0512. The components of an autopilot system are the ;
- 1. actuators.
  - 2. mode control panel.
  - 3. EFIS control panel.
  - 4. mode annunciator panel.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2 , 3 , 4.
- c) 1 , 2 , 4.
- d) 3 , 4.

Answer ;

c) 1 , 2 , 4.

---

0513. The components of an autopilot system are the ;
- 1. actuators.
  - 2. mode control panel.
  - 3. mode annunciator panel.
  - 4. computer.

The combination that regroups all of the correct statements is ;

- a) 1 , 4.
- b) 1 , 2 , 3 , 4.
- c) 1 , 2 , 4.
- d) 1 , 2.

Answer ;

b) 1 , 2 , 3 , 4.

---

**0514.** The compressibility correction to CAS to give EAS ;

1. may be positive.
2. is always negative.
3. depends on Mach number only.
4. depends on pressure altitude only.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 2 , 3.
- c) 1 , 3.
- d) 1 , 4.

**Answer ;**

**b) 2 , 3.**

---

**0515.** The compressibility error must be taken into account only for aeroplane with ;

- a) Mach number greater or equal to 1.
- b) TAS greater than approximately 100 km/h.
- c) TAS greater than approximately 200 kt.
- d) Mach number greater than 0.8.

**Answer ;**

**c) TAS greater than approximately 200 kt.**

---

**0516.** The computer of the autopilot system uses , among others , the following parameters ;

- a) CAS , altitude , temperature , inertial position , attitude.
- b) altitude , vertical speed , heading , attitude , GPS position.
- c) CAS , altitude , vertical speed , heading , attitude.
- d) altitude , heading , temperature , fuel flow , attitude.

**Answer ;**

**c) CAS , altitude , vertical speed , heading , attitude.**

---

**0517.** The computer of the autopilot system uses , among others , input signals from the ;

1. attitude reference system.
2. mode annunciator panel.
3. ADC.
4. mode control panel.

The combination that regroups all of the correct statements is ;

- a) 1 , 3 , 4.
- b) 1 , 2 , 3 , 4.
- c) 1 , 4.
- d) 2 , 3.

**Answer ;**

**a) 1 , 3 , 4.**

---

**0518.** The control law of a transport airplane autopilot control channel may be defined as the relationship between the ;

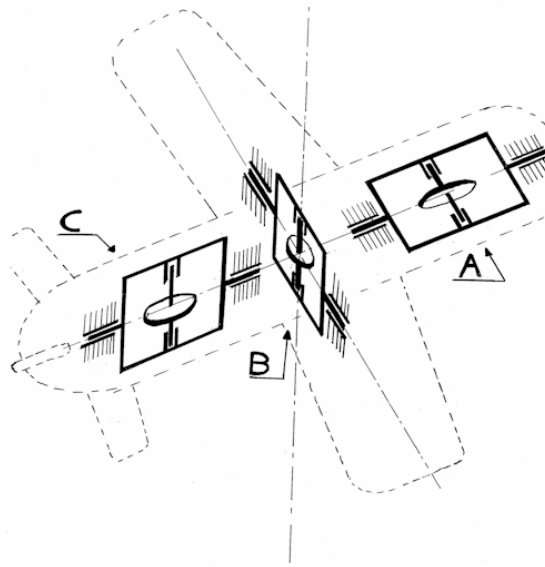
- a) crew inputs to the computer and the detector responses ( returned to the airplane ).
- b) input and output signals at the amplifier level respectively control deviation data and control deflection signals.
- c) computer input deviation data and the output control deflection signals.
- d) computer input deviation data and the signals received by the servoactuators.

**Answer ;**

**c) computer input deviation data and the output control deflection signals.**

---

0519. ( The diagram on annex 022-648A shows three gyro assemblies )



**Annex 022-648A**

- A , B and C. Among these gyros ,
- one is a roll gyro ( noted 1 )
  - one is a pitch gyro ( noted 2 )
  - one is a yaw gyro ( noted 3 )

The correct matching of gyros and assemblies is ;

- a) 1C , 2B , 3A.
- b) 1B , 2C , 3A.
- c) 1B , 2A , 3C.
- d) 1A , 2B , 3C.

**Answer ;**

- b) 1B , 2C , 3A.**

---

0520. The CPDLC function consists in exchanging messages relative to ;

- 1. Route modifications.
- 2. Crossing constraints.
- 3. Transfer of ATC center.
- 4. Speed changes.

The combination which regroups all of the correct statements is ;

- a) 2 , 4.
- b) 1 , 3 , 4.
- c) 3.
- d) 1 , 2 , 3 , 4.

**Answer ;**

- d) 1 , 2 , 3 , 4.**

---

0521. The CPDLC is a FANS application which consists in ;

- a) transmitting datalink formatted messages between the pilot and ATC controller.
- b) connecting the aircraft to the appropriate ATC center.
- c) sending automatically aircraft surveillance data to the air traffic controller.
- d) establishing air/ground connection to verify if the data link communication can be performed.

**Answer ;**

- a) transmitting datalink formatted messages between the pilot and ATC controller.**
-

0522. The CPDLC messages may concerne ;
1. Route modifications.
  2. Speed changes.
  3. Voice contact request.
  4. Emergency messages.

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 1 , 2 , 3 , 4.
- c) 1 , 2 , 4.
- d) 3 , 4.

Answer ;

- b) 1 , 2 , 3 , 4.**
- 

0523. The CS 25 gives the following definition : “Where the pilot has the ability to make inputs to the automatic pilot by movement of the normal control wheel”. The corresponding mode is ;

- a) nose wheel steering ( NWS ).
- b) control wheel steering ( CWS ).
- c) automatic wheel steering ( AWS ).
- d) alternate wheel steering ( AWS ).

Answer ;

- b) control wheel steering ( CWS ).**
- 

0524. The CVR ( Cockpit Voice Recorder ) includes ;

1. a microphone.
2. a recorder in compliance with the shock and fire resistance standards.
3. an independent battery.
4. a flight data recorder.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 4.
- c) 1 , 2.
- d) 2 , 4.

Answer ;

- c) 1 , 2.**
- 

0525. The data output from the ADC are ;

1. Barometric altitude.
2. Mach number.
3. CAS.
4. TAS.
5. SAT.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4 , 5.
- b) 1 , 4.
- c) 1 , 3 , 5.
- d) 2 , 3 , 4.

Answer ;

- a) 1 , 2 , 3 , 4 , 5.**
- 

0526. The data supplied by a radio altimeter ;

- a) is used by the automatic pilot in the altitude hold mode.
- b) concerns only the decision height.
- c) indicates the distance between the ground and the aircraft.
- d) is used only by the radio altimeter indicator.

Answer ;

- c) indicates the distance between the ground and the aircraft.**
- 

0527. The data that needs to be inserted into an Inertial Reference System in order to enable the system to make a succesful alignment for navigation is ;

- a) aircraft position in latitude and longitude or airport ICAO identifier.
- b) aircraft heading.
- c) the position of an in-range DME.
- d) the navigation database reference.

Answer ;

- a) aircraft position in latitude and longitude or airport ICAO identifier.**
-

0528. The datalink communications between the aircraft and the ground can be performed by the following system ;

- a) ACARS.
- b) CVR.
- c) EGPWS.
- d) TCAS.

Answer ;

a) ACARS.

0529. The Decision Height ( DH ) warning light comes on when an aircraft ;

- a) passes over the ILS inner marker.
- b) passes over the outer marker.
- c) descends below a pre-set radio altitude.
- d) descends below a pre-set barometric altitude.

Answer ;

c) descends below a pre-set radio altitude.

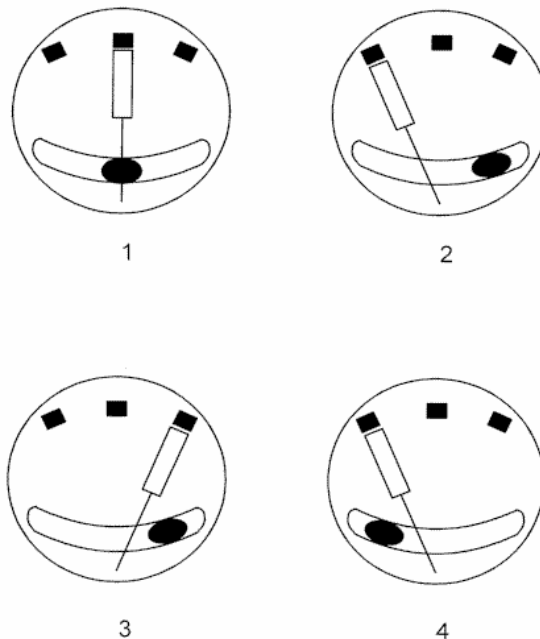
0530. The density altitude is ;

- a) the altitude in the standard atmosphere on which the density is equal to the actual density of the atmosphere.
- b) the pressure altitude corrected for the relative density prevailing at this point.
- c) the temperature altitude corrected for the difference between the real temperature and the standard temperature.
- d) the pressure altitude corrected for the density of air at this point.

Answer ;

a) the altitude in the standard atmosphere on which the density is equal to the actual density of the atmosphere.

0531. ( For this question use annex 022-11532A )



**Annex 022-11532A**

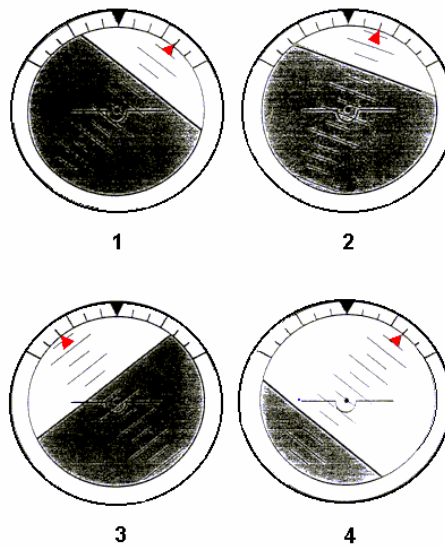
The diagram representing a left turn with insufficient rudder is ;

- a) 3.
- b) 4.
- c) 1.
- d) 2.

Answer ;

b) 4.

0532. ( For this question use annex 022-11567A )



**Annex 022-11567A**

The diagram which shows a 40° left bank and 15° nose down attitude is n° ;

- a) 3.
- b) 4.
- c) 1.
- d) 2.

Answer ;

**c) 1.**

0533. The different types of ADS contracts can be ;

- 1. periodic : at periodic time intervals.
- 2. on demand : when asked for by the ATC.
- 3. on event : whenever a specified event occurs.

The combination which regroupes all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2.
- c) 2 , 3.
- d) 1 , 3.

Answer ;

**a) 1 , 2 , 3.**

0534. The direct reading magnetic compass is no more reliable when approaching ;

- 1. the magnetic poles.
- 2. the magnetic equator with a east or west heading.
- 3. the magnetic equator with north or south heading.

The combination that regroupes all of the correct statements is ;

- a) 1.
- b) 1 , 2 , 3.
- c) 1 , 3.
- d) 1 , 2.

Answer ;

**a) 1.**

0535. The disadvantage of an electronic rpm indicator is the ;

- a) generation of spurious signals at the commutator.
- b) influence of temperature on the indication.
- c) necessity of providing a power supply source.
- d) high influence of line resistance on the indication.

Answer ;

**c) necessity of providing a power supply source.**

0536. The disadvantages of a float type fuel gauge are ;
1. the design is complex.
  2. the indications are influenced by the aircraft attitude variations.
  3. the indications are influenced by the accelerations.
  4. the indications are influenced by temperature variations.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3 , 4.

Answer ;

**c) 2 , 3 , 4.**

---

0537. The disadvantages of a single-phase AC generator tachometer are ;
1. the presence of spurious signals due to a DC generator commutator.
  2. the importance of line resistance on the information value.
  3. the influence of temperature on the tachometer information.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 2.
- c) 1 , 2 , 3.
- d) 1 , 2.

Answer ;

**b) 2.**

---

0538. The disadvantages of an “electric” fuel ( float ) gauge are ;
1. the design is complex.
  2. the indications are influenced by the airplane attitude variations.
  3. the indications are influenced by the accelerations.
  4. the indications are influenced by temperature variations.
  5. that an alternative current supply is necessary.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4 , 5.
- b) 2 , 3 , 4.
- c) 1 , 2 , 3 , 4.
- d) 1.

Answer ;

**b) 2 , 3 , 4.**

---

0539. The drift of the gyroscopes of a stand alone inertial system ;
- a) is not an error source.
  - b) induces a bounded position error.
  - c) induces a position error that decreases along the flight.
  - d) is the main error source.

Answer ;

**d) is the main error source.**

---

0540. The duration of a FMS navigation database loaded before expiring is ;
- a) 3 months.
  - b) 28 days.
  - c) 15 days.
  - d) 2 months.

Answer ;

**b) 28 days.**

---

0541. The EAS is obtained from the CAS by correcting for ;
- a) position error.
  - b) instrument error.
  - c) density error.
  - d) compressibility error.

Answer ;

**d) compressibility error.**

---

**0542.** The EAS is obtained from the CAS by correcting for the following errors ;

1. position
2. compressibility.
3. instrument.
4. density.

The combination that regroups all of the correct statements is ;

- a) 2 , 4.
- b) 1 , 2 , 3 , 4.
- c) 4.
- d) 2.

**Answer ;**

**d) 2.**

---

**0543.** The EAS is obtained from the IAS by correcting for the following errors ;

1. instrument.
2. position.
3. density.
4. compressibility.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 2 , 3 , 4.
- c) 1 , 2.
- d) 1 , 2 , 3.

**Answer ;**

**a) 1 , 2 , 4.**

---

**0544.** The EGPWS ;

- a) is an enhanced GPWS which warns the crew if the airplane is not in the appropriate configuration for take-off.
- b) is an enhanced GPWS which is able to provide resolution advisories in the lateral plane.
- c) is an enhanced GPWS which has a its own world terrain database.
- d) has nothing to do with the GPWS.

**Answer ;**

**c) is an enhanced GPWS which has a its own world terrain database.**

---

**0545.** The EGPWS may propose the following functions ;

1. Abnormal Take-off Configuration ( ATC ).
2. Terrain Clearance Floor ( TCF ).
3. Predictive Wind Shear ( PWS ).
4. Terrain Look Ahead Alerting.
5. Terrain Alerting and Display ( TAD ).

The combination regrouping all of the correct statements is ;

- a) 2 , 4 , 5.
- b) 1 , 2 , 3 , 5.
- c) 1 , 3 , 4.
- d) 2 , 3 , 4 , 5.

**Answer ;**

**a) 2 , 4 , 5.**

---

**0546.** The EGPWS may propose the Terrain Look-Ahead Alerting function , this function uses ;

- a) an electronic map of the world giving ground elevation.
- b) the same caution and warning envelopes than the mode 2 GPWS ( excessive terrain closure rate ).
- c) a specific radar to detect any high ground in conflict with the flight path of the aircraft.
- d) the weather radar to detect any high ground in conflict with the flight path of the aircraft.

**Answer ;**

**a) an electronic map of the world giving ground elevation.**

---



0547. The electric float gauge ;
1. gives a mass information.
  2. gives information independent of aircraft's manoeuvres and attitude changes.
  3. gives information all the more accurate as the tank is full.
  4. is typically a DC powered system.

The combination regrouping all the correct statements is ;

- a) 1.
- b) 4.
- c) 1 , 3 , 4.
- d) 1 , 2 , 3 , 4.

Answer ;

**b) 4.**

---

0548. The electronic tachometer sensor is composed of ;
- a) a notched wheel rotating in front of an electro-magnet.
  - b) the rotor of a three-phase A.C. generator.
  - c) the rotor of a single phase A.C. generator.
  - d) a circular magnet with four poles.

Answer ;

**a) a notched wheel rotating in front of an electro-magnet.**

---

0549. The elements which take part in the local vertical alignment of an inertial strap-down unit are ;
- a) the flow inductors.
  - b) the accelerometers.
  - c) the gyroscopes.
  - d) the accelerometers and gyroscopes.

Answer ;

**b) the accelerometers.**

---

0550. The energy required to operate a strapdown inertial system is supplied by ;
- a) the hydraulic system.
  - b) the bleed air system.
  - c) a dedicated pneumatic system.
  - d) the electrical system.

Answer ;

**d) the electrical system.**

---

0551. The engine instrument utilising an aneroid pressure diaphragm is the ;
- a) fuel pressure gauge.
  - b) oil temperature gauge.
  - c) manifold pressure gauge.
  - d) oil pressure gauge.

Answer ;

**c) manifold pressure gauge.**

---

0552. The Enhanced GPWS ( EGPWS ) terrain display uses with the following colours ;
- a) green , amber , red , magenta.
  - b) blue , red , magenta.
  - c) white , amber , red.
  - d) magenta , red , flashing red.

Answer ;

**a) green , amber , red , magenta.**

---

0553. The EPR ( Engine Pressure Ratio ) is ;
- a) the ratio of the compressor outlet total pressure to the compressor inlet total pressure.
  - b) the difference between the compressor inlet total pressure and the turbine outlet total pressure.
  - c) the ratio of the turbine outlet total pressure to the ambient total pressure.
  - d) the ratio of the turbine outlet total pressure to the compressor inlet total pressure.

Answer ;

**d) the ratio of the turbine outlet total pressure to the compressor inlet total pressure.**

---

0554. The EPR ( Engine Pressure Ratio ) is computed by ;
- dividing compressor discharge pressure by turbine discharge pressure.
  - multiplying compressor inlet pressure by turbine discharge pressure.
  - multiplying compressor discharge pressure by turbine inlet pressure.
  - dividing turbine discharge pressure by compressor inlet pressure.

Answer ;

- d) dividing turbine discharge pressure by compressor inlet pressure.
- 

0555. The error in altimeter readings caused by the variation of the static pressure near the source is known as ;

- position error.
- instrument error.
- barometric error.
- hysteresis effect.

Answer ;

- a) position error.
- 

0556. The error induced by the location of the static pressure source is known as the ;

- instrument error.
- position error.
- hysteresis effect.
- barometric error.

Answer ;

- b) position error.
- 

0557. The essential components of a flight director are ;

- a computer.
- an automatic pilot.
- an auto-throttle.
- command bars.

The combination regrouping all the correct statements is ;

- 2 , 3.
- 2 , 4.
- 1 , 2.
- 1 , 4.

Answer ;

- d) 1 , 4.
- 

0558. The FADEC ( Full Authority Digital Engine Control ) ;

- provides fully automatic engine starting.
- modifies aircraft airspeed in order to allow for the lowest fuel consumption.
- counters any yaw movement in case of engine failure.
- provides thrust reverser control.

The combination that regroups all of the correct statements is ;

- 1 , 2 , 3 , 4.
- 1 , 4.
- 1 , 2 , 4.
- 2.

Answer ;

- b) 1 , 4.
- 

0559. The FADEC ( Full Authority Digital Engine Control ) provides ;

- engine automatic shut-down if maximum N1 is exceeded.
- engine automatic shut-down if maximum EGT is exceeded.
- automatic thrust rating control.
- fully automatic engine starting.

The combination that regroups all of the correct statements is ;

- 3 , 4.
- 1 , 3.
- 1 , 3 , 4.
- 2 , 3 , 4.

Answer ;

- a) 3 , 4.
-

0560. The FANS concept includes ;
- a) an enhanced detection of the nearby terrain.
  - b) an enhanced detection of the intruding traffic.
  - c) an improvement in the accuracy of the navigation systems aiming to carry out RNAV approaches.
  - d) a datalink communication between the aircraft and the ATC centers to replace the voice communication.

Answer ;

- d) a datalink communication between the aircraft and the ATC centers to replace the voice communication.
- 

0561. The fields affecting a magnetic compass originate from ;
- 1. magnetic masses.
  - 2. ferrous metal masses.
  - 3. non ferrous metal masses.
  - 4. electrical currents.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 2 , 3.
- c) 1 , 3 , 4.
- d) 1 , 2 , 3 , 4.

Answer ;

- a) 1 , 2 , 4.
- 

0562. The flexible take-off mode ;
- 1. can be used only if the engines are recent.
  - 2. reduces engine wear.
  - 3. can be used in situations where take-off can be executed without the need for full engine power.
  - 4. can only be used with an auto-throttle.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 1 , 2 , 3.
- c) 1 , 4.
- d) 2 , 3.

Answer ;

- d) 2 , 3.
- 

0563. The flight director command bars provide the pilot with corrections to apply to ;
- a) pitch and/or roll attitude.
  - b) pitch and/or roll attitude , and yaw.
  - c) pitch attitude only.
  - d) roll attitude only.

Answer ;

- a) pitch and/or roll attitude.
- 

0564. The flight director computer continuously ;
- a) compares the current attitude with the computed attitude.
  - b) computes the required attitude for the autopilot synchronization.
  - c) compares the computed attitude with the EGPWS signals.
  - d) compares the computed attitude with the ACAS signals.

Answer ;

- a) compares the current attitude with the computed attitude.
- 

0565. The flight director indicates the ;
- a) path permitting reaching a selected radial in minimum time.
  - b) optimum path at the moment it is entered to reach a selected radial.
  - c) path permitting reaching a selected radial over a minimum distance.
  - d) optimum instantaneous path to reach selected radial.

Answer ;

- d) optimum instantaneous path to reach selected radial.
-

**0566.** The flight director is engaged in the heading select mode ( HDG SEL ) , heading 180° selected. When heading is 160° , the vertical bar of the FD ;

- a) is cantered if the aircraft has a 20° right drift.
- b) is cantered if the aircraft has a 20° left drift.
- c) cannot be cantered.
- d) is cantered if the bank angle of the aircraft is equal to the bank angle computed by the flight director calculator.

**Answer ;**

- d)** is cantered if the bank angle of the aircraft is equal to the bank angle computed by the flight director calculator.
- 

**0567.** The flight director provides information for the pilot ;

- a) to join a desired track with a 45° intercept angle.
- b) to join to a desired path with the optimum attitude.
- c) to join a desired track with a constant bank angle of 25°.
- d) to remain within the flight envelope.

**Answer ;**

- b)** to join to a desired path with the optimum attitude.
- 

**0568.** The flight envelope protection function(s) consist(s) in ;

- 1. alerting the flight crew in case of dangerous proximity with the ground.
- 2. avoiding midair collision.
- 3. preventing the aircraft from exceeding some aerodynamic limits.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 3.
- c) 1.
- d) 1 , 3.

**Answer ;**

- b)** 3.
- 

**0569.** The flight envelope protection function(s) consist(s) in ;

- 1. automatically performing an evasion manoeuvre if necessary.
- 2. preventing the aircraft from exceeding the limits for specific flight parameters.
- 3. alerting the flight crew in case of dangerous proximity with a threatening traffic.
- 4. preventing any incursion beyond an ATC clearance.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 2.
- c) 1 , 2.
- d) 1 , 3.

**Answer ;**

- b)** 2.
- 

**0570.** The flight envelope protection system prevents the aircraft from exceeding the limits of the following parameters ( among others ) ;

- 1. cabin altitude.
- 2. bank angle.
- 3. angle of attack.
- 4. speed.
- 5. pitch attitude .

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4 , 5.
- b) 1 , 2 , 4.
- c) 2 , 3 , 4 , 5.
- d) 2 , 5.

**Answer ;**

- c)** 2 , 3 , 4 , 5.
-

**0571.** The flight envelope protection system prevents the aircraft from exceeding the limits of the following parameters ( among others ) ;

1. N1.
2. angle of attack.
3. speed.
4. pitch attitude.
5. flight level.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4 , 5.
- b) 2.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3 , 4 , 5.

**Answer ;**

**c) 2 , 3 , 4.**

---

**0572.** The flight warning system ( FWS ) ;

1. draws the attention of the crew to the existence of an abnormal condition.
2. gives indications to the crew to identify an abnormal condition.
3. transmits automatically to ATC urgency messages.
4. can not generate alerts in case of engine malfunctions.
5. prioritises warnings.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 4.
- c) 1 , 2 , 5.
- d) 2 , 4 , 5.

**Answer ;**

**c) 1 , 2 , 5.**

---

**0573.** The flight warning system ( FWS ) ;

1. increases the situation awareness of the crew.
2. transmits automatically to ATC distress messages.
3. gives suitable indications to the crew of the action necessary to avoid impending danger.
4. prioritises warnings.
5. can not generate alerts in case of aerodynamic limits exceeding.

The combination that regroups all of the correct statements is ;

- a) 1 , 4 , 5.
- b) 1 , 3 , 4.
- c) 1 , 3 , 5.
- d) 2 , 3 , 4 , 5.

**Answer ;**

**b) 1 , 3 , 4.**

---

**0574.** The flight warning system ( FWS ) generates a caution message if ;

- a) crew awareness is required and corrective or compensatory action by the crew is immediately required.
- b) immediate crew awareness is required and subsequent crew action will be required.
- c) immediate recognition and corrective or compensatory action by the crew is required.
- d) crew awareness is required and subsequent crew action may be required.

**Answer ;**

**d) crew awareness is required and subsequent crew action may be required.**

---

**0575.** The flight warning system ( FWS ) generates a warning message if ;

- a) crew awareness is required and subsequent crew action may be required.
- b) immediate crew awareness is required and subsequent crew action will be required.
- c) immediate crew awareness is required and and corrective or compensatory action by the crew may be required.
- d) immediate recognition and corrective or compensatory action by the crew is required.

**Answer ;**

**d) immediate recognition and corrective or compensatory action by the crew is required.**

---

0576. The flight warning system ( FWS ) generates an advisory message if ;
- immediate crew awareness is required and subsequent crew action will be required.
  - immediate recognition and corrective or compensatory action by the crew is required.
  - crew awareness is required and corrective or compensatory action by the crew is required.
  - crew awareness is required and subsequent crew action may be required.

Answer ;

- d) crew awareness is required and subsequent crew action may be required.
- 

0577. The float type fuel gauges provide information on ;
- mass whose indication is independent of the temperature of the fuel.
  - mass whose indication varies with the temperature of the fuel.
  - volume whose indication is independent of the temperature of the fuel.
  - volume whose indication varies with the temperature of the fuel.

Answer ;

- d) volume whose indication varies with the temperature of the fuel.
- 

0578. The flux valve of a gyromagnetic compass ;
- feeds the error detector.
  - feeds the direct indicating compass.
  - gives the earth's magnetic field direction.

The combination regrouping all of the correct statements is ;

- 3.
- 2 , 3.
- 1 , 2.
- 1 , 3.

Answer ;

- d) 1 , 3.
- 

0579. The FMC determines and update present aircraft position from the following systems ;
- GPS.
  - IRS.
  - Navigation radios.
  - ACARS.

The combination that regroups all of the correct statements is ;

- 3.
- 1 , 2 , 4.
- 1 , 2 , 3.
- 1 , 2 , 3 , 4.

Answer ;

- c) 1 , 2 , 3.
- 

0580. The FMC determines and update present aircraft position from the following systems ;
- SATCOM.
  - GPS.
  - DME.
  - IRS.

The combination that regroups all of the correct statements is ;

- 1 , 2 , 4.
- 2 , 3 , 4.
- 1 , 3.
- 2 , 4.

Answer ;

- b) 2 , 3 , 4.
- 

0581. The FMS cross track ( XTK ) is ;
- the angular distance error , to the left or right from the desired track ( DTK ) to the aircraft track ( TK ).
  - the distance error between the FMS computed position and the IRS computed position.
  - the abeam distance error , to the left or right from the desired flight plan leg to the aircraft position.
  - the distance error between the FMS computed position and the GPS computed position.

Answer ;

- c) the abeam distance error , to the left or right from the desired flight plan leg to the aircraft position.
-

**0582.** The FMS enables o fly an optimum flight profile. For this , the FMC ( Flight Management Computer ) uses ;

1. flight-crew entered flight plan data.
2. data from ADC.
3. aircraft position.
4. a memorised relief world data base.
5. data from the FMC navigation data base.

The combination that regroups all of the correct statements is ;

- a) 1 , 3 , 5.
- b) 2 , 3 , 4.
- c) 1 , 2 , 3 , 5.
- d) 1 , 2 , 5.

**Answer ;**

**c) 1 , 2 , 3 , 5.**

---

**0583.** The FMS FLIGHT PLAN or LEG page displays the following parameters relative to the flight plan legs or waypoints ;

1. aircraft position ( Long / Lat ).
2. speed.
3. distance.
4. track.

The combination that regroups all of the correct statements is ;

- a) 1 , 2.
- b) 1 , 3 , 4.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3 , 4.

**Answer ;**

**c) 2 , 3 , 4.**

---

**0584.** The FMS FLIGHT PLAN or LEG page displays the following parameters relative to the flight plan legs or waypoints ;

1. track.
2. magnetic variation.
3. waypoint elevation.
4. speed.

The combination that regroups all of the correct statements is ;

- a) 3 , 4.
- b) 1 , 2 , 4.
- c) 1 , 4.
- d) 1 , 3.

**Answer ;**

**c) 1 , 4.**

---

**0585.** The FMS is approved for Localizer approaches ;

- a) if the GPS only is used as navigation sensor.
- b) if the GPS and DMEs only are used as combined navigation sensors.
- c) if the Localizer signals are used by the FMS.
- d) if the DMEs only are used as navigation sensors.

**Answer ;**

**c) if the Localizer signals are used by the FMS.**

---

**0586.** The FMS is approved to provide guidance for the following approaches ;

1. RNAV.
2. PAR.
3. VOR/DME , VOR.
4. MLS.

The combination which regroups all of the correct statements is ;

- a) 2 , 4.
- b) 1 , 3 , 4.
- c) 1 , 3.
- d) 1 , 2.

**Answer ;**

**c) 1 , 3.**

---

0587. The FMS is approved to provide guidance for the following approaches ;

1. RNAV.
2. ILS.
3. MLS.
4. VOR , NDB.

The combination which regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 2 , 4.
- c) 1 , 3 , 4.
- d) 1 , 4.

Answer ;

d) 1 , 4.

---

0588. The FMS is approved to provide guidance for the following approaches ;

- a) non precision approaches.
- b) non precision approaches and ILS CAT I precision approaches only.
- c) non precision and precision approaches.
- d) precision approaches limited to CAT II.

Answer ;

a) non precision approaches.

---

0589. The FMS lateral offset function consists in ;

- a) flying along the flight plan legs with a constant right or left offset manually entered on the FMS CDU.
- b) displaying the lateral cross track deviation ( XTK ) of the aircraft according to the active flight plan leg.
- c) creating a new waypoint using a reference flight plan waypoint and a distance from this waypoint along the flight plan legs.
- d) flying a FMS selected lateral pattern used for search and rescue operations.

Answer ;

a) flying along the flight plan legs with a constant right or left offset manually entered on the FMS CDU.

---

0590. The FMS navigation database includes the following data ;

1. obstacles.
2. waypoints.
3. SID , STAR.
4. terrain cells.
5. magnetic variation.

The combination which regroups all of the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 3 , 5.
- c) 2 , 3 , 5.
- d) 1 , 2 , 3.

Answer ;

c) 2 , 3 , 5.

---

0591. The FMS navigation database includes the following data ;

1. airports.
2. obstacles.
3. nav aids.
4. airways.
5. relief.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 2 , 5.
- c) 1 , 3 , 4 , 5.
- d) 1 , 3 , 4.

Answer ;

d) 1 , 3 , 4.

---



**0592.** The FMS navigation database includes the following data ;

1. airports.
2. take off speeds.
3. nav aids.
4. relief.
5. runways.

The combination that regroups all of the correct statements is ;

- a) 1 , 3 , 5.
- b) 3 , 4 , 5.
- c) 1 , 2 , 5.
- d) 1 , 2 , 3.

**Answer ;**

**a) 1 , 3 , 5.**

---

**0593.** The FMS navigation database includes the following data ;

1. obstacles.
2. waypoints.
3. SID , STAR.
4. relief.
5. magnetic variation.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 5.
- b) 1 , 3 , 5.
- c) 1 , 2 , 4.
- d) 1 , 2 , 3.

**Answer ;**

**a) 2 , 3 , 5.**

---

**0594.** The FMS navigation database includes the following data ;

1. obstacles.
2. nav aids.
3. SID , STAR and approach procedures.
4. waypoints.
5. airways.

The combination which regroups all of the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 2 , 3 , 4 , 5.
- c) 1 , 3 , 5.
- d) 2 , 3 , 4 , 5.

**Answer ;**

**d) 2 , 3 , 4 , 5.**

---

**0595.** The FMS navigation database includes the following data ;

1. airports.
2. take-off speeds.
3. nav aids.
4. terrain cells.
5. runways.

The combination which regroups all of the correct statements is ;

- a) 1 , 3 , 5.
- b) 1 , 2 , 3.
- c) 3 , 4 , 5.
- d) 1 , 2 , 5.

**Answer ;**

**a) 1 , 3 , 5.**

---

0596. The FMS navigation database includes the following data ;

1. airports.
2. obstacles.
3. nav aids.
4. airways.
5. terrain cells.

The combination which regroups all of the correct statements is ;

- a) 2 , 5.
- b) 1 , 3 , 4.
- c) 1 , 2 , 3.
- d) 1 , 3 , 4 , 5.

Answer ;

**b)** 1 , 3 , 4.

---

0597. The FMS navigation database processing should include the following check(s) ;

- a) at the loading phase into the FMS , the check is performed by the FMS.
- b) at each phase of the process , from the reception of each supplier's data to the distribution and loading of the formatted database.
- c) at the reception of each supplier's data and after the assembly of those data collected.
- d) at the end of the FMS formatting phase of all of the data collected and assembled.

Answer ;

**b)** at each phase of the process , from the reception of each supplier's data to the distribution and loading of the formatted database.

---

0598. The FMS Overfly function consists in ;

- a) manually selecting a flight plan to fly over each of the waypoints at the transitions along the route.
- b) selecting the secondary flight plan making it active to fly over the legs.
- c) manually selecting a flight plan waypoint to hold over for a selected time.
- d) manually selecting a flight plan waypoint to fly over when sequencing it instead of flying by at the transition.

Answer ;

**d)** manually selecting a flight plan waypoint to fly over when sequencing it instead of flying by at the transition.

---

0599. The FMS provides the following functions ;

1. aid for fuel management.
2. lateral flight plan management.
3. check-list completion.
4. aircraft position computation.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 2 , 3.
- c) 3 , 4.
- d) 1 , 2 , 4.

Answer ;

**d)** 1 , 2 , 4.

---

0600. The FMS provides the following functions ;

1. lateral and vertical flight plan management.
2. de icing management.
3. aircraft position computation.
4. terrain awareness and warning.

The combination that regroups all of the correct statements is ;

- a) 4.
- b) 1 , 3 , 4.
- c) 1 , 3.
- d) 1 , 2 , 3.

Answer ;

**c)** 1 , 3.

---

- 0601.** The FMS provides the following functions ;
1. traffic advisories emission.
  2. resolution advisories emission.
  3. aid for fuel management.
  4. lateral flight plan management.

The combination that regroups all of the correct statements is ;

- a) 1, 2, 3.
- b) 3, 4.
- c) 1, 3, 4.
- d) 1, 4.

**Answer ;**

**b) 3, 4.**

---

- 0602.** The FMS provides the following functions ;
1. vertical flight plan management.
  2. fuel management.
  3. lateral flight plan management.
  4. terrain awareness and warning.

The combination which regroups all of the correct statements is ;

- a) 1, 2, 3.
- b) 2, 3, 4.
- c) 1, 2, 4.
- d) 1, 3, 4.

**Answer ;**

**a) 1, 2, 3.**

---

- 0603.** The FMS provides the following functions ;
1. fuel management.
  2. lateral flight plan management.
  3. de-icing management.
  4. aircraft position computation.

The combination which regroups all of the correct statements is ;

- a) 1, 2, 3.
- b) 1, 2, 4.
- c) 2, 3.
- d) 3, 4.

**Answer ;**

**b) 1, 2, 4.**

---

- 0604.** The FMS provides the following functions ;
1. aircraft position computation.
  2. traffic alerts.
  3. lateral flight plan management.
  4. fuel management.

The combination which regroups all of the correct statements is ;

- a) 1, 2, 4.
- b) 2, 3, 4.
- c) 1, 3, 4.
- d) 1, 2, 3.

**Answer ;**

**c) 1, 3, 4.**

---

- 0605.** The FMS provides the following functions ;
1. radio tuning.
  2. fuel management.
  3. lateral flight plan management.
  4. traffic alerts.

The combination which regroups all of the correct statements is ;

- a) 1, 2, 4.
- b) 2, 3, 4.
- c) 3, 4.
- d) 1, 2, 3.

**Answer ;**

**d) 1, 2, 3.**

---

- 0606.** The FMS Required Time of Arrival ( RTA ) function can provide ;
- a) a time slot computed for the arrival time at destination , using the current aircraft speed and speed constraints along the flight plan.
  - b) a time prediction at the active TO waypoint complying with the wind computation.
  - c) a speed target to satisfy a time constraint entered at a flight plan waypoint.
  - d) a time prediction at the flight plan waypoints based on the current speed and speed constraints along the flight plan.

**Answer ;**

- c)** a speed target to satisfy a time constraint entered at a flight plan waypoint.
- 

- 0607.** The FMS vertical navigation management is generally performed based on ;
- a) a mix of baro and GPS altitudes.
  - b) the baro altitude input from the ADC.
  - c) the GPS altitude computed by the GPS receiver.
  - d) the geometric altitude input from the Terrain Awareness and Warning System ( TAWS ).

**Answer ;**

- b)** the baro altitude input from the ADC.
- 

- 0608.** The FMS vertical navigation management is generally performed based on ;
- a) the GPS altitude computed by the GPS receiver.
  - b) the baro altitude input from the air data system.
  - c) a mix of baro and GPS altitudes.
  - d) the geometric altitude input from the Terrain Awareness and Warning System ( TAWS ).

**Answer ;**

- b)** the baro altitude input from the air data system.
- 

- 0609.** The following ATC clearances can be received via the datalink application ;
- a) Departure , Oceanic.
  - b) Take off , En route , Landing.
  - c) Departure , En route , Landing.
  - d) Departure , Take off , En route.

**Answer ;**

- a)** Departure , Oceanic.
- 

- 0610.** The Fuel management performed by most FMS along the flight plan is considered as ;
- a) an accurate function which can be considered as the prime source to determine the remaining fuel quantity along the flight plan.
  - b) an accurate and very reliable function providing that the fuel on board quantity has been properly initialized by the crew before start up.
  - c) a function which helps the crew to estimate the remaining fuel quantity along the flight plan but should not be considered as an accurate and reliable source.
  - d) the prime source to manage the fuel consumption along the flight.

**Answer ;**

- c)** a function which helps the crew to estimate the remaining fuel quantity along the flight plan but should not be considered as an accurate and reliable source.
- 

- 0611.** The fuel management performed by most FMS along the flight plan is considered as ;
- a) the prime mean to manage the Fuel consumption along the flight.
  - b) an accurate and very reliable function providing that the Fuel on board quantity has been properly initialized by the crew before start up.
  - c) an accurate function which can be considered as the prime mean to determine the remaining Fuel quantity along the flight plan.
  - d) a function which helps the crew to estimate the remaining Fuel quantity along the flight plan but should not be considered as an accurate and reliable mean.

**Answer ;**

- d)** a function which helps the crew to estimate the remaining Fuel quantity along the flight plan but should not be considered as an accurate and reliable mean.
- 

- 0612.** The functions of an autopilot ( basic modes ) consist of ;
- a) guiding the airplane path.
  - b) stabilizing and monitoring the movement around the airplane centre of gravity.
  - c) stabilizing and monitoring the movement around the airplane aerodynamic centre.
  - d) monitoring the movement of the airplane centre of gravity.

**Answer ;**

- b)** stabilizing and monitoring the movement around the airplane centre of gravity.
-

- 0613.** The functions of the altitude alerting system is to alert the flight crew ;
1. upon approaching a pre-selected altitude.
  2. upon approaching a pre-selected altitude , during climb only.
  3. of a loss of altitude during take-off or missed approach.
  4. of a wrong landing configuration.
  5. when deviating from the selected altitude.

The combination regrouping all the correct statements is ;

- a) 3 , 4.
- b) 1 , 5.
- c) 1 , 3 , 4.
- d) 2 , 5.

**Answer ;**

- b) 1 , 5.**
- 

- 0614.** The gauge indicating the quantity of fuel measured by a capacity gauging system can be graduated directly in weight units because the dielectric constant of fuel is ;

- a) twice that of air and varies inversely with density.
- b) twice that of air and varies directly with density.
- c) the same as that of air and varies directly with density.
- d) the same as that of air and varies inversely with density.

**Answer ;**

- b) twice that of air and varies directly with density.**
- 

- 0615.** The GPWS automatically provides distinctive warning to the flight crew in case of ;

1. impending stall.
2. excessive descent rate.
3. altitude loss after take-off or go-around.
4. unsafe terrain clearance with flaps not in landing configuration.
5. dangerous ground proximity.
6. downward glide-slope deviation.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4 , 5 , 6.
- b) 1 , 3 , 4 , 5.
- c) 1 , 2 , 6.
- d) 2 , 3 , 4 , 5.

**Answer ;**

- a) 2 , 3 , 4 , 5 , 6.**
- 

- 0616.** The GPWS calculator is able to operate in the following modes ;

1. excessive descent rate.
2. excessive rate of terrain closure.
3. excessive angle of attack.
4. too high descent attitude.
5. loss of altitude after take-off.
6. abnormal gear/flaps configuration.
7. excessive glidepath deviation.

The combination regrouping all the correct statements is ;

- a) 3 , 4 , 5 , 6.
- b) 1 , 2 , 4 , 6 , 7.
- c) 1 , 2 , 5 , 6 , 7.
- d) 2 , 3 , 5 , 7.

**Answer ;**

- c) 1 , 2 , 5 , 6 , 7.**
-

- 0617.** The GPWS can warn the crew in case of ;
1. excessive climb rate.
  2. unsafe terrain clearance when not in landing configuration.
  3. descent path angle greater than 5 degrees.
  4. windshear.

The combination that regroups all of the correct statements is ;

- a) 2 , 4.
- b) 3.
- c) 1 , 2 , 3.
- d) 2.

**Answer ;**

**a) 2 , 4.**

---

- 0618.** The GPWS can warn the crew in case of ;
1. excessive deviation below selected altitude.
  2. windshear.
  3. excessive terrain closing rate.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 2 , 3.
- c) 3.
- d) 1.

**Answer ;**

**b) 2 , 3.**

---

- 0619.** The GPWS computer receives the following signals ;
1. vertical speed.
  2. radio altimeter height.
  3. pressure altitude.
  4. glidepath deviation.
  5. gear and flaps position.
  6. flight path angle.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4 , 6.
- b) 1 , 2 , 4 , 5.
- c) 1 , 3 , 4 , 5 , 6.
- d) 1 , 2 , 5 , 6.

**Answer ;**

**b) 1 , 2 , 4 , 5.**

---

- 0620.** The GPWS CPU ( Central Processing Unit ) is able to detect ;
1. excessive descent rate.
  2. excessive rate of terrain closure.
  3. excessive angle of attack.
  4. too high descent attitude.
  5. loss of altitude after take-off.
  6. abnormal gear/flaps configuration.
  7. excessive glidepath deviation.

The combination that regroups all of the correct statements is ;

- a) 3 , 4 , 5 , 6.
- b) 2 , 3 , 5 , 7.
- c) 1 , 2 , 5 , 6 , 7.
- d) 1 , 2 , 4 , 6 , 7.

**Answer ;**

**c) 1 , 2 , 5 , 6 , 7.**

---

**0621.** The GPWS generates a warning in the following cases ;

1. excessive descent rate.
2. excessive terrain closing rate.
3. altitude loss after take-off go-around.
4. unsafe terrain clearance with abnormal gear/flaps configuration.
5. excessive deviation under the glidepath.
6. abnormal airbrakes configuration.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 5.
- b) 1 , 2 , 3 , 4 , 5.
- c) 1 , 3 , 4.
- d) 2 , 4.

**Answer ;**

- b)** 1 , 2 , 3 , 4 , 5.
- 

**0622.** The GPWS ( Ground Proximity Warning System ) releases a warning in the following cases ;

1. excessive rate of descent.
2. excessive ground proximity rate.
3. loss of altitude after take-off or go-around.
4. abnormal gear/flaps configuration.
5. excessive deviation under the glidepath.
6. abnormal airbrakes configuration.

The combination regrouping all the correct statements is ;

- a) 3 , 4 , 5 , 6.
- b) 1 , 2 , 3 , 4 , 5.
- c) 2 , 4 , 5 , 6.
- d) 1 , 2 , 3 , 4 , 5 , 6.

**Answer ;**

- b)** 1 , 2 , 3 , 4 , 5.
- 

**0623.** The GPWS is able to detect ;

1. excessive descent rate.
2. excessive terrain closing rate.
3. excessive angle of attack.
4. excessive descent pitch attitude.

The combination that regroups all of the correct statements is ;

- a) 1 , 2.
- b) 1 , 2 , 3.
- c) 2 , 4.
- d) 1 , 4.

**Answer ;**

- a)** 1 , 2.
- 

**0624.** The GPWS receives data from the following systems ;

1. landing gear and flaps systems.
2. engine control computer ( FADEC or ECU ).
3. radio altimeter.
4. TCAS.

The combination that regroups all of the correct statements is ;

- a) 1 , 2.
- b) 3 , 4.
- c) 1 , 3.
- d) 1 , 2 , 3 , 4.

**Answer ;**

- c)** 1 , 3.
-

- 0625.** The GPWS warns the crew in case of ;
1. excessive descent rate.
  2. excessive terrain closure rate.
  3. potential midair collision threat.
  4. serious midair collision threat.
  5. unsafe terrain clearance with landing gear not down.

The combination that regroups all of the correct statements is ;

- a) 4.
- b) 1 , 2 , 3 , 5.
- c) 1 , 2 , 5.
- d) 1 , 2 , 4 , 5.

**Answer ;**

- c) 1 , 2 , 5.**
- 

- 0626.** The GPWS warns the crew in case of ;
1. deviation above or below the selected altitude.
  2. deviation below the selected altitude.
  3. unsafe terrain clearance with flaps not in landing configuration.
  4. unsafe terrain clearance with landing gear not down.

The combination that regroups all of the correct statements is ;

- a) 3 , 4.
- b) 2 , 3 , 4.
- c) 1 , 4.
- d) 2 , 3.

**Answer ;**

- a) 3 , 4.**
- 

- 0627.** The GPWS warns the crew in case of ;
1. an altitude at a lower level than the one shown in the flight plan entered in the FMS.
  2. a dangerous ground proximity.
  3. a loss of altitude during take-off or missed approach.
  4. a wrong landing configuration.
  5. a descent below glidepath.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4.
- b) 2 , 3 , 4 , 5.
- c) 2.
- d) 2 , 5.

**Answer ;**

- b) 2 , 3 , 4 , 5.**
- 

- 0628.** The Ground Proximity Warning System ( GPWS ) generates the following sound signal or signals when the aircraft is sinking after a take-off or a go-around ;

- a) WHOOP WHOOP PULL UP repetitive only.
- b) DON'T SINK repetitive only.
- c) DON'T SINK always followed by WHOOP WHOOP PULL UP.
- d) DON'T SINK followed by WHOOP WHOOP PULL UP if the sink rate overshoots a second level.

**Answer ;**

- b) DON'T SINK repetitive only.**
- 

- 0629.** The Ground Proximity Warning System ( GPWS ) is a system working according to a height span ranging from ;

- a) 50 ft to 2 500 ft.
- b) 30 ft to 5 000 ft.
- c) the ground to 1 000 ft.
- d) the ground to 500 ft.

**Answer ;**

- a) 50 ft to 2 500 ft.**
-



0630. The ground routing of the ATC datalink communications is performed ;
- a) by the airline ground network interconnected to the ATC network.
  - b) by the GSM ( mobile phone ) providers interconnected to provide continuity of transmissions.
  - c) by each ATC local network of the FIR airspace where the aircraft flies into.
  - d) by service providers ( SITA , ARINC ) that can be interconnected to provide continuity of the transmissions.

Answer ;

- d) by service providers ( SITA , ARINC ) that can be interconnected to provide continuity of the transmissions.
- 

0631. The “guidance” functions of a autopilot consist in ;
- a) monitoring the movements of the aerodynamic centre in the three dimensions of space.
  - b) stabilizing and monitoring the movements around the centre of gravity.
  - c) stabilizing and monitoring the movements around the aerodynamic centre.
  - d) monitoring the movements of the centre of gravity in the three dimensions of space.

Answer ;

- d) monitoring the movements of the centre of gravity in the three dimensions of space.
- 

0632. The gyro axis of an electric artificial horizon is tied to the ;
- a) earth’s vertical by four pendulous vanes.
  - b) earth’s horizontal by two mercury level switches and two torque motors.
  - c) earth’s vertical by two mercury level switches and two torque motors.
  - d) earth’s horizontal by four pendulous vanes.

Answer ;

- c) earth’s vertical by two mercury level switches and two torque motors.
- 

0633. The gyromagnetic compass torque motor ;
- a) causes the heading indicator to precess.
  - b) feeds the error detector system.
  - c) is fed by the flux valve.
  - d) causes the directional gyro unit to precess.

Answer ;

- d) causes the directional gyro unit to precess.
- 

0634. The gyroscope of a turn indicator has ;
- NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.
- a) 2 degrees of freedom.
  - b) 0 degree of freedom.
  - c) 1 degree of freedom.
  - d) 3 degrees of freedom.

Answer ;

- c) 1 degree of freedom.
- 

0635. The gyroscope used in an attitude indicator has a spin axis which is ;
- a) horizontal , perpendicular to the yaw axis.
  - b) horizontal , parallel to the longitudinal axis.
  - c) horizontal , perpendicular to the longitudinal axis.
  - d) vertical.

Answer ;

- d) vertical.
- 

0636. The heading information originating from the gyromagnetic compass flux valve is sent to the ;
- a) erector system.
  - b) error detector.
  - c) amplifier.
  - d) heading indicator.

Answer ;

- b) error detector.
-

**0637.** The heading read on the dial of a directional gyro is subject to errors , one of which is due to the movement of the aircraft. This error ;

- a) is at its greatest value when the aircraft follows a meridional track.
- b) is , in spite of this , insignificant and may be neglected.
- c) is dependent on the ground speed of the aircraft , its true track and the latitude of the flight.
- d) shows itself by an apparent rotation of the horizontal axis of the gyroscope which seems to turn at 15° per hour to the right in the northern hemisphere.

**Answer ;**

- c)** is dependent on the ground speed of the aircraft , its true track and the latitude of the flight.
- 

**0638.** The horizontal command bar of a flight director ;

- a) gives information only about the direction of the corrections to be applied on the pitch of the aircraft.
- b) gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.
- c) repeats the position information given by the ILS in the horizontal plane.
- d) repeats the position information given by the ILS in the vertical plane.

**Answer ;**

- b)** gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.
- 

**0639.** The horizontal command bar of a flight director ;

- 1. repeats the position information given by the ILS in the horizontal plane.
- 2. repeats the position information given by the ILS in the vertical plane.
- 3. gives information about the direction and the amplitude of the corrections to be applied on the pitch of the aircraft.

The combination regrouping all the correct statements is ;

- a) 3.
- b) 2 , 3.
- c) 2.
- d) 1 , 3.

**Answer ;**

- a)** 3.
- 

**0640.** The indicated Mach number is independent from ;

- a) dynamic pressure.
- b) total pressure.
- c) temperature.
- d) static pressure.

**Answer ;**

- c)** temperature.
- 

**0641.** The indication of a fuel float gauge varies with ;

- 1. aircraft attitude.
- 2. accelerations.
- 3. atmospheric pressure.
- 4. temperature.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 2 , 4.
- c) 1 , 2.
- d) 4.

**Answer ;**

- b)** 1 , 2 , 4.
-

**0642.** The indication of a fuel float gauge varies with ;

1. aircraft attitude.
2. accelerations.
3. fuel temperature.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 3.
- c) 1 , 2.
- d) 1 , 2 , 3.

**Answer ;**

**d) 1 , 2 , 3.**

---

**0643.** The indication of the directional gyro is valid only for a limited period of time. The causes of this inaccuracy are ;

1. rotation of the earth.
2. longitudinal accelerations.
3. aircraft's moving over the surface of the earth.
4. vertical components of the earth's magnetic field.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 3 , 4.
- c) 1 , 3.
- d) 1 , 2 , 3.

**Answer ;**

**c) 1 , 3.**

---

**0644.** The indication of the directional gyro is valid only for a limited period of time. The causes of this inaccuracy are ;

1. rotation of the earth.
2. longitudinal accelerations.
3. aircraft's moving over the surface of the earth.
4. mechanical imperfections.
5. vertical components of the earth's magnetic field.
6. geometry of the gimbal system.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 4 , 6.
- b) 1 , 3 , 4 , 6.
- c) 2 , 3 , 5.
- d) 1 , 2 , 3 , 4 , 6.

**Answer ;**

**b) 1 , 3 , 4 , 6.**

---

**0645.** The indications on a directional gyroscope are subject to errors. The most significant are ;

1. apparent wander due to earth rotation.
2. apparent wander due to change of aircraft position.
3. gimbaling errors.
4. north change.
5. mechanical defects.

The combination regrouping the correct statement is ;

- a) 1 , 2 , 4 , 5.
- b) 2 , 3 , 5.
- c) 3 , 4 , 5.
- d) 1 , 2 , 3 , 5.

**Answer ;**

**d) 1 , 2 , 3 , 5.**

---

**0646.** The inertia of a gyroscope is greater when ;

- a) its rotation speed is lower and the mass of the spinning wheel is located further from the axis of rotation.
- b) its rotation speed is lower and the mass of the spinning wheel is closer to the axis of rotation.
- c) its rotation speed is higher and the mass of the spinning wheel is located further from the axis of rotation.
- d) its rotation speed is higher and the mass of the spinning wheel is closer to the axis of rotation.

**Answer ;**

**c) its rotation speed is higher and the mass of the spinning wheel is located further from the axis of rotation.**

---

0647. The initiation of an automatic go-around can be ;
- at  $V_Y \pm 5$  kt only.
  - in a range of approach airspeeds.
  - at  $70 \text{ kt} \pm 5 \text{ kt}$  only.
  - with no limit of approach airspeeds.

Answer ;

**b)** in a range of approach airspeeds.

---

0648. The input data of an ADC are ;
- OAT.
  - TAT.
  - static pressure.
  - total pressure.

The combination regrouping the correct statement is ;

- 1, 3, 4.
- 1, 2, 4.
- 1, 2, 3, 4.
- 2, 3, 4.

Answer ;

**d)** 2, 3, 4.

---

0649. The input data to the GPWS originate from the ;
- transponder.
  - angle of attack sensor.
  - auto throttle system.
  - ADC.

The combination that regroups all of the correct statements is ;

- 2.
- 4.
- 1, 4.
- 3, 4.

Answer ;

**b)** 4.

---

0650. The input signal of the amplifier of the gyromagnetic compass resetting device originates from the ;
- error detector.
  - flux valve.
  - directional gyro erection device.
  - directional gyro unit.

Answer ;

**a)** error detector.

---

0651. The input(s) of a  $V_{MO}/M_{MO}$  warning system is(are) ;
- static pressure and aircraft configuration.
  - static pressure and SAT.
  - static pressure only.
  - static pressure and total pressure.

Answer ;

**d)** static pressure and total pressure.

---

0652. The inputs to the GPWS ( Ground Proximity Warning System ) , are ;
- Air Data Computer ( Mach number and Vertical Speed ).
  - Radio Altimeter.
  - NAV/ILS ( Glide Slope ).
  - NAV/VOR.
  - Flap ( position ).
  - Weight of the aircraft.
  - Landing Gear ( position ).

The combination of correct statement is ;

- 1, 2, 3, 5, 7.
- 2, 4, 5, 6.
- 2, 5, 6.
- 1, 4, 7.

Answer ;

**a)** 1, 2, 3, 5, 7.

---

- 0653.** The inputs to the GPWS ( Ground Proximity Warning System ) , are ;
1. Air Data Computer ( Mach number and Vertical Speed ).
  2. Radio Altimeter.
  3. NAV/ILS ( Glide Slope ).
  4. NAV/VOR.
  5. Flap ( position ).
  6. Angle of Attack.
  7. Landing Gear ( position ).

The combination of correct statement is ;

- a) 1 , 2 , 5 , 6 , 7.
- b) 1 , 2 , 3 , 6 , 7.
- c) 1 , 2 , 3 , 5 , 7.
- d) 2 , 3 , 4 , 5 , 7.

**Answer ;**

- c) 1 , 2 , 3 , 5 , 7.**
- 

- 0654.** The lateral flight path modes of an autopilot system are ;

1. Heading hold.
2. Speed hold.
3. FMS lateral navigation.
4. TAS hold.
5. Localizer intercept and track.

The combination that regroups all of the correct statements is ;

- a) 2 , 5.
- b) 1 , 3 , 4.
- c) 1 , 2 , 3 , 5.
- d) 1 , 3 , 5.

**Answer ;**

- d) 1 , 3 , 5.**
- 

- 0655.** The lateral flight path modes of an autopilot system are ;

1. Speed hold.
2. Localiser intercept and track.
3. Track hold.
4. FMS lateral navigation.
5. Pitch attitude hold.

The combination that regroups all of the correct statements is ;

- a) 1 , 3 , 5.
- b) 2 , 4.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3 , 4.

**Answer ;**

- c) 2 , 3 , 4.**
- 

- 0656.** The latitude at which the apparent wander of a directional gyro is equal to 0 is ;

- a) latitude 45°.
- b) the Equator.
- c) latitude 30°.
- d) the North pole.

**Answer ;**

- b) the Equator.**
- 

- 0657.** The limit speed maximum operating (  $V_{MO}$  ) is a speed expressed as ;

- a) true airspeed ( TAS ).
- b) calibrated airspeed ( CAS ).
- c) computed airspeed ( COAS ).
- d) equivalent airspeed ( EAS ).

**Answer ;**

- b) calibrated airspeed ( CAS ).**
-

0658. The limits of the green scale of an airspeed indicator are ;

- a)  $V_{S0}$  and  $V_{NE}$ .
- b)  $V_{S1}$  and  $V_{NO}$ .
- c)  $V_{S1}$  and  $V_{NE}$ .
- d)  $V_{S1}$  and  $V_{MO}$ .

Answer ;

- b)  $V_{S1}$  and  $V_{NO}$ .
- 

0659. The limits of the green scale of an airspeed indicator are ;

- a)  $V_{S1}$  for the lower limit and  $V_{LO}$  for the upper limit.
- b)  $V_{S1}$  for the lower limit and  $V_{NO}$  for the upper limit.
- c)  $V_{S0}$  for the lower limit and  $V_{NO}$  for the upper limit.
- d)  $V_{S1}$  for the lower limit and  $V_{NE}$  for the upper limit.

Answer ;

- b)  $V_{S1}$  for the lower limit and  $V_{NO}$  for the upper limit.
- 

0660. The limits of the white scale of an airspeed indicator are ;

- a)  $V_{S0}$  for the lower limit and  $V_{FE}$  for the upper limit.
- b)  $V_{S0}$  for the lower limit and  $V_{LE}$  for the upper limit.
- c)  $V_{S1}$  for the lower limit and  $V_{FE}$  for the upper limit.
- d)  $V_{S1}$  for the lower limit and  $V_{LE}$  for the upper limit.

Answer ;

- a)  $V_{S0}$  for the lower limit and  $V_{FE}$  for the upper limit.
- 

0661. The limits of the yellow scale of an airspeed indicator are ;

- a)  $V_{LE}$  for the lower limit and  $V_{NE}$  for the upper limit.
- b)  $V_{FE}$  for the lower limit and  $V_{NE}$  for the upper limit.
- c)  $V_{NO}$  for the lower limit and  $V_{NE}$  for the upper limit.
- d)  $V_{LO}$  for the lower limit and  $V_{NE}$  for the upper limit.

Answer ;

- c)  $V_{NO}$  for the lower limit and  $V_{NE}$  for the upper limit.
- 

0662. The low-altitude radio altimeters used in precision approaches ;

1. operate in the 1540-1660 MHz range.
2. are of the pulsed type.
3. are of the frequency modulation type.
4. have an operating range of 0 to 5000 ft.
5. have a precision of +/- 2 feet between 0 and 500 ft and +/- 1.5% whichever is the greatest.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4.
- b) 1 , 2 , 5.
- c) 3 , 5.
- d) 3 , 4.

Answer ;

- c) 3 , 5.
- 

0663. The Low Altitude Radio Altimeter uses the following wavelengths ;

- a) decimetric.
- b) metric.
- c) myriametric.
- d) centimetric.

Answer ;

- d) centimetric.
- 

0664. The Mach number is ;

- a) a direct function of temperature , it varies in proportion to the square root of the absolute temperature.
- b) the ratio of the indicated airspeed to the sonic velocity at the altitude considered.
- c) the ratio of the aircraft true airspeed to the sonic velocity at the altitude considered.
- d) the ratio of the aircraft conventional airspeed to the sonic velocity at the altitude considered.

Answer ;

- c) the ratio of the aircraft true airspeed to the sonic velocity at the altitude considered.
-

0665. The mach number is proportional to the ratio ;  
NB : “a” indicates the local speed of sound.

- a) EAS/a
- b) TAS/a
- c) IAS/a
- d) CAS/a

Answer ;

**b) TAS/a**

---

0666. The mach number is the ;

- a) equivalent airspeed ( EAS ) divided by the local speed of sound.
- b) corrected airspeed ( CAS ) divided by the local speed of sound.
- c) indicated airspeed ( IAS ) divided by the local speed of sound.
- d) true airspeed ( TAS ) divided by the local speed of sound.

Answer ;

**d) true airspeed ( TAS ) divided by the local speed of sound.**

---

0667. The Mach trim system allows to ;

- a) trim the pitch-up tendency at a high Mach number.
- b) search for the ideal CG location by transferring the fuel into the horizontal stabilizer.
- c) increase the longitudinal static stability of the aircraft by changing the horizontal stabilizer according to the Mach number.
- d) interlock the operation of the stick shaker at the oncoming of the high speed stall.

Answer ;

**c) increase the longitudinal static stability of the aircraft by changing the horizontal stabilizer according to the Mach number.**

---

0668. The machmeter is subject to position error. This error concerns ;

- a) pitot tubes and static ports.
- b) pitot tubes only.
- c) alternate static sources only.
- d) static ports only.

Answer ;

**a) pitot tubes and static ports.**

---

0669. The Machmeter is subject to position error. This error results from ;

- a) non standard atmospheric conditions.
- b) imperfect elasticity of the capsules.
- c) incorrect altimeter setting.
- d) incorrect pressure sensing caused by disturbed airflow around the pitot tube and/or static ports.

Answer ;

**d) incorrect pressure sensing caused by disturbed airflow around the pitot tube and/or static ports.**

---

0670. The machmeter is subject to position error. This error varies according to ;

- 1. angle of attack.
- 2. OAT.
- 3. TAS.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 3.
- c) 3.
- d) 1.

Answer ;

**b) 1 , 3.**

---

0671. The machmeter is subject to position error. This error varies according to ;

- a) TAS only.
- b) TAS and OAT.
- c) OAT only.
- d) TAS and angle of attack.

Answer ;

**d) TAS and angle of attack.**

---

0672. The magnetic heading can be derived from the true heading by means of a ;
- map showing the isogon lines.
  - compass swinging curve.
  - deviation correction curve.
  - map showing the isoclinic lines.

Answer ;

a) map showing the isogon lines.

---

0673. The main function(s) of a TCAS is to ;
1. alert the crew to ground proximity.
  2. alert the crew to possible conflicting traffic.
  3. provide terrain alerting and display.
  4. automatically resolve conflict when autopilot engaged.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 3.
- c) 2.
- d) 2 , 4.

Answer ;

c) 2.

---

0674. The main input data to the Stall Warning Annunciator System are ;

1. Mach Meter indication.
2. Angle of Attack.
3. Indicate Airspeed ( IAS ).
4. Aircraft configuration ( Flaps/Slats ).

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 1 , 4.
- c) 2 , 3.
- d) 2 , 4.

Answer ;

d) 2 , 4.

---

0675. The main inputs to the flight envelope protection system are ;

1. GPWS signals.
2. ACAS signals.
3. angle of attack.
4. bank angle.

The combination that regroups all of the correct statements is ;

- a) 3 , 4.
- b) 1 , 2 , 3 , 4.
- c) 2 , 3 , 4.
- d) 1 , 2.

Answer ;

a) 3 , 4.

---

0676. The maximum directional gyro error due to the earth rotation is ;

- a) 15°/hour.
- b) 180°/hour.
- c) 90°/hour.
- d) 5°/hour.

Answer ;

a) 15°/hour.

---

0677. The maximum TAS is obtained at ;

- a) all the Flight Level(s) where  $CAS = V_{MO}$ .
- b) all the Flight Level(s) where  $M = M_{MO}$ .
- c) the maximum Flight Level.
- d) the Flight Level at which simultaneously  $CAS = V_{MO}$  and  $M = M_{MO}$ .

Answer ;

d) the Flight Level at which simultaneously  $CAS = V_{MO}$  and  $M = M_{MO}$ .

---



0678. The maximum  $V_{MO}$  is expressed in ;

- a) CAS or TAS.
- b) TAS only.
- c) TAS or EAS.
- d) CAS or EAS.

Answer ;

d) CAS or EAS.

---

0679. The measurement of the turbine temperature or of the EGT ( Exhaust Gas Temperature ) is carried out at the ;

- a) high pressure chamber intake.
- b) combustion chamber intake.
- c) combustion chamber outlet.
- d) high pressure turbine outlet.

Answer ;

d) high pressure turbine outlet.

---

0680. The mode selector of an inertial unit comprises the OFF - STBY - ALIGN - NAV - ATT positions ;

- 1. on "STBY" , the unit aligns on the local geographic trihedron.
- 2. the "ATT" position is used in automatic landing ( mode LAND ).
- 3. on "NAV" the coordinates of the start position can be entered.
- 4. the platform is levelled before azimuth alignment.
- 5. in cruise , the unit can only be used in "NAV" mode.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 5.
- b) 4.
- c) 2 , 5.
- d) 1 , 2 , 4.

Answer ;

b) 4.

---

0681. The mode selector of an inertial unit comprises the OFF - STBY - ALIGN - NAV - ATT positions ;

- 1. on "STBY" , the unit aligns on the local geographic trihedron.
- 2. the "ATT" position is used in automatic landing ( mode LAND ).
- 3. on "NAV" the coordinates of the start position can be entered.
- 4. the horizontal alignment precedes the unit orientation calculation.
- 5. in cruise , the unit can only be used in "NAV" mode.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 5.
- b) 1 , 2 , 4.
- c) 4.
- d) 2 , 5.

Answer ;

c) 4.

---

0682. The most common sensors interfacing a FMS to compute the aircraft position along the flight plan are ;

- 1. DME.
- 2. GPS.
- 3. LOCALIZER.
- 4. NDB.

The combination which regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2.
- c) 1 , 3 , 4.
- d) 2 , 3.

Answer ;

b) 1 , 2.

---

**0683.** The most common sensors interfacing a FMS to compute the aircraft position along the flight plan are ;

1. GPS.
2. NDB.
3. DME.
4. LOCALIZER.

The combination which regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 4.
- c) 1 , 3.
- d) 1 , 2 , 3.

**Answer ;**

**c) 1 , 3.**

---

**0684.** The most common sensors interfacing a FMS to compute the aircraft position along the flight plan are ;

1. IRS.
2. DME.
3. NDB.
4. GPS.

The combination which regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 2 , 4.
- c) 2 , 3 , 4.
- d) 1 , 3.

**Answer ;**

**b) 1 , 2 , 4.**

---

**0685.** The most common sensors interfacing a FMS to compute the aircraft position along the flight plan are ;

1. MLS.
2. GPS.
3. VOR.
4. IRS.

The combination which regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 1 , 4.
- c) 1 , 2 , 4.
- d) 1 , 2.

**Answer ;**

**a) 2 , 3 , 4.**

---

**0686.** The most common system used to monitor turbine gas exhaust temperature is the ;

- a) hot junction Tungsten/copper system.
- b) hot and cold junction , alumel/chromel system.
- c) flame switch.
- d) fixed junction mercurial oxide/chromium system.

**Answer ;**

**b) hot and cold junction , alumel/chromel system.**

---

**0687.** The most favourable to apply the flexible take-off procedure are ;

1. high take-off mass.
2. low take-of mass.
3. high outside temperature.
4. low outside temperature.
5. high atmospheric pressure.
6. low atmospheric pressure.

The combination grouping all the correct statements is ;

- a) 1 , 3 , 5.
- b) 2 , 3 , 6.
- c) 2 , 4 , 6.
- d) 2 , 4 , 5.

**Answer ;**

**d) 2 , 4 , 5.**

---

**0688.** The most significant advantages of an air data computer ( ADC ) are ;

1. Position error correction.
2. Hysteresis error correction.
3. Remote data transmission capability.
4. Gimballing errors correction.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2 , 4.
- c) 1 , 3.
- d) 2 , 3 , 4.

**Answer ;**

**c) 1 , 3.**

---

**0689.** The Navigation Display ( ND ) modes can be ;

1. ARC or MAP , covering 45 degrees on either side of the instantaneous track.
2. ROSE or MAP CENTERED : rose with current heading up.
3. PLAN : map orientated to true north.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 2 , 3.
- c) 1 , 2 , 3.
- d) 1 , 2.

**Answer ;**

**c) 1 , 2 , 3.**

---

**0690.** The Navigation Display ( ND ) of an EFIS equipped aircraft can display the following data ;

- a) terrain map , navaid bearings , flight director active modes.
- b) intruding traffics , altitude , autopilot active modes , weather radar.
- c) flight plan , weather radar , terrain map , intruding traffic.
- d) flight plan , engine failure , nav aids , resolution advisories.

**Answer ;**

**c) flight plan , weather radar , terrain map , intruding traffic.**

---

**0691.** The navigation precision of a stand alone inertial system decreases along the flight , due to ;

- a) the accelerations of the aircraft.
- b) the motion of the aircraft.
- c) the drift of the gyroscopes.
- d) the meteorological conditions.

**Answer ;**

**c) the drift of the gyroscopes.**

---

**0692.** The Notification phase ( LOG ON ) is a FANS application which consists in ;

- a) transmitting datalink messages between the pilot and ATC controller.
- b) sending automatically aircraft surveillance data to the ATS facility.
- c) establishing air/ground connection to verify if the datalink communication can be performed.
- d) requesting to transfer datalink communication to the next ATC center on the route.

**Answer ;**

**c) establishing air/ground connection to verify if the datalink communication can be performed.**

---

**0693.** The Notification phase ( LOG ON ) is a FANS application which consists in transmitting aircraft information and ;

- a) aircraft position to request the datalink clearance to enter an airspace.
- b) aircraft position to the airport ground controller to request the datalink taxi clearance.
- c) park stand to the airport pre-flight controller to request the datalink pre-departure clearance ( PDC ).
- d) associated datalink capability , prior to operating any datalink communications with ATC.

**Answer ;**

**d) associated datalink capability , prior to operating any datalink communications with ATC.**

---

0694. The on-board communication devices to transmit and receive datalink communication can be ;

1. VHF COM.
2. HF COM.
3. SATCOM.

The combination which regroups all of the correct statements is ;

- a) 3.
- b) 1 , 2 , 3.
- c) 1.
- d) 1 , 2.

Answer ;

**b) 1 , 2 , 3.**

---

0695. The open-ended tube parallel to the longitudinal axis of the aircraft senses the ;

- a) total pressure.
- b) dynamic pressure.
- c) static pressure.
- d) total pressure plus static pressure.

Answer ;

**a) total pressure.**

---

0696. The operating frequency range of a low altitude radio altimeter is ;

- a) 4200 MHz to 4400 MHz.
- b) 5 GHz.
- c) 2700 MHz to 2900 MHz.
- d) 5400 MHz or 9400 MHz.

Answer ;

**a) 4200 MHz to 4400 MHz.**

---

0697. The operating principle of an “electronic” tachometer is to measure the ;

- a) frequency of the electric impulse created by a notched wheel rotating in a magnetic field.
- b) electromotive force ( EMF ) produced by a dynamo or an alternator.
- c) rotation speed of an asynchronous motor energized by an alternator.
- d) magnetic field produced by a dynamo or an alternator.

Answer ;

**a) frequency of the electric impulse created by a notched wheel rotating in a magnetic field.**

---

0698. The operating principle of an inductive probe tachometer is to measure the ;

- a) frequency of the electric impulse created by a notched wheel rotating in a magnetic field.
- b) magnetic field produced by a dynamo or an alternator.
- c) electromotive force produced by a dynamo or an alternator.
- d) rotation speed of an asynchronous motor energized by an alternator.

Answer ;

**a) frequency of the electric impulse created by a notched wheel rotating in a magnetic field.**

---

0699. The operating principle of an inertial system consists in ;

- a) measuring the position of the aircraft and performing integrations to elaborate the ground speed and acceleration.
- b) measuring the earth rotation and performing integrations to elaborate the aircraft ground speed and position.
- c) measuring the acceleration , speed and position of the aircraft.
- d) measuring the acceleration of the aircraft and performing integrations to elaborate the ground speed and the position.

Answer ;

**d) measuring the acceleration of the aircraft and performing integrations to elaborate the ground speed and the position.**

---

0700. The operating principle of the “induction” type of tachometer is to measure the ;

- a) frequency of the electric impulse created by a notched wheel rotating in a magnetic field.
- b) magnetic field produced by a dynamo or an alternator.
- c) rotation speed of an asynchronous motor energized by an alternator.
- d) electromotive force ( EMF ) produced by a dynamo or an alternator.

Answer ;

**c) rotation speed of an asynchronous motor energized by an alternator.**

---

- 0701.** The operating principle of the vertical speed indicator ( VSI ) is based on the measurement of the rate of change of ;
- a) Dynamic pressure.
  - b) Static pressure.
  - c) Kinetic pressure.
  - d) Total pressure.

Answer ;

**b)** Static pressure.

---

- 0702.** The operation of the GPWS ( Ground Proximity Warning System ) is governed by laws taking the aircraft height into account as well as ;

- 1. the descent rate.
- 2. the climb rate.
- 3. the aircraft configuration.
- 4. the selected engine rpm.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 3.
- c) 2 , 4.
- d) 2 , 3.

Answer ;

**b)** 1 , 3.

---

- 0703.** The operation of the radio altimeter of a modern aircraft is based on ;

- a) frequency modulation of the carrier wave.
- b) amplitude modulation of the carrier wave.
- c) a combination of frequency modulation and pulse modulation.
- d) pulse modulation of the carrier wave.

Answer ;

**a)** frequency modulation of the carrier wave.

---

- 0704.** The output data from the ADC are used by ;

- 1. Transponder.
- 2. EFIS.
- 3. Automatic Flight Control System ( AFCS ).

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 3.
- c) 1 , 2.
- d) 2 , 3.

Answer ;

**a)** 1 , 2 , 3.

---

- 0705.** The output data of an IRS include ;

- 1. angle of attack.
- 2. altitude.
- 3. ground speed.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 3.
- c) 1 , 2.
- d) 2.

Answer ;

**b)** 3.

---

**0706.** The output data of an IRS include ;

1. attitude.
2. altitude.
3. present position ( lat , long ).
4. static air temperature.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 1 , 2 , 3.
- c) 1 , 3 , 4.
- d) 1 , 2 , 3 , 4.

**Answer ;**

**a) 1 , 3.**

---

**0707.** The output data of an IRS include ;

1. number of satellites tracked.
2. mach number.
3. ground speed.
4. true track.

The combination regrouping all the correct statements is ;

- a) 1 , 4.
- b) 2 , 3.
- c) 1 , 2 , 3 , 4.
- d) 3 , 4.

**Answer ;**

**d) 3 , 4.**

---

**0708.** The output data of an IRS include ;

1. present position ( lat , long ).
2. altitude.
3. ground speed.
4. true heading.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 2 , 3.
- c) 1 , 4.
- d) 1 , 3 , 4.

**Answer ;**

**d) 1 , 3 , 4.**

---

**0709.** The output data of an IRS include ;

1. present position ( lat , long ).
2. TAS.
3. attitude.
4. ground speed.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2 , 4.
- c) 1 , 3 , 4.
- d) 2 , 3 , 4.

**Answer ;**

**c) 1 , 3 , 4.**

---

**0710.** The output data of an IRS include ;

1. present position ( lat , long ).
2. total pressure.
3. static air temperature.
4. true heading.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 1 , 2 , 4.
- c) 2 , 3.
- d) 1 , 4.

**Answer ;**

**d) 1 , 4.**

---

0711. The output data of an IRS include ;
1. satellites status.
  2. altitude.
  3. drift angle.
  4. present position ( lat , long ).

The combination regrouping all the correct statements is ;

- a) 3 , 4.
- b) 2 , 4.
- c) 1 , 3 , 4.
- d) 4.

Answer ;

- a) 3 , 4.
- 

0712. The output data of the flight director computer are ;

- a) three channels ; pitch , roll and yaw.
- b) two channels ; pitch and roll.
- c) two channels ; pitch and yaw.
- d) three channels ; pitch , roll and sideslipping.

Answer ;

- b) two channels ; pitch and roll.
- 

0713. The output from an engine vibration transducer is ;

- a) fed directly to the indicator in the cockpit without amplification or filtering.
- b) directly proportional to engine speed.
- c) inversely proportional to engine speed.
- d) always filtered to remove unwanted frequencies.

Answer ;

- d) always filtered to remove unwanted frequencies.
- 

0714. The “overfly” symbol related to a waypoint on an FMS page indicates that ;

- a) a time estimate is given for this waypoint.
- b) a turn anticipation is permitted.
- c) a fuel prediction is given for this waypoint.
- d) the aircraft is required to pass directly over this waypoint.

Answer ;

- d) the aircraft is required to pass directly over this waypoint.
- 

0715. The parameter that determines the relationship between EAS and TAS is ;

- a) Mach number.
- b) OAT.
- c) density altitude.
- d) pressure altitude.

Answer ;

- c) density altitude.
- 

0716. The parameters taken into account by the flight director computer in the altitude holding mode ( ALT HOLD ) are ;

1. altitude deviation.
2. engine rpm.
3. ground speed.
4. pitch attitude.

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 1 , 4.
- c) 1 , 3.
- d) 1.

Answer ;

- b) 1 , 4.
-

0717. The parameters taken into account by the flight director computer in the altitude holding mode ( ALT HOLD ) are ;
1. altitude deviation.
  2. roll.
  3. bank angle.
  4. pitch angle.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 1.
- c) 1 , 4.
- d) 1 , 2.

Answer ;

c) 1 , 4.

---

0718. The position data ( lat , long ) computed by an IRS can be used by the ;

- a) ILS receiver.
- b) ADC.
- c) TCAS.
- d) FMS.

Answer ;

d) FMS.

---

0719. The position error of a stand alone inertial system ;

- a) increases up to 2 NM due to the drift error of the gyroscopes , then stabilizes.
- b) increases along the time.
- c) remains constant.
- d) is sinusoidal.

Answer ;

b) increases along the time.

---

0720. The position error of a stand alone inertial system is ;

- a) constant along the flight with an accuracy depending on the accuracy of the accelerometers.
- b) large a few minutes after initialisation and reduces along the flight.
- c) small a few minutes after initialisation and increases along the flight.
- d) small and constant along the flight.

Answer ;

c) small a few minutes after initialisation and increases along the flight.

---

0721. The position error of a stand alone inertial system , is approximately ;

- a) 8 to 10 NM per hour.
- b) 6 to 8 NM per hour.
- c) 0.01 to 0.2 NM per hour.
- d) 0.5 to 2 NM per hour.

Answer ;

d) 0.5 to 2 NM per hour.

---

0722. The position error of the static vent on which the altimeter is connected varies substantially with the ;

- a) Mach number of the aircraft.
- b) flight time at high altitude.
- c) static temperature.
- d) deformation of the aneroid capsule.

Answer ;

a) Mach number of the aircraft.

---

0723. The position of a Flight Director command bars ;

- a) only displays information relating to radio-electric deviation.
- b) indicates the manoeuvres to execute , to achieve or maintain a flight situation.
- c) repeats the ADI and HSI information.
- d) enables the measurement of deviation from a given position.

Answer ;

b) indicates the manoeuvres to execute , to achieve or maintain a flight situation.

---



0724. The position of the command bars of a flight director enables the pilot to know ;
1. the direction and the amplitude of the corrections to apply on the controls.
  2. only the direction of the corrections to apply on the controls.
  3. the attitude of the aircraft.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 1.
- c) 2.
- d) 2 , 3.

Answer ;

- b) 1.**
- 

0725. The position of the command bars of a flight director enables the pilot to know ;
- a) the direction an the amplitude of the corrections to apply on the controls.
  - b) the position of the aircraft.
  - c) only the direction of the corrections to apply on the controls.
  - d) the attitude of the aircraft.

Answer ;

- a) the direction an the amplitude of the corrections to apply on the controls.**
- 

0726. The pressure altitude is the altitude corresponding ;
- a) in standard atmosphere , to the pressure Ps prevailing at this point.
  - b) in standard atmosphere , to the reference pressure Ps.
  - c) in ambient atmosphere , to the reference pressure Ps.
  - d) in ambient atmosphere , to the pressure Ps prevailing at this point.

Answer ;

- a) in standard atmosphere , to the pressure Ps prevailing at this point.**
- 

0727. The pressure capsule of an airspeed indicator is sensitive to the difference ;
- a) ( Dynamic Pressure – Total Pressure ) , called Static Pressure.
  - b) ( Total Pressure – Dynamic Pressure ) , called Static Pressure.
  - c) ( Total Pressure – Static Pressure ) , called Dynamic Pressure.
  - d) ( Dynamic Pressure – Static Pressure ) , called Total Pressure.

Answer ;

- c) ( Total Pressure – Static Pressure ) , called Dynamic Pressure.**
- 

0728. The Primary Flight Display ( PFD ) displays information dedicated to ;
- a) engines and alarms.
  - b) weather situation.
  - c) systems.
  - d) piloting.

Answer ;

- d) piloting.**
- 

0729. The Primary Flight Display ( PFD ) of an EFIS equipped aircraft can display information relative to the following conditions ;

- a) altitude capture , braking system , autopilot and flight director mode changes.
- b) low engine oil pressure , terrain alerts , windshear detection.
- c) pressurization failure , altitude capture , TCAS resolution advisory.
- d) altitude capture , TCAS resolution advisory , autopilot and flight director mode changes.

Answer ;

- d) altitude capture , TCAS resolution advisory , autopilot and flight director mode changes.**
- 

0730. The Primary Flight Display ( PFD ) of an EFIS equipped aircraft displays the following parameters ;
- a) attitude , heading , altitude , systems information.
  - b) altitude , attitude , heading , engine parameters.
  - c) attitude , heading , IAS , navigation map in Plan layout mode.
  - d) IAS , attitude , altitude , heading.

Answer ;

- d) IAS , attitude , altitude , heading.**
-

**0731.** The Primary Flight Display ( PFD ) of an EFIS equipped aircraft displays the following parameters ;

1. radio height.
2. IAS.
3. Localizer and Glide slope deviation pointers.
4. flight director modes.
5. autopilot modes.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 5.
- b) 2 , 5.
- c) 3 , 4.
- d) 1 , 2 , 3 , 4 , 5.

**Answer ;**

**d) 1 , 2 , 3 , 4 , 5.**

---

**0732.** The principle of a laser gyro is based on ;

- a) two rotating cavities provided with mirrors.
- b) a gyroscope associated with a laser compensating for gimballing errors.
- c) frequency difference between two laser beams rotating in opposite direction.
- d) a gyroscope associated with a laser compensating for apparent wander due to the rotation of the earth.

**Answer ;**

**c) frequency difference between two laser beams rotating in opposite direction.**

---

**0733.** The principle of capacity gauges is based on the ;

- a) capacitance variation by the volume measurement carried out on the sensor.
- b) current variation in the Wheatstone bridge.
- c) flow rate and torque variation occurring in a supply line.
- d) capacitance variation of a given capacitor with the type of dielectric.

**Answer ;**

**d) capacitance variation of a given capacitor with the type of dielectric.**

---

**0734.** The principle of capacity gauges is based on the capacitance variation of ;

- a) two dissimilar metals joined together at one end only.
- b) two dissimilar metals joined together at their ends.
- c) liquids with the variations in temperature.
- d) a capacitor with the type of dielectric.

**Answer ;**

**d) a capacitor with the type of dielectric.**

---

**0735.** The principle of the Mach indicator is based on the computation of the ratio ;

- a)  $P_t/P_s$
- b)  $(P_t - P_s)/P_t$
- c)  $(P_t + P_s)/P_s$
- d)  $(P_t - P_s)/P_s$

**Answer ;**

**d)  $(P_t - P_s)/P_s$**

---

**0736.** The principle of the Schuler pendulum is used in the design of a ;

- a) strapdown inertial system.
- b) artificial horizon control system.
- c) stabilised platform inertial system.
- d) directional gyro control system.

**Answer ;**

**c) stabilised platform inertial system.**

---

**0737.** The principle of the TCAS makes use of ;

- a) transponders fitted in the aircraft.
- b) air traffic control radar systems.
- c) airborne weather radar system.
- d) FMS.

**Answer ;**

**a) transponders fitted in the aircraft.**

---

0738. The principle of the TCAS ( Traffic Collision Avoidance Systems ) is based on the use of ;
- a) air traffic control radar systems.
  - b) FMS ( Flight Management System ).
  - c) airborne weather radar system.
  - d) transponders fitted in the aircraft.

Answer ;

d) transponders fitted in the aircraft.

---

0739. The probe used to measure the air intake pressure of a gas turbine engine powerplant is ;
- a) a differential capsule.
  - b) a bellows sensor.
  - c) a Bourdon tube.
  - d) an aneroid capsule.

Answer ;

d) an aneroid capsule.

---

0740. The properties of a gyroscope are ;
- 1. rigidity in space.
  - 2. rigidity on earth.
  - 3. precession.
  - 4. Schuler oscillations.

The combination regrouping all the correct statements is ;

- a) 1 , 4.
- b) 1 , 3.
- c) 2 , 3.
- d) 2 , 4.

Answer ;

b) 1 , 3.

---

0741. The properties of a single-phase electrical tachometer are ;
- 1. indication depending on the line resistance.
  - 2. independence from the aircraft airborne power supply.
  - 3. simple design.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 1 , 2 , 3.
- c) 1 , 2.
- d) 2 , 3.

Answer ;

b) 1 , 2 , 3.

---

0742. The purpose of a flight director is to ;
- a) reduce the pilots workload by presenting data in the form of control commands.
  - b) provide an automatic landing system function.
  - c) convey air traffic control information to the pilot.
  - d) automatically steer the aircraft to waypoints selected on the CDU.

Answer ;

a) reduce the pilots workload by presenting data in the form of control commands.

---

0743. The purpose of an airplane automatic trim system is to trim out the hinge moment of the ;
- a) elevator(s).
  - b) elevator(s) and rudder(s).
  - c) elevator(s) , rudder(s) and ailerons.
  - d) rudder(s).

Answer ;

a) elevator(s).

---

0744. The purpose of Auto Throttle is ;
- a) automatic shut down of one engine at too high temperature.
  - b) to deactivate manual throttles and transfer engine control to Auto Pilot.
  - c) to maintain constant engine power or airplane speed.
  - d) to synchronize engines to avoid “yawing”.

Answer ;

c) to maintain constant engine power or airplane speed.

---

0745. The purpose of Auto Trim function in autopilot is to ;
- a) control elevator trim tab in order to relieve elevator load.
  - b) trim throttles to obtain smooth engine power variation.
  - c) help Auto Pilot compensate for crosswind influence.
  - d) tell the pilot when elevator trimming is required.

Answer ;

- a) control elevator trim tab in order to relieve elevator load.
- 

0746. The purpose of magnetic chip detectors is to ;
- a) perform the function of a micron filter.
  - b) warn of impending failure.
  - c) increase lubricating oil adhesion to main surfaces.
  - d) remove large items of debris from the system.

Answer ;

- b) warn of impending failure.
- 

0747. The purpose of the altitude alert system is to generate a visual and aural warning to the pilot when the ;
- a) airplane altitude differs from a selected altitude.
  - b) airplane altitude is equal to the decision altitude.
  - c) proximity to the ground becomes dangerous.
  - d) altimeter setting differs from the standard setting above the transition altitude.

Answer ;

- a) airplane altitude differs from a selected altitude.
- 

0748. The purpose of the autopilot Control Wheel Steering ( CWS ) mode is ;
- a) to control the nose wheel steering during automatic landing.
  - b) to control the nose wheel steering during low visibility take off and landing.
  - c) to consider as target parameters , the current pitch and roll angles at the time the mode becomes active.
  - d) to capture and hold the altitude selected with the control wheel on the mode control panel.

Answer ;

- c) to consider as target parameters , the current pitch and roll angles at the time the mode becomes active.
- 

0749. The purpose of the FMS temperature compensation function is ;
- a) to provide compensated temperatures at the waypoints along the vertical approach profile.
  - b) to provide the destination airport air temperature.
  - c) to provide compensated altitudes for temperatures different from standard atmosphere along the vertical approach profile.
  - d) to provide the destination airport or runway elevation.

Answer ;

- c) to provide compensated altitudes for temperatures different from standard atmosphere along the vertical approach profile.
- 

0750. The purpose of the FMS temperature compensation function is ;
- a) to provide compensated temperatures at the waypoints along the vertical approach profile.
  - b) to provide compensated altitudes for temperatures different from ISA along the vertical approach profile.
  - c) to provide the destination airport or runway elevation.
  - d) to provide the destination airport air temperature.

Answer ;

- b) to provide compensated altitudes for temperatures different from ISA along the vertical approach profile.
- 

0751. The purpose of the Navigation Display ( ND ) is to display the following information ;
- a) terrain map , navaid bearings , flight director active modes.
  - b) flight plan , engine failure , nav aids , resolution advisor ( RA ).
  - c) intruding traffic , TCAS symbols , weather radar.
  - d) flight plan , weather radar , terrain map , TCAS symbols.

Answer ;

- d) flight plan , weather radar , terrain map , TCAS symbols.
- 

0752. The purpose of the vibrating device of an altimeter is to ;
- a) reduce the hysteresis effect.
  - b) reduce the effect of friction in the linkages.
  - c) allow damping of the measurement in the unit.
  - d) inform the crew of a failure of the instrument.

Answer ;

- b) reduce the effect of friction in the linkages.
-

0753. The purpose(s) of the flight director system is(are) to ;
1. give the position of the aircraft according to radioelectric axis.
  2. give the position of the aircraft according to waypoints.
  3. to aid the pilot when flying manually.

The combination that regroups all of the correct statements is ;

- a) 3.
- b) 2 , 3.
- c) 1 , 3.
- d) 1 , 2 , 3.

Answer ;

- a) 3.
- 

0754. The QNH is by definition the value of the ;

- a) altimeter setting so that the altimeter , on the apron of the aerodrome for which it is given , reads the elevation.
- b) atmospheric pressure at the sea level of the location for which it is given.
- c) atmospheric pressure at the level of the ground overflown by the aircraft.
- d) altimeter setting so that the altimeter , on the apron of the aerodrome for which it is given , reads zero.

Answer ;

- a) altimeter setting so that the altimeter , on the apron of the aerodrome for which it is given , reads the elevation.
- 

0755. The quadrantal deviation of the magnetic compass is due to the action of ;

- a) the soft iron pieces influenced by the geomagnetic field.
- b) the hard iron pieces influenced by the mild iron pieces.
- c) the hard iron pieces and the soft iron pieces influenced by the hard iron pieces.
- d) the hard iron pieces influenced by the geomagnetic field.

Answer ;

- a) the soft iron pieces influenced by the geomagnetic field.
- 

0756. The radio altimeter ;

1. operates in the 1600-1660 kHz range.
2. operates in the 4200-4400 MHz range.
3. measures a frequency difference.
4. measures an amplitude difference.

The combination regrouping all the correct statements is ;

- a) 1 and 3.
- b) 1 and 4.
- c) 2 and 4.
- d) 2 and 3.

Answer ;

- d) 2 and 3.
- 

0757. The radio altimeter is required to indicate zero height AGL as the main wheels touch down on the runway. For this reason , it is necessary to ;

- a) adjust the gross height according to the aircraft instantaneous pitch.
- b) have a specific radio altimeter dedicated to automatic landing.
- c) compensate for residual height and cable length.
- d) change the display scale in short final , in order to have a precise readout.

Answer ;

- c) compensate for residual height and cable length.
- 

0758. The radio altimeter supplies data to the following system(s) ;

1. altitude alert system.
2. TCAS.
3. GPWS.
4. automatic landing system.

The combination that regroups all the correct statements is ;

- a) 3 , 4.
- b) 3.
- c) 2 , 3 , 4.
- d) 1 , 4.

Answer ;

- c) 2 , 3 , 4.
-

0759. The radio altimeter uses the following wavelengths ;

- a) myriametric.
- b) millimetric.
- c) metric.
- d) centimetric.

Answer ;

d) centimetric.

---

0760. The range of a low altitude radio altimeter is ;

- a) 500 ft.
- b) greater than 10 000 ft.
- c) 10 000 ft.
- d) 2 500 ft.

Answer ;

d) 2 500 ft.

---

0761. The rate of turn given by the rate of turn indicator is valid ;

- a) for all airspeeds.
- b) for the airspeed range defined during the calibration of the instrument.
- c) for the cruising speed.
- d) with flaps retracted only.

Answer ;

b) for the airspeed range defined during the calibration of the instrument.

---

0762. The rate of turn indicator uses a gyroscope ;

- 1. the spinning wheel axis of which is parallel to the yawing axis.
- 2. the spinning wheel axis of which is parallel to the pitch axis.
- 3. the spinning wheel axis of which is parallel to the roll axis.
- 4. with one degree of freedom.
- 5. with two degrees of freedom

The combination regrouping all the correct statements is ;

NB : the degree(s) of freedom of a gyro does not take into account its rotor spin axis.

- a) 3 , 5.
- b) 1 , 5.
- c) 2 , 4.
- d) 3 , 4.

Answer ;

c) 2 , 4.

---

0763. The rate-of-turn is the ;

- a) pitch rate in a turn.
- b) yaw rate in a turn.
- c) change-of-heading rate of the aircraft.
- d) aircraft speed in a turn.

Answer ;

c) change-of-heading rate of the aircraft.

---

0764. The reading of a Mach indicator is independent of ;

- a) the total pressure.
- b) the differential pressure measurement.
- c) the outside temperature.
- d) the static pressure.

Answer ;

c) the outside temperature.

---

**0765.** The requirement to carry a GPWS ( Ground Proximity Warning System ) concerns aeroplanes which are , depending on their age , weight and passenger capacity ;

1. turboprop-powered.
2. piston-powered.
3. jet-powered.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 1.
- c) 3.
- d) 1 , 3.

**Answer ;**

**d) 1 , 3.**

---

**0766.** The response time of a vertical speed detector may be decreased by adding a ;

- a) bimetallic strip.
- b) second calibrated port.
- c) return spring.
- d) correction based on an accelerometer sensor.

**Answer ;**

**d) correction based on an accelerometer sensor.**

---

**0767.** The role of the FMS is to ;

1. aid the flight crew with navigation.
2. shut down the engine in case of a malfunction.
3. automatically avoid conflicting traffic when autopilot engaged.
4. reduce crew workload.
5. aid fuel efficiency.

The combination that regroups all of the correct statements is ;

- a) 1 , 3 , 4.
- b) 2 , 3 , 4 , 5.
- c) 1 , 2 , 3.
- d) 1 , 4 , 5.

**Answer ;**

**d) 1 , 4 , 5.**

---

**0768.** The role of the FMS is to aid the flight crew with ;

1. immediate actions in case of emergency procedure.
2. navigation.
3. in-flight performance optimization.
4. electronic check-lists.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 3 , 4.
- c) 1 , 2.
- d) 2 , 3.

**Answer ;**

**d) 2 , 3.**

---

**0769.** The RPM indicator ( or tachometer ) of a piston engine can include a small red arc within the arc normally used ( green arc ). In the RPM range corresponding to this small red arc the ;

- a) propeller generates vibration , continuous rating is forbidden.
- b) rating is the maximum possible in continuous mode.
- c) rating is the minimum usable in cruise.
- d) propeller efficiency is minimum at this rating.

**Answer ;**

**a) propeller generates vibration , continuous rating is forbidden.**

---

0770. The sensor(s) feeding the EPR-indicator is(are) ;

- a) temperature probes , one located upstream from the compressor inlet , and an other downstream from the turbine outlet.
- b) tachometer located on the shaft of the N2 compressor.
- c) pressure probes , one located upstream from the compressor inlet , and the other downstream from the turbine outlet.
- d) tachometer located on the shaft of the N1 compressor.

Answer ;

- c) pressure probes , one located upstream from the compressor inlet , and the other downstream from the turbine outlet.
- 

0771. The sensors used to measure the exhaust gas temperature on an aircraft equipped with turbojets are ;

- a) thermocouples.
- b) based on metallic conductors whose resistance increases linearly with temperature.
- c) based on metallic parts whose expansion/contraction is measured.
- d) capacitors whose capacity varies proportionally with temperature.

Answer ;

- a) thermocouples.
- 

0772. The sequence of the automatic landing comprises several phases ( from final approach to touch-down ) actuated by ;

- a) the DME ( Distance Measuring Equipment ) of the ILS ( Instrument Landing System ).
- b) the altimeter set to the QNH.
- c) the distance left before the touch down zone.
- d) the radio altimeter.

Answer ;

- d) the radio altimeter.
- 

0773. The sideslip indication displayed on the PFD ( Primary Flight Display ) is generated by ;

- a) the inertial system.
- b) the stall protection system.
- c) the Automatic Flight Control System ( AFCS ).
- d) the yaw damper.

Answer ;

- a) the inertial system.
- 

0774. The sideslip indication displayed on the PFD ( Primary Flight Display ) is generated by the ;

- a) compass.
- b) ADC.
- c) inertial system.
- d) yaw damper.

Answer ;

- c) inertial system.
- 

0775. The signal supplied by a transmitter fitted with a 3-phase AC generator , connected to RPM indicator , is ;

- a) an AC voltage , the frequency of which varies with the RPM , the indicator converts the signal into square pulses which are then counted.
- b) a three-phase voltage , the frequency of which varies with the RPM , the indicator is provided with a motor which drives a magnetic tachometer.
- c) an AC voltage varying with the RPM , the indicator rectifies the signal via a diode bridge and is provided with a voltmeter.
- d) a DC voltage varying with the RPM , the indicator is a plain voltmeter with a rev/min. scale.

Answer ;

- b) a three-phase voltage , the frequency of which varies with the RPM , the indicator is provided with a motor which drives a magnetic tachometer.
-



0776. The signal supplied by a transmitter fitted with a magnetic sensor , connected to an RPM indicator is ;
- a) an AC voltage , the frequency of which varies with the RPM , the indicator converts the signal into square pulses which are then counted.
  - b) an AC voltage varying with the RPM , the indicator rectifies the signal via a diode bridge and is provided with a voltmeter.
  - c) a three-phase voltage frequency varies with the RPM , the indicator is provided with a motor which drives a magnetic tachometer.
  - d) a DC voltage varying with the RPM , the indicator is a simple voltmeter with a rev/min. scale.

Answer ;

- a) an AC voltage , the frequency of which varies with the RPM , the indicator converts the signal into square pulses which are then counted.
- 

0777. The signal transmitted by a radio altimeter is ;
- a) a frequency modulated carrier wave.
  - b) a combination of frequency modulation and pulse modulation.
  - c) a pulse modulated carrier wave.
  - d) an amplitude modulated carrier wave.

Answer ;

- a) a frequency modulated carrier wave.
- 

0778. The spin axis of the turn indicator gyro is aligned along the ;
- a) vertical axis of the aircraft.
  - b) longitudinal axis of the aircraft.
  - c) longitudinal axis of flight.
  - d) lateral axis of the aircraft.

Answer ;

- d) lateral axis of the aircraft.
- 

0779. The spin axis of the turn indicator gyroscope is parallel to the ;
- a) yaw axis.
  - b) pitch axis.
  - c) roll axis.
  - d) longitudinal axis.

Answer ;

- b) pitch axis.
- 

0780. The stall warning computer of a large aeroplane uses the following data ;
- 1. mass of the aeroplane.
  - 2. angle of attack.
  - 3. wing flap deflection.
  - 4. position of the landing gear.
  - 5. TAT.
  - 6. pressure altitude.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 2 , 3 , 5.
- c) 1 , 2 , 3 , 4 , 5 , 6.
- d) 1 , 2 , 3 , 4.

Answer ;

- a) 2 , 3.
- 

0781. The stall warning computer of a large aeroplane uses the following data ;
- 1. pitch attitude.
  - 2. angle of attack.
  - 3. configuration ( slats / flaps ).

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 3.
- c) 1 , 2.
- d) 2 , 3.

Answer ;

- d) 2 , 3.
-

- 0782.** The stall warning is inhibited ;
1. when the flaps are retracted.
  2. when the aeroplane is on the ground.
  3. when encountering a windshear.
  4. upon receiving a GPWS alert.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 2.
- c) 2 , 3.
- d) 1 , 2.

**Answer ;**

**b) 2.**

---

- 0783.** The stall warning system of a large aeroplane includes ;
1. an angle of attack sensor.
  2. a computer.
  3. an independent pitot probe.
  4. a transmitter of the flap/slat position indicating system.

The combination that regroupes all of the correct statements is ;

- a) 2 , 4.
- b) 1 , 2 , 3.
- c) 1 , 2 , 4.
- d) 1 , 3.

**Answer ;**

**c) 1 , 2 , 4.**

---

- 0784.** The stall warning system of a large transport airplane includes ;
1. an angle of attack sensor.
  2. a computer.
  3. a transmitter originating from the anemometer.
  4. an independent pitot probe.
  5. a transmitter of the flap/slat position indicating system.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 5.
- b) 1 , 2 , 4.
- c) 1 , 4.
- d) 1 , 2 , 4 , 5.

**Answer ;**

**a) 1 , 2 , 5.**

---

- 0785.** The stall warning system receives information about the ;
1. airplane angle of attack.
  2. airplane speed.
  3. airplane bank angle.
  4. airplane configuration.
  5. load factor on the airplane.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 5.
- b) 1 , 4.
- c) 2 , 3 , 4 , 5.
- d) 1 , 2 , 3 , 4 , 5.

**Answer ;**

**b) 1 , 4.**

---

- 0786.** The Static Air temperature ( SAT ) is ;
- a) the TAT divided by the recovery factor.
  - b) the ambient outside air temperature.
  - c) the outside air temperature measured by the pitot probe.
  - d) the temperature resulting from the aircraft motion in the air.

**Answer ;**

**b) the ambient outside air temperature.**

---

**0787.** The stick shaker calculator receives the following information ;

1. mass of the airplane.
2. angle of attack.
3. wing flap deflection.
4. position of the landing gear.
5. total air temperature.
6. pressure altitude.

The combination regrouping all the correct statements is ;

- a) 2 , 3.
- b) 2 , 3 , 5.
- c) 1 , 2 , 3 , 4 , 5 , 6.
- d) 1 , 2 , 3 , 4.

**Answer ;**

**a) 2 , 3.**

---

**0788.** The systems that can be connected to the Communication Management Unit ( CMU ) , are ;

1. EGPWS.
2. HF Communication Unit.
3. Multipurpose Control and Display Unit ( MCDU ).
4. VHF Communication Unit.
5. Satcom.

The combination which regroups all of the correct statements is ;

- a) 1 , 3 , 4.
- b) 1 , 3.
- c) 2 , 3 , 4 , 5.
- d) 2 , 4 , 5.

**Answer ;**

**c) 2 , 3 , 4 , 5.**

---

**0789.** The systems that can be connected to the Communication Management Unit ( CMU ) , are ;

1. Flight Director.
2. FMS.
3. Multipurpose Control and Display Unit ( MCDU ).
4. communication unit ( VHF , HF , Satcom ).

The combination which regroups all of the correct statements is ;

- a) 2 , 3.
- b) 2 , 4.
- c) 2 , 3 , 4.
- d) 1 , 3 , 4.

**Answer ;**

**c) 2 , 3 , 4.**

---

**0790.** The TAS is equal to the EAS only if ;

- a)  $P = 1013.25 \text{ hPa}$  ,  $OAT = 15^\circ\text{C}$  and  $TAS < 200 \text{ kt}$ .
- b)  $P = 1013.25 \text{ hPa}$  and  $OAT = 273^\circ\text{K}$ .
- c)  $P = 1013.25 \text{ hPa}$  ,  $OAT = 15^\circ\text{C}$  and  $TAS > 200 \text{ kt}$ .
- d)  $P = 1013.25 \text{ hPa}$  and  $OAT = 15^\circ\text{C}$ .

**Answer ;**

**d)  $P = 1013.25 \text{ hPa}$  and  $OAT = 15^\circ\text{C}$ .**

---

**0791.** The TAS is obtained from the CAS by correcting for the following errors ;

1. instrument.
2. compressibility.
3. position.
4. density.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 2 , 4.
- c) 2.
- d) 4.

**Answer ;**

**b) 2 , 4.**

---

**0792.** The TAS is obtained from the IAS by correcting for the following errors ;

1. instrument.
2. position.
3. compressibility.
4. density.

The combination that regroups all of the correct statements is ;

- a) 1 , 2.
- b) 1 , 3 , 4.
- c) 3 , 4.
- d) 1 , 2 , 3 , 4.

**Answer ;**

**d) 1 , 2 , 3 , 4.**

---

**0793.** The TCAS ( Traffic Collision Avoidance System ) computer receives information ;

1. about the aircraft true airspeed.
2. about the airplane configuration.
3. about the pressure altitude through the mode C transponder.
4. from the radio altimeter.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 2 , 3 , 4.
- c) 1 , 2 , 4.
- d) 1 , 3 , 4.

**Answer ;**

**b) 2 , 3 , 4.**

---

**0794.** The TCAS 2 data display devices can be in the form of ;

1. a specific dedicated screen.
2. a screen combined with the weather radar.
3. a variometer represented on a liquid crystal screen which allows the display of Traffic Advisory ( TA ) and Resolution Advisory ( RA ).
4. an EFIS ( Electronic Flight Instrument System ) screen.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 and 4.
- b) 1 and 3.
- c) 3 and 4.
- d) 1 , 2 and 3.

**Answer ;**

**a) 1 , 2 , 3 and 4.**

---

**0795.** The TCAS 2 has inputs from the radio altimeter in order to ;

- a) to alert the crew from a dangerous proximity of the ground.
- b) determinethe relative height of the intruder.
- c) stop the TCAS operation below 2500 AGL.
- d) gradually inhibit the resolution advisories ( RAs ) when getting closer to the ground.

**Answer ;**

**d) gradually inhibit the resolution advisories ( RAs ) when getting closer to the ground.**

---

**0796.** The TCAS 2 ( Traffic Collision Avoidance System ) gives avoidance resolutions ;

- a) in horizontal and vertical planes.
- b) based on speed control.
- c) only in the vertical plane.
- d) only in the horizontal plane.

**Answer ;**

**c) only in the vertical plane.**

---

0797. The TCAS 2 ( Traffic Collision Avoidance System ) provides ;
1. traffic information ( TA : Traffic Advisory ).
  2. horizontal resolution ( RA : Resolution Advisory ).
  3. vertical resolution ( RA : Resolution Advisory ).
  4. ground proximity warning.

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 1 , 2 , 3 , 4.
- c) 1 , 3.
- d) 1 , 2 , 3.

Answer ;

- c) 1 , 3.
- 

0798. The time for a normal alignment ( not a quick alignment ) of a strapdown inertial system is ;

- a) 1 to 2 minutes.
- b) 15 to 20 minutes.
- c) 3 to 10 minutes.
- d) less than 1 minute.

Answer ;

- c) 3 to 10 minutes.
- 

0799. The Total Air Temperature ( TAT ) is ;

- a) the temperature resulting from the aircraft motion in the air.
- b) the average temperature resulting from the temperature measure of the pitot and TAT probes.
- c) the static air temperature ( SAT ) multiplied by the recovery factor.
- d) the impact air temperature measured by the pitot probe.

Answer ;

- a) the temperature resulting from the aircraft motion in the air.
- 

0800. The total pressure head comprises a mast which moves its port to a distance from the aircraft skin in order ;

- a) not to disturb the aerodynamic flow around the aircraft.
- b) to locate it outside the boundary layer.
- c) it is easily accessible during maintenance checks.
- d) it is protected from icing.

Answer ;

- b) to locate it outside the boundary layer.
- 

0801. The total pressure probe ( pitot tube ) comprises a mast which moves its port to a distance from the aircraft skin in order ;

- a) it is easily accessible during maintenance checks.
- b) to locate it outside the boundary layer.
- c) not to disturb the aerodynamic flow around the aircraft.
- d) it is protected from icing.

Answer ;

- b) to locate it outside the boundary layer.
- 

0802. The total pressure probe ( pitot tube ) is mounted at a distance from the aeroplane skin such that ;

- a) it is located outside the boundary layer.
- b) it is easily accessible during maintenance checks.
- c) it does not disturb the aerodynamic flow around the aircraft.
- d) it is protected from icing.

Answer ;

- a) it is located outside the boundary layer.
- 

0803. The transmitter of RPM indicator may consist of ;

1. a magnetic sensor supplying an induced AC voltage.
2. a DC generator supplying a DC voltage.
3. a single-phase AC generator supplying an AC voltage.
4. a three-phase AC generator supplying a three-phase voltage.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 2 , 3.
- c) 1 , 4.
- d) 2 , 3 , 4.

Answer ;

- a) 1 , 2 , 3 , 4.
-

0804. The turning error of a direct reading magnetic compass ;
- a) decreases when the magnetic latitude increases.
  - b) increases when the magnetic latitude increases.
  - c) does not depend on the magnetic latitude.
  - d) decreases when the magnetic longitude increases.

Answer ;

- b) increases when the magnetic latitude increases.
- 

0805. The turning errors of a direct reading magnetic compass are ;

- a) maximum at the magnetic equator.
- b) maximum at the magnetic poles.
- c) minimum at the magnetic poles.
- d) minimum at a latitude of 45°.

Answer ;

- b) maximum at the magnetic poles.
- 

0806. The two main sources of information used to calculate turbojet thrust are the ;

- a) fan rotation speed ( or N1 ) or the EPR ( Engine Pressure Ratio ).
- b) fan rotation speed ( or N1 ) or the total pressure at the high pressure compressor outlet.
- c) fan rotation speed ( or N1 ) or the total pressure at the low pressure turbine outlet.
- d) high pressure turbine rotation speed or the EPR ( Engine Pressure Ratio ).

Answer ;

- a) fan rotation speed ( or N1 ) or the EPR ( Engine Pressure Ratio ).
- 

0807. The unit used to measure the capacitance of a capacitor is the ;

- a) Joule.
- b) Farad.
- c) Hz.
- d) Watt.

Answer ;

- b) Farad.
- 

0808. The upper antenna of the TCAS II is ;

- a) directional to improve the surveillance of intruders.
- b) omnidirectional because it is merged with the transponder antenna.
- c) directional because it is merged with the transponder antenna.
- d) omnidirectional to improve the surveillance of intruders.

Answer ;

- a) directional to improve the surveillance of intruders.
- 

0809. The use of an alternate static source fed from within the cabin results that the static pressure sensed is likely to be ;

- a) higher than ambient pressure due to position error.
- b) higher than ambient pressure due to aerodynamic suction.
- c) lower than ambient pressure due to aerodynamic suction.
- d) lower than ambient pressure due to position error.

Answer ;

- c) lower than ambient pressure due to aerodynamic suction.
- 

0810. The validity period of the “permanent” database of aeronautical information stored in the database of an FMS is ;

- a) 3 calendar months.
- b) 14 days.
- c) one calendar month.
- d) 28 days.

Answer ;

- d) 28 days.
- 

0811. The velocity of sound at the sea level in a standard atmosphere is ;

- a) 644 kt.
- b) 332 kt.
- c) 661 kt.
- d) 1059 kt.

Answer ;

- c) 661 kt.
-

0812. The vertical command bar of a flight director ;

- a) gives information about the direction and the amplitude of the corrections to be applied on the control commands.
- b) repeats the position information given by the VOR.
- c) gives information only about the direction of the corrections to be applied on the bank of the aircraft.
- d) repeats the position information given by the EHSL.

Answer ;

- a) gives information about the direction and the amplitude of the corrections to be applied on the control commands.
- 

0813. The vertical command bar of a flight director ;

- a) repeats the position information given by the ILS in the horizontal plane.
- b) gives information about the direction and the amplitude of the corrections to be applied on the control commands.
- c) gives information only about the direction of the corrections to be applied on the bank of the aircraft.
- d) repeats the position information given by the ILS in the vertical plane.

Answer ;

- b) gives information about the direction and the amplitude of the corrections to be applied on the control commands.
- 

0814. The vertical command bar of a flight director ;

- a) repeats the position information given by the EHSL.
- b) repeats the position information given by the VOR.
- c) gives information about the direction and the amplitude of the corrections to be applied on the bank of the aircraft.
- d) gives information only about the direction of the corrections to be applied on the bank of the aircraft.

Answer ;

- b) gives information about the direction and the amplitude of the corrections to be applied on the bank of the aircraft.
- 

0815. The vertical command bar of a flight director ;

- 1. repeats the position information given by the ILS in the horizontal plane.
- 2. repeats the position information given by the ILS in the vertical plane.
- 3. gives information about the direction and the amplitude of the corrections to be applied on the bank of the aircraft.

The combination regrouping all the correct statements is ;

- a) 2 , 3.
- b) 3.
- c) 1 , 3.
- d) 1.

Answer ;

- b) 3.
- 

0816. The vertical command bar of a flight director ;

- 1. repeats the position information given by the EHSL.
- 2. repeats the position information given by the VOR.
- 3. gives information about the direction and the amplitude of the corrections to be applied on the bank of the aircraft.

The combination regrouping all the correct statements is ;

- a) 2 , 3.
- b) 1 , 3.
- c) 1 , 2 , 3.
- d) 3.

Answer ;

- d) 3.
-

- 0817.** The vertical flight path modes of an autopilot system are ;
1. Pitch attitude hold.
  2. Altitude hold.
  3. Track hold.
  4. Glide slope intercept and track.
- The combination that regroups all of the correct statements is ;
- a) 2 , 4.
  - b) 1 , 2 , 4.
  - c) 1 , 2 , 3.
  - d) 1.

**Answer ;**

**a) 2 , 4.**

---

- 0818.** The vertical speed indicator ( VSI ) is fed by ;
- a) dynamic pressure.
  - b) total pressure.
  - c) differential pressure.
  - d) static pressure.

**Answer ;**

**d) static pressure.**

---

- 0819.** The yaw damper affects ;
- a) ailerons only.
  - b) ailerons and rudder.
  - c) all control surfaces in a coordinated way.
  - d) rudder only.

**Answer ;**

**d) rudder only.**

---

- 0820.** The Yaw Damper system ;
1. counters any wrong pilot action on the rudder pedals.
  2. counters dutch roll.
  3. is active only when autopilot is engaged.
- The combination regrouping all the correct statements are ;
- a) 1 , 2.
  - b) 2.
  - c) 1 , 2 , 3.
  - d) 2 , 3.

**Answer ;**

**b) 2.**

---

- 0821.** The yaw damper system acts on ;
- a) the ailerons without moving the roll trim.
  - b) the ailerons and simultaneously moves the roll trim.
  - c) the rudder without moving the rudder pedals.
  - d) the rudder and simultaneously moves the rudder pedals.

**Answer ;**

**c) the rudder without moving the rudder pedals.**

---

- 0822.** The yaw damper system acts on ;
- a) the rudder only.
  - b) the rudder and the speed brakes if necessary.
  - c) the rudder and the roll trim if necessary.
  - d) the rudder and the ailerons if necessary.

**Answer ;**

**a) the rudder only.**

---

- 0823.** The yaw damper system controls ;
- a) the rudder , with the angular rate about the yaw axis as the input signal.
  - b) the ailerons , with Mach Number as the input signal.
  - c) the ailerons , with the angular rate about the yaw axis as the input signal.
  - d) the rudder , with Mach Number as the input signal.

**Answer ;**

**a) the rudder , with the angular rate about the yaw axis as the input signal.**

---



0824. The yaw damper system is operative ;
- a) during manual or automatic flight.
  - b) only if the flight director is engaged.
  - c) only if the autopilot is engaged.
  - d) only when flying manual.

Answer ;

a) during manual or automatic flight.

---

0825. The yaw damper system sends a motion order to the rudder if the yawing rate of the aircraft ;
- a) is not constant.
  - b) is  $> 0$  only.
  - c) is constant.
  - d) is  $> 1$  only.

Answer ;

a) is not constant.

---

0826. To allow the coupling of a dual channel flight director ;
- a) both VHF NAV channels must be serviceable.
  - b) only one autopilot channel must be selected ON.
  - c) both VHF Comms channels must be serviceable.
  - d) both autopilot channels must be selected ON.

Answer ;

d) both autopilot channels must be selected ON.

---

0827. To be as accurate as possible , an anemometer must be calibrated according to the following formula ;
- a) Bernoulli , taking into account the air compressibility.
  - b) Saint-Venant , taking into account the air compressibility.
  - c) Bernoulli , considering the air as an incompressible fluid.
  - d) Saint-Venant , considering the air as an incompressible fluid.

Answer ;

b) Saint-Venant , taking into account the air compressibility.

---

0828. To indicate a temperature , a thermocouple requires ;
- a) no power supply.
  - b) battery power.
  - c) alternating current.
  - d) direct current.

Answer ;

a) no power supply.

---

0829. To obtain the instantaneous position from the accelerations ;
- a) it is necessary to integrate twice the acceleration in time , and to know the initial position only.
  - b) integrating twice acceleration in time is sufficient.
  - c) it is necessary to integrate twice the acceleration in time , and to know the initial position and the initial speed.
  - d) it is necessary to integrate twice the acceleration in time , and to know the initial speed only.

Answer ;

c) it is necessary to integrate twice the acceleration in time , and to know the initial position and the initial speed.

---

0830. To obtain the instantaneous position from the accelerations , it is necessary to ;
1. integrate twice the acceleration in time.
  2. know the initial position.
  3. know the initial speed.

The combination that regroups all of the correct statements is ;

- a) 1 , 2.
- b) 1 , 3.
- c) 1 , 2 , 3.
- d) 1.

Answer ;

c) 1 , 2 , 3.

---

0831. To obtain the instantaneous speed from the accelerations ;
- a) integrating the acceleration once in time is sufficient.
  - b) it is necessary to integrate the acceleration once in time , and to know the initial speed and the initial position.
  - c) it is necessary to integrate the acceleration once in time , and to know the initial speed only.
  - d) it is necessary to integrate the acceleration once in time , and to know the initial position only.

Answer ;

- c) it is necessary to integrate the acceleration once in time , and to know the initial speed only.
- 

0832. To obtain the instantaneous speed from the accelerations , it is necessary to ;

- 1. integrate once the acceleration in time.
- 2. know the initial position.
- 3. know the initial speed.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 1.
- c) 1 , 2.
- d) 1 , 3.

Answer ;

- d) 1 , 3.
- 

0833. To permit turbine exit temperatures to be measured , gas turbines are equipped with thermometers which work on the following principle ;

- a) liquid expansion.
- b) gas pressure.
- c) bi-metallic strip.
- d) thermocouple.

Answer ;

- d) thermocouple.
- 

0834. Torque can be determined by measuring the ;

- a) quantity of light passing through a rack-wheel attached to a transmission shaft.
- b) phase difference between 2 impulse tachometers attached to a transmission shaft.
- c) frequency of an impulse tachometer attached to a transmission shaft.
- d) oil pressure at the fixed crown of an epicycloidal reducer of the main engine gearbox.

Answer ;

- d) oil pressure at the fixed crown of an epicycloidal reducer of the main engine gearbox.
- 

0835. Total Air Temperature ( TAT ) is ;

- a) higher or equal to Static Air Temperature ( SAT ) , depending on mach number and SAT.
- b) lower than Static Air Temperature ( SAT ) , depending on altitude and SAT.
- c) higher or equal to Static Air Temperature ( SAT ) , depending on altitude and SAT.
- d) lower than Static Air Temperature ( SAT ) , depending on mach number and SAT.

Answer ;

- a) higher or equal to Static Air Temperature ( SAT ) , depending on mach number and SAT.
- 

0836. True Air Speed ( TAS ) is ;

- a) Equivalent Air Speed ( EAS ) corrected for compressibility and density errors.
- b) Calibrated Air Speed ( CAS ) corrected for compressibility and density errors.
- c) Equivalent Air Speed ( EAS ) corrected for density error only.
- d) Calibrated Air Speed ( CAS ) corrected for density error only

Answer ;

- b) Calibrated Air Speed ( CAS ) corrected for compressibility and density errors.
- 

0837. True Air Speed ( TAS ) is ;

- a) Equivalent Air Speed ( EAS ) corrected for compressibility error.
- b) Calibrated Air Speed ( CAS ) corrected for density error.
- c) Equivalent Air Speed ( EAS ) corrected for density error.
- d) Calibrated Air Speed ( CAS ) corrected for compressibility error.

Answer ;

- c) Equivalent Air Speed ( EAS ) corrected for density error.
-

- 0838.** True Air Speed ( TAS ) is ;
- Indicated Air Speed ( IAS ) corrected for instrument , position , compressibility and density errors.
  - Calibrated Air Speed ( CAS ) corrected for instrument , compressibility and density errors.
  - Indicated Air Speed ( IAS ) corrected for compressibility and density errors only.
  - Calibrated Air Speed ( CAS ) corrected for instrument , position , compressibility and density errors.

**Answer ;**

- a)** Indicated Air Speed ( IAS ) corrected for instrument , position , compressibility and density errors.
- 

- 0839.** True Air Speed ( TAS ) is equal to Equivalent Air Speed ( EAS ) only if ;

- P = 1013.25 hPa , OAT = 15°C and TAS > 200 kt.
- P = 1013.25 hPa and OAT = 273° K.
- P = 1013.25 hPa and OAT = 15°C.
- P = 1013.25 hPa , OAT = 15°C and TAS < 200 kt.

**Answer ;**

- c)** P = 1013.25 hPa and OAT = 15°C.
- 

- 0840.** True Air Speed ( TAS ) is obtained from Calibrated Air Speed ( CAS ) by correcting for the following errors ;

- instrument.
- compressibility.
- position.
- density.

The combination regrouping all the correct statements is ;

- 2 , 4.
- 4.
- 1 , 2 , 3 , 4.
- 2.

**Answer ;**

- a)** 2 , 4.
- 

- 0841.** True Air Speed ( TAS ) is obtained from Equivalent Air Speed ( EAS ) by correcting for ;

- instrument error.
- position and instrument errors.
- density error.
- compressibility error.

**Answer ;**

- c)** density error.
- 

- 0842.** True Air Speed ( TAS ) is obtained from Indicated Air Speed ( IAS ) by correcting for the following errors ;

- instrument.
- position.
- compressibility.
- density.

The combination regrouping all the correct statements is ;

- 1 , 3 , 4.
- 1 , 2.
- 1 , 2 , 3 , 4.
- 3 , 4.

**Answer ;**

- c)** 1 , 2 , 3 , 4.
- 

- 0843.** Under normal operating conditions , when an aircraft is in a banked turn , the rate-of turn indicator indicates ;

- the angular velocity of the aircraft about the yaw axis.
- the bank of the aircraft.
- the direction of the aircraft turn.
- the angular velocity of the aircraft about the real vertical.

The combination regrouping all the correct statements is ;

- 1 , 2.
- 3 , 4.
- 1 , 3.
- 2 , 4.

**Answer ;**

- c)** 1 , 3.
-

0844.  $V_{FE}$  is the maximum speed ;
- a) with the flaps extended in a given position.
  - b) with the flaps extended in landing position.
  - c) at which the flaps can be operated in turbulence.
  - d) with the flaps extended in take-off position.

Answer ;

- a) with the flaps extended in a given position.
- 

0845.  $V_{FE}$  is the maximum speed ;
- a) with the flaps extended for each approved flap position.
  - b) at which the flaps can be operated in turbulence.
  - c) with the flaps extended in the take-off position.
  - d) with the flaps extended in the landing position.

Answer ;

- a) with the flaps extended for each approved flap position.
- 

0846.  $V_{LO}$  is the maximum ;
- a) speed at which the landing gear can be operated with full safety.
  - b) flight speed with landing gear down.
  - c) cruising speed not to be exceeded except in still air with caution.
  - d) speed with flaps extended in a given position.

Answer ;

- a) speed at which the landing gear can be operated with full safety.
- 

0847.  $V_{NE}$  is the maximum speed ;
- a) not to be exceeded except in still air and with caution.
  - b) which must never be exceeded.
  - c) with flaps extended in landing position.
  - d) at which the flight controls can be fully deflected.

Answer ;

- b) which must never be exceeded.
- 

0848.  $V_{NO}$  is the maximum speed ;
- a) not to be exceeded except in still air and with caution.
  - b) with flaps extended in landing position.
  - c) at which the flight controls can be fully deflected.
  - d) which must never be exceeded.

Answer ;

- a) not to be exceeded except in still air and with caution.
- 

0849. When a TCAS is operating , a failure of the active transponder will cause the TCAS to ;
- a) no longer operate normally.
  - b) operate normally.
  - c) operate in the Traffic Advisory ( TA ) mode only.
  - d) operate in the Resolution Advisory ( RA ) mode only.

Answer ;

- a) no longer operate normally.
- 

0850. When a TCAS Resolution Advisory ( RA ) climb instruction is generated , the required vertical speed range displayed on the vertical speed indicator ;

- a) does not take into account the stall margin.
- b) takes into account a 1.3  $V_S$  stall margin.
- c) always takes into account the stall margin.
- d) takes into account a 1.1  $V_S$  stall margin.

Answer ;

- a) does not take into account the stall margin.
- 

0851. When an aircraft has turned 90 degrees with a constant attitude and bank , the pilot observes the following on a classic artificial horizon ;

- a) too much nose-up and bank too low.
- b) attitude and bank correct.
- c) too much nose-up and bank too high.
- d) too much nose-up and bank correct.

Answer ;

- a) too much nose-up and bank too low.
-

**0852.** When an aircraft has turned 360 degrees with a constant attitude and bank , the pilot observes the following on a classic artificial horizon ;

- a) too much nose-up and bank too high.
- b) too much nose-up and bank correct.
- c) attitude and bank correct.
- d) too much nose-up and bank too low.

**Answer ;**

**c) attitude and bank correct.**

---

**0853.** When an aircraft , operating in the VOR coupled mode , approaches the “cone of confusion” over a VOR station , the roll channel of the autopilot ;

- a) remains always coupled to the selected VOR radial.
- b) temporarily maintains a heading.
- c) is temporarily disconnected.
- d) is damped by a trim input signal from the lateral trim system.

**Answer ;**

**b) temporarily maintains a heading.**

---

**0854.** When available , the yaw damper indicator supplies the pilot with information regarding the ;

- a) rudder displacement by the rudder pedals.
- b) rudder position.
- c) yaw damper action on the rudder.
- d) yaw damper action only on the ground.

**Answer ;**

**c) yaw damper action on the rudder.**

---

**0855.** When being engaged , and without selecting a particular mode , an automatic pilot enables ;

- a) aeroplane piloting and guidance functions.
- b) a constant speed on track , wings horizontal.
- c) aeroplane stabilisation with attitude hold or maintaining vertical speed and possibly automatic trim.
- d) all aeroplane piloting and guidance functions except maintaining radio-navigation course lines.

**Answer ;**

**c) aeroplane stabilisation with attitude hold or maintaining vertical speed and possibly automatic trim.**

---

**0856.** When climbing at a constant CAS ;

- a) EAS does not depend on altitude.
- b) EAS increases.
- c) EAS decreases.
- d) EAS remains constant.

**Answer ;**

**c) EAS decreases.**

---

**0857.** When climbing at a constant CAS ;

- a) Mach number increases.
- b) the difference between surrounding conditions and ISA must be known to deduce the Mach number variation.
- c) Mach number remains constant.
- d) Mach number decreases.

**Answer ;**

**a) Mach number increases.**

---

**0858.** When climbing at a constant CAS in a standard atmosphere ;

- a) TAS remains constant.
- b) TAS first decreases , then remains constant above the tropopause.
- c) TAS increases.
- d) TAS decreases.

**Answer ;**

**c) TAS increases.**

---

0859. When climbing at a constant CAS in a standard atmosphere ;
1. TAS decreases.
  2. TAS increases.
  3. Mach number increases.
  4. Mach number decreases.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 2 , 3.
- c) 1 , 4.
- d) 2 , 4.

Answer ;

- b) 2 , 3.
- 

0860. When climbing at a constant CAS through an isothermal layer , the Mach number ;

- a) remains constant.
- b) increases.
- c) decreases if OAT is lower than the standard temperature.
- d) decreases.

Answer ;

- b) increases.
- 

0861. When climbing at a constant Mach number ;

- a) CAS increases.
- b) difference between surrounding conditions and ISA must be known to deduce the CAS variation.
- c) CAS decreases.
- d) CAS remains constant.

Answer ;

- c) CAS decreases.
- 

0862. When climbing at a constant Mach number through an isothermal layer , the CAS ;

- a) remains constant.
- b) decreases if OAT is lower than the standard temperature , increases if higher.
- c) decreases.
- d) increases.

Answer ;

- c) decreases.
- 

0863. When compared with the volumetric fuel flowmeter , the mass fuel flowmeter takes into account the fuel ;

- a) temperature.
- b) pressure.
- c) turbulent flow in the line.
- d) density.

Answer ;

- d) density.
- 

0864. When comparing a TCAS Traffic Advisory ( TA ) and a Resolution Advisory ( RA ) , which of the following statements is correct ?

- a) A TA indicates the relative position of the intruding traffic , an RA provides a vertical traffic avoidance manoeuvre.
- b) A TA provides the display of the traffic on the Navigation Display and the red arc on the vertical speed indicator , an RA provides the voice alerts.
- c) An RA indicates the relative position of the intruding traffic , a TA provides a vertical traffic avoidance manoeuvre.
- d) An RA generates the intruders colour codes on the Navigation Display according to the threat , a TA manages the other TCAS functions.

Answer ;

- a) A TA indicates the relative position of the intruding traffic , an RA provides a vertical traffic avoidance manoeuvre.
-

0865. When cruising , the autothrottle system can be engaged in the following mode(s) ; holding of constant ;

1. TAS.
2. IAS.
3. Mach number.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 2.
- c) 2 , 3.
- d) 1 , 2.

Answer ;

c) 2 , 3.

---

0866. When descending at a constant CAS ;

- a) EAS decreases.
- b) EAS does not depend on altitude.
- c) EAS increases.
- d) EAS remains constant.

Answer ;

c) EAS increases.

---

0867. When descending at a constant CAS ;

- a) the difference between surrounding conditions and ISA must be known to deduce the Mach number variation.
- b) Mach number remains constant.
- c) Mach number increases.
- d) Mach number decreases.

Answer ;

d) Mach number decreases.

---

0868. When descending at a constant CAS in a standard atmosphere ;

1. TAS increases.
2. TAS decreases.
3. Mach number increases.
4. Mach number decreases.

The combination regrouping all the correct statements is ;

- a) 1 , 3.
- b) 2 , 4.
- c) 1 , 4.
- d) 2 , 3.

Answer ;

b) 2 , 4.

---

0869. When descending at a constant CAS through an isothermal layer , the Mach number ;

- a) increases if OAT is lower than the standard temperature , decreases if higher.
- b) increases.
- c) remains constant.
- d) decreases.

Answer ;

d) decreases.

---

0870. When descending at a constant Mach number ;

- a) CAS increases.
- b) CAS decreases.
- c) CAS remains constant.
- d) the difference between surrounding conditions and ISA must be known to deduce the CAS variation.

Answer ;

a) CAS increases.

---

0871. When disengaging the autopilot , the function providing a smooth “hand-over” is the ;

- a) automatic pitch trim function.
- b) automatic synchronisation function.
- c) automatic CWS ( Control Wheel Steering ) function.
- d) Mach trim function.

Answer ;

a) automatic pitch trim function.

---

0872. When engaged in the FMS lateral navigation mode ( LNAV ) the autopilot uses ;

- a) the roll or heading command computed by the FMS.
- b) the FMS active ( TO ) waypoint coordinates.
- c) the FMS computation of the aircraft position and the FMS active ( TO ) waypoint bearing.
- d) the path angle command computed by the FMS.

Answer ;

- a) the roll or heading command computed by the FMS.
- 

0873. When engaged in the FMS lateral navigation mode ( LNAV ) the autopilot uses the command provided by the ;

- a) track selector.
- b) VOR receiver.
- c) VOR or Localizer receiver.
- d) FMS.

Answer ;

- d) FMS.
- 

0874. When engaged in the lateral navigation mode ( LNAV ) the autopilot uses ;

- a) the FMS active ( TO ) waypoint coordinates.
- b) the FMS path angle command.
- c) the FMS computation of the aircraft position and the FMS active ( TO ) waypoint bearing.
- d) the FMS roll or heading command.

Answer ;

- d) the FMS roll or heading command.
- 

0875. When engaged in the pitch hold mode , the autopilot uses data issued by the ;

- a) Inertial Vertical Speed Indicator.
- b) Flight Management Computer ( FMC ).
- c) ADC.
- d) attitude reference system.

Answer ;

- d) attitude reference system.
- 

0876. When engaging the autopilot , the function providing a smooth “take-over” is the ;

- a) automatic CWS ( Contol Wheel Steering ) function.
- b) automatic synchronisation function.
- c) automatic pitch trim function.
- d) Mach trim funcion.

Answer ;

- b) automatic synchronisation function.
- 

0877. When Enhanced GPWS ( EGPWS ) terrain is displayed , if the computed aircraft position becomes less accurate ;

- a) the terrain display will diverge from the real terrain environment around the aircraft position.
- b) the EGPWS will perform a correction of position thus the terrain display accuracy won't be degraded.
- c) the EGPWS will fail the terrain display function and the terrain will be removed from the navigation display.
- d) the EGPWS has its own position sensors thus the terrain display accuracy won't be degraded.

Answer ;

- a) the terrain display will diverge from the real terrain environment around the aircraft position.
- 

0878. When flying in cold air ( colder than standard atmosphere ) , indicated altitude is ;

- a) the same as the true altitude.
- b) equal to the standard altitude.
- c) higher than the true altitude.
- d) lower than the true altitude.

Answer ;

- c) higher than the true altitude.
- 

0879. When flying in cold air ( colder than standard atmosphere ) , the altimeter will ;

- a) underestimate.
- b) be just as correct as before.
- c) show the actual height above ground.
- d) overestimate.

Answer ;

- d) overestimate.
-



0880. When flying in warm air ( warmer than standard atmosphere ) , indicated altitude is ;

- a) equal to the standard altitude.
- b) higher than the true altitude.
- c) lower than the true altitude.
- d) the same as the true altitude.

Answer ;

c) lower than the true altitude.

---

0881. When in flight , the needle and ball of a needle-and-ball indicator are on the left , the aircraft is ;

- a) turning right with too much bank.
- b) turning left with too much bank.
- c) turning left with not enough bank.
- d) turning right with not enough bank.

Answer ;

b) turning left with too much bank.

---

0882. When in flight , the needle and ball of a needle-and-ball indicator are on the right , the aircraft is ;

- a) turning left with not enough bank.
- b) turning right with not enough bank.
- c) turning right with too much bank.
- d) turning left with too much bank.

Answer ;

c) turning right with too much bank.

---

0883. When in flight , the needle of a needle-and-ball indicator is on the right and the ball on the left , the aircraft is ;

- a) turning right with not enough bank.
- b) turning right with too much bank.
- c) turning left with too much bank.
- d) turning left with not enough bank.

Answer ;

a) turning right with not enough bank.

---

0884. When in flight , the needle of a needle-and-ball indicator is on the left and the ball on the right , the aircraft is ;

- a) turning right with too much bank.
- b) turning left with not enough bank.
- c) turning left with too much bank.
- d) turning right with not enough bank.

Answer ;

b) turning left with not enough bank.

---

0885. When pressing the push button EVENT on the unit control of flight data recorder ;

- a) the recording is rewound so that whatever happens after this event is recorded.
- b) the recording is automatically stopped upon this event.
- c) a mark is set on the recording , enabling this event to be found rapidly at a subsequent analysis.
- d) an event mark is set on the recording , after that the recording is automatically stopped.

Answer ;

c) a mark is set on the recording , enabling this event to be found rapidly at a subsequent analysis.

---

0886. When required , the ground proximity warning system must automatically provide distinctive warning to the flight crew of ;

1. impeding stall.
2. excessive sink rate.
3. altitude loss after take-off or go-around.
4. incorrect landing configuration.
5. dangerous ground proximity.
6. downward glide-slope deviation.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4 , 5 , 6.
- b) 1 , 4 , 5 , 6.
- c) 1 , 2 , 4 , 5.
- d) 3 , 5 , 6.

Answer ;

a) 2 , 3 , 4 , 5 , 6.

---

0887. When sent , a MAYDAY datalink message to the ATC has the following effect on ADS ;
- a) the ADS current contract reporting rate is unchanged but includes more data.
  - b) no change to the current contract in process.
  - c) the ADS current contract is stopped.
  - d) the ADS contract is switched to high periodic reporting rate.

Answer ;

- d) the ADS contract is switched to high periodic reporting rate.
- 

0888. When the auto-pilot is engaged , the role of the automatic trim is to ;
- a) react to altitude changes in Altitude Hold mode.
  - b) relieve the pressure on the control column and return , the aircraft in-trim at AP disconnect.
  - c) synchronize the longitudinal loop.
  - d) relieve the AP servo motor and return the aircraft in-trim at AP disconnect.

Answer ;

- d) relieve the AP servo motor and return the aircraft in-trim at AP disconnect.
- 

0889. When the flight warning system ( FWS ) identifies an overspeed condition ( airspeed exceeding  $V_{MO}/M_{MO}$  ) , it generates ;
- a) an advisory message.
  - b) a call message.
  - c) a warning message.
  - d) a caution message.

Answer ;

- c) a warning message.
- 

0890. When the intruding aircraft is equipped with a serviceable mode C transponder , the TCAS 2 ( Traffic Collision Avoidance System ) generates a ;
- a) “traffic advisory” and horizontal “resolution advisory”.
  - b) “traffic advisory” and vertical “resolution advisory”.
  - c) “traffic advisory” , vertical and horizontal “resolution advisory”.
  - d) vertical “traffic advisory” and an horizontal “resolution advisory”.

Answer ;

- b) “traffic advisory” and vertical “resolution advisory”.
- 

0891. When the intruding aircraft is equipped with a transponder without altitude reporting capability , the TCAS ( Traffic Collision Avoidance System ) issues a ;
- a) “traffic advisory” and horizontal “resolution advisory”.
  - b) “traffic advisory” and vertical “resolution advisory”.
  - c) “traffic advisory” , vertical and horizontal “resolution advisory”.
  - d) “traffic advisory” only.

Answer ;

- d) “traffic advisory” only.
- 

0892. When the rotary knob on the INS control panel is set to “NAV” mode , it is ;
- a) the navigation mode allowing use of all the functions of the system except attitude.
  - b) the alignment function in flight.
  - c) the normal operating mode allowing use of all the functions of the system.
  - d) the navigation mode allowing use of all the functions of the system except heading.

Answer ;

- c) the normal operating mode allowing use of all the functions of the system.
- 

0893. When the TCAS 2 ( Traffic alert and Collision Avoidance System ) generates a Resolution Advisory ( RA ) , the associated intruder appears on TCAS display as a ;
- a) solid amber square.
  - b) solid amber circle.
  - c) solid red square.
  - d) solid red circle.

Answer ;

- c) solid red square.
-

0894. When the yaw damper system sends motion orders to the rudder ;
- a) a feedback is provided on the rudder pedals only.
  - b) a feedback is provided on the roll trim only.
  - c) a feedback is provided on the rudder pedals and roll trim.
  - d) no feedback is provided on the rudder pedals.

Answer ;

- d) no feedback is provided on the rudder pedals.
- 

0895. When two waypoints are entered on the FMS flight plan page , a track between the two fixes is computed and can be displayed on the Navigation map display ( ND ). This leg created is ;

- a) a rhumb-line.
- b) a great circle arc.
- c) two great circle arcs joined by a straight segment.
- d) two rhumb-lines joined by a straight segment.

Answer ;

- b) a great circle arc.
- 

0896. When using the autopilot , the function of the pitch channel automatic trim is to ;

- 1. cancel the hinge moment of the elevator.
- 2. ease as much as possible the load of the servo-actuator.
- 3. restore to the pilot a correctly trimmed airplane during the autopilot disengagement.

The combination regrouping all the correct statements is ;

- a) 1 and 3.
- b) 1 and 2.
- c) 1 , 2 and 3.
- d) 3.

Answer ;

- c) 1 , 2 and 3.
- 

0897. Which of the following statements about hard and soft iron in relation to magnetism is correct ?

- a) Hard iron magnetism is of a permanent nature and soft iron is a non-permanent nature.
- b) Both hard and soft iron are of a permanent nature.
- c) Both hard and soft iron are of a non-permanent nature.
- d) Hard iron is a non-permanent nature and soft iron is a permanent nature.

Answer ;

- a) Hard iron magnetism is of a permanent nature and soft iron is a non-permanent nature.
- 

0898. Which of the following statements are correct for an aeroplane cruising at FL 60 with a true airspeed ( TAS ) of 100 kt in standard atmospheric conditions ?

- 1. The TAS is approximately 10% higher than the IAS.
- 2. The difference between the equivalent airspeed ( EAS ) and the calibrated airspeed ( CAS ) is negligible.
- 3. The speed displayed on the airspeed indicator is a calibrated airspeed ( CAS ) if the position error and instrument error are zero.

- a) 2 and 3 are correct.
- b) 1 and 2 are correct.
- c) 1 and 3 are correct.
- d) 1 , 2 and 3 are all correct.

Answer ;

- d) 1 , 2 and 3 are all correct.
- 

0899. Which of these statements about the activation of the take-off warning when a take-off is initiated are correct or incorrect ?

- I. An aural warning is given when the elevator is not in a safe position for take-off.
- II. An aural warning is given when the brake pressure is too low.

- a) I is correct , II is correct.
- b) I is incorrect , II is correct.
- c) I is correct , II is incorrect.
- d) I is incorrect , II is incorrect.

Answer ;

- d) I is incorrect , II is incorrect.
-

- 0900.** Which of these statements about the activation of the take-off warning when a take-off is initiated are correct or incorrect ?
- I. An aural warning is given when the elevator is not in a safe position for take-off.
  - II. An aural warning is given when the parking brake is still ON.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is correct , II is incorrect.

**Answer ;**

- a) I is incorrect , II is correct.
- 

- 0901.** Which of these statements about the activation of the take-off warning when a take-off is initiated are correct or incorrect ?
- I. An aural warning is given when the stabiliser is not in a safe position for take-off
  - II. An aural warning is given when the parking brake is still ON.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

**Answer;**

- b) I is correct , II is correct.
- 

- 0902.** Which of these statements about the activation of the take-off warning when a take-off is initiated are correct or incorrect ?
- I. An aural warning is given when the stabiliser is not in a safe position for take-off.
  - II. An aural warning is given when the brake pressure is too low.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is correct , II is incorrect.

**Answer ;**

- d) I is correct , II is incorrect.
- 

- 0903.** Which one of the following statements is true with regard to the operation of a Mach trim system ;
- a) It only operates above a pre-determined Mach number.
  - b) It operates to counteract the larger than normal forward movements of the wing centre of pressure at high subsonic airspeeds.
  - c) It only operates when the autopilot is engaged.
  - d) It operates over the full aircraft speed range.

**Answer ;**

- a) It only operates above a pre-determined Mach number.
- 

- 0904.** While inertial platform system is operating on board an aircraft , it is necessary to use a device with the following characteristics , in order to keep the vertical line with a pendulous system ;
- a) with damping and a period of about 84 minutes.
  - b) with damping and a period of 84 seconds.
  - c) without damping and a period of about 84 seconds.
  - d) without damping and a period of about 84 minutes.

**Answer ;**

- a) with damping and a period of about 84 minutes.
- 

- 0905.** With a TCAS 2 , when a corrective resolution is generated ;
- a) the vertical speed must be effectively modified without delay.
  - b) no action is required , vertical speed , heading and IAS can remained unchanged.
  - c) the heading must be effectively modified without delay.
  - d) the IAS must be effectively modified without delay.

**Answer ;**

- a) the vertical speed must be effectively modified without delay.
- 

- 0906.** With CAS and pressure altitude , we can deduce ;
- a) TAS.
  - b) EAS.
  - c) EAS and TAS.
  - d) nothing.

**Answer ;**

- b) EAS.
-

0907. With constant weight and configuration , an aeroplane always takes off at the same ;

- a) TAS.
- b) ground speed.
- c) IAS.
- d) EAS.

Answer ;

d) EAS.

---

0908. With constant weight and configuration , an aircraft always takes off at the same ;

- a) indicated airspeed.
- b) equivalent airspeed.
- c) ground speed.
- d) true airspeed.

Answer ;

b) equivalent airspeed.

---

0909. With EAS and density altitude (  $Z_d$  ) , we can deduce ;

- a) CAS.
- b) CAS and TAS.
- c) IAS
- d) TAS.

Answer ;

d) TAS.

---

0910. With EAS and pressure altitude (  $Z_p$  ) , we can deduce ;

- a) TAS.
- b) IAS.
- c) CAS and TAS.
- d) CAS.

Answer ;

d) CAS.

---

0911. Your aircraft and an intruding aircraft both are TCAS II equipped. If the transponder of the intruder is switched off or unserviceable ;

- a) the intruding aircraft remains visible to your TCAS equipment , but only RAs ( resolution advisories ) can be generated.
- b) the intruding aircraft remains visible to your TCAS equipment , but only TAs ( traffic advisories ) can be generated.
- c) the information available to your TCAS equipment is two dimensional only.
- d) the intruding aircraft is invisible to your TCAS equipment.

Answer ;

d) the intruding aircraft is invisible to your TCAS equipment.

---

0912. Your aircraft and an intruding traffic are both TCAS II equipped. The information available to your TCAS equipment is ;

- a) three dimensional ; your TCAS can generate both TAs ( Traffic Advisories ) and RAs ( Resolution Advisories ).
- b) two dimensional only ; your TCAS can generate both TAs ( Traffic Advisories ) and Ras ( Resolution Advisories ).
- c) two dimensional only ; avoidance manoeuvres between both aircraft cannot be coordinated.
- d) three dimensional ; avoidance manoeuvres between both aircraft cannot be coordinated.

Answer ;

a) three dimensional ; your TCAS can generate both TAs ( Traffic Advisories ) and RAs ( Resolution Advisories ).

---

0913. Your aircraft and an intruding traffic are both TCAS II equipped. Your TCAS ;

- a) cannot generate coordinated resolution advisories.
- b) can generate coordinated resolution advisories.
- c) can generate coordinated resolution advisories by interrogating the radar ground station.
- d) can generate coordinated resolution advisories by activating the on-board radar system.

Answer ;

b) can generate coordinated resolution advisories.

---

**0914.** Your aircraft and an intruding traffic are both TCAS II equipped. Your TCAS can generate ;

- a) resolution advisories coordinated with the ATC instructions.
- b) traffic advisories coordinated with the ATC instructions.
- c) coordinated traffic advisories.
- d) coordinated and complementary resolution advisories.

**Answer ;**

- d)** coordinated and complementary resolution advisories.
- 

**0915.** Your aircraft and an intruding traffic both are TCAS II equipped. Your TCAS determines the relative height of the intruding aircraft by ;

- a) interrogating the ATC ground station.
- b) using the range function of the on board weather radar.
- c) comparing the GPS heights of the two aircraft.
- d) comparing the altitudes of two aircraft.

**Answer ;**

- d)** comparing the altitudes of two aircraft.
- 

**0916.** Your aircraft and an intruding aircraft both are TCAS 2 ( Traffic alert and Collision Avoidance System ) equipped. Your TCAS determines the range of the intruding aircraft by ;

- a) measuring the time lapse between the transmission of an interrogation signal and the reception of a reply signal from the transponder of the intruder.
- b) comparing the GPS positions of two aircraft.
- c) measuring the time lapse between the transmission of an interrogation signal and the reception of a reply signal from the ground station.
- d) using the range function of the on-board weather radar.

**Answer ;**

- a)** measuring the time lapse between the transmission of an interrogation signal and the reception of a reply signal from the transponder of the intruder.
- 

**0917.** Your aircraft and an intruding aircraft both are TCAS 2 ( Traffic alert and Collision Avoidance System ) equipped. Your TCAS determines the bearing of the intruding aircraft by ;

- a) measuring the time lapse between the transmission of an interrogation signal and the reception of a reply signal.
- b) using the bearing function of the on-board weather radar.
- c) using a specific directional antenna.
- d) comparing the GPS positions of two aircraft.

**Answer ;**

- c)** using a specific directional antenna.
- 

**0918.** Your aircraft and an intruding traffic are TCAS 2 equipped. Your TCAS can generate ;

- a) coordinated traffic advisories.
- b) resolution advisories coordinated with the ATC instructions.
- c) traffic advisories coordinated with the ATC instructions.
- d) coordinated resolution advisories.

**Answer ;**

- d)** coordinated resolution advisories.
- 

**0919.** Your aircraft is TCAS 2 equipped , an intruding traffic only has a mode A transponder. The information available to your TCAS equipment is ;

- a) three dimensional ; your TCAS can generate both TAs ( Traffic Advisories ) and RAs ( Resolution Advisories ).
- b) two dimensional only ; your TCAS cannot generate RAs ( Resolution Advisories ).
- c) two dimensional only ; your TCAS cannot generate TAs ( Traffic Advisories ).
- d) three dimensional ; your TCAS cannot generate TAs ( Traffic Advisories ).

**Answer ;**

- b)** two dimensional only ; your TCAS cannot generate RAs ( Resolution Advisories ).
-

**0920.** Your aircraft is TCAS 2 equipped , an intruding traffic only has a mode C transponder. The information available to your TCAS equipment is ;

- a) two dimensional only ; your TCAS cannot generate TAs ( Traffic Advisories ).
- b) three dimensional ; your TCAS can generate both TAs ( Traffic Advisories ) and RAs ( Resolution Advisories ).
- c) three dimensional ; your TCAS cannot generate RAs ( Resolution Advisories ).
- d) two dimensional only ; your TCAS cannot generate RAs ( Resolution Advisories ).

**Answer ;**

- b)** three dimensional ; your TCAS can generate both TAs ( Traffic Advisories ) and RAs ( Resolution Advisories ).
- 

**0921.** Your aircraft is TCAS 2 equipped. To be able to generate a resolution advisory ( RA ) , the intruder must be at least equipped with ;

- a) a TCAS 2.
- b) a transponder mode C.
- c) a transponder mode S.
- d) a transponder mode A.

**Answer ;**

- b)** a transponder mode C.
- 

**0922.** Your aircraft is TCAS 2 equipped. To be able to generate a traffic advisory ( TA ) , the intruder must be at least equipped with ;

- a) a transponder mode A.
- b) a TCAS 2.
- c) a transponder mode S.
- d) a transponder mode C.

**Answer ;**

- a)** a transponder mode A.
- 

~ SON ~

0001. A 3 phase AC generator has 3 separate stator windings spaced at ;

- a) 45°.
- b) 90°.
- c) 60°.
- d) 120°.

Answer ;

d) 120°.

---

0002. A booster or LP compressor in a twin spool turbofan ;

- a) rotates at a rotor speed different from both fan , HP compressor and HP turbine.
- b) rotates at the same rotor speed as the fan.
- c) rotates at the same rotor speed as the HP compressor.
- d) rotates at the same rotor speed as the HP turbine.

Answer ;

b) rotates at the same rotor speed as the fan.

---

0003. A bus-bar is ;

- a) a device which may only be used in DC circuits.
- b) a distribution point for electrical power.
- c) the stator of a moving coil instrument.
- d) a device permitting operation of two or more switches together.

Answer ;

b) a distribution point for electrical power.

---

0004. A cabin pressure controller maintains a pre-set cabin altitude by regulating the ;

- a) position of the outflow valve(s).
- b) position of the duct relief valve(s).
- c) mass air flow into the cabin.
- d) position of the inward relief valve.

Answer ;

a) position of the outflow valve(s).

---

0005. A cabin rate of descent ;

- a) is not possible at constant aeroplane altitudes.
- b) is always the same as the aeroplane's rate of descent.
- c) results in a cabin pressure decrease.
- d) results in a cabin pressure increase.

Answer ;

d) results in a cabin pressure increase.

---

0006. A capacitor in parallel with breaker points ;

- a) induces a very high current across the primary windings.
- b) induces a very high voltage across the secondary windings.
- c) permits arcing across the points.
- d) induces a very high current across the secondary windings.

Answer ;

b) induces a very high voltage across the secondary windings.

---

0007. A cantilever wing is ;

- a) a wing attached to the fuselage at the wing root only.
- b) a low wing configuration.
- c) a high wing configuration.
- d) a wing planform other than rectangular.

Answer ;

a) a wing attached to the fuselage at the wing root only.

---

0008. A chip detector in the oil system of an engine/gearbox is to indicate that ;

- a) the piston rings are worn.
- b) to indicate that seals are worn.
- c) there are metal particles in the oil.
- d) the oil temperature is too high.

Answer ;

c) there are metal particles in the oil.

---



0009. A composite structural component consists of ;
- a) two metal sheets bonded together.
  - b) aluminium alloy with a covering layer of pure aluminium.
  - c) a matrix and fibres.
  - d) two thin metal sheets and a light core material.

Answer ;

c) a matrix and fibres.

---

0010. A condenser in parallel with breaker points will ;
- a) assist in negative feedback to secondary coil.
  - b) assist in collapse of secondary winding..
  - c) permit arcing across points.
  - d) intensify current in secondary winding.

Answer ;

d) intensify current in secondary winding.

---

0011. A Constant Speed Drive aims at ensuring ;
- a) that the CSD remains at a constant RPM not withstanding the generator RPM.
  - b) that the electric generator produces a constant frequency.
  - c) equal AC voltage from all generators.
  - d) that the starter-motor maintains a constant RPM not withstanding the acceleration of the engine.

Answer ;

b) that the electric generator produces a constant frequency.

---

0012. A Constant Speed Drive ( CSD ) that has been disconnected in flight ;
- a) can be reset in flight using the reset mechanism.
  - b) automatically resets on engine shut-down.
  - c) automatically resets in flight providing engine rpm is below a given value.
  - d) can be reset only on the ground after engine shut-down.

Answer ;

d) can be reset only on the ground after engine shut-down.

---

0013. A continuous loop detector system is a ;
- a) fire fighting system.
  - b) smoke detection system.
  - c) carbon dioxide warning system.
  - d) overheat detection system.

Answer ;

d) overheat detection system.

---

0014. A CSD of an AC generator may be disconnected in flight. The primary reason(s) for disconnection are ;
- a) slight variation about the normal operating frequency.
  - b) excessive variation of voltage and kVAR.
  - c) illumination of the CSD disconnect warning light.
  - d) low oil pressure and/or high oil temperature of the generator drive.

Answer ;

d) low oil pressure and/or high oil temperature of the generator drive.

---

0015. A CSD unit is normally provided with means of monitoring during flight the ;
- a) oil temperature and synchronous speed.
  - b) oil temperature and pressure.
  - c) oil temperature and quantity only.
  - d) output voltage and oil pressure.

Answer ;

b) oil temperature and pressure.

---

0016. A CSD unit which has been disconnected in flight ;
- a) automatically resets at engine shut-down.
  - b) may be reset in flight using the reset mechanism.
  - c) may be reset on the ground only , after engine shut-down.
  - d) automatically resets in flight providing engine rpm is below a given value.

Answer ;

c) may be reset on the ground only , after engine shut-down.

---

0017. A current limiter fuse in a DC generation system is used to ;
- a) allow a short term overload before rupturing.
  - b) limit the current in the armature.
  - c) limit the current in the field circuit.
  - d) instantaneously rupture to limit the current in the load.

Answer ;

- a) allow a short term overload before rupturing.
- 

0018. A diffuser in a gas turbine engine is a device in which the ;
- a) velocity , pressure and temperature rise.
  - b) velocity rises and pressure falls.
  - c) pressure rises and velocity falls.
  - d) pressure rises at a constant velocity.

Answer ;

- c) pressure rises and velocity falls.
- 

0019. A diluter demand oxygen regulator selected to “100%” ;
- a) delivers oxygen flow when inhaling.
  - b) mixes air and oxygen in a passenger oxygen mask.
  - c) is only recommended for use with smoke in the cockpit.
  - d) delivers oxygen flow only above FL 100.

Answer ;

- a) delivers oxygen flow when inhaling.
- 

0020. A FADEC on a gas turbine engine ;
- a) is a three channel system.
  - b) is a single channel system.
  - c) is single fault tolerant.
  - d) consists of two identical channels cross linked through a non engine related aircraft computer.

Answer ;

- c) is single fault tolerant.
- 

0021. A “fan” stage of a ducted-fan turbine engine is driven by ;
- a) airflow drawn across it by the high pressure compressor.
  - b) the high pressure compressor through reduction gearing.
  - c) the low pressure turbine.
  - d) the high pressure turbine.

Answer ;

- c) the low pressure turbine.
- 

0022. A fault protection circuit in a fire detection system will ;
- a) activate the fire detection system when the detection line is connected to ground.
  - b) inhibit the fire detector when the detection line is connected to ground.
  - c) automatically initiate APU shutdown and fire extinguisher striker activation in the event of fire.
  - d) activate an alarm in the cockpit and in the landing gear bay for ground crew.

Answer ;

- b) inhibit the fire detector when the detection line is connected to ground.
- 

0023. A feeder fault on a direct current circuit results from a flux unbalance between the ;
- a) shunt exciter and the series winding turn.
  - b) generator and the series winding turn.
  - c) voltage coil and the series winding.
  - d) voltage coil and the series winding turn.

Answer ;

- d) voltage coil and the series winding turn.
- 

0024. A flat rated turbofan engine has a constant maximum rated thrust level ;
- a) up to a specific pressure altitude.
  - b) above a specific pressure altitude.
  - c) above a specific OAT.
  - d) up to a specific OAT.

Answer ;

- d) up to a specific OAT.
-

0025. A flat rated turbofan will be temperature limited if ;
- a) OAT is lower than flat rated temperature.
  - b) ambient pressure is higher than standard sea level ambient pressure.
  - c) OAT is higher than flat rated temperature.
  - d) ambient pressure is lower than standard sea level ambient pressure.

Answer ;

- c) OAT is higher than flat rated temperature.
- 

0026. A flight control surface actuator is said to be “irreversible” when ;
- a) the pilot does not feel any force when moving that control surface in flight..
  - b) there is no feedback to the pilot’s controls of the aerodynamic forces acting on the control surface.
  - c) there is a need to lock the flight controls on the ground.
  - d) the flight control system has an alternate means of control in case of a control jam.

Answer ;

- b) there is no feedback to the pilot’s controls of the aerodynamic forces acting on the control surface.
- 

0027. A flight control surface actuator is said to be “reversible” when ;
- a) there is feedback to the pilot’s controls of the aerodynamic forces acting on the control surface.
  - b) the flight control system has an alternate means of control in case of a control jam.
  - c) the pilot does not feel any force when moving that flight control surface in flight.
  - d) there is a need to have an artificial feel system.

Answer ;

- a) there is feedback to the pilot’s controls of the aerodynamic forces acting on the control surface.
- 

0028. A fuel strainer when fitted to a carburettor will be positioned ;
- a) between the metering jet and the discharge nozzle.
  - b) upstream of the needle valve.
  - c) between the needle valve and the metering jet.
  - d) downstream of the discharge nozzle.

Answer ;

- b) upstream of the needle valve.
- 

0029. A gas turbine engine is run at ground idle for a period of time prior to shutdown to ;
- a) prevent seizure of the engine bearings.
  - b) prevent vapour lock in the fuel control and/or fuel lines.
  - c) allow the engine oil to cool.
  - d) prevent seizure of the rotor blades in their seals.

Answer ;

- d) prevent seizure of the rotor blades in their seals.
- 

0030. A gas turbine engine turbine section may employ active clearance control ;
- a) as a method of attaching the stators and rotors to the hub of the disc.
  - b) to provide enhanced clearances between the blade tips and the casing.
  - c) in order to reduce “creep” in the blades.
  - d) in order to cool the disc.

Answer ;

- b) to provide enhanced clearances between the blade tips and the casing.
- 

0031. A gaseous fire loop system is tested by ;
- a) checking the wiring harness for faults but not the sensor.
  - b) checking the continuity of the system.
  - c) checking the sensor with external pressurised gas.
  - d) heating up the sensor.

Answer ;

- d) heating up the sensor.
- 

0032. A gaseous sensor/responder tube fire sensor is tested by ;
- a) heating up the sensor with test power connection.
  - b) checking the continuity of the system with a test switch.
  - c) checking the wiring harness for faults but not the sensor.
  - d) checking the sensor with pressurized gas.

Answer ;

- a) heating up the sensor with test power connection.
-

0033. A hung ( abortive ) start of a gas turbine engine may be caused by ;
- a) the starter cutting out early in the starting sequence before the engine has accelerated to the required rpm for ignition.
  - b) failure of the fuel to ignite in the starting sequence after the engine has been accelerated to the required rpm by the starter.
  - c) compressor surging.
  - d) a too low pneumatic starter pressure.

Answer ;

d) a too low pneumatic starter pressure.

---

0034. A “hung start” is the failure of an engine to accelerate to its normal idle speed. It may be caused by ;
- a) failure of the fuel to ignite in the starting sequence after the engine has been accelerated to the required rpm by the starter.
  - b) an attempt to ignite the fuel before the engine has been accelerated sufficiently by the starter.
  - c) the starter cutting out early in the starting sequence before the engine has accelerated to the required rpm for ignition.
  - d) compressor surging.

Answer ;

b) an attempt to ignite the fuel before the engine has been accelerated sufficiently by the starter.

---

0035. A hydraulic low pressure alert is the first indication of ;
- a) the reservoir level being at a minimum acceptable level.
  - b) a leak in the reservoir return line.
  - c) the pump output pressure being insufficient.
  - d) the hydraulic system accumulator becoming deflated.

Answer ;

c) the pump output pressure being insufficient.

---

0036. A hydraulic shuttle valve ;
- a) is a non-return valve.
  - b) enables an alternate supply to operate an actuator.
  - c) regulates pump delivery pressure.
  - d) allows two units to be operated by one pressure source.

Answer ;

b) enables an alternate supply to operate an actuator.

---

0037. A Krueger flap is normally located at the ;
- a) trailing edge close to the wing tip.
  - b) leading edge.
  - c) trailing edge.
  - d) trailing edge close to the wing root.

Answer ;

b) leading edge.

---

0038. A magnetic circuit-breaker is ;
- a) a protection system that has a quick tripping response.
  - b) permits an overcurrent limited in time.
  - c) can be reset without any danger even when fault remains.
  - d) is a system with a slow response time.

Answer ;

a) a protection system that has a quick tripping response.

---

0039. A magnetic plug in an engine oil system can be used to ;
- a) collect static electricity.
  - b) collect ferrous particles reaching the sump.
  - c) prevent metallic particles from entering the oil system.
  - d) collect carbon found in the oil.

Answer ;

b) collect ferrous particles reaching the sump.

---

0040. A manifold pressure gauge of a piston engine measures ;
- a) vacuum in the carburettor.
  - b) fuel pressure leaving the carburettor.
  - c) absolute pressure in intake system near the inlet valve.
  - d) absolute air pressure entering the carburettor.

Answer ;

c) absolute pressure in intake system near the inlet valve.

---

0041. A manual inflation handle ;
- a) operates a hand pump for manual inflation of a slide.
  - b) serves to inflate a life jacket when the normal inflation function fails.
  - c) is generally not applied on slides.
  - d) serves to actuate inflation of a slide when automatic inflation fails.

Answer ;

d) serves to actuate inflation of a slide when automatic inflation fails.

---

0042. A mixture setting richer than best power has to be used during climb segments. This results in a ;
- a) lower cylinder head temperature.
  - b) higher torque.
  - c) increase of power.
  - d) higher efficiency.

Answer ;

a) lower cylinder head temperature.

---

0043. A modern Auxiliary Power Unit ( APU ) is designed to provide power for ground starting of an engine. It also supplies both in the air ( subject to certification limitations ) and on the ground ;
- a) air conditioning and thrust in the event of engine failure.
  - b) air conditioning and electrical services.
  - c) air conditioning and electrical services ( on the ground ) electrical and hydraulic back-up services ( in the air ).
  - d) either air conditioning or electrical services , but never both at the same time.

Answer ;

b) air conditioning and electrical services.

---

0044. A non-cantilever wing is ;
- a) a wing platform other than rectangular.
  - b) a low wing configuration.
  - c) a wing supported by braces or a strut connected to the fuselage.
  - d) a high wing configuration.

Answer ;

c) a wing supported by braces or a strut connected to the fuselage.

---

0045. A passenger emergency mask is a ;
- a) continuous flow mask and can be used if there is smoke in the cabin.
  - b) mask with flow on request and can be used if there is smoke.
  - c) continuous flow mask and cannot be used if there is smoke in the cabin.
  - d) mask with flow on request and cannot be used if there is smoke in the cabin.

Answer ;

c) continuous flow mask and cannot be used if there is smoke in the cabin.

---

0046. A passenger oxygen mask is ;
- a) a continuous flow mask and must be used if there is smoke in the cabin.
  - b) a continuous flow mask and in principle should not be used if there is smoke in the cabin.
  - c) an on demand type mask and in principle should not be used if there is smoke in the cabin.
  - d) an on demand type mask and must be used if there is smoke in the cabin.

Answer ;

b) a continuous flow mask and in principle should not be used if there is smoke in the cabin.

---

0047. A pilot can actuate the feathering system by ;
- a) pulling the power levers rearwards.
  - b) pulling the propeller control lever rearwards.
  - c) pushing the propeller control lever forwards.
  - d) pushing the power lever forwards.

Answer ;

b) pulling the propeller control lever rearwards.

---

0048. A pilot can determine the amount of oxygen in a bottle by observing its ;

- a) level.
- b) pressure.
- c) temperature.
- d) volume.

Answer ;

b) pressure.

---

0049. A piston engine compression ratio is the ratio of the ;

- a) total volume to the swept volume.
- b) clearance volume to the swept volume.
- c) swept volume to the clearance volume.
- d) total volume to the clearance volume.

Answer ;

d) total volume to the clearance volume.

---

0050. A piston engine may use a fuel of a different grade than the recommended ;

- a) provided that it is an aeronautical petrol.
- b) provided that the grade is lower.
- c) never.
- d) provided that the grade is higher.

Answer ;

d) provided that the grade is higher.

---

0051. A pressure regulator is used in a hydraulic system ;

- a) to ensure that an equal pressure flow is delivered to critical components such as servo actuators.
- b) in conjunction with a constant delivery type pump.
- c) in conjunction with a variable delivery type pump.
- d) as an interface between the system and the cockpit indicators.

Answer ;

b) in conjunction with a constant delivery type pump.

---

0052. A pressure relief valve in an oil system that does not seat correctly could result in ;

- a) low oil temperature.
- b) low oil pressure.
- c) high oil pressure.
- d) excessive oil consumption.

Answer ;

b) low oil pressure.

---

0053. A pressure relief valve that does not fit on its seat properly would result in ;

- a) high oil pressure.
- b) low oil temperature.
- c) low oil pressure.
- d) excessive oil consumption.

Answer ;

c) low oil pressure.

---

0054. A pressurized aeroplane is flying at FL 370 and experiences a rapid decompression. Which of the following statements is correct about the oxygen system ?

- a) Manual override of the automatic presentation of passenger oxygen masks is , generally speaking , not possible.
- b) The oxygen masks are automatically presented to cabin crew members and passengers.
- c) If the automatic mask presentation has been activated , the oxygen will flow within the first 3 minutes.
- d) the oxygen masks are automatically presented to flight crew members.

Answer ;

b) The oxygen masks are automatically presented to cabin crew members and passengers.

---

**0055.** A pressurized aeroplane is operated at FL 300. It undergoes a rapid decompression so that the pressure in the cabin goes quickly down to the outside pressure value. What happens concerning the oxygen system ?

- a) If the automatic mask presentation has been activated , the oxygen will flow within the first 3 minutes.
- b) The oxygen masks are automatically presented to cabin crew members and passengers.
- c) Manual override of the automatic presentation of passenger oxygen masks is , generally speaking , not possible.
- d) The oxygen masks are automatically presented to flight crew members.

**Answer ;**

- b)** The oxygen masks are automatically presented to cabin crew members and passengers.
- 

**0056.** A propeller blade is twisted , so as to ;

- a) keep the local Angle of Attack constant along the blade.
- b) decrease the blade tangential velocity from the blade root to the tip.
- c) allow a higher mechanical stress.
- d) avoid the appearance of sonic phenomena.

**Answer ;**

- a)** keep the local Angle of Attack constant along the blade.
- 

**0057.** A radio signal loses strength as range from the transmitter increases , this is called ;

- a) propagation.
- b) attenuation.
- c) refraction.
- d) ducting.

**Answer ;**

- b)** attenuation.
- 

**0058.** A red or amber light on an undercarriage position indicator signifies ;

- a) the landing gear has been selected down using the emergency extension system.
- b) all wheels are down and locked.
- c) all wheels are up and locked.
- d) at least one wheel is in the travelling or unlocked condition.

**Answer ;**

- d)** at least one wheel is in the travelling or unlocked condition.
- 

**0059.** A relay is ;

- a) a magnetically operated switch.
- b) a device that is used to increase electrical power.
- c) a unit that is used to convert electrical energy to heat energy.
- d) another name for a solenoid valve.

**Answer ;**

- a)** a magnetically operated switch.
- 

**0060.** A relay is ;

- a) an electromagnetically operated switch.
- b) an electrical energy conversion unit.
- c) an electrical security switch.
- d) a switch specially designed for AC circuits.

**Answer ;**

- a)** an electromagnetically operated switch.
- 

**0061.** A reverse thrust alert on the flight deck occurs when ;

1. the reverser doors are unlocked with the reverse lever in the stowed position.
2. the reverser doors are locked with the reverse lever in the stowed position.
3. the reverser doors remain stowed with the reverse lever in the deployed position.
4. the reverser doors have moved to the reverse thrust position with the reverse lever in the deployed position.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 4.
- c) 1 , 3.
- d) 2 , 4.

**Answer ;**

- c)** 1 , 3.
-

0062. A reverse thrust door warning light on the flight deck instrument panel illuminates when ;
- a) the reverser doors have moved to the reverse thrust position.
  - b) the reverser doors are locked.
  - c) reverse has been selected but the doors have remained locked.
  - d) the reverser doors are unlocked.

Answer ;

- d) the reverser doors are unlocked.
- 

0063. A rich mixture setting has to be used during climb segments. This result in a ;
- a) lower cylinder head temperature.
  - b) higher torque.
  - c) increase of power.
  - d) higher efficiency.

Answer ;

- a) lower cylinder head temperature.
- 

0064. A safe-life aircraft structural component ;
- a) has parallel load paths.
  - b) may be used during a declared number of cycles or flight hours.
  - c) should have enough strength during the whole lifetime of an aircraft.
  - d) is so strong that it never will fail during a declared time period.

Answer ;

- b) may be used during a declared number of cycles or flight hours.
- 

0065. A sandwich structural part ;
- a) always uses honeycomb as core material.
  - b) consists of two thin sheets separated by a light core material.
  - c) is a so-called integral construction.
  - d) is unsuitable for fuel tanks.

Answer ;

- b) consists of two thin sheets separated by a light core material.
- 

0066. A sandwich structural part is ;
- a) composed of two thin sheets and a light core material.
  - b) composed of resin and fibres.
  - c) well suited for absorbing point concentrated loads.
  - d) a so-called integral construction.

Answer ;

- a) composed of two thin sheets and a light core material.
- 

0067. A sandwich structural part is unsuitable for absorbing ;
- a) bending loads.
  - b) concentrated loads.
  - c) shear loads.
  - d) torsional loads.

Answer ;

- b) concentrated loads.
- 

0068. A sandwich type structure is often used in aircraft because of its ;
- a) high temperature resistance.
  - b) low mass and low stiffness.
  - c) low mass and high stiffness.
  - d) ease of deformation under load.

Answer ;

- c) low mass and high stiffness.
- 

0069. A scissor is a component found on landing gears. Its function is to ;
- a) create the wheel pitch on bogie gears.
  - b) make the body gears pivot when the nose wheel is turned through more than 20°.
  - c) transform the translational movement of the rudder pedals into the rotational movement of the nose wheel.
  - d) prevent any rotation of the oleo strut in the undercarriage shock absorber.

Answer ;

- d) prevent any rotation of the oleo strut in the undercarriage shock absorber.
-



0070. A semi-monocoque aircraft fuselage structure usually consists of ;

- a) frames , fittings , stringers.
- b) skin , frames , stringers.
- c) ribs , spars , skin.
- d) ribs , front spar , rear spar.

Answer ;

b) skin , frames , stringers.

---

0071. A single action actuator ;

- a) is a one shot actuator used for emergency systems only.
- b) is powered in one direction only by hydraulic power , the return movement being under spring force.
- c) travels one direction under one application of hydraulic power and in the opposite direction under a second application of hydraulic power.
- d) cannot be used for variable position operations as it is designed to lock in the extremities of travel.

Answer ;

b) is powered in one direction only by hydraulic power , the return movement being under spring force.

---

0072. A smoke hood is a device covering ;

- a) only the nose and the mouth and with a continuous oxygen flow.
- b) only the nose and the mouth and with an oxygen/air mix.
- c) the whole head and with an oxygen flow only on demand.
- d) the whole head and with a continuous oxygen flow.

Answer ;

d) the whole head and with a continuous oxygen flow.

---

0073. A smoke mask is a ;

- a) continuous flow mask and covers only the nose and the mouth.
- b) mask with flow on request and covers only the nose and the mouth.
- c) mask with flow on request and covers the whole face.
- d) continuous flow mask and covers the whole face.

Answer ;

c) mask with flow on request and covers the whole face.

---

0074. A stage in an axial compressor ;

- a) has a compression ration in the order of 0.8.
- b) consists of a row of stator blades followed by a rotor disc.
- c) consists of a rotor disc followed by a row of stator blades.
- d) has a compression ratio in the order of 2.1.

Answer ;

c) consists of a rotor disc followed by a row of stator blades.

---

0075. A static converter is powered by ;

- a) alternating current on input , direct current on output.
- b) alternating current on input , alternating current on output.
- c) direct current on input , direct current on output.
- d) direct current on input , alternating current on output.

Answer ;

d) direct current on input , alternating current on output.

---

0076. A static inverter is powered by ;

- a) alternating current and produces a direct current on output.
- b) direct current and produces a direct current on output.
- c) alternating current and produces an alternating current on output.
- d) direct current and produces an alternating current output.

Answer ;

d) direct current and produces an alternating current output.

---

0077. A structure in which the skin takes all of the load is ;

- a) a monocoque structure.
- b) a semi-braced structure.
- c) a semi-monocoque structure.
- d) a box structure.

Answer ;

a) a monocoque structure.

---

0078. A substance which may never be used in the vicinity or on parts of an oxygen installation is ;

- a) Water.
- b) Nitrogen.
- c) Halon.
- d) Grease.

Answer ;

d) Grease.

---

0079. A test to assess the state of charge of a lead-acid battery would involve ;

- a) checking the battery voltage “off-load”.
- b) checking the discharge current of the battery “on-load”.
- c) checking the level of the electrolyte.
- d) comparing the “on-load” and “off-load” battery voltages.

Answer ;

d) comparing the “on-load” and “off-load” battery voltages.

---

0080. A thermal circuit breaker ;

- a) uses electromagnetic trip system to prevent over heating.
- b) can be re-engaged without any danger even if the fault remains.
- c) protects the system in the event of overload/overheating.
- d) limits any overcurrent.

Answer ;

c) protects the system in the event of overload/overheating.

---

0081. A torsion link assembly is installed on the landing gear to ;

- a) control the wheels.
- b) avoid rotation of the piston rod relative to the gear oleo strut.
- c) absorb the spring tension.
- d) lock the landing gear.

Answer ;

b) avoid rotation of the piston rod relative to the gear oleo strut.

---

0082. A “trip-free” type circuit breaker is a circuit protection device which ;

- a) can be reset at any time.
- b) is free from the normal CB tripping characteristic.
- c) will allow the contacts to be held closed in order to clear a fault in the circuit.
- d) will not allow the contacts to be held closed while a current fault exists in the circuit.

Answer ;

d) will not allow the contacts to be held closed while a current fault exists in the circuit.

---

0083. A tubeless tyre is a tyre ;

1. which requires solid or branched wheels.
2. whose valve can be sheared in sudden accelerations.
3. whose mounting rim must be flawless.
4. which requires no rim protection between rim flange and tire removing device.
5. which does not burst in the event of a tire puncture.
6. which eliminates internal friction between the tube and the tire.

The combination regrouping all the correct statements is ;

- a) 1 , 5 , 6.
- b) 1 , 2 , 5.
- c) 2 , 3 , 6.
- d) 3 , 4 , 5.

Answer ;

a) 1 , 5 , 6.

---

**0084.** A turbo compressor air conditioning system ( bootstrap system ) includes two heat exchangers ; the primary exchanger ( P ) and the secondary exchanger ( S ). The functions of these heat exchangers are as follows ;

- a) P : warms up engine bleed air ,  
S : recirculates the cabin air , reducing its temperature.
- b) P : precools the engine bleed air ,  
S : cools air behind the pack's compressor.
- c) P : pre-cools the engine bleed air ,  
S : increases the temperature of the air used for air-conditioning of cargo compartment ( animals ).
- d) P : warms up engine bleed air ,  
S : increases the temperature of air originating from the compressor of the pack.

**Answer ;**

- b)** P : precools the engine bleed air ,  
S : cools air behind the pack's compressor.
- 

**0085.** A turbo compressor air conditioning system ( bootstrap system ) will ;

- a) cause a pressure drop as well as an associated temperature drop in the charge air.
- b) not affect the charge air pressure.
- c) decrease charge air pressure whilst causing the temperature to rise in the heat exchanger.
- d) increase charge air pressure whilst causing the temperature to drop in the heat exchanger.

**Answer ;**

- a)** cause a pressure drop as well as an associated temperature drop in the charge air.
- 

**0086.** A turbocharger consists of a ;

- a) compressor and turbine mounted on a common shaft.
- b) turbine driving a compressor via a reduction gear.
- c) compressor and turbine on individual shafts.
- d) compressor driving a turbine via a reduction gear.

**Answer ;**

- a)** compressor and turbine mounted on a common shaft.
- 

**0087.** A turbocharger system in a reciprocating engine is normally driven by ;

- a) an electrically activated hydraulically powered clutch.
- b) an hydraulic motor.
- c) an electric motor.
- d) the exhaust system.

**Answer ;**

- d)** the exhaust system.
- 

**0088.** A turbocharger system is normally driven by ;

- a) an electrically activated hydraulically powered clutch.
- b) an electric motor.
- c) the exhaust system.
- d) an hydraulic motor.

**Answer ;**

- c)** the exhaust system.
- 

**0089.** A turbofan engine with a bypass air mass flow of 200 kg/s and a HP compressor mass flow of 20 kg/s has a bypass ratio of ;

- a) 9
- b) 1.1
- c) 10
- d) 11

**Answer ;**

- c)** 10

**Çözüm ;**

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 200 : 20 = 10

---

**0090.** A turbofan engine with a bypass air mass flow of 250 kg/s and a HP compressor mass flow of 50 kg/s has a bypass ratio of ;

- a) 5
- b) 6
- c) 1.2
- d) 4

**Answer ;**

- a)** 5

**Çözüm ;**

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 250 : 50 = 5

---

0091. A turbofan engine with a bypass air mass flow of 888 kg/s and a HP compressor mass flow of 111 kg/s has a bypass ratio of ;
- a) 9
  - b) 1.12
  - c) 7
  - d) 8

Answer ;

d) 8

Çözüm ;

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 888 : 111 = 8

---

0092. A turbofan engine with an inlet air mass flow of 200 kg/s and a HP compressor mass flow of 20 kg/s has a bypass ratio of ;
- a) 5
  - b) 10
  - c) 9
  - d) 11

Answer ;

c) 9

Çözüm ;

Bypass air mass flow = Inlet air mass flow – Compressor air mass flow = 200 – 20 = 180 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 180 : 20 = 9

---

0093. A turbofan engine with an inlet air mass flow of 220 kg/s and a bypass mass flow of 200 kg/s has a bypass ratio of ;
- a) 11
  - b) 10
  - c) 9
  - d) 1.1

Answer ;

b) 10

Çözüm ;

Compressor air mass flow = Inlet air mass flow – Bypass air mass flow = 220 – 200 = 20 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 200 : 20 = 10

---

0094. A turbofan engine with an inlet air mass flow of 220 kg/s and a HP compressor mass flow of 20 kg/s has a bypass ratio of ;
- a) 11
  - b) 10
  - c) 1.1
  - d) 12

Answer ;

b) 10

Çözüm ;

Bypass air mass flow = Inlet air mass flow – Compressor air mass flow = 220 – 20 = 200 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 200 : 20 = 10

---

0095. A turbofan engine with an inlet air mass flow of 300 kg/s and a bypass mass flow of 250 kg/s has a bypass ratio of ;
- a) 6
  - b) 5
  - c) 2.2
  - d) 1.2

Answer ;

b) 5

Çözüm ;

Compressor air mass flow = Inlet air mass flow – Bypass air mass flow = 300 – 250 = 50 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 250 : 50 = 5

---

0096. A turbofan engine with an inlet air mass flow of 300 kg/s and a HP compressor mass flow of 50 kg/s has a bypass ratio of ;
- a) 6
  - b) 5
  - c) 7
  - d) 1.2

Answer ;

b) 5

Çözüm ;

Bypass air mass flow = Inlet air mass flow – Compressor air mass flow = 300 – 50 = 250 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 250 : 50 = 5

---

- 0097.** A turbofan engine with an inlet air mass flow of 1000 kg/s and a by-pass mass flow of 889 kg/s has a by-pass ratio of ;
- a) 1.12
  - b) 17.02
  - c) 2.12
  - d) 8.01

**Answer ;**

**d) 8.01**

**Çözüm ;**

Compressor air mass flow = Inlet air mass flow – Bypass air mass flow = 1000 – 889 = 111 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 889 : 111 = 8.009 ≈ **8.01**

---

- 0098.** A turbofan engine with an inlet air mass flow of 1000 kg/s and a HP compressor mass flow of 111 kg/s has a bypass ratio of ;
- a) 1.12
  - b) 10.01
  - c) 9.01
  - d) 8.01

**Answer ;**

**d) 8.01**

**Çözüm ;**

Bypass air mass flow = Inlet air mass flow – Compressor air mass flow = 1000 – 111 = 889 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 889 : 111 = 8.009 ≈ **8.01**

---

- 0099.** A twin-spool engine with a bypass mass flow of 250 kg/s and a HP compressor mass flow of 250 kg/s has a bypass ratio of ;
- a) 0
  - b) 1
  - c) 0.5
  - d) 2

**Answer ;**

**b) 1**

**Çözüm ;**

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 250 : 250 = **1**

---

- 0100.** A twin-spool engine with an inlet air mass flow of 220 kg/s and a HP compressor mass flow of 220 kg/s has a bypass ratio of ;
- a) 1
  - b) 10
  - c) 0
  - d) 1.1

**Answer ;**

**c) 0**

**Çözüm ;**

Bypass air mass flow = Inlet air mass flow – Compressor air mass flow = 220 – 220 = 0 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 0 : 220 = **0**

---

- 0101.** A twin-spool engine with an inlet air mass flow of 500 kg/s and a bypass mass flow of 250 kg/s has a bypass ratio ( BPR ) of ;
- a) 0
  - b) 3
  - c) 2
  - d) 1

**Answer ;**

**d) 1**

**Çözüm ;**

Compressor air mass flow = Inlet air mass flow – Bypass air mass flow = 500 – 250 = 250 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 250 : 250 = **1**

---

**0102.** A twin-spool engine with an inlet air mass flow of 500 kg/s and a HP compressor mass flow of 250 kg/s has a bypass ratio of ;

- a) 2
- b) 3
- c) 0
- d) 1

**Answer ;**

**d) 1**

**Çözüm ;**

Bypass air mass flow = Inlet air mass flow – Compressor air mass flow = 500 – 250 = 250 kg/s

Bypass ratio = Bypass air mass flow : Compressor air mass flow = 250 : 250 = 1

---

**0103.** A twin spool gas turbine engine is one in which the ;

- a) turbines are free to rotate at speeds that differ from their associated compressors.
- b) LP turbine drives the LP compressor and the HP turbine drives the HP compressor.
- c) LP compressor drives the HP compressor.
- d) LP turbine drives both the LP compressor and the HP compressor.

**Answer ;**

**b) LP turbine drives the LP compressor and the HP turbine drives the HP compressor.**

---

**0104.** A unit that converts electrical DC into AC is ;

- a) a thermistor.
- b) an AC generator.
- c) a transformer rectifier unit.
- d) an inverter.

**Answer ;**

**d) an inverter.**

---

**0105.** A warning device alerts the crew in case of an excessive cabin altitude. This warning must be triggered on reaching the following altitude ;

- a) 10.000 ft ( approx. 3.000 m ).
- b) 14.000 ft ( approx. 4.200 m ).
- c) 12.000 ft ( approx. 3.600 m ).
- d) 8.000 ft ( approx. 2.400 m ).

**Answer ;**

**a) 10.000 ft ( approx. 3.000 m ).**

---

**0106.** A wing spar typically consists of ;

- a) frames and webs.
- b) ribs and stiffeners.
- c) a web and girders.
- d) ribs and frames.

**Answer ;**

**c) a web and girders.**

---

**0107.** A “Zener” diode is used for ;

- a) rectification.
- b) digital displays.
- c) reverse current protection.
- d) voltage stabilization.

**Answer ;**

**d) voltage stabilization.**

---

**0108.** About a twin spool compressor ;

- a) the low pressure spool runs at a lower rpm than the high pressure spool.
- b) both spools run at the same rpm.
- c) the low pressure spool runs at a very high rpm , thus preventing the onset of compressor stall.
- d) when operating at very high rpm's , the later stages within the high pressure spool will have a large negative angle of attack.

**Answer ;**

**a) the low pressure spool runs at a lower rpm than the high pressure spool.**

---

**0109.** AC generators operated in parallel must have the same ;

- a) voltage and amperage.
- b) frequency and amperage.
- c) voltage and phase.
- d) amperage and kVAR.

**Answer ;**

**c) voltage and phase.**

---

**0110.** According JAR/CS 25 the allowable average failure probability per flight hour for a minor failure should be on the order of ( ^ means to the power of ) ;

- a) between  $10^{-7}$  and  $10^{-9}$ . ( extremely remote )
- b) between  $10^{-3}$  and  $10^{-5}$ . ( probable )
- c) less than  $10^{-9}$ . ( extremely improbable )
- d) between  $10^{-5}$  and  $10^{-7}$ . ( remote )

**Answer ;**

**b) between  $10^{-3}$  and  $10^{-5}$ . ( probable )**

---

**0111.** According JAR/CS 25 the allowable average failure probability per flight hour probability for a major failure should be on the order of ( ^ means to the power of ) ;

- a) between  $10^{-5}$  and  $10^{-7}$ . ( remote )
- b) between  $10^{-7}$  and  $10^{-9}$ . ( extremely remote )
- c) between  $10^{-3}$  and  $10^{-5}$ . ( probable )
- d) less than  $10^{-9}$ . ( extremely improbable )

**Answer ;**

**a) between  $10^{-5}$  and  $10^{-7}$ . ( remote )**

---

**0112.** According JAR/CS 25 the allowable quantitative average failure probability per flight hour for a catastrophic failure should be on the order of ( ^ means to the power of ) ;

- a) between  $10^{-7}$  and  $10^{-9}$ . ( extremely remote )
- b) less than  $10^{-9}$ . ( extremely improbable )
- c) between  $10^{-3}$  and  $10^{-5}$ . ( probable )
- d) between  $10^{-5}$  and  $10^{-7}$ . ( remote )

**Answer ;**

**b) less than  $10^{-9}$ . ( extremely improbable )**

---

**0113.** According JAR/CS 25 the allowable quantitative average failure probability per flight hour for a hazardous failure should be on the order of ( ^ means to the power of ) ;

- a) less than  $10^{-9}$ . ( extremely improbable )
- b) between  $10^{-7}$  and  $10^{-9}$ . ( extremely remote )
- c) between  $10^{-5}$  and  $10^{-7}$ . ( remote )
- d) between  $10^{-3}$  and  $10^{-5}$ . ( probable )

**Answer ;**

**b) between  $10^{-7}$  and  $10^{-9}$ . ( extremely remote )**

---

**0114.** According JAR/CS 25 the worst effect of a catastrophic failure on the aeroplane could be ;

- a) hull loss.
- b) large reduction in functional capabilities of safety margins.
- c) slight reduction in functional capabilities of safety margins.
- d) significant reduction in functional capabilities of safety margins.

**Answer ;**

**a) hull loss.**

---

**0115.** According JAR/CS 25 the worst effect of a catastrophic failure on the flight crew could be ;

- a) physical distress or excessive workload , impairs ability to perform tasks.
- b) fatalities or incapacitation.
- c) a slight increase in workload.
- d) physical discomfort or a significant increase in workload.

**Answer ;**

**b) fatalities or incapacitation.**

---

**0116.** According JAR/CS 25 the worst effect of a catastrophic failure on the occupants of an aeroplane excluding flight crew could be ;

- a) physical discomfort.
- b) serious or fatal injury to a small number of passengers or cabin crew.
- c) multiple fatalities.
- d) physical distress , possibly including injuries.

**Answer ;**

**c) multiple fatalities.**

---

**0117.** According JAR/CS 25 the worst effect of a hazardous failure on the aeroplane could be ;

- a) slight reduction in functional capabilities or safety margins.
- b) large reduction in functional capabilities or safety margins.
- c) significant reduction in functional capabilities or safety margins.
- d) hull loss.

**Answer ;**

**b) large reduction in functional capabilities or safety margins.**

---

**0118.** According JAR/CS 25 the worst effect of a hazardous failure on the flight crew could be ;

- a) fatalities or incapacitation.
- b) physical discomfort or a significant increase in workload.
- c) a slight increase in workload.
- d) physical distress or excessive workload , impairs ability to perform tasks.

**Answer ;**

**d) physical distress or excessive workload , impairs ability to perform tasks.**

---

**0119.** According JAR/CS 25 the worst effect of a hazardous failure on the occupants of an aeroplane excluding flight crew could be ;

- a) serious or fatal injury to a small number of passengers or cabin crew.
- b) physical distress , possibly including injuries.
- c) multiple fatalities.
- d) physical discomfort.

**Answer ;**

**a) serious or fatal injury to a small number of passengers or cabin crew.**

---

**0120.** According JAR/CS 25 the worst effect of a major failure on the aeroplane could be ;

- a) no effect on operational capabilities or safety.
- b) significant reduction in functional capabilities or safety margins.
- c) large reduction in functional capabilities or safety margins.
- d) slight reduction in functional capabilities or safety margins.

**Answer ;**

**b) significant reduction in functional capabilities or safety margins.**

---

**0121.** According JAR/CS 25 the worst effect of a major failure on the flight crew could be ;

- a) physical discomfort or a significant increase in workload.
- b) a slight increase in workload.
- c) no effect on flight crew.
- d) physical distress or excessive workload , impairs ability to perform tasks.

**Answer ;**

**a) physical discomfort or a significant increase in workload.**

---

**0122.** According JAR/CS 25 the worst effect of a major failure on the occupants of an aeroplane excluding flight crew could be ;

- a) serious or fatal injury to a small number of passengers or cabin crew.
- b) physical discomfort
- c) physical distress , possibly including injuries.
- d) inconvenience.

**Answer ;**

**c) physical distress , possibly including injuries.**

---

**0123.** According JAR/CS 25 the worst effect of a minor failure on the aeroplane could be ;

- a) no effect on operational capabilities or safety.
- b) slight reduction in functional capabilities or safety margins.
- c) large reduction in functional capabilities or safety margins.
- d) significant reduction in functional capabilities or safety margins.

**Answer ;**

**b) slight reduction in functional capabilities or safety margins.**

---



0124. According JAR/CS 25 the worst effect of a minor failure on the flight crew could be ;
- a) a slight increase in workload.
  - b) physical distress or excessive workload , impairs ability to perform tasks.
  - c) no effect on flight crew.
  - d) physical discomfort or a significant increase in workload.

Answer ;

a) a slight increase in workload.

---

0125. According JAR/CS 25 the worst effect of a minor failure on the occupants of an aeroplane excluding flight crew could be ;
- a) inconvenience.
  - b) physical discomfort.
  - c) serious or fatal injury to a small number of passengers or cabin crew.
  - d) physical distress , possibly including injuries.

Answer ;

b) physical discomfort.

---

0126. After a successful start of a high bypass turbofan engine , the air starter ;
- a) is disengaged by using a clutch that is controlled by the main engine control.
  - b) is disengaged by the coupling between engine N2 spool and starter by the use of centrifugal forces.
  - c) is disengaged by the coupling between engine N1 spool and starter by the use of centrifugal forces.
  - d) remains engaged and rotates continuously with the N2 spool.

Answer ;

b) is disengaged by the coupling between engine N2 spool and starter by the use of centrifugal forces.

---

0127. After air has passed through the compressor of a gas turbine engine the ;
- a) velocity will be higher than the inlet velocity.
  - b) velocity will be the same as the inlet velocity.
  - c) pressure will be the same as the inlet pressure.
  - d) temperature will be higher than the inlet temperature.

Answer ;

d) temperature will be higher than the inlet temperature.

---

0128. Aircraft fuel tanks should be checked for water ;
- a) before each flight.
  - b) immediately after every refuelling.
  - c) before the first flight of the day.
  - d) during refuelling.

Answer ;

c) before the first flight of the day.

---

0129. Aircraft magneto's receive electrical energy from ;
- a) rotating permanent magneto's.
  - b) generators.
  - c) batteries.
  - d) condensers.

Answer ;

a) rotating permanent magneto's.

---

0130. Alternating current can be derived from direct current by ;
- a) an inverter.
  - b) an alternating current motor.
  - c) the use of relays.
  - d) a series wound motor.

Answer ;

a) an inverter.

---

- 0131.** Among the different types of aircraft structures , the shell structures efficiently transmit the ;
1. normal bending stresses.
  2. tangent bending stresses.
  3. torsional moment.
  4. shear stresses.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 3 , 4.
- c) 2 , 3 , 4.
- d) 1 , 2 , 4.

**Answer ;**

- a)** 1 , 2 , 3.
- 

- 0132.** An abortive ( hung ) start is the term used to describe an attempt to start in which the engine ;

- a) fails to light up.
- b) lights up , but fails to accelerate sufficiently.
- c) will not rotate.
- d) commences initial rotation , but fails to receive any fuel.

**Answer ;**

- b)** lights up , but fails to accelerate sufficiently.
- 

- 0133.** An AC generator driven by a CSD unit ;

- a) requires a voltage controller to maintain constant voltage under load.
- b) requires a voltage controller to maintain constant frequency.
- c) does not need a voltage controller since an AC generator voltage cannot alter under load.
- d) does not need a voltage controller since the CSD will ensure constant voltage.

**Answer ;**

- a)** requires a voltage controller to maintain constant voltage under load.
- 

- 0134.** An accumulator in a hydraulic system will ;

- a) reduce fluid temperature only.
- b) increase pressure surges within the system.
- c) reduce fluid temperature and pressure.
- d) store fluid under pressure.

**Answer ;**

- d)** store fluid under pressure.
- 

- 0135.** An aeroplane equipped with fully powered flight controls ( irreversible type ) ;

- a) remains normally controllable in case of total loss of power to the flight control system.
- b) must be equipped with control locks.
- c) does not require the use of an artificial feel system.
- d) requires the use of an artificial feel system.

**Answer ;**

- d)** requires the use of an artificial feel system.
- 

- 0136.** An aeroplane equipped with irreversible flight controls ;

- a) must be equipped with control locks.
- b) does not require an artificial feel system.
- c) requires an artificial feel system.
- d) may be equipped with simple spring type fuel units on all flight controls.

**Answer ;**

- c)** requires an artificial feel system.
- 

- 0137.** An aeroplane equipped with irreversible flight controls ;

- a) must have a mechanical back-up control system.
- b) may be equipped with simple spring type fuel units on all flight controls.
- c) does not require an artificial feel system.
- d) need not be equipped with a separate gust lock system.

**Answer ;**

- d)** need not be equipped with a separate gust lock system.
-

- 0138.** An aeroplane equipped with reversible flight controls ;
- a) does not require an artificial feel system.
  - b) is equipped with simple spring type feel units.
  - c) does not have mechanical back-up.
  - d) need not be equipped with a separate gust lock system.

**Answer ;**

**a)** does not require an artificial feel system.

---

- 0139.** An air cycle machine ( air conditioning pack ) ;
- a) decreases bleed air pressure whilst causing the temperature to rise in the heat exchanger.
  - b) does not affect bleed air.
  - c) causes a pressure and temperature drop in the bleed air.
  - d) increases outlet pressure whilst causing the temperature to drop in the heat exchanger.

**Answer ;**

**c)** causes a pressure and temperature drop in the bleed air.

---

- 0140.** An aircraft flying in conditions of continuous heavy rain and high ambient temperatures may require the precautionary use of ;

- a) both engine intake anti-icing and airframe de-icing.
- b) engine igniters.
- c) airframe de-icing only.
- d) engine intake anti-icing only.

**Answer ;**

**b)** engine igniters.

---

- 0141.** An aircraft magneto is switched off by ;
- a) opening the secondary circuit.
  - b) opening the primary circuit.
  - c) grounding the secondary circuit.
  - d) grounding the primary circuit.

**Answer ;**

**d)** grounding the primary circuit.

---

- 0142.** An aircraft magneto is switched off by ;
- 1. grounding the primary circuit.
  - 2. opening the primary circuit.
  - 3. opening the secondary circuit.
  - 4. grounding the secondary circuit.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 2 , 3.
- c) 1 , 4.
- d) 2 , 4.

**Answer ;**

**c)** 1 , 4.

---

- 0143.** An aircraft whose maximum approved configuration for passenger seats is 200 seats must be equipped with ;
- a) 5 hand fire extinguishers in the passenger compartment.
  - b) 3 hand fire extinguishers in the compartment.
  - c) 4 hand fire extinguishers in the passenger compartment.
  - a) 7 hand fire extinguishers in the passenger compartment.

**Answer ;**

**b)** 3 hand fire extinguishers in the compartment.

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

**0144.** An aircraft with a pressurized cabin flies at FL 310 and , following a malfunction of the pressure controller , the outflow valve runs to the open position. Given ;

CAB V/S : Cabin rate of climb indication

CAB ALT : Cabin pressure altitude

DELTA P : Differential pressure

This will result in a ;

- a) CAB V/S decrease  
CAB ALT increase  
DELTA P decrease
- b) CAB V/S increase  
CAB ALT increase  
DELTA P decrease
- c) CAB V/S decrease  
CAB ALT decrease  
DELTA P increase
- d) CAB V/S increase  
CAB ALT increase  
DELTA P increase

Answer ;

- b) CAB V/S increase  
CAB ALT increase  
DELTA P decrease

**0145.** An aircraft with a pressurized cabin is settled at its cruise level. During the flight , a malfunction of the pressure controller is detected by the crew and the cabin rate indicator reads –200ft/min.Given that ;

DELTA P : Differential pressure

Zc : Cabin altitude

- a) The crew has to intermittently cut off the incoming air flow in order to maintain a zero Zc.
- b) The aircraft has to climb to a higher flight level in order to reduce Zc to its initial value.
- c) A descent must be initiated to prevent the oxygen masks dropping when Zc reaches 14.000 ft.
- d) DELTA P will rise up to its maximum value , thus causing the safety relief valves to open.

Answer ;

- d) DELTA P will rise up to its maximum value , thus causing the safety relief valves to open.

**0146.** An airplane whose maximum approved passenger seating configuration is 7 to 30 seats must be equipped with at least ;

- a) 3 hand fire-extinguishers conveniently located in the passenger compartment.
- b) 4 hand fire-extinguishers conveniently located in the passenger compartment.
- c) 2 hand fire-extinguishers conveniently located in the passenger compartment.
- d) 1 hand fire-extinguishers conveniently located in the passenger compartment.

Answer ;

- d) 1 hand fire-extinguishers conveniently located in the passenger compartment.

Açıklama ;

| Onaylı maksimum yolcu oturma düzenlemesi | Söndürücü sayısı |
|------------------------------------------|------------------|
| 7 – 30 arası                             | 1                |
| 31 – 60 arası                            | 2                |
| 61 – 200 arası                           | 3                |
| 201 – 300 arası                          | 4                |
| 301 – 400 arası                          | 5                |
| 401 – 500 arası                          | 6                |
| 501 – 600 arası                          | 7                |
| 601 ve daha fazla                        | 8                |

**0147.** An airplane whose maximum approved passenger seating configuration is 31 to 60 seats must be equipped with at least ;

- a) 2 hand fire-extinguishers conveniently located in the passenger compartment.
- b) 5 hand fire-extinguishers conveniently located in the passenger compartment.
- c) 3 hand fire-extinguishers conveniently located in the passenger compartment.
- d) 4 hand fire-extinguishers conveniently located in the passenger compartment.

Answer ;

- a) 2 hand fire-extinguishers conveniently located in the passenger compartment.

Açıklama ;

Soru 0146'nın açıklamasına bakınız.

- 0148.** An airplane whose maximum approved passenger seating configuration is 61 to 200 seats must be equipped with at least ;
- a) 4 hand fire-extinguishers conveniently located in the passenger compartment.
  - b) 5 hand fire-extinguishers conveniently located in the passenger compartment.
  - c) 3 hand fire-extinguishers conveniently located in the passenger compartment.
  - d) 2 hand fire-extinguishers conveniently located in the passenger compartment.

**Answer ;**

**c) 3 hand fire-extinguishers conveniently located in the passenger compartment.**

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

- 0149.** An airplane whose maximum approved passenger seating configuration is 201 to 300 seats must be equipped with at least ;
- a) 6 hand fire-extinguishers conveniently located in the passenger compartment.
  - b) 4 hand fire-extinguishers conveniently located in the passenger compartment.
  - c) 3 hand fire-extinguishers conveniently located in the passenger compartment.
  - d) 5 hand fire-extinguishers conveniently located in the passenger compartment.

**Answer ;**

**b) 4 hand fire-extinguishers conveniently located in the passenger compartment.**

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

- 0150.** An airplane whose maximum approved passenger seating configuration is 301 to 400 seats must be equipped with at least ;
- a) 6 hand fire-extinguishers conveniently located in the passenger compartment.
  - b) 5 hand fire-extinguishers conveniently located in the passenger compartment.
  - c) 4 hand fire-extinguishers conveniently located in the passenger compartment.
  - d) 3 hand fire-extinguishers conveniently located in the passenger compartment.

**Answer ;**

**b) 5 hand fire-extinguishers conveniently located in the passenger compartment.**

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

- 0151.** An airplane whose maximum approved passenger seating configuration is 401 to 500 seats must be equipped with at least ;
- a) 5 hand fire-extinguishers conveniently located in the passenger compartment.
  - b) 6 hand fire-extinguishers conveniently located in the passenger compartment.
  - c) 8 hand fire-extinguishers conveniently located in the passenger compartment.
  - d) 7 hand fire-extinguishers conveniently located in the passenger compartment.

**Answer ;**

**b) 6 hand fire-extinguishers conveniently located in the passenger compartment.**

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

- 0152.** An airplane whose maximum approved passenger seating configuration is 501 to 600 seats must be equipped with at least ;
- a) 8 hand fire-extinguishers conveniently located in the passenger compartment.
  - b) 6 hand fire-extinguishers conveniently located in the passenger compartment.
  - c) 5 hand fire-extinguishers conveniently located in the passenger compartment.
  - d) 7 hand fire-extinguishers conveniently located in the passenger compartment.

**Answer ;**

**d) 7 hand fire-extinguishers conveniently located in the passenger compartment.**

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

- 0153.** An airplane whose maximum approved passenger seating configuration is greater than 60 seats must be equipped with at least ;

- a) 1 HALON 1211 fire-extinguisher.
- b) 3 HALON 1211 fire-extinguisher.
- c) 4 HALON 1211 fire-extinguisher.
- d) 2 HALON 1211 fire-extinguisher.

**Answer ;**

**b) 3 HALON 1211 fire-extinguisher.**

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

**0154.** An airplane whose maximum approved passenger seating configuration is greater than 600 seats must be equipped with at least ;

- a) 7 hand fire-extinguishers conveniently located in the passenger compartment.
- b) 6 hand fire-extinguishers conveniently located in the passenger compartment.
- c) 9 hand fire-extinguishers conveniently located in the passenger compartment.
- d) 8 hand fire-extinguishers conveniently located in the passenger compartment.

**Answer ;**

**d) 8 hand fire-extinguishers conveniently located in the passenger compartment.**

**Açıklama ;**

Soru 0146'nın açıklamasına bakınız.

---

**0155.** An artificial feel system ;

- a) is mounted in parallel on a spring tab.
- b) functions in parallel to an irreversible servo-control unit.
- c) is necessary in a reversible flight control actuator unit.
- d) functions in series to an irreversible servo-control unit.

**Answer ;**

**b) functions in parallel to an irreversible servo-control unit.**

---

**0156.** An artificial feel unit is necessary in the pitch channel when ;

- a) the elevators are fitted with servo-tabs or trim tabs.
- b) the elevators are actuated by reversible servo-control units.
- c) the elevators are actuated by irreversible servo-control units.
- d) there is a trimmable stabilizer.

**Answer ;**

**c) the elevators are actuated by irreversible servo-control units.**

---

**0157.** An asymmetric loading ( p-factor ) on the propeller exists ;

- a) only for counterrotating propeller.
- b) if there is an unbalanced propeller.
- c) if the aeroplane has a large angle of attack.
- d) only if the "constant speed propeller" mechanism is broken.

**Answer ;**

**c) if the aeroplane has a large angle of attack.**

---

**0158.** An axial compressor , when operating below its design speed , has ;

- a) a tendency to stall in the centre stages.
- b) a tendency to stall in the rear stages.
- c) no tendency to stall.
- d) a tendency to stall in the front stages.

**Answer ;**

**d) a tendency to stall in the front stages.**

---

**0159.** An EGT ( Exhaust Gas Temperature ) indicator for a piston engine is used to ;

- a) control the cylinder head temperature.
- b) control the carburettor inlet air flow.
- c) assist the pilot to set the correct mixture.
- d) control the fuel temperature.

**Answer ;**

**c) assist the pilot to set the correct mixture.**

---

**0160.** An electrically heated windscreen is manufactured from ;

- a) a glass and polycarbonate laminate.
- b) a boron aluminide and glass laminate.
- c) a perspex and polycarbonate laminate with gold heating element.
- d) triple glass sheets with the grain laid at 60° to each other.

**Answer ;**

**a) a glass and polycarbonate laminate.**

---

**0161.** An engine fire in a large transport aeroplane is indicated by a(n) ;

- a) bell.
- b) visual warning only.
- c) aural alert only.
- d) visual warning and an aural alert.

**Answer ;**

**d) visual warning and an aural alert.**

---

**0162.** An engine fire is indicated by a(n) ;

- a) aural alert only.
- b) bell.
- c) warning only.
- d) warning and an aural alert.

**Answer ;**

**d) warning and an aural alert.**

---

**0163.** An engine pressure ratio ( EPR ) can be defined as the ratio of ;

- a) jet pipe total pressure to compressor inlet total pressure.
- b) compressor outlet pressure to compressor inlet total pressure.
- c) combustion chamber pressure to compressor inlet total pressure.
- d) jet pipe total pressure to combustion chamber pressure.

**Answer ;**

**a) jet pipe total pressure to compressor inlet total pressure.**

---

**0164.** An engine pressure ratio ( EPR ) gauge reading normally shows the ratio of ;

- a) jet pipe pressure to combustion chamber pressure.
- b) jet pipe pressure to compressor inlet pressure.
- c) compressor outlet pressure to compressor inlet pressure.
- d) combustion chamber pressure to compressor inlet pressure.

**Answer ;**

**b) jet pipe pressure to compressor inlet pressure.**

---

**0165.** An excessively rich mixture can be detected by ;

- a) high cylinder head temperatures.
- b) black smoke from exhaust.
- c) a long purple flame from exhaust.
- d) white smoke from exhaust.

**Answer ;**

**b) black smoke from exhaust.**

---

**0166.** An impulse coupling does not function at such speeds above those encountered in starting. Its engaging pawls are prevented from operating at higher speeds by ;

- a) engine oil pressure.
- b) centrifugal force.
- c) a coil spring.
- d) electro-magnetic action of operating magneto.

**Answer ;**

**b) centrifugal force.**

---

**0167.** An impulse coupling is inhibited at higher than normal rotational starting speeds by ;

- a) a coil spring.
- b) engine oil pressure.
- c) centrifugal force.
- d) electro-magnetic action of operating magneto.

**Answer ;**

**c) centrifugal force.**

---

**0168.** An impulse coupling is installed on a magneto of a piston engine to ;

- a) provide a stronger spark on top dead centre for engine starting.
- b) facilitate quick removal and installation.
- c) advance the ignition timing.
- d) absorb starting loads.

**Answer ;**

**a) provide a stronger spark on top dead centre for engine starting.**

---

0169. An impulse magneto coupling ;
- a) reduces magneto speed during engine warm-up.
  - b) gives a retarded spark at starting.
  - c) advances ignition timing and gives a hotter spark at starting.
  - d) gives an automatic spark increase during high speed operation.

Answer ;

- b) gives a retarded spark at starting.
- 

0170. An inverter is a ;
- a) unit used to convert DC into AC.
  - b) device for reversing the polarity of the static charge.
  - c) filter against radio interference.
  - d) static discharger.

Answer ;

- a) unit used to convert DC into AC.
- 

0171. An oxygen regulator has 3 controls ;
- a power lever : ON/OFF
  - an "O<sub>2</sub>" lever : NORMAL/100%
  - an emergency lever : ON/OFF

Among the following statements , the correct proposition is ;

- a) with the EMERGENCY lever on OFF , in an emergency situation , one cannot use the oxygen mask to breathe.
- b) the "O<sub>2</sub>" lever on ON enables breathing of the over-pressure oxygen at a constant flow rate.
- c) the power lever on ON , and , the "O<sub>2</sub>" lever on NORMAL allows the oxygen to enter the regulator and enables breathing of a mixture of air/oxygen according to altitude.
- d) the EMERGENCY lever on ON enables breathing of pure oxygen at ambient pressure.

Answer ;

- c) the power lever on ON , and , the "O<sub>2</sub>" lever on NORMAL allows the oxygen to enter the regulator and enables breathing of a mixture of air/oxygen according to altitude.
- 

0172. An oxygen supply system must be kept absolutely free of oil or grease as ;
- a) these substances could block the oxygen masks filters.
  - b) these substances catch fire spontaneously in the presence of oxygen.
  - c) the breathing oxygen would be contaminated by these substances.
  - d) these substances mixed with oxygen could catch fire in the presence of a spark.

Answer ;

- b) these substances catch fire spontaneously in the presence of oxygen.
- 

0173. An under-inflated tyre will ;
- a) be more subject to viscous aquaplaning.
  - b) experience increased wear at the shoulders.
  - c) have an increased hydroplaning speed.
  - d) experience reduced wear at the shoulders.

Answer ;

- b) experience increased wear at the shoulders.
- 

0174. Apart from flight into known icing conditions , the intake system of a diesel engine ;
- a) is never heated because a diesel engine is a injection engine.
  - b) needs to be heated at low power settings , due to vaporisation heat.
  - c) is never heated because diesel engines operate at very high temperatures.
  - d) needs to be heated to prevent ice on the air filter.

Answer ;

- a) is never heated because a diesel engine is a injection engine.
- 

0175. As altitude increases , the mixture ratio of a piston engine should be adjusted to ;
- a) reduce the fuel flow in order to compensate for the increasing air density.
  - b) increase the mixture ratio.
  - c) increase the fuel flow in order to compensate for the decreasing air pressure and density.
  - d) reduce the fuel flow in order to compensate for the decreasing air density.

Answer ;

- d) reduce the fuel flow in order to compensate for the decreasing air density.
-

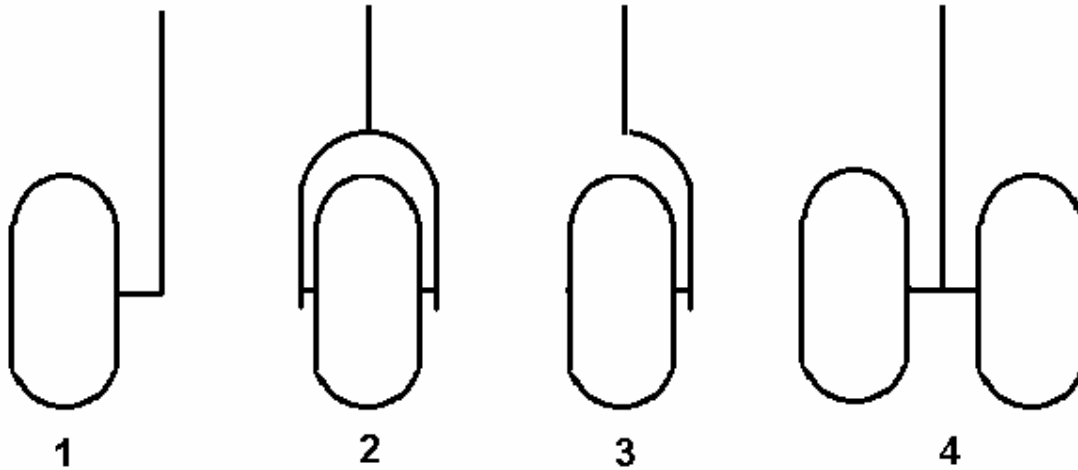


- 0176.** As the flight altitude increases , if no leaning is made with the mixture control ;
- the density of air entering the carburettor decreases and the amount of fuel remains constant.
  - the volume of air entering the carburettor remains constant and the amount of fuel decreases.
  - the volume of air entering the carburettor decreases and the amount of fuel decreases.
  - the density of air entering the carburettor decreases and the amount of fuel increases.

**Answer ;**

- a)** the density of air entering the carburettor decreases and the amount of fuel remains constant.

- 0177.** ( For this question use annex 021-11494A )



**Annex 021-6736A**

Associate the correct legend to each of the numbered diagrams ;

- 1- cantilever , 2- dual wheels , 3- half fork , 4- fork.
- 1- cantilever , 2- fork , 3- half fork , 4- dual wheels.
- 1- half-fork , 2- single trace , 3- cantilever , 4- dual wheels.
- 1- half fork , 2- fork , 3- cantilever , 4- tandem.

**Answer ;**

- b)** 1- cantilever , 2- fork , 3- half fork , 4- dual wheels.

- 0178.** Assume a constant speed DC generator providing a constant output voltage. If the electrical load increases , the voltage regulator will ;

- change the direction of the excitation current.
- maintain the intensity of the excitation current constant.
- increase the intensity of the excitation current.
- decrease the intensity of the excitation current.

**Answer ;**

- c)** increase the intensity of the excitation current.

- 0179.** Assume an initial condition at a high cruise altitude with a constant speed propeller. What will happen if the altitude is decreased while the throttle remains fully open and the waste gate is seized in the cruise position ?

- The manifold absolute pressure ( MAP ) value will stay constant.
- The blade angle may reach the full fine limit.
- The manifold absolute pressure ( MAP ) value may exceed the maximum allowed value.
- The power of the engine will decrease.

**Answer ;**

- c)** The manifold absolute pressure ( MAP ) value may exceed the maximum allowed value.

- 0180.** Assume that during cruise flight with air-conditioning packs ON , the outflow valve(s) would close. The result would be that ;

- the pressure differential would go to the maximum value.
- the cabin pressure would become equal to the ambient outside air pressure.
- the air supply would automatically be stopped.
- the skin of the cabin would be overstressed.

**Answer ;**

- a)** the pressure differential would go to the maximum value.

**0181.** Assuming a CSD fault is indicated , the CSD should be disconnected ;

- a) at flight idle engine rpm only.
- b) on the ground only.
- c) when the AC generator voltage is outside limits.
- d) provided the engine is running.

**Answer ;**

- d)** provided the engine is running.
- 

**0182.** Assuming a hydraulic accumulator is pre-charged with air to 1000 psi. If the hydraulic system is then pressurised to its operating pressure of 3000 psi , the indicated pressure on the air side of the accumulator should be ;

- a) 4000 psi.
- b) 1000 psi.
- c) 2000 psi.
- d) 3000 psi.

**Answer ;**

- d)** 3000 psi.
- 

**0183.** Assuming an accumulator is pre-charged with air to 1000 psi and the hydraulic system is pressurised to 1500 psi , the accumulator gauge will read ;

- a) 1500 psi.
- b) 500 psi.
- c) 1000 psi.
- d) 2500 psi.

**Answer ;**

- a)** 1500 psi.
- 

**0184.** Assuming cabin differential pressure has attained the required value in normal flight conditions , if flight altitude is maintained ;

- a) a constant mass air flow is permitted through the cabin.
- b) the pressurisation system must be controlled manually.
- c) the pressurisation system ceases to function until leakage reduces the pressure.
- d) the outflow valves will move to the fully open position.

**Answer ;**

- a)** a constant mass air flow is permitted through the cabin.
- 

**0185.** Assuming cabin differential pressure has reached the required value in normal flight conditions , if flight altitude and air conditioning system setting are maintained ;

- a) the pressurisation system ceases to function until leakage reduces the pressure.
- b) the outflow valves will move to the fully open position.
- c) the mass air flow through the cabin is constant.
- d) the outflow valves will move to the fully closed position.

**Answer ;**

- c)** the mass air flow through the cabin is constant.
- 

**0186.** Assuming cabin pressure decreases , the cabin rate of climb indicator should indicate ;

- a) zero.
- b) a rate of descent of approximately 300 feet per minutes.
- c) a rate of descent dependent upon the cabin differential pressure.
- d) a rate of climb.

**Answer ;**

- d)** a rate of climb.
- 

**0187.** Assuming the initiating cause is removed , which of these statements about resetting are correct or incorrect ?

- I. A fuse is not resettable
- II. A circuit breaker is not resettable.
- a) I is correct , II is incorrect.
- b) I is correct , II is correct.
- c) I is incorrect , II is correct.
- d) I is incorrect , II is incorrect.

**Answer ;**

- a)** I is correct , II is incorrect.
-

**0188.** Assuming the initiating cause is removed , which of these statements about resetting are correct or incorrect ?

- I. A fuse is not resettable
- II. A circuit breaker is resettable.
- a) I is correct , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is incorrect.
- d) I is incorrect , II is correct.

**Answer ;**

- a) I is correct , II is correct.
- 

**0189.** Assuming the initiating cause is removed , which of these statements about resetting are correct or incorrect ?

- I. A fuse is resettable
- II. A circuit breaker is not resettable.
- a) I is correct , II is correct.
- b) I is correct , II is incorrect.
- c) I is incorrect , II is correct.
- d) I is incorrect , II is incorrect.

**Answer ;**

- d) I is incorrect , II is incorrect.
- 

**0190.** Assuming the initiating cause is removed , which of these statements about resetting are correct or incorrect ?

- I. A fuse is resettable
- II. A circuit breaker is resettable.
- a) I is incorrect , II is incorrect.
- b) I is correct , II is correct.
- c) I is correct , II is incorrect.
- d) I is incorrect , II is correct.

**Answer ;**

- d) I is incorrect , II is correct.
- 

**0191.** Assuming the jet pipe is not choked , the equation to calculate jet engine thrust  $T$  ( with mass flow  $m$  , airspeed  $V_v$  and jet velocity  $V_j$  ) is ;

- a)  $T = m ( V_v + V_j )$
- b)  $T = m ( V_v - V_j )$
- c)  $T = m ( V_j - V_v )$
- d)  $T = m V_j$

**Answer ;**

- c)  $T = m ( V_j - V_v )$
- 

**0192.** Assuming the same swept volume and no turbo charger , diesel engines compared to petrol engines ;

- a) operate at higher exhaust gas temperatures.
- b) produce less maximum power output.
- c) show a higher fuel flow at the same power output.
- d) produce a higher maximum power output.

**Answer ;**

- b) produce less maximum power output.
- 

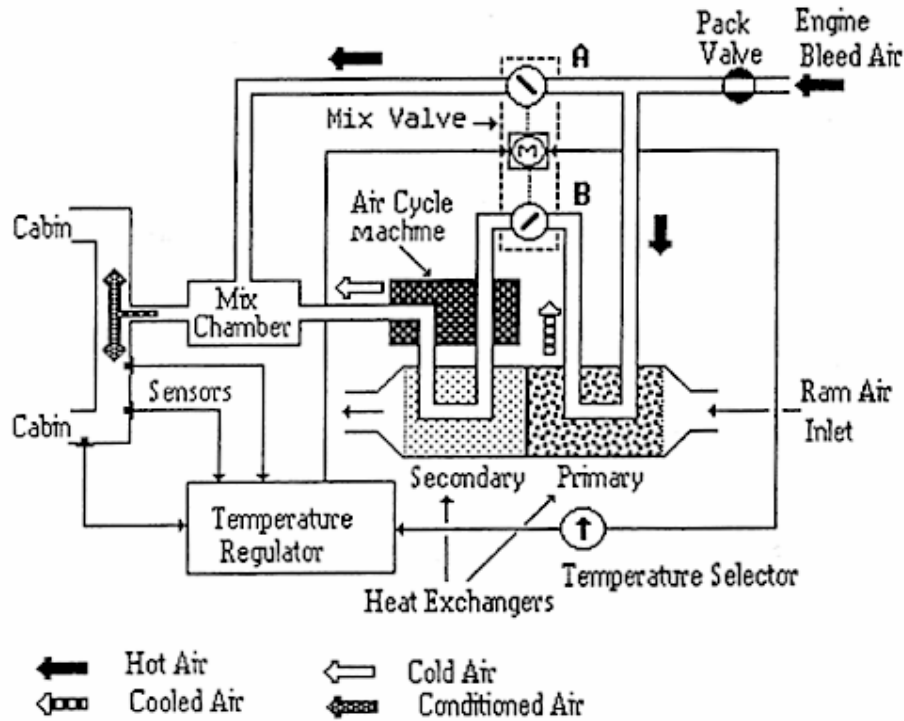
**0193.** At constant fuel flow , if engine compressor air is bled-off for engine anti-icing or a similar system , the turbine temperature ;

- a) may rise or fall depending on which stage of the compressor is used for the bleed and the rpm of the engine at the moment of selection.
- b) will fall.
- c) will be unchanged.
- d) will rise.

**Answer ;**

- d) will rise.
-

0194. ( For this question use annex 021-4960A )



Air Conditioning System

Annex 021-4960A

Automatic temperature control of the system as shown , would be accomplished by:

- a) the temperature selector in conjunction with cabin sensors and the temperature regulator , modulating the mix valve.
- b) the temperature selector only modulating the mix valve.
- c) the cabin sensors only modulating the mix valve.
- d) automatic control of the ram air.

Answer ;

- a) the temperature selector in conjunction with cabin sensors and the temperature regulator , modulating the mix valve.

0195. Axial piston pumps are often used in hydraulic systems due to ;

- a) the safety feature of the quill drive shearing due to pump seizure and the ease of repair by replacement of pump and shaft only.
- b) their ability to produce high pressure combined with constant high flow rate.
- c) their low cost , simplicity and durability.
- d) their ability to produce high pressure when required but can be off loaded to reduce power consumption.

Answer ;

- d) their ability to produce high pressure when required but can be off loaded to reduce power consumption.

0196. Because of the connection in parallel of electrical equipments on an AC bus bar , isolation of individual equipments ;

- a) increases the bus bar voltage.
- b) decreases the bus bar voltage.
- c) increases the bus bar current consumption.
- d) decreases the bus bar current consumption.

Answer ;

- d) decreases the bus bar current consumption.

0197. Because of the input/output relationship of an OR gate , it is often referred to as the ;

- a) “all or nothing” gate.
- b) “any or all” gate.
- c) “state indicator” gate.
- d) “inhibited” or “negated” gate.

Answer ;

b) “any or all” gate.

---

0198. Because of the input/output relationship of an AND gate , it is often referred to as the ;

- a) “all or nothing” gate.
- b) “any or all” gate.
- c) “inhibited” or “negated” gate.
- d) “state indicator” gate.

Answer ;

a) “all or nothing” gate.

---

0199. Below its design speed an axial compressor ;

- a) has a tendency to surge in the front stages.
- b) has a tendency to surge in the centre stages.
- c) has a tendency to surge in the rear stages.
- d) has no tendency to surge.

Answer ;

a) has a tendency to surge in the front stages.

---

0200. Bleeding air from the HP compressor of a turbofan , whilst maintaining thrust , increases ;

- a) HP compressor discharge pressure and N2.
- b) EGT and fuel flow.
- c) EGT and N1.
- d) N2 and fuel flow.

Answer ;

b) EGT and fuel flow.

---

0201. Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a gas turbine are correct or incorrect ?

- I. The process in a gas turbine engine is continuous.
  - II. Theoretically the combustion occurs at constant pressure.
- a) I is correct , II is incorrect.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

d) I is correct , II is correct.

---

0202. Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a gas turbine are correct or incorrect ?

- I. The process in a gas turbine engine is continuous.
  - II. Theoretically the combustion occurs at constant volume.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

b) I is correct , II is incorrect.

---

0203. Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a gas turbine are correct or incorrect ?

- I. The process in a gas turbine engine is intermittent.
  - II. Theoretically the combustion occurs at constant pressure.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

c) I is incorrect , II is correct.

---

**0204.** Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a gas turbine are correct or incorrect ?

- I. The process in a gas turbine engine is intermittent.
- II. Theoretically the combustion occurs at constant volume.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is correct , II is correct.
- d) I is incorrect , II is incorrect.

**Answer ;**

- d) I is incorrect , II is incorrect.**
- 

**0205.** Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a piston engine are correct or incorrect ?

- I. The process in a piston engine is continuous.
- II. Theoretically the combustion occurs at constant pressure.
- a) I is correct , II is correct.
- b) I is incorrect , II is correct.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is incorrect.

**Answer ;**

- c) I is incorrect , II is incorrect.**
- 

**0206.** Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a piston engine are correct or incorrect ?

- I. The process in a piston engine is continuous.
- II. Theoretically the combustion occurs at constant volume.
- a) I is correct , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is incorrect , II is correct.
- d) I is correct , II is incorrect.

**Answer ;**

- c) I is incorrect , II is correct.**
- 

**0207.** Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a piston engine are correct or incorrect ?

- I. The process in a piston engine is intermittent.
- II. Theoretically the combustion occurs at constant pressure.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is correct , II is correct.
- d) I is incorrect , II is incorrect.

**Answer ;**

- a) I is correct , II is incorrect.**
- 

**0208.** Both gas turbine and piston engines use a cycle made up of induction , compression , combustion and exhaust phases. Which of these statements about a piston engine are correct or incorrect ?

- I. The process in a piston engine is intermittent.
- II. Theoretically the combustion occurs at constant volume.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is correct.

**Answer ;**

- d) I is correct , II is correct.**
- 

**0209.** Both gas turbine and piston engines utilise a cycle of induction , compression , combustion and exhaust. However in the gas turbine these processes are (i) ..... and combustion occurs at (ii) ..... . Respectively (i) and (ii) are ;

- a) (i) intermittent , (ii) constant volume.
- b) (i) intermittent , (ii) constant pressure.
- c) (i) continuous , (ii) constant volume.
- d) (i) continuous , (ii) constant pressure.

**Answer ;**

- d) (i) continuous , (ii) constant pressure.**
-

0210. By-pass ratio in a turbine engine is the ratio of the ;
- a) cold air mass flow to the hot air mass flow.
  - b) tertiary air mass flow to the primary air mass flow.
  - c) speed of the combusted air to the speed of the by-pass air.
  - d) intake air pressure to the turbine delivery air pressure.

Answer ;

a) cold air mass flow to the hot air mass flow.

---

0211. By-pass turbine engines are mainly used in commercial aviation , because ;
- a) they are lighter than straight jet engines.
  - b) twin spool or triple spool jet engines require a high by pass ratio.
  - c) they produce less noise.
  - d) at high subsonic airspeeds they have a better propulsive efficiency than propeller or straight jet engines.

Answer ;

d) at high subsonic airspeeds they have a better propulsive efficiency than propeller or straight jet engines.

---

0212. Cabin air for a large airplane , during flight ;
- a) has an increased oxygen ratio.
  - b) is temperature controlled.
  - c) has a constant oxygen ratio regulated to a preset value.
  - d) has a reduced oxygen ratio.

Answer ;

b) is temperature controlled.

---

0213. Cabin air for JAR 25 aeroplane is usually taken from ;
- a) the high pressure compressor.
  - b) the second fan stage.
  - c) the high pressure compressor and from the low pressure compressor if necessary.
  - d) the low pressure compressor.

Answer ;

c) the high pressure compressor and from the low pressure compressor if necessary.

---

0214. Cabin air for modern aircraft is usually taken from ;
- a) the low pressure compressor and from the high pressure compressor if necessary.
  - b) the second fan stage.
  - c) the high pressure compressor.
  - d) the low pressure compressor.

Answer ;

a) the low pressure compressor and from the high pressure compressor if necessary.

---

0215. Cabin altitude is the ;
- a) cabin pressure expressed as altitude.
  - b) difference in height between the cabin floor and ceiling.
  - c) flight level at which the aeroplane is flying.
  - d) altitude at maximum differential pressure.

Answer ;

a) cabin pressure expressed as altitude.

---

0216. Cabin altitude means the ;
- a) cabin pressure expressed as altitude.
  - b) flight level the aircraft is flying at.
  - c) flight level altitude at maximum differential pressure.
  - d) difference in height between the cabin floor and ceiling.

Answer ;

a) cabin pressure expressed as altitude.

---

0217. Cabin differential pressure means the pressure difference between ;
- a) cockpit and passenger cabin.
  - b) actual cabin pressure and selected pressure.
  - c) cabin pressure and ambient air pressure at MSL.
  - d) cabin pressure and ambient air pressure.

Answer ;

d) cabin pressure and ambient air pressure.

---

0218. Cabin heating in a large jet transport aeroplane is obtained from ;
- a) hot air bled from the compressors.
  - b) hot air bled from the turbines.
  - c) an electrical heater system.
  - d) a fuel heater system.

Answer ;

a) hot air bled from the compressors.

---

0219. Cabin pressure is controlled by ;
- a) controlling the flow of air into the cabin with a constant outflow.
  - b) the cabin air mass flow control inlet valve(s).
  - c) delivering a substantially constant flow of air into the cabin and controlling the outflow.
  - d) the cabin air re-circulation system.

Answer ;

c) delivering a substantially constant flow of air into the cabin and controlling the outflow.

---

0220. Cabin pressurization is controlled by the ;
- a) cabin outflow valve.
  - b) cabin inlet airflow.
  - c) engine's RPM.
  - d) engine's bleed valves.

Answer ;

a) cabin outflow valve.

---

0221. Cabin pressurization is regulated by the ;
- a) engine's bleed valves.
  - b) cabin outflow valve(s).
  - c) engine's RPM.
  - d) cabin inlet valve(s).

Answer ;

b) cabin outflow valve(s).

---

0222. Chemical oxygen generators are used to furnish oxygen to the ;
- a) cabin only.
  - b) cockpit and the cabin.
  - c) toilets only.
  - d) cockpit only.

Answer ;

a) cabin only.

---

0223. Chemical oxygen generators are used to furnish oxygen to the ;
1. cockpit.
  2. cabin.
  3. toilets.
  4. smoke hood.

The combination that regroups all of the correct statements is ;

- a) 1 , 4.
- b) 1 , 2.
- c) 2 , 3 , 4.
- d) 1 , 3.

Answer ;

c) 2 , 3 , 4.

---

0224. Circuit breakers protecting circuits may be ;
- a) used in AC and DC circuits.
  - b) used only in AC circuits.
  - c) used only in DC circuits.
  - d) reset at any time.

Answer ;

a) used in AC and DC circuits.

---



0225. Compared to a tyre fitted with an inner tube , a tubeless tyre presents the following characteristics ;

1. high heating.
2. valve fragility.
3. lower risk of bursting.
4. better adjustment to wheels.

The combination containing all the correct statements is ;

- a) 2 , 4.
- b) 2 , 3.
- c) 1 , 2 , 3 , 4.
- d) 3 , 4.

Answer ;

d) 3 , 4.

---

0226. Compressor stall in a gas turbine engine ;

- a) can be caused by an axial airflow velocity that is low relative to the rotor speed.
- b) is unlikely to occur in a compressor with many stages.
- c) only occurs during engine start-up.
- d) does not occur with well-designed compressor blades.

Answer ;

a) can be caused by an axial airflow velocity that is low relative to the rotor speed.

---

0227. Compressor stall is due to ;

- a) an excessively high axial velocity.
- b) an inadequate fuel flow.
- c) an excessively low angle of attack of the rotor blades.
- d) an excessively high angle of attack of the rotor blades.

Answer ;

d) an excessively high angle of attack of the rotor blades.

---

0228. Compressor surge may be controlled by means of ;

- a) a variable metering orifice.
- b) fixed inlet guide vanes.
- c) bleed valves.
- d) a pressure drop control orifice.

Answer ;

c) bleed valves.

---

0229. Concerning a constant speed propeller of a twin engine aeroplane controlled by a single-acting pitch control unit ;

- a) oil pressure turns the propeller blades towards smaller pitch angle.
- b) spring force turns the propeller blades towards smaller pitch angle.
- c) aerodynamic force turns the propeller blades towards higher pitch angle.
- d) oil pressure turns the propeller blades towards higher pitch angle.

Answer ;

a) oil pressure turns the propeller blades towards smaller pitch angle.

---

0230. Concerning electrically powered ice protection devices , the only true statement is ;

- a) on modern aeroplanes , electrically powered thermal devices are used as de-icing devices for pitot-tubes , static ports and windshield.
- b) on modern aeroplanes , electrically powered thermal devices are used to prevent icing on small surfaces ( pitot-static , windshield... ).
- c) on modern aeroplanes , electrically powered thermal devices are very efficient , therefore they only need little energy.
- d) on modern aeroplanes , electrical power supply being available in excess , this system is very often used for large surfaces de-icing.

Answer ;

b) on modern aeroplanes , electrically powered thermal devices are used to prevent icing on small surfaces ( pitot-static , windshield... ).

---

0231. Concerning the fuel system high-pressure shut-off valve ( HPSOV ) of a gas turbine engine ;

- a) the HPSOV is located between the HP fuel pump and the fuel control unit.
- b) ignition is activated before fuel is delivered by the HPSOV to the fuel nozzles.
- c) manual HPSOV control from the cockpit is not possible with FADEC equipped engines.
- d) the HPSOV is mounted in the wing front spar near the pylon.

Answer ;

b) ignition is activated before fuel is delivered by the HPSOV to the fuel nozzles.

---

0232. Concerning the sequential pneumatic impulses used in certain leading edge de-icing devices , one can affirm that ;
1. They prevent ice formation.
  2. They are triggered from the flight deck after icing has become visible.
  3. A cycle lasts more than ten seconds.
  4. There are more than ten cycles per second.

The combination which regroups all the correct statements is ;

- a) 1 , 4.
- b) 2 , 3.
- c) 2 , 4.
- d) 1 , 3.

Answer ;

b) 2 , 3.

---

0233. Concerning the starting procedure of a gas turbine engine ;
- a) the selected igniters are activated after fuel is delivered to the fuel nozzles.
  - b) if the HP fuel valve is opened at a relatively high compressor rpm , this may result in a hot start.
  - c) if the air starter seizes , the engine can still be started by means of another air bleed source.
  - d) the higher the starter air pressure , the higher the EGT during starting.

Answer ;

a) the selected igniters are activated after fuel is delivered to the fuel nozzles.

---

0234. "Conditioned air" is air that has ;
- a) been controlled in respect of temperature and pressure.
  - b) had the oxygen content increased.
  - c) had the oxygen content reduced.
  - d) had any moisture removed from it.

Answer ;

a) been controlled in respect of temperature and pressure.

---

0235. Connecting two 12 volt 40 ampere-hour capacity batteries in series will result in a total voltage and capacity respectively of ;
- a) 12 volts , 80 ampere-hours.
  - b) 12 volts , 40 ampere-hours.
  - c) 24 volts , 80 ampere-hours.
  - d) 24 volts , 40 ampere-hours.

Answer ;

d) 24 volts , 40 ampere-hours.

---

0236. Consider a jet engine whose control is based on the Engine Pressure Ratio ( EPR ) ;
1. With a constant EPR , the thrust decreases when the altitude increases.
  2. With a constant EPR , the thrust is independent of the Mach number.
  3. At same environmental conditions , a given EPR setting maintains the thrust irrespective of engine wear due to ageing.
  4. The EPR is determined by the impact pressure difference between the turbine outlet and the compressor inlet.
  5. On take-off , in the event of icing not detected by the crew , the indicated EPR is lower than the real EPR.

The combination regrouping all the correct statements is ;

- a) 1 , 5.
- b) 3 , 4 , 5.
- c) 2 , 3 , 4.
- d) 1 , 3.

Answer ;

d) 1 , 3.

---

**0237.** Consider the flight deck oxygen supply system. The purpose of the oxygen regulator ( as a function of demand and altitude ) is to ;

1. decrease oxygen pressure from 1800 psi ( in the bottles ) down to about 50-75 psi ( low pressure system ).
2. supply pure oxygen.
3. supply diluted oxygen.
4. supply oxygen at normal pressure.
5. supply oxygen at emergency/positive pressure.
6. trigger the continuous cabin altitude warning at 10000 ft cabin altitude.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 4 , 6.
- b) 1 , 2 , 3 , 4.
- c) 2 , 3 , 4 , 5.
- d) 3 , 4 , 5 , 6.

**Answer ;**

**c) 2 , 3 , 4 , 5.**

---

**0238.** Consider the variable-pitch propeller of a turbo-prop. During deceleration ;

- a) with propeller windmilling , the thrust is zero and the propeller supplies engine power.
- b) when feathered , the propeller produces thrust and absorbs no engine power.
- c) at zero power , the propeller thrust is zero and the engine power absorbed is nil.
- d) when braking , the propeller supplies negative thrust and absorbs engine power.

**Answer ;**

**d) when braking , the propeller supplies negative thrust and absorbs engine power.**

---

**0239.** Continuous loop fire detector systems operate on the principle that an increase in temperature produces ;

- a) a decrease in pressure.
- b) a decrease in resistance.
- c) a decrease in the reference current.
- d) an increase in resistance.

**Answer ;**

**b) a decrease in resistance.**

---

**0240.** Control surface flutter can be avoided by ;

1. a high torsional stiffness of the structure.
2. a low torsional stiffness of the structure.
3. locating a balancing mass in front of the control hinge.
4. locating a balancing mass behind the control hinge.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 1 , 4.
- c) 2 , 4.
- d) 2 , 3.

**Answer ;**

**a) 1 , 3.**

---

**0241.** Cooling air for a reciprocating engine can be obtained by means of ;

- a) ram air.
- b) a supercharger.
- c) a pneumatic system.
- d) a turbocharger.

**Answer ;**

**a) ram air.**

---

**0242.** Creep in a gas turbine engine is ;

- a) the elastic increase in length of metal parts due to tensile stress and high temperature.
- b) crack formation in metal parts due to cyclic mechanical loads and high temperatures.
- c) the permanent increase in length of metal parts due to a combination of tensile stress and high temperature over a prolonged period.
- d) expansion of metal parts due to temperature increase.

**Answer ;**

**c) the permanent increase in length of metal parts due to a combination of tensile stress and high temperature over a prolonged period.**

---

0243. Creep of turbine blades is caused by ;
- a) bending stresses set up by gas pressure.
  - b) high blade temperature whilst under centrifugal loading.
  - c) prolonged idling at low rpm.
  - d) static imbalance of the blades.

Answer ;

b) high blade temperature whilst under centrifugal loading.

---

0244. Derated or reduced take-offs save engine life primarily due to ;
- a) reduced corrosion of the HP turbine blades.
  - b) reduced low cycle fatigue damage.
  - c) the maximum oil temperature not being exceeded.
  - d) reduced creep damage due to higher EGT levels during take-off.

Answer ;

b) reduced low cycle fatigue damage.

---

0245. Diesel engines always produce a certain amount of soot , because ;
- a) the fuel droplets are too small.
  - b) the fuel droplets coming from the atomiser do not burn completely.
  - c) the intake air is too cold.
  - d) the mixture is always too rich.

Answer ;

b) the fuel droplets coming from the atomiser do not burn completely.

---

0246. Diesel engines , compared to petrol engines have ;
- a) a lower compression ratio.
  - b) a variable compression ratio.
  - c) a higher compression ratio.
  - d) the same compression ratio.

Answer ;

c) a higher compression ratio.

---

0247. Direct current generators are connected ;
- a) in series to provide maximum voltage.
  - b) in parallel to provide maximum power.
  - c) in series to provide maximum power.
  - d) in parallel to provide maximum voltage.

Answer ;

b) in parallel to provide maximum power.

---

0248. Dual ignition provides a factor of reliability and ;
- a) saves wear caused by using one magneto constantly.
  - b) aids in starting.
  - c) provides more voltage.
  - d) improves combustion efficiency.

Answer ;

d) improves combustion efficiency.

---

0249. During a climb in a standard atmosphere with constant Manifold Absolute Pressure ( MAP ) and RPM indications and at a constant mixture setting , the power output of a piston engine ;
- a) only stays constant if the propeller lever is pushed.
  - b) decreases.
  - c) increases.
  - d) stays constant.

Answer ;

c) increases.

---

0250. During a normal pressurised climb after take-off ;
- a) the cabin differential pressure is maintained constant.
  - b) absolute cabin pressure increases to compensate for the fall in pressure outside the aircraft.
  - c) the pressurisation system is inoperative until an altitude of 10 000 feet is reached.
  - d) cabin pressure decreases more slowly than atmospheric pressure.

Answer ;

d) cabin pressure decreases more slowly than atmospheric pressure.

---

0251. During a power change on an engine equipped with a constant speed propeller , a wrong combination of manifold pressure and RPM values results in excessive pressures in the cylinders. This is the case when one simultaneously selects a ;

- a) high manifold pressure and high RPM.
- b) low manifold pressure and low RPM.
- c) low manifold pressure and high RPM.
- d) high manifold pressure and low RPM.

Answer ;

d) high manifold pressure and low RPM.

---

0252. During climb with constant throttle and RPM lever setting ( mixture being constant ) the ;

- a) Manifold Absolute Pressure ( MAP ) increases.
- b) Manifold Absolute Pressure ( MAP ) decreases.
- c) RPM decreases.
- d) RPM increases.

Answer ;

b) Manifold Absolute Pressure ( MAP ) decreases.

---

0253. During flight , the wing anti-ice system has to protect ;

- a) the whole leading edge and the whole upper wing surface.
- b) a part of the whole leading edge.
- c) slats and the leading edge flaps only.
- d) the whole upper wing surface and the flaps.

Answer ;

b) a part of the whole leading edge.

---

0254. During hydroplaning , the friction coefficient between tyre and runway surface is approximately ;

- a) 0
- b) 1
- c) 0.25
- d) 0.5

Answer ;

a) 0

---

0255. During level flight at a constant cabin pressure altitude ( which could be decreased , even at this flight level ) , the cabin outflow valves are ;

- a) fully closed until the cabin climbs to a selected altitude.
- b) at the pre-set position for take-off.
- c) partially open.
- d) fully closed until the cabin descends to a selected altitude.

Answer ;

c) partially open.

---

0256. During re-fuelling the automatic shut-off valves will switch off the fuel supply system when ;

- a) the surge vent tank is filled.
- b) fuelling system has reached a certain pressure.
- c) there is fire.
- d) the fuel has reached a predetermined volume or mass.

Answer ;

d) the fuel has reached a predetermined volume or mass.

---

0257. During start of an engine fitted with a brushless AC generator with no commutator rings , the generator is activated by ;

- a) a set of permanent magnets.
- b) the stabiliser winding jointly with the voltage regulator.
- c) the auxiliary winding.
- d) the main field winding.

Answer ;

a) a set of permanent magnets.

---

0258. During take-off , the angle of attack of the blades of a fixed pitch propeller , optimised for cruise condition is ;

- a) zero.
- b) relatively high.
- c) negative.
- d) relatively small.

Answer ;

b) relatively high.

---

0259. During take-off , the fan in high by-pass ratio turbofan engine produces ;

- a) all the thrust.
- b) the greater part of the thrust.
- c) half the thrust.
- d) the lesser part of the thrust.

Answer ;

b) the greater part of the thrust.

---

0260. During the ram recovery process in a gas turbine engine subsonic intake , the ;

- a) flow velocity decreases.
- b) total pressure increases.
- c) total temperature increases.
- d) static temperature decreases.

Answer ;

a) flow velocity decreases.

---

0261. Electrical bonding of an aircraft is used to ;

- 1. protect the aircraft against lightning effects.
- 2. reset the electrostatic potential of the aircraft to a value approximating 0 volt.
- 3. reduce radio interference on radio communication systems.
- 4. set the aircraft to a single potential.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3.
- b) 2 , 4.
- c) 1 , 3 , 4.
- d) 3 , 4.

Answer ;

c) 1 , 3 , 4.

---

0262. Electrical powered ice protection devices on aircraft are ;

- a) used primarily because they are very efficient.
- b) used as de-icing devices for pitot-tubes , static ports and windshield.
- c) used for large surfaces only.
- d) used as anti-icing devices for pitot-tubes , static ports and windshield.

Answer ;

d) used as anti-icing devices for pitot-tubes , static ports and windshield.

---

0263. Emergency extension of landing gear systems can be performed ;

- 1. with compressed CO<sub>2</sub>.
- 2. with compressed nitrogen.
- 3. with compressed oxygen.
- 4. by mechanical/manual means.
- 5. by freefall.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 1 , 3 , 4.
- c) 1 , 2 , 5.
- d) 2 , 4 , 5.

Answer ;

d) 2 , 4 , 5.

---

0264. Engine auxiliary gearboxes ;

- a) always provide drives for fuel pumps , oil pumps , speed sensing and torque sensing.
- b) never provide a torque sensing facility.
- c) provide the drives for a wide variety of services and may even provide for the support of the engine in the airframe.
- d) are driven by the output from the free power turbine.

Answer ;

c) provide the drives for a wide variety of services and may even provide for the support of the engine in the airframe.

---

0265. Engine bleed air used for air conditioning and pressurization in turbo-jet aeroplanes is usually taken from the ;
- a) by-pass ducting.
  - b) fan section.
  - c) compressor section.
  - d) turbine section.

Answer ;

c) compressor section.

---

0266. Engine compartment decking and firewalls are manufactured from ;
- a) aluminium alloy sheet.
  - b) stainless steel or titanium sheet.
  - c) asbestos blankets.
  - d) composite materials such as carbon , kevlar and fibre glass.

Answer ;

b) stainless steel or titanium sheet.

---

0267. Engine temperature limitations of a gas turbine engine are usually imposed due to temperature limitations on the ;
- a) jet pipe.
  - b) turbine section.
  - c) casing of the combustion chamber.
  - d) compressor section.

Answer ;

b) turbine section.

---

0268. EPR can be defined as the ratio of ;
- a) the total turbine outlet pressure to the total compressor inlet pressure.
  - b) the total turbine inlet pressure to the total compressor inlet pressure.
  - c) the total turbine inlet pressure to the total compressor outlet pressure.
  - d) the total turbine outlet pressure to the total compressor outlet pressure.

Answer ;

a) the total turbine outlet pressure to the total compressor inlet pressure.

---

0269. Evacuation slide inflation is ensured by ;
- a) a pressurized gas canister combined with the slide itself.
  - b) a manual pump , used when needed by the cabin crew.
  - c) pressurized air from the air conditioning system.
  - d) the aircraft's general pneumatic circuit.

Answer ;

a) a pressurized gas canister combined with the slide itself.

---

0270. Excessive pressure in the cylinders of an engine equipped with a constant speed propeller ;
- a) low manifold pressure and low RPM.
  - b) high manifold pressure and high RPM.
  - c) high manifold pressure and low RPM.
  - d) low manifold pressure and high RPM.

Answer ;

c) high manifold pressure and low RPM.

---

0271. Excessive priming of a piston engine should be avoided because ;
- 1. it drains the carburettor float chamber.
  - 2. the risk of engine fire.
  - 3. the risk of flooding the engine.
  - 4. it fouls the spark plugs.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 1 , 2 , 3.
- c) 1 , 3 , 4.
- d) 1 , 2 , 4.

Answer ;

a) 2 , 3 , 4.

---

0272. Exhaust gas temperature sensing is usually taken ;
- a) between the gas generator turbine and the free power turbine.
  - b) at the outlet from the combustion chamber.
  - c) after the free power turbine.
  - d) between the stages of the gas generator turbine.

Answer ;

- a) between the gas generator turbine and the free power turbine.
- 

0273. "Fail safe construction" is ;
- a) a type of construction for small aircraft only.
  - b) a construction which is suitable for aerobatic flight.
  - c) a simple and cheap type of construction.
  - d) a type of construction in which the load is carried by other components if a part of the structure fails.

Answer ;

- d) a type of construction in which the load is carried by other components if a part of the structure fails.
- 

0274. Filters in hydraulic systems often incorporate pop out indicators to ;
- a) warn of a hydraulic system overheat.
  - b) indicate that the filter is clogged and unfiltered oil is passing around the system.
  - c) warn of an impending by-pass situation.
  - d) indicate that the filter is due maintenance.

Answer ;

- c) warn of an impending by-pass situation.
- 

0275. Filtration in a hydraulic system is usually ensured by ;
- a) the use of sealed containers only during replenishment.
  - b) a filter on the return line only.
  - c) a filter in the pressure line only.
  - d) filters in both the pressure and return lines.

Answer ;

- d) filters in both the pressure and return lines.
- 

0276. Fire precautions to be observed before refuelling are ;
- a) Ground Power Units ( GPU ) are not to be operated.
  - b) Aircraft must be more than 10 metres from radar or HF radio equipment under test.
  - c) All bonding and connections to the earth terminal between ground equipment and the aircraft should be made before filler caps are removed.
  - d) Passengers may be boarded ( traversing the refuelling zone ) providing suitable fire extinguishers are readily available.

Answer ;

- c) All bonding and connections to the earth terminal between ground equipment and the aircraft should be made before filler caps are removed.
- 

0277. Fixed oxygen systems in pressurised aeroplanes are used to provide oxygen ;
- 1. in the event of depressurisation.
  - 2. in the event of any passenger indisposition.
  - 3. during a normal flight.
  - 4. in the event of smoke or toxic fumes in the cockpit.

The combination of correct statements is ;

- a) 1 , 2 , 4.
- b) 1 , 4.
- c) 3.
- d) 2 , 3.

Answer ;

- b) 1 , 4.
- 

0278. Fixed-pitch propellers are usually designed for maximum efficiency at ;
- a) idling.
  - b) take-off.
  - c) full throttle.
  - d) cruising speed.

Answer ;

- d) cruising speed.
-



0279. Flat rated jet engines are designed to provide constant maximum take-off thrust up to a certain level of ;

- a) ambient pressure.
- b) ambient temperature.
- c) pressure altitude.
- d) engine pressure ratio

Answer ;

b) ambient temperature.

---

0280. Flutter results from two deformation modes which are ;

- a) torsion and bending.
- b) torsion and shearing.
- c) shearing and elongation.
- d) bending an elongation.

Answer ;

a) torsion and bending.

---

0281. For a fan jet engine , the by-pass ratio is the ;

- a) external airflow mass divided by the internal airflow mass.
- b) internal airflow mass divided by the external airflow mass.
- c) fuel flow mass divided by the internal airflow mass.
- d) internal airflow mass divided by the fuel flow mass.

Answer ;

a) external airflow mass divided by the internal airflow mass.

---

0282. For a fan jet engine , the by-pass ratio is the ;

- a) cold air mass flow divided by the hot air mass flow.
- b) hot air mass flow divided by the cold air mass flow.
- c) fuel flow divided by hot air mass flow.
- d) hot air mass flow divided by the fuel flow.

Answer ;

a) cold air mass flow divided by the hot air mass flow.

---

0283. For a free power turbine engine ;

- a) pilot control of the engine is limited to the selection of “OFF” , “GROUND IDLE” and “FLIGHT” , leaving the engine to automatically respond to rotor demands.
- b) the turbine driving the main rotor transmission is mechanically independent from the gas generator.
- c) the compressor driving the main transmission is mechanically independent from the power turbine.
- d) the gas generator is free to run at a constant speed while the power turbine changes speed in response to changes in drag on the main rotor blades.

Answer ;

b) the turbine driving the main rotor transmission is mechanically independent from the gas generator.

---

0284. For a given type of oil , the oil viscosity depends on the ;

- a) quantity of oil.
- b) oil pressure.
- c) oil temperature.
- d) outside pressure.

Answer ;

c) oil temperature.

---

0285. For a JAR25 aeroplane , spoilers are ;

- a) lower wing surface devices , their deflection is always asymmetrical.
- b) lower wing surface devices , their deflection can be symmetrical or asymmetrical.
- c) upper wing surface devices , their deflection can be symmetrical or asymmetrical.
- d) upper wing surface devices , their deflection is always asymmetrical.

Answer ;

c) upper wing surface devices , their deflection can be symmetrical or asymmetrical.

---

0286. For a piston engine , the chemically correct fuel/air ratio of the mixture is ;

- a) 1: 9
- b) 1:15
- c) 1:10
- d) 1:12

Answer ;

b) 1:15

---

0287. For a reciprocating engine fuel/air ratio or mixture is the ratio between the ;
- a) mass of fuel and mass of air entering the carburettor.
  - b) volume of fuel and volume of air entering the carburettor.
  - c) volume of fuel and volume of air entering the cylinder.
  - d) mass of fuel and mass of air entering the cylinder.

Answer ;

- d) mass of fuel and mass of air entering the cylinder.
- 

0288. For a subsonic airflow , air passing through a divergent duct ;
- a) decreases in pressure and velocity.
  - b) increases in pressure and velocity.
  - c) decreases in velocity and increases in pressure.
  - d) increases in velocity and decreases in pressure.

Answer ;

- c) decreases in velocity and increases in pressure.
- 

0289. For a subsonic airflow , in a divergent duct the ;
- a) pressure decreases , velocity increases and temperature increases.
  - b) pressure decreases , velocity increases and temperature decreases.
  - c) pressure increases , velocity decreases and temperature decreases.
  - d) pressure increases , velocity decreases and temperature increases.

Answer ;

- d) pressure increases , velocity decreases and temperature increases.
- 

0290. For a turbine engine , the self-sustaining speeds relates to the minimum RPM at which the engine ;
- a) will continue to operate without any starter motor assistance.
  - b) operates most efficiently in the cruise.
  - c) is designed to idle after starting.
  - b) will enable the generators to supply bus-bar voltage.

Answer ;

- a) will continue to operate without any starter motor assistance.
- 

0291. For a turbine engine , the term self-sustaining speed relates to the lowest rpm range at which the engine ;
- a) operates most efficiently in the cruise.
  - b) will enable the generators to supply bus-bar voltage.
  - c) is designed to idle after starting.
  - d) will continue to run without any further starter assistance.

Answer ;

- d) will continue to run without any further starter assistance.
- 

0292. For a turbine engine , the term self-sustaining speed relates to the speed at which the engine ;
- a) will enable the generators to supply bus-bar voltage.
  - b) operates most efficiently in the cruise.
  - c) will run without any external assistance.
  - b) is designed to idle after starting.

Answer ;

- c) will run without any external assistance.
- 

0293. For an aeroplane hydraulic supply circuit , the correct statement is ;
- a) the regulation system deals only with emergency operation and is not applied to all hydraulic services but only those considered as essential.
  - b) the reservoir constitutes a reserve of hydraulic fluid maintained under pressure by a pneumatic back pressure ( air or nitrogen ) and destined to serve as a fluid or pressure reserve.
  - c) the security components comprise the filters , the pressure relief valves , the by-passes , and the fire shut-off valve.
  - d) the pumps are always electric due to the high pressures which they must deliver ( 140 to 210 kg/cm<sup>2</sup> ).

Answer ;

- c) the security components comprise the filters , the pressure relief valves , the by-passes , and the fire shut-off valve.
-

0294. For fail safe designed structural components ;

1. there is more than one load carrying component.
2. one load carrying component is sufficient , provided it is strong enough.
3. the component is removed at the end of the calculated life time or number of cycles.
4. the design is based on the principle of redundancy of components.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 3.
- c) 1 , 4.
- d) 2 , 4.

Answer ;

- c) 1 , 4.
- 

0295. For internal cooling , reciprocating engines are especially dependent on ;

- a) a properly functioning thermostat.
- b) a lean fuel/air mixture.
- c) a rich fuel/air mixture.
- d) the circulation of lubricating oil.

Answer ;

- d) the circulation of lubricating oil.
- 

0296. For most large aeroplanes , spoilers are ;

- a) lower wing surface devices and their deflection is always asymmetrical.
- b) upper wing surface devices and their deflection can be symmetrical or asymmetrical.
- c) lower wing surface devices and their deflection can be symmetrical or asymmetrical.
- d) upper wing surface devices and their deflection is always asymmetrical.

Answer ;

- b) upper wing surface devices and their deflection can be symmetrical or asymmetrical.
- 

0297. For piston engines , mixture ratio is the ratio between the ;

- a) mass of fuel and mass of air entering the cylinder.
- b) volume of fuel and volume of air entering the carburettor.
- c) mass of fuel and volume of air entering the carburettor.
- d) volume of fuel and volume of air entering the cylinder.

Answer ;

- a) mass of fuel and mass of air entering the cylinder.
- 

0298. For safe life designed structural components ;

1. there is more than one load carrying component.
2. one load carrying component is sufficient for a given load , provided it is strong enough.
3. the component is removed at the end of the calculated life time or number of cycles.
4. the design is based on the principle of redundancy of components.

The combination that regroups all of the correct statements is ;

- a) 1 , 3.
- b) 2 , 3.
- c) 2 , 3.
- d) 1 , 4.

Answer ;

- d) 1 , 4.
- 

0299. For take-off , the correct combination of propeller pitch (1) , and propeller lever position (2) at brake release is ;

- a) (1) high (2) aft.
- b) (1) low (2) forward.
- c) (1) high (2) forward.
- d) (1) low (2) aft.

Answer ;

- b) (1) low (2) forward.
- 

0300. "Frequency wild" in relation to a AC generation system means the generator ;

- a) output frequency is too high.
- b) output frequency varies with engine speed.
- c) output frequency is too low.
- d) voltage regulator is out of adjustment.

Answer ;

- b) output frequency varies with engine speed.
-

**0301.** From the cruise , with all the parameters correctly set , if the altitude is reduced , to maintain the same mixture the fuel flow should ;

- a) increase.
- b) remain the same.
- c) increase or decrease , depending on the engine type.
- d) decrease.

**Answer ;**

**a)** increase.

---

**0302.** Fuel/air ratio is the ratio between the ;

- a) volume of fuel and volume of air entering the carburettor.
- b) volume of fuel and volume of air entering the cylinder.
- c) mass of fuel and mass of air entering the cylinder.
- d) mass of fuel and mass of air entering the carburettor.

**Answer ;**

**c)** mass of fuel and mass of air entering the cylinder.

---

**0303.** Fuel flow information for gas turbine engines is ;

- a) measured in the line between the fuel control unit and the engine burners.
- b) not used on aircraft.
- c) measured at the outlet of the high pressure fuel pump.
- d) measured in the line between the booster pump and the engine.

**Answer ;**

**a)** measured in the line between the fuel control unit and the engine burners.

---

**0304.** Fuel for gas turbine engines is usually heated in order to ;

- a) prevent icing-up of the LP filter.
- b) improve atomisation at low temperature.
- c) improve thermal efficiency.
- d) improve the specific fuel consumption.

**Answer ;**

**a)** prevent icing-up of the LP filter.

---

**0305.** Fuel pumps submerged in the fuel tanks of a multi-engine aircraft are ;

- a) low pressure variable swash plate pumps.
- b) centrifugal low pressure type pumps.
- c) centrifugal high pressure pumps.
- d) high pressure variable swash plate pumps.

**Answer ;**

**b)** centrifugal low pressure type pumps.

---

**0306.** Generally , for large aeroplanes , electrical heating for ice protection is used on ;

- a) pitot tubes.
- b) elevator leading edges.
- c) slat leading edges.
- d) fin leading edges.

**Answer ;**

**a)** pitot tubes.

---

**0307.** Generally , on modern jet transport aircraft , how can the landing gear be extended if there is a complete hydraulic system failure ?

- a) By hydraulic accumulators.
- b) Electrically.
- c) Mechanically.
- d) Pneumatically.

**Answer ;**

**c)** Mechanically.

---

0308. Generally , when the fire handle of the fire-extinguishing system on an aeroplane is pulled , the effects are ;
1. opening of the bleed air valves and HP valves on the engine concerned.
  2. arming of the extinguishing system.
  3. immediate discharge of extinguishing agent.
  4. deactivation of the electric generators.

The combination that regroups all of the correct statements is ;

- a) 1 , 4.
- b) 2 , 4.
- c) 1 , 3.
- d) 2 , 3.

Answer ;

b) 2 , 4.

---

0309. Generally , when the fire handle of the fire-extinguishing system on an aircraft is pulled , the effects are ;

1. closing of the LP valve of the fuel system.
2. opening of the air bleed valves and HP valves on the engine concerned.
3. setting of extinguishing systems.
4. closing of the isolation and de-icing valves.
5. isolation of the associated electric current generators.
6. immediate discharge of extinguishing agent.

The combination regrouping all the correct statements is ;

- a) 2 , 3 , 4 , 5.
- b) 1 , 3 , 4 , 5.
- c) 1 , 3 , 4.
- d) 1 , 2 , 5 , 6.

Answer ;

b) 1 , 3 , 4 , 5.

---

0310. Generators , when connected to the same bus bar , are usually connected ;

- a) in parallel mode.
- b) dependent on the type of engine.
- c) dependent on the type of generator.
- d) in series mode.

Answer ;

a) in parallel mode.

---

0311. Given ;

$p_0$  = static ambient pressure  
 $t_0$  = static ambient temperature  
 $m$  = air mass flow  
 $V_v$  = True Air Speed  
 $V_j$  = jet efflux ( exhaust ) velocity  
 $p_j$  = jet efflux ( exhaust ) static pressure  
 $t_j$  = jet efflux ( exhaust ) static temperature  
 $A$  = exhaust cross-sectional area

The thrust  $T$  of a jet engine is ;

- a)  $T = m ( V_v - V_j ) + A ( p_j - p_0 )$
- b)  $T = m ( V_j - V_v ) + A ( p_j - p_0 )$
- c)  $T = m ( V_v - V_j ) + A ( t_j - t_0 )$
- d)  $T = m ( V_j - V_v ) + A ( t_j - t_0 )$

Answer ;

b)  $T = m ( V_j - V_v ) + A ( p_j - p_0 )$

---

0312. Given a conventional transport aeroplane with irreversible flight controls on the ground with engines running. Which of these statements about rudder trim actuation is correct ?

- a) The rudder trim tab moves and the rudder pedals do not move.
- b) The rudder moves , the rudder pedals do not move.
- c) The rudder moves , the rudder pedals move in the corresponding direction.
- d) The “zero force point” of the artificial feel system changes , the rudder does not move.

Answer ;

c) The rudder moves , the rudder pedals move in the corresponding direction.

---

**0313.** Given an aeroplane with irreversible primary flight controls , how is control maintained if one hydraulic system is lost due to a hydraulic leak ?

- a) The remaining systems will take over control.
- b) By switching to manual back-up mode.
- c) By switching the flight control system to the reversible mode.
- d) Sufficient reserve hydraulic fluid is available to compensatethe effects of the leak.

**Answer ;**

- a) The remaining systems will take over control.

---

**0314.** Given the following information about an aeroplane with a turbojet engine ;

- mass air flow 50 kg/s
- TAS 90 m/s
- exhaust nozzle gas velocity 150 m/s
- exhaust nozzle static pressure 1050 hPa
- ambient static pressure 1000 hPa
- cross-sectional area of the nozzle 0.10 m.

Net engine thrust equals ;

- a) 4000 N.
- b) 3500 N.
- c) 2500 N.
- d) 3000 N.

**Answer ;**

- b) 3500 N.

**Çözüm ;**

---

**0315.** Given the following statements about diesel engines ;

- 1. power is regulated by the throttle valve.
- 2. there is no throttle valve.
- 3. power is set by the fuel flow.
- 4. thermal efficiency is lower than that for a petrol engine.
- 5. diesel fuel is less inflammable than petrol.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3.
- b) 2 , 3 , 5.
- c) 1 , 3 , 5.
- d) 2 , 4 , 5.

**Answer ;**

- b) 2 , 3 , 5.

---

**0316.** Given the following statements about diesel engines ;

- 1. power is set by the mixture control.
- 2. there is no mixture control.
- 3. the amount of power is determined by the fuel flow only.
- 4. thermal efficiency is higher than that for a petrol engine.
- 5. diesel fuel is more inflammable than petrol.

The combination that regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 2 , 4 , 5.
- c) 1 , 3 , 5.
- d) 1 , 2 , 3.

**Answer ;**

- a) 2 , 3 , 4.

---

**0317.** Halon is used as a fire extinguishing agent because it ;

- a) is an electrical conductor.
- b) uses the cooling effect created by the venturi during discharge.
- c) is highly volatile.
- d) acts as a very effective flame inhibitor.

**Answer ;**

- d) acts as a very effective flame inhibitor.

0318. Hydraulic fluid , entering the hydraulic pump , is slightly pressurised to ;
- a) ensure sufficient pump output.
  - b) prevent overheating of the pump.
  - c) prevent vapour locking.
  - d) prevent cavitation in the pump.

Answer ;

d) prevent cavitation in the pump.

---

0319. Hydraulic fluids ;
- a) do not require special care.
  - b) cause high fire risk.
  - c) are irritating to eyes and skin and cause high fire risk.
  - d) are irritating to eyes and skin.

Answer ;

d) are irritating to eyes and skin.

---

0320. Hydraulic fluids must have the following characteristics ;
- 1. thermal stability.
  - 2. low emulsifying characteristics.
  - 3. corrosion resistant.
  - 4. good resistance to combustion.
  - 5. high compressibility.
  - 6. high volatility.
  - 7. high viscosity.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 5 , 7.
- b) 1 , 2 , 3 , 4.
- c) 2 , 3 , 4 , 5.
- d) 1 , 3 , 4 , 6.

Answer ;

b) 1 , 2 , 3 , 4.

---

0321. Hydraulic fluids used in systems of large modern airliners are ;
- a) Phosphate ester base fluids.
  - b) Mineral base fluids.
  - c) Water base fluids.
  - d) Vegetable base fluids.

Answer ;

a) Phosphate ester base fluids.

---

0322. Hydraulic power is a function of ;
- a) pump RPM only.
  - b) system pressure and volume flow.
  - c) system pressure and tank capacity.
  - d) pump size and volume flow.

Answer ;

b) system pressure and volume flow.

---

0323. Ice formation on turbofan engine intakes is usually ;
- a) prevented by using compressor bleed air.
  - b) prevented by using turbine bleed air.
  - c) removed by using electrical heating.
  - d) removed by using pneumatic boots

Answer ;

a) prevented by using compressor bleed air.

---

0324. If a current is passed through a conductor which is positioned in a magnetic field ;
- a) a force will be exerted on the conductor.
  - b) the current will increase.
  - c) the intensity of the magnetic field will decrease.
  - d) there will be no effect unless the conductor is moved.

Answer ;

a) a force will be exerted on the conductor.

---

**0325.** If a gas turbine engine fuel/oil heat exchanger is located downstream of the HP fuel pump , internal leakage of the heat exchanger will cause the oil level to ;

- a) rise and the fuel nozzles eventually to become clogged.
- b) fall and the oil jets eventually to become clogged.
- c) fall.
- d) rise.

**Answer ;**

**d) rise.**

---

**0326.** If a twin spool gas turbine engine is controlled by a conventional N2 control system , rupture of the LP shaft may result in ;

- a) N2 and EPR increase.
- b) overspeed of the LP turbine.
- c) N1 and N2 increase.
- d) N1 decrease and EPR increase.

**Answer ;**

**b) overspeed of the LP turbine.**

---

**0327.** If air is tapped from a gas turbine HP compressor ( engine N2 controlled ) , the effect on the engine pressure ratio ( EPR ) and the exhaust gas temperature ( EGT ) is that ;

- a) EPR decreases and EGT remains constant.
- b) EPR remains constant and EGT increases.
- c) both EPR and EGT decrease.
- d) EPR decreases and EGT increases.

**Answer ;**

**d) EPR decreases and EGT increases.**

---

**0328.** If an abortive start ( hung start ) occurs when starting a gas turbine engine , the correct action is to ;

- a) move the thrust lever forward to increase fuel flow.
- b) select the other igniter system.
- c) activate both igniter systems.
- d) shut down the engine.

**Answer ;**

**d) shut down the engine.**

---

**0329.** If an engine detonates during climb-out , the normal corrective action would be to ;

- a) retard the throttle.
- b) lean the mixture.
- c) increase the rate of climb.
- d) apply carburettor heat.

**Answer ;**

**a) retard the throttle.**

---

**0330.** If an engine fails to stop with the magneto switch in OFF position , the cause may be ;

- a) defective condenser.
- b) switch wire grounded.
- c) excessive carbon formation in cylinder head.
- d) fouled spark plugs.

**Answer ;**

**c) excessive carbon formation in cylinder head.**

---

**0331.** If engine compressor air is bled off for engine anti-icing or a similar system , the turbine temperature ;

- a) will rise.
- b) may rise or fall depending on which stage of the compressor is used for the bleed and the rpm of the engine at the moment of selection.
- c) will fall.
- d) will be unchanged.

**Answer ;**

**a) will rise.**

---



0332. If one of the 12 cells of a lead-acid battery is dead , the battery ;
- a) has 1/12 less voltage, but can still be used.
  - b) has 1/12 less voltage and less capacity, but can still be used.
  - c) is unserviceable.
  - d) has 1/12 less capacity , but can still be used.

Answer ;

c) is unserviceable.

---

0333. If the cabin altitude rises ( aircraft in level flight ) , the differential pressure ;
- a) decreases.
  - b) increases.
  - c) may exceed the maximum permitted differential unless immediate preventative action is taken.
  - d) remains constant.

Answer ;

a) decreases.

---

0334. If the electrical load of an AC generator providing a constant output voltage increases , the voltage regulator will ;
- a) decrease the intensity of the excitation current.
  - b) maintain the intensity of the excitation current constant.
  - c) increase the intensity of the excitation current.
  - d) change the direction of the excitation current.

Answer ;

c) increase the intensity of the excitation current.

---

0335. If the engine , with a fixed pitch propeller , is detonating during climb-out after take-off , the normal corrective action would be to ;
- a) reduce the throttle.
  - b) lean the mixture.
  - c) increase the rate of climb.
  - d) apply carburettor heat.

Answer ;

a) reduce the throttle.

---

0336. If the ground wire between the magneto and the ignition switch becomes disconnected , the most noticeable result will be that the engine ;
- a) will not operate at the left magneto.
  - b) cannot be started with the switch in the ON position.
  - c) will not operate at the right magneto.
  - d) cannot be shut down by turning the switch to the OFF position.

Answer ;

d) cannot be shut down by turning the switch to the OFF position.

---

0337. If the ground wire between the magnetos and the ignition switch becomes disconnected , the most noticeable result will be that ;
- a) the engine cannot be shut down by turning the ignition switch to the “OFF” position.
  - b) the engine cannot be started with the ignition switch in the “ON” position.
  - c) a still operating engine will run down.
  - d) the power developed by the engine will be strongly reduced.

Answer ;

a) the engine cannot be shut down by turning the ignition switch to the “OFF” position.

---

0338. If the main oil filter of a gas turbine engine becomes clogged , oil to the main bearings is still assured ;
- a) if breather pressure in the main compartments is increased to ensure positive lubrication of the bearings.
  - b) if a higher engine rpm is selected to increase oil pressure.
  - c) because fuel upstream of the oil filter will be mixed with oil to allow easier passage through the filter.
  - d) because the oil filter bypass valve opens.

Answer ;

d) because the oil filter bypass valve opens.

---

0339. If the manifold pressure is increased , the blade angle of a constant speed propeller ;
- a) will increase and after a short time will reduce.
  - b) will increase.
  - c) will remain the same.
  - d) will reduce so that the engine RPM can increase.

Answer ;

b) will increase.

---

0340. If the maximum operating altitude of an aeroplane is limited by the pressurized cabin, this limitation is due to the maximum ;
- a) positive cabin differential pressure at maximum operating ceiling.
  - b) negative cabin differential pressure at maximum operating ceiling.
  - c) positive cabin differential pressure at maximum cabin altitude.
  - d) negative differential pressure at maximum cabin altitude.

Answer ;

c) positive cabin differential pressure at maximum cabin altitude.

---

0341. If the maximum operating pressure of the oxygen system is exceeded the ;
- a) oxygen bottles will explode.
  - b) oxygen is discharged overboard via a safety plug.
  - c) oxygen becomes unusable for the passengers.
  - d) passenger oxygen masks will drop down.

Answer ;

b) oxygen is discharged overboard via a safety plug.

---

0342. If the pressure in the cabin tends to become lower than the outside ambient air pressure the ;
- a) negative pressure relief valve will close.
  - b) outflow valve open completely.
  - c) air cycle machine will stop.
  - d) negative pressure relief valve will open.

Answer ;

d) negative pressure relief valve will open.

---

0343. If there is a single input data failure on a FADEC gas turbine engine ;
- a) the engine stops.
  - b) the engine continues to operate normally.
  - c) the engine is set at a fail-safe thrust.
  - d) the electronic engine control unit uses a constant data value for computation.

Answer ;

b) the engine continues to operate normally.

---

0344. If two constant frequency AC generators are operating independently , then the phase relation between both generators ;
- a) must be synchronised.
  - b) must be 120° out of phase.
  - c) must be 240° out of phase.
  - d) is unimportant.

Answer ;

d) is unimportant.

---

0345. If , when the magneto selector switch is set to the OFF position , a piston engine continues to run normally , the most probable cause is that ;

- a) on a magneto , the grounding wire is broken.
- b) a wire from the magneto is in contact with a metallic part of the engine.
- c) there are local hot points in the engine ( probably due to overheating of the cylinder heads ).
- d) there is a carbon deposit on the spark plugs electrodes.

Answer ;

a) on a magneto , the grounding wire is broken.

---

0346. If , when the magneto selector switch is set to the OFF position , a piston engine continues to run normally , the most probable cause is that ;

- a) there are local hot points in the engine
- b) a grounding wire of one of the magnetos is in contact with a metallic part of the engine.
- c) the grounding wire of one of the magnetos is broken.
- d) there is a carbon deposit on the spark plugs.

Answer ;

c) the grounding wire of one of the magnetos is broken.

---

0347. Ignition occurs in each cylinder of a four stroke engine ( TDC = Top Dead Centre ) ;
- a) before TDC at each crankshaft revolution.
  - b) before TDC at each second crankshaft revolution.
  - c) behind TDC at each second crankshaft revolution.
  - d) behind TDC at each crankshaft revolution.

Answer ;

b) before TDC at each second crankshaft revolution.

---

0348. Ignition systems of a running piston engine receive electrical energy from ;
- a) capacitors.
  - b) rotating permanent magnets.
  - c) generators.
  - d) batteries.

Answer ;

b) rotating permanent magnets.

---

0349. Immediately after starting engine(s) with no other electrical services switched on , an ammeter showing a high charge rate to the battery ;
- a) indicates a generator failure , thus requiring the engine to be shut down immediately.
  - b) indicates a battery failure since there should be no immediate charge.
  - c) would be normal and is only cause for concern if the high charge rate persists.
  - d) indicates a faulty reverse current relay.

Answer ;

c) would be normal and is only cause for concern if the high charge rate persists.

---

0350. In a “bootstrap” cooling system , the charge air is first compressed in the cold air unit to ;
- a) ensure an adequate pressure and temperature drop across the cooling turbine.
  - b) ensure an adequate charge air flow across the inter-cooler heat exchanger.
  - c) maintain a constant cabin mass air flow.
  - d) increase the cabin air supply pressure when the charge pressure is too low.

Answer ;

a) ensure an adequate pressure and temperature drop across the cooling turbine.

---

0351. In a bootstrap cooling system the supply air is first ;
- a) used to increase the cabin air supply pressure when the charge pressure is too low.
  - b) compressed , passed through a secondary heat exchanger , and then across an expansion turbine.
  - c) passed across an expansion turbine , then compressed and passed through a secondary heat exchanger.
  - d) passed across an expansion turbine , then directly to the heat exchanger.

Answer ;

b) compressed , passed through a secondary heat exchanger , and then across an expansion turbine.

---

0352. In a cabin air conditioning system , equipped with a bootstrap , the mass air flow is routed via the ;
- a) secondary heat exchanger outlet to the turbine inlet of the cold air unit.
  - b) turbine outlet of the cold air unit to the primary heat exchanger inlet.
  - c) compressor outlet of the cold air unit to the primary heat exchanger inlet.
  - d) secondary heat exchanger outlet to the compressor inlet of the cold air unit.

Answer ;

a) secondary heat exchanger outlet to the turbine inlet of the cold air unit.

---

0353. In a commercial transport aircraft the landing gear operating system is usually ;
- a) Electrically driven.
  - b) Hydraulically driven.
  - c) Mechanically driven.
  - d) Pneumatically driven.

Answer ;

b) Hydraulically driven.

---

0354. In a compensated capacitance type quantity indicating system , the contents gauge of a half-full fuel tank indicates a fuel mass of 8000 kg. If a temperature rise increased the volume of fuel by 5% , the indicated fuel mass would ;
- a) remain the same.
  - b) increase by 5%.
  - c) decrease by 5%.
  - d) increase by 10%.

Answer ;

a) remain the same.

---

0355. In a compressor stage , if the axial air velocity increases more than the rotor blade velocity the ;
- a) pressure ratio will be too high.
  - b) stall margin decreases.
  - c) relative air velocity becomes supersonic.
  - d) stall margin increases.

Answer ;

d) stall margin increases.

---

0356. In a compressor stage of a jet engine , the sequence is ;

- a) stator - rotor
- b) rotor - stator
- c) stator - stator - rotor
- d) rotor - rotor - stator

Answer ;

b) rotor - stator

---

0357. In a fire detection system with single-loop continuous components ( with no fault protection ) , if the line is accidentally grounded ;

- a) the engine fire extinguisher striker is automatically activated.
- b) the fire alarm is triggered.
- c) the power supply is cut off automatically.
- d) there will be no effect on the system.

Answer ;

b) the fire alarm is triggered.

---

0358. In a four-stroke piston engine , the only “driving” stroke is ;

- a) induction.
- b) exhaust.
- c) power.
- d) compression.

Answer ;

c) power.

---

0359. In a free turbine engine ;

- a) there is no mechanical connection between the compressor and the power output shaft.
- b) the air enters the compressor via the input turbine.
- c) its shaft may be connected to either a compressor or another turbine.
- d) the compressor and power output shaft are mechanically connected.

Answer ;

a) there is no mechanical connection between the compressor and the power output shaft.

---

0360. In a free turbine engine , the energy delivered by the free turbine stages ;

- a) is not connected with the rotation speed of the generator.
- b) is dependent on the rotational speed of the gas generator.
- c) is used to drive the compressor only.
- d) is used to drive the compressor and the main rotor gearbox.

Answer ;

b) is dependent on the rotational speed of the gas generator.

---

0361. In a gas turbine combustion chamber ;

- a) the gas temperature is highest at the chamber outlet.
- b) secondary air is required for cooling of the inner casing.
- c) the gas pressure increases in an axial ( downstream ) direction.
- d) the axial velocity of the gas stream is greatest behind the swirl vanes.

Answer ;

b) secondary air is required for cooling of the inner casing.

---

0362. In a gas turbine engine , compressor blades , which are not rigidly fixed in position when the engine is stationary , take up a rigid position when the engine is running due to ;

- a) the resultant of aerodynamic and centrifugal forces.
- b) thermal expansion.
- c) oil pressure.
- d) blade creep.

Answer ;

a) the resultant of aerodynamic and centrifugal forces.

---

0363. In a gas turbine engine lubrication system , the oil to fuel heat exchanger provides ;
- a) automatic controlled fuel heating by engine oil to prevent icing in the fuel filter.
  - b) fuel cooling to prevent vapour locking interrupting the fuel supply to the nozzles.
  - c) fuel heating as required whenever fuel filter clogging is detected.
  - d) oil cooling through thermal exchange with the fuel.

Answer ;

- d) oil cooling through thermal exchange with the fuel.
- 

0364. In a gas turbine engine the creep phenomenon is the most severe in the ;
- a) combustor liner.
  - b) turbine rotors.
  - c) compressor blades.
  - d) combustor casing.

Answer ;

- b) turbine rotors.
- 

0365. In a gas turbine engine , the maximum gas temperature is attained ;
- a) in the cooling airflow around the flame tube.
  - b) across the turbine.
  - c) within the combustion chamber.
  - d) at the entry to the exhaust unit.

Answer ;

- c) within the combustion chamber.
- 

0366. In a gas turbine engine , the power changes are normally made by controlling the amount of ;
- a) air entering the compressor and fuel entering the combustion chambers.
  - b) fuel supplied.
  - c) air entering the compressor.
  - d) air leaving the compressor by the opening or closing of bleed valves.

Answer ;

- b) fuel supplied.
- 

0367. In a gas turbine engine , the propelling jet is generated by ;
- a) expansion of hot gases resulting in a conversion of potential energy ( pressure ) into ( mechanical ) work.
  - b) compression of hot gases resulting in a conversion of potential energy ( pressure ) into kinetic energy ( velocity ).
  - c) compression of hot gases resulting in a conversion of kinetic energy ( pressure ) into ( mechanical ) work.
  - d) expansion of hot gases resulting in a conversion of potential energy ( pressure ) into kinetic energy ( velocity ).

Answer ;

- d) expansion of hot gases resulting in a conversion of potential energy ( pressure ) into kinetic energy ( velocity ).
- 

0368. In a hydraulic braking system , an accumulator is pre-charged to 1200 psi. An electrically driven hydraulic pump is started and provides a system pressure of 3000 psi. The hydraulic pressure gauge which is connected to the gas section of the accumulator , reads ;

- a) 4200 psi.
- b) 3000 psi.
- c) 1200 psi.
- d) 1800 psi.

Answer ;

- b) 3000 psi.
- 

0369. In a hydraulic system overheat detectors are mostly installed ;
- a) at the pumps.
  - b) at the actuators.
  - c) in the reservoirs.
  - d) at the coolers.

Answer ;

- a) at the pumps.
-

0370. In a hydraulic system , the reservoir is pressurized in order to ;

- a) reduce fluid combustibility.
- b) seal the system.
- c) keep the hydraulic fluid at optimum temperature.
- d) prevent pump cavitation.

Answer ;

d) prevent pump cavitation.

---

0371. In a jet engine the propelling nozzle ;

- a) must be designed to obtain the correct balance of pressure and velocity to prevent flow separation.
- b) directs the hot gas stream onto the turbine at the optimum angle of attack.
- c) decreases the velocity and increases the pressure in the jet pipe to provide greater thrust.
- d) reduces noise.

Answer ;

a) must be designed to obtain the correct balance of pressure and velocity to prevent flow separation.

---

0372. In a modern hydraulic system , “hydraulic fuses” can be found. Their function is ;

- a) to isolate a part of the system and protect it against accidental pollution.
- b) to switch to the secondary system in case of a leak in the primary brake system.
- c) to allow by-passing of a hydraulic pump in case it is subject to excessive pressure , without further damage to the system.
- d) to prevent total system loss in case of a leaking hydraulic line.

Answer ;

d) to prevent total system loss in case of a leaking hydraulic line.

---

0373. In a multi-spool turbofan engine , the fan is driven by ;

- a) the rearmost turbine.
- b) the intermediate turbine.
- c) all three turbines since they are on a common shaft with the compressor.
- d) the foremost turbine.

Answer ;

a) the rearmost turbine.

---

0374. In a piston engine if the ratio of air to fuel , by weight , is approximately 9:1 , the mixture is said to be ;

- a) normal
- b) too weak to support combustion
- c) weak
- d) rich

Answer ;

d) rich

---

0375. In a piston engine , magnetos are used to produce the spark which ignites the fuel/air mixture. The operating principle of magnetos consists in ;

- a) breaking the primary current in order to induce a low amp high volt current which is distributed to the spark plugs.
- b) accumulating in a condenser a low volt current from the battery , reconstitute it as high voltage current at the moment the spark is generated.
- c) obtaining a high amp low volt current in order to generate the spark.
- d) creating a brief high intensity magnetic field which will be sent through the distributor at the appropriate time.

Answer ;

a) breaking the primary current in order to induce a low amp high volt current which is distributed to the spark plugs.

---

0376. In a piston engine , the purpose of an altitude mixture control is to ;

- a) prevent a weak cut when the throttle is opened rapidly at altitude.
- b) correct for variations in the fuel/air ratio due to decreased air density at altitude.
- c) weaken the mixture strength because of reduced exhaust back pressure at altitude.
- d) enrich the mixture strength due to decreased air density at altitude.

Answer ;

b) correct for variations in the fuel/air ratio due to decreased air density at altitude.

---

0377. In a piston engine , turbocharger boost pressure may be monitored by ;
- a) a manifold pressure gauge only.
  - b) a cylinder head temperature gauge ( CHT ) , a manifold pressure gauge , and engine rpm readings.
  - c) both a CHT gauge and manifold pressure gauge.
  - d) both engine rpm readings and a manifold pressure gauge.

Answer ;

a) a manifold pressure gauge only.

---

**B. 747 TU**  
**AIR FRANCE DO.NT**

**Conduif du vol**

**Performances generalites**

**62. 20. 05**  
**FEV. 84**

**ATMOSPHERE STANDARD**

| ALTITUDE<br>Pieds | TEMPERATURE   |        | PRESSION    |       |          |          | RAPPORT<br>de<br>PRESSION<br>$\delta = P / P_0$ | IDENSITE<br>RELATIVE<br>$\delta = \rho / \rho_0$ | $\sqrt{\text{DENSITE}}$ | VITESSE<br>du<br>SON (a)<br>kt | ALTITUDE<br>Metres |
|-------------------|---------------|--------|-------------|-------|----------|----------|-------------------------------------------------|--------------------------------------------------|-------------------------|--------------------------------|--------------------|
|                   | °C            | °F     | mb          | P S I | In<br>Hg | mm<br>Hg |                                                 |                                                  |                         |                                |                    |
| <b>45.000</b>     | <b>- 56,5</b> | - 69,7 | <b>147</b>  | 2,14  | 4,36     | 110,7    | 0,1415                                          | 0,1936                                           | 0,440                   | 574                            | <b>13. 716</b>     |
| <b>44.000</b>     | <b>- 56,5</b> | - 69,7 | <b>155</b>  | 2,24  | 4,57     | 116,0    | 0,1527                                          | 0,2031                                           | 0,451                   | 574                            | <b>13. 411</b>     |
| <b>43.000</b>     | <b>- 56,5</b> | - 69,7 | <b>162</b>  | 2,35  | 4,79     | 121,7    | 0,1602                                          | 0,2131                                           | 0,462                   | 574                            | <b>13. 106</b>     |
| <b>42.000</b>     | <b>- 56,5</b> | - 69,7 | <b>170</b>  | 2,47  | 5,03     | 127,8    | 0,1681                                          | 0,2236                                           | 0,473                   | 574                            | <b>12. 802</b>     |
| <b>41.000</b>     | <b>- 56,5</b> | - 69,7 | <b>179</b>  | 2,19  | 5,28     | 134,1    | 0,1764                                          | 0,2346                                           | 0,484                   | 574                            | <b>12. 497</b>     |
| <b>40.000</b>     | <b>- 56,5</b> | - 69,7 | <b>188</b>  | 2,72  | 5,54     | 140,7    | 0,1851                                          | 0,2462                                           | 0,496                   | 574                            | <b>12. 192</b>     |
| <b>39.000</b>     | <b>- 56,5</b> | - 69,7 | <b>197</b>  | 2,81  | 5,81     | 147,6    | 0,1942                                          | 0,2583                                           | 0,508                   | 574                            | <b>11. 887</b>     |
| <b>38.000</b>     | <b>- 56,5</b> | - 69,7 | <b>206</b>  | 2,99  | 6,10     | 154,9    | 0,2018                                          | 0,2710                                           | 0,521                   | 574                            | <b>11. 582</b>     |
| <b>37.000</b>     | <b>- 56,5</b> | - 69,7 | <b>217</b>  | 3,14  | 6,40     | 162,6    | 0,2138                                          | 0,2843                                           | 0,533                   | 574                            | <b>11. 278</b>     |
| <b>36.000</b>     | <b>- 56,3</b> | - 69,4 | <b>227</b>  | 3,30  | 6,71     | 170,4    | 0,2243                                          | 0,2981                                           | 0,546                   | 574                            | <b>10. 973</b>     |
| <b>35.000</b>     | <b>- 54,3</b> | - 65,8 | <b>238</b>  | 3,46  | 7,04     | 178,8    | 0,2353                                          | 0,3099                                           | 0,557                   | 576                            | <b>10. 668</b>     |
| <b>34.000</b>     | <b>- 52,4</b> | - 62,3 | <b>250</b>  | 3,63  | 7,38     | 187,5    | 0,2467                                          | 0,3220                                           | 0,567                   | 579                            | <b>10. 363</b>     |
| <b>33.000</b>     | <b>- 50,4</b> | - 58,7 | <b>262</b>  | 3,80  | 7,74     | 196,6    | 0,2586                                          | 0,3345                                           | 0,578                   | 582                            | <b>10. 058</b>     |
| <b>32.000</b>     | <b>- 48,4</b> | - 55,1 | <b>274</b>  | 3,98  | 8,11     | 206,0    | 0,2709                                          | 0,3473                                           | 0,589                   | 584                            | <b>9. 754</b>      |
| <b>31.000</b>     | <b>- 46,4</b> | - 51,6 | <b>287</b>  | 4,17  | 8,49     | 215,6    | 0,2837                                          | 0,3605                                           | 0,600                   | 587                            | <b>9. 449</b>      |
| <b>30.000</b>     | <b>- 44,4</b> | - 48,0 | <b>301</b>  | 4,36  | 8,89     | 225,8    | 0,2970                                          | 0,3741                                           | 0,611                   | 589                            | <b>9. 144</b>      |
| <b>29.000</b>     | <b>- 42,5</b> | - 44,4 | <b>315</b>  | 4,57  | 9,30     | 236,2    | 0,3107                                          | 0,3881                                           | 0,623                   | 591                            | <b>8. 839</b>      |
| <b>28.000</b>     | <b>- 40,5</b> | - 40,9 | <b>329</b>  | 4,78  | 9,73     | 247,1    | 0,3250                                          | 0,4025                                           | 0,634                   | 594                            | <b>8. 534</b>      |
| <b>27.000</b>     | <b>- 38,5</b> | - 37,3 | <b>344</b>  | 4,99  | 10,17    | 258,3    | 0,3398                                          | 0,4173                                           | 0,646                   | 597                            | <b>8. 230</b>      |
| <b>26.000</b>     | <b>- 36,5</b> | - 33,7 | <b>360</b>  | 5,22  | 10,63    | 270,0    | 0,3552                                          | 0,4325                                           | 0,658                   | 599                            | <b>7. 925</b>      |
| <b>25.000</b>     | <b>- 34,5</b> | - 30,2 | <b>376</b>  | 5,45  | 11,10    | 281,9    | 0,3711                                          | 0,4481                                           | 0,669                   | 602                            | <b>7. 620</b>      |
| <b>24.000</b>     | <b>- 32,5</b> | - 26,6 | <b>393</b>  | 5,70  | 11,60    | 294,6    | 0,3876                                          | 0,4642                                           | 0,681                   | 604                            | <b>7. 315</b>      |
| <b>23.000</b>     | <b>- 30,6</b> | - 23,0 | <b>410</b>  | 5,95  | 12,11    | 307,6    | 0,4047                                          | 0,4806                                           | 0,693                   | 607                            | <b>7. 010</b>      |
| <b>22.000</b>     | <b>- 28,6</b> | - 19,5 | <b>428</b>  | 6,21  | 12,64    | 321,1    | 0,4223                                          | 0,4976                                           | 0,705                   | 609                            | <b>6. 706</b>      |
| <b>21.000</b>     | <b>- 26,6</b> | - 15,9 | <b>446</b>  | 6,47  | 13,18    | 334,8    | 0,4406                                          | 0,5150                                           | 0,718                   | 612                            | <b>6. 401</b>      |
| <b>20.000</b>     | <b>- 24,6</b> | - 12,3 | <b>466</b>  | 6,75  | 13,75    | 349,3    | 0,4596                                          | 0,5328                                           | 0,730                   | 614                            | <b>6. 096</b>      |
| <b>19.000</b>     | <b>- 22,6</b> | - 8,8  | <b>485</b>  | 7,04  | 14,34    | 364,2    | 0,4791                                          | 0,5511                                           | 0,742                   | 617                            | <b>5. 791</b>      |
| <b>18.000</b>     | <b>- 20,7</b> | - 5,2  | <b>506</b>  | 7,34  | 14,94    | 379,5    | 0,4994                                          | 0,5699                                           | 0,755                   | 619                            | <b>5. 486</b>      |
| <b>17.000</b>     | <b>- 18,7</b> | - 1,6  | <b>527</b>  | 7,65  | 15,57    | 395,5    | 0,5203                                          | 0,5892                                           | 0,768                   | 622                            | <b>5. 182</b>      |
| <b>16.000</b>     | <b>- 16,7</b> | + 1,9  | <b>549</b>  | 7,97  | 16,22    | 412,0    | 0,5420                                          | 0,6089                                           | 0,780                   | 624                            | <b>4. 877</b>      |
| <b>15.000</b>     | <b>- 14,7</b> | + 5,5  | <b>572</b>  | 8,29  | 16,89    | 429,0    | 0,5644                                          | 0,6292                                           | 0,793                   | 626                            | <b>4. 572</b>      |
| <b>14.000</b>     | <b>- 12,7</b> | + 9,1  | <b>595</b>  | 8,63  | 17,58    | 446,1    | 0,5875                                          | 0,6500                                           | 0,806                   | 629                            | <b>4. 267</b>      |
| <b>13.000</b>     | <b>- 10,7</b> | + 12,6 | <b>619</b>  | 8,99  | 18,29    | 464,6    | 0,6113                                          | 0,6713                                           | 0,819                   | 631                            | <b>3. 962</b>      |
| <b>12.000</b>     | <b>- 8,8</b>  | + 16,2 | <b>644</b>  | 9,35  | 19,03    | 483,4    | 0,6360                                          | 0,6932                                           | 0,833                   | 634                            | <b>3. 658</b>      |
| <b>11.000</b>     | <b>- 6,8</b>  | + 19,8 | <b>670</b>  | 9,72  | 19,79    | 502,7    | 0,6614                                          | 0,7155                                           | 0,846                   | 636                            | <b>3. 353</b>      |
| <b>10.000</b>     | <b>- 4,8</b>  | + 23,3 | <b>697</b>  | 10,11 | 20,58    | 522,7    | 0,6877                                          | 0,7385                                           | 0,859                   | 638                            | <b>3. 048</b>      |
| <b>9.000</b>      | <b>- 2,8</b>  | + 26,9 | <b>724</b>  | 10,50 | 21,39    | 543,3    | 0,7148                                          | 0,7619                                           | 0,873                   | 641                            | <b>2. 743</b>      |
| <b>8.000</b>      | <b>- 0,8</b>  | + 30,5 | <b>753</b>  | 10,92 | 22,23    | 564,6    | 0,7428                                          | 0,7860                                           | 0,887                   | 643                            | <b>2. 438</b>      |
| <b>7.000</b>      | <b>+ 1,1</b>  | + 34,0 | <b>782</b>  | 11,34 | 23,09    | 586,5    | 0,7716                                          | 0,8106                                           | 0,900                   | 645                            | <b>2. 134</b>      |
| <b>6.000</b>      | <b>+ 3,1</b>  | + 37,6 | <b>812</b>  | 11,78 | 23,98    | 609,1    | 0,8014                                          | 0,8358                                           | 0,914                   | 648                            | <b>1. 829</b>      |
| <b>5.000</b>      | <b>+ 5,1</b>  | + 41,2 | <b>843</b>  | 12,23 | 24,90    | 632,5    | 0,8321                                          | 0,8616                                           | 0,928                   | 650                            | <b>1. 524</b>      |
| <b>4.000</b>      | <b>+ 7,1</b>  | + 44,7 | <b>875</b>  | 12,69 | 25,84    | 656,3    | 0,8637                                          | 0,8881                                           | 0,942                   | 652                            | <b>1. 219</b>      |
| <b>3.000</b>      | <b>+ 9,1</b>  | + 48,3 | <b>908</b>  | 13,17 | 26,82    | 681,2    | 0,8962                                          | 0,9151                                           | 0,957                   | 655                            | <b>914</b>         |
| <b>2.000</b>      | <b>+ 11,0</b> | + 51,9 | <b>942</b>  | 13,66 | 27,82    | 706,6    | 0,9298                                          | 0,9427                                           | 0,971                   | 657                            | <b>610</b>         |
| <b>1.000</b>      | <b>+ 13,0</b> | + 55,4 | <b>977</b>  | 14,17 | 28,86    | 733,0    | 0,9644                                          | 0,9710                                           | 0,985                   | 659                            | <b>305</b>         |
| <b>0</b>          | <b>+ 15,0</b> | + 59,0 | <b>1013</b> | 14,70 | 29,92    | 760,0    | 1,0000                                          | 1,0000                                           | 1,000                   | 661                            | <b>0</b>           |
| <b>1.000</b>      | <b>+ 17,0</b> | + 62,5 | <b>1050</b> | 15,23 | 31,02    | 787,9    | 1,0366                                          | 1,0295                                           | 1,015                   | 664                            | <b>-305</b>        |

**Annex 021-786A**

In a pressurized aircraft whose cabin altitude is 8000 ft , a crack in a cabin window makes it necessary to reduce the differential pressure to 5 psi. The flight level to be maintained in order to keep the same cabin altitude is ;

- a) FL 340
- b) FL 230
- c) FL 180
- d) FL 280

Answer ;

**b) FL 230**



0379. In a pressurized transport aircraft , the protective breathing equipment ;
- a) protects all the occupants against the effects of accidental depressurization.
  - b) gives medical assistance to certain passengers with respiratory disorders.
  - c) protects the members of the crew against smoke and noxious gases.
  - d) protects the members of the crew against the effects of accidental depressurization.

Answer ;

- c) protects the members of the crew against smoke and noxious gases.
- 

0380. In a reciprocating four-stroke engine , the only “driving” stroke is the ;

- a) induction stroke.
- b) compression stroke.
- c) power stroke.
- d) exhaust stroke.

Answer ;

- c) power stroke.
- 

0381. In a single spool gas turbine engine , the compressor rpm is ;

- a) less than turbine rpm.
- b) the same as turbine rpm.
- c) greater than turbine rpm.
- d) independent of turbine rpm.

Answer ;

- b) the same as turbine rpm.
- 

0382. In a subsonic convergent duct the ;

- a) pressure increases , velocity decreases and temperature decreases.
- b) pressure increases , velocity decreases and temperature increases.
- c) temperature increases , velocity increases and pressure decreases.
- d) velocity increases , pressure decreases and temperature decreases.

Answer ;

- d) velocity increases , pressure decreases and temperature decreases.
- 

0383. In a theoretical gas turbine cycle , combustion takes place at constant ;

- a) energy.
- b) volume.
- c) temperature.
- d) pressure.

Answer ;

- d) pressure.
- 

0384. In a turbo compressor air conditioning system ( bootstrap system ) , the main water separation unit is ;

- a) before the cooling turbine.
- b) after the cooling turbine.
- c) just after the heat exchangers.
- d) before the heat exchangers.

Answer ;

- b) after the cooling turbine.
- 

0385. In a turbo compressor air conditioning system ( bootstrap system ) , the purpose of the heat exchangers is to ;

- a) cool the bleed air in front and behind the compressor of the air cycle machine.
- b) allow a homogeneous temperature by mixing air flows from various air conditioning groups in operation.
- c) allow a steady compressor outlet temperature.
- d) cool bleed air before entering the complete pneumatic system.

Answer ;

- a) cool the bleed air in front and behind the compressor of the air cycle machine.
- 

0386. In a turbo compressor air conditioning system ( bootstrap system ) , the supply of air behind the primary heat exchanger is ;

- a) passed across an expansion turbine , then directly to the heat exchanger.
- b) compressed , passed through a heat exchanger , and then across an expansion turbine.
- c) compressed , then passed across an expansion turbine and finally across a heat exchanger.
- d) passed across an expansion turbine , compressed and then passed through a heat exchanger.

Answer ;

- b) compressed , passed through a heat exchanger , and then across an expansion turbine.
-

0387. In a turbo-jet , the purpose of the turbine is to ;
- a) drive the compressor by using part of the energy from the exhaust gases.
  - b) drive devices like pumps , regulator, generator.
  - c) compress the air in order to provide a better charge of the combustion chamber.
  - d) clear the burnt gases , the expansion of which provide the thrust.

Answer ;

- a) drive the compressor by using part of the energy from the exhaust gases.
- 

0388. In a typical bootstrap cooling system the supply air is ;
- a) compressed , passed through a secondary heat exchanger , and then across an expansion turbine.
  - b) passed across an expansion turbine , then through a secondary heat exchanger and then across a compressor.
  - c) passed across an expansion turbine , then compressed and passed through a secondary heat exchanger.
  - d) passed across an expansion turbine , then directly to the heat exchanger.

Answer ;

- a) compressed , passed through a secondary heat exchanger , and then across an expansion turbine.
- 

0389. In accordance with JAR-OPS , 100 % of the passengers in a non-pressurised airplane must have an oxygen supply reserve for the entire flight time when the cabin altitude pressure is greater than ;

- a) 13.000 ft.
- b) 10.000 ft but not exceeding 13.000 ft.
- c) 10.000 ft.
- d) 10.000 ft but not exceeding 13.000 ft minus 30 minutes.

Answer ;

- d) 10.000 ft but not exceeding 13.000 ft minus 30 minutes.
- 

0390. In accordance with JAR-OPS , 100 % of the passengers in a non-pressurised airplane must have an oxygen supply reserve for the entire flight time at pressure altitudes greater than ;

- a) 15.000 ft.
- b) 10.000 ft.
- c) 14.000 ft.
- d) 13.000 ft.

Answer ;

- d) 13.000 ft.
- 

0391. In addition to providing “customer” bleed for aeroplane systems , air may be bled from compressors of a turbofan engine to increase stall margin ;

- a) at high rotor speed.
- b) and recover from stall at high rotor speed.
- c) at low rotor speed.
- d) and prevent stall at high rotor speed.

Answer ;

- c) at low rotor speed.
- 

0392. In addition to the fire hazard introduced , excessive priming should be avoided because ;

- a) it washes the lubricant of cylinder walls.
- b) it fouls the spark plugs.
- c) it drains the carburettor float chamber.
- d) the gasoline dilutes the oil and necessitates changing oil.

Answer ;

- a) it washes the lubricant of cylinder walls.
- 

0393. In an AC power generation system , the Constant Speed Drive ( CSD ) ;

- 1. can be disconnected from the drive shaft.
- 2. can be disconnected from the generator.
- 3. is a hydro-mechanical system.
- 4. is an electronic system.
- 5. can not be disconnected during flight.
- 6. can be disconnected during flight.

The combination regrouping all the correct statements is ;

- a) 1 , 4 , 5.
- b) 1 , 3 , 6.
- c) 2 , 3 , 4.
- d) 1 , 2 , 5.

Answer ;

- b) 1 , 3 , 6.
-

0394. In an aeroplane fitted with a constant frequency AC power supply , DC power is obtained from a ;

- a) transformer rectifier unit ( TRU ).
- b) 3 phase current transformer unit.
- c) rotary converter.
- d) static inverter.

Answer ;

- a) transformer rectifier unit ( TRU ).
- 

0395. In an air cycle machine ;

- a) the temperature drop across the turbine is the main contributor to the cooling effect of the air cycle machine.
- b) the turbine drives the compressor which provides pressurisation.
- c) the turbine drives the compressor of the machine which causes a higher temperature , and so increases the second heat exchanger efficiency.
- d) the turbine increases the pressure of the air supply to the cabin.

Answer ;

- c) the turbine drives the compressor of the machine which causes a higher temperature , and so increases the second heat exchanger efficiency.
- 

0396. In an aircraft air conditioning system the air cannot be treated for ;

- a) pressure.
- b) purity.
- c) temperature.
- d) humidity.

Answer ;

- d) humidity.
- 

0397. In an aircraft electrical system where AC generators are not paralleled mounted , the changeover relay allows ;

- a) power supply to the faulty AC generators busbar.
- b) connection of the ground power truck to its distribution busbar.
- c) connection of the AC generator to its distribution busbar.
- d) connection of the Auxiliary Power Unit ( APU ) to its main busbar.

Answer ;

- a) power supply to the faulty AC generators busbar.
- 

0398. In an aircraft equipped with a DC main power system , AC for instrument operation may be obtained from ;

- a) a rectifier.
- b) an inverter.
- c) a TRU.
- d) a contactor.

Answer ;

- b) an inverter.
- 

0399. In an alternator rotor coil you can find ;

- a) Only induced current.
- b) Three-phase AC.
- c) AC.
- d) DC.

Answer ;

- c) AC.
- 

0400. In an engine equipped with a float-type carburettor , the low temperature that causes carburettor ice is normally the result of ;

- a) vaporization of fuel and expansion of the air in the carburettor.
- b) low volatility of aviation fuel.
- c) compression of air at the carburettor venturi.
- d) freezing temperature of the air entering the carburettor.

Answer ;

- a) vaporization of fuel and expansion of the air in the carburettor.
-

0401. In an ignition system during a normal start of a gas turbine engine , igniter activation ;
- a) should commence after fuel has entered the combustion chamber.
  - b) commences upon starter motor engagement , and is deactivated at a rotor speed of approximately 15% below idle rpm.
  - c) should commence prior to fuel entering the combustion chamber.
  - d) should commence prior to starter motor engagement to prevent a hot start.

Answer ;

c) should commence prior to fuel entering the combustion chamber.

---

0402. In an on-board computer , the software is ;
- a) resident in a random access memory ( RAM )
  - b) resident in a read-only memory ( REEPROM , EEPROM , ... )
  - c) loaded from a hard disk
  - d) loaded from a floppy disk

Answer ;

b) resident in a read-only memory ( REEPROM , EEPROM , ... )

---

0403. In aviation , the reflection on ionosphere layers phenomenon is used in the following frequencies ;
- a) HF.
  - b) UHF.
  - c) VLF.
  - d) VHF.

Answer ;

a) HF.

---

0404. In case of engine failure during flight the blades of a constant speed propeller in a single engine aeroplane , not fitted with feathering system ;

- a) move in a certain pitch position depending on windmilling RPM.
- b) move in the lowest pitch position by the centrifugal force and/or the spring force.
- c) move in the highest pitch position by the aerodynamic force.
- d) move in low pitch position by oil pressure created by the windmilling propeller.

Answer ;

b) move in the lowest pitch position by the centrifugal force and/or the spring force.

---

0405. In case of smoke in the cockpit , the crew oxygen regulator must be set to ;
- a) 100%.
  - b) on demand.
  - c) emergency.
  - d) normal.

Answer ;

a) 100%.

---

0406. In computer technology , a storage peripheral is a ;
- a) key board.
  - b) screen unit.
  - c) hard disk drive.
  - d) printer.

Answer ;

c) hard disk drive.

---

0407. In computer technology , an input peripheral is a ;
- a) diskette drive.
  - b) hard disk drive.
  - c) printer.
  - d) keyboard.

Answer ;

d) keyboard.

---

0408. In computer technology , an output peripheral is a ;
- a) keyboard.
  - b) hard disk drive.
  - c) diskette drive.
  - d) screen unit.

Answer ;

d) screen unit.

---

**0409.** In flight , a cantilever wing of an airplane containing fuel is subjected to vertical loads that produce a bending moment which is ;

- a) equal to half the weight of the aircraft multiplied by the semi-span.
- b) highest at the wing root.
- c) lowest at the wing root.
- d) equal to the zero-fuel weight multiplied by the span.

**Answer ;**

**b)** highest at the wing root.

---

**0410.** In flight , if the constant speed drive ( CSD ) temperature indicator is in the red arc the ;

- a) pilot can disconnect it to allow it to cool and use it again.
- b) pilot has to throttle back.
- c) pilot must disconnect it and manually control the alternator.
- d) pilot must disconnect it , and the generator is not available for the rest of flight.

**Answer ;**

**d)** pilot must disconnect it , and the generator is not available for the rest of flight.

---

**0411.** In flight , the cabin air for modern airplanes with jet engines is usually supplied by ;

- a) ram air intakes.
- b) engine compressors.
- c) piston compressors.
- d) single radial compressors.

**Answer ;**

**b)** engine compressors.

---

**0412.** In flight , the most commonly used anti-icing method for the wings of modern commercial aircraft fitted with turbo-jet units is ;

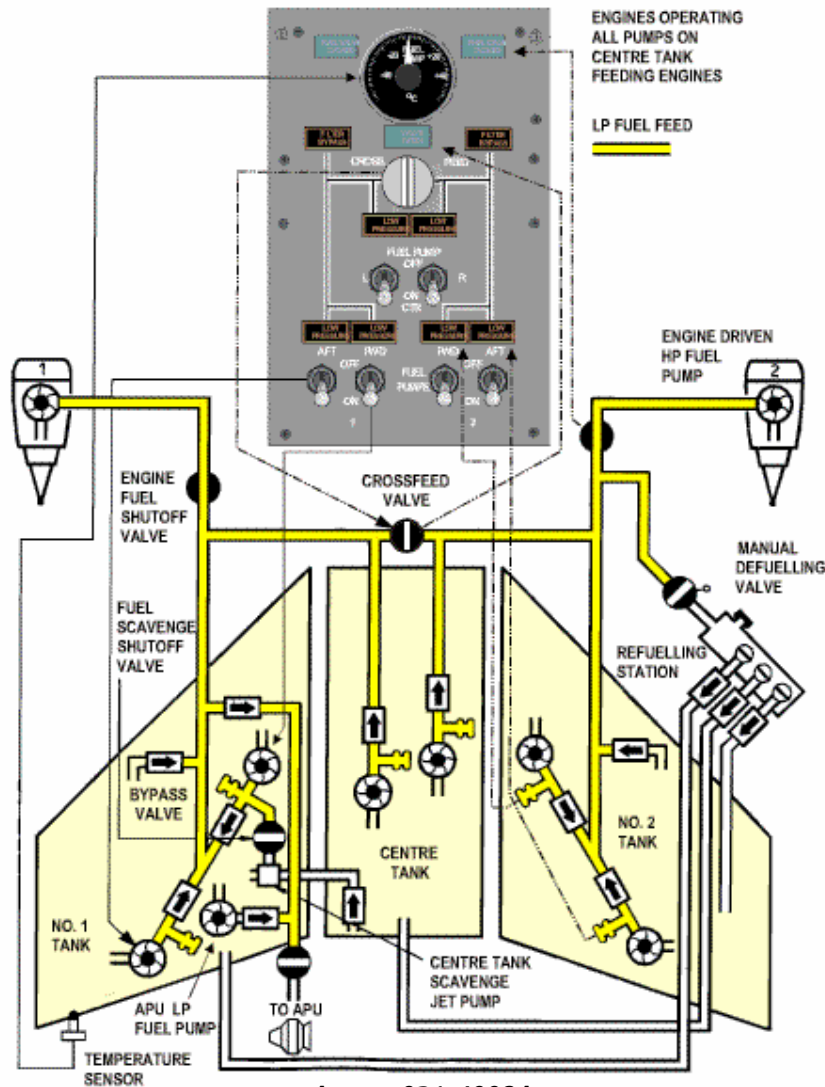
- a) electrical ( electrical resistances ).
- b) physical/chemical ( glycol-based liquid ).
- c) thermal ( use of hot air ).
- d) mechanical ( pneumatic source which acts by deforming the profiles of the leading edge ).

**Answer ;**

**c)** thermal ( use of hot air ).

---

0413. ( For this question use annex 021-4008A )



Annex 021-4008A

In flight , with centre tank empty and APU operating , a fuel unbalance is detected ( quantity in tank 1 < quantity in tank 2 ). Rebalancing of the two tanks is ;

- a) impossible because there is no fuel in centre tank.
- b) possible with "CROSSFEED" open and tank 2 pumps "OFF".
- c) impossible without causing the APU to stop.
- d) possible with "CROSSFEED" open and tank 1 pumps "OFF" and tank 2 pumps "ON".

Answer ;

- d) possible with "CROSSFEED" open and tank 1 pumps "OFF" and tank 2 pumps "ON".

0414. In hydraulic system , a shuttle valve ;

- a) allows two units to be operated by one pressure source.
- b) regulates pump delivery pressure.
- c) is a non-return valve.
- d) enables an alternate system to operate the same actuators in case of normal system failure.

Answer ;

- d) enables an alternate system to operate the same actuators in case of normal system failure.

0415. In hydraulic systems of large modern transport category aircraft the fluids used are ;

- a) Vegetable oil.
- b) Mineral oil.
- c) Water and glycol.
- d) Synthetic oil.

Answer ;

- d) Synthetic oil.

0416. In jet aeroplanes the “thermal anti-ice system” is primarily supplied by ;

- a) ram air , heated via a heat exchanger.
- b) bleed air from the engines.
- c) turbo compressors.
- d) the APU.

Answer ;

b) bleed air from the engines.

---

0417. In jet transport aeroplanes , oxygen for the flight deck oxygen system is stored as a ;

- a) chemical compound.
- b) gas.
- c) chemical compound or gas.
- d) liquid.

Answer ;

b) gas.

---

0418. In jet transport aircraft , breathing oxygen for the cockpit is stored in the following state ;

- a) gaseous.
- b) chemical.
- c) liquid.
- d) chemical or gaseous.

Answer ;

a) gaseous.

---

0419. In large modern aircraft , in the air conditioning system , reduction of air temperature and pressure is achieved by ;

- a) an evaporator.
- b) a condenser.
- c) a compressor.
- d) an expansion turbine.

Answer ;

d) an expansion turbine.

---

0420. In order that DC generators will achieve equal load sharing when operating in parallel , it is necessary to ensure that ;

- a) adequate voltage differences exists.
- b) the synchronising bus-bar is disconnected from the busbar system.
- c) equal loads are connected to each generator busbar before paralleling.
- d) their voltages are almost equal.

Answer ;

d) their voltages are almost equal.

---

0421. In order to get the optimum efficiency of a piston engine , the positions of the intake and exhaust valve at the end of the power stroke are ;

- a) exhaust valve closed and intake valve open.
- b) intake valve closed and exhaust valve open.
- c) both valves open.
- d) both valves closed.

Answer ;

c) both valves open.

---

0422. In order to produce an alternating voltage of 400 Hz , the number of pairs of poles required in an AC generator running at 6000 rpm is ;

- a) 4.
- b) 12.
- c) 24.
- d) 8.

Answer ;

a) 4.

---

0423. In regard to passengers briefing in public transport aeroplane , information about oxygen must be given to passengers through a demonstration. If a flight is to be carried out at FL 290 , this demonstration must be completed before ;

- a) take-off.
- b) the aircraft reaches FL 100.
- c) the aircraft reaches FL 250.
- d) the aircraft reaches FL 140.

Answer ;

- a) take-off.

---

0424. In respect of a constant speed propeller of a twin engine aeroplane controlled by a single-acting pitch control unit ;

- a) spring force turns the propeller blades towards smaller pitch angle.
- b) oil pressure turns the propeller blades towards higher pitch angle.
- c) aerodynamic force turns the propeller blades towards higher pitch angle.
- d) oil pressure turns the propeller blades towards smaller pitch angle.

Answer ;

- d) oil pressure turns the propeller blades towards smaller pitch angle.

---

0425. In some aircraft , there is a protection device to avoid the landing gear being inadvertently retracted on the ground. It consists of ;

- a) an aural warning horn.
- b) a bolt.
- c) a warning light which is activated by the WOW ( Weight On Wheels ) sensor system.
- d) a latch located in the landing gear lever.

Answer ;

- d) a latch located in the landing gear lever.

---

0426. In straight and level flight the most important loads on a cantilever wing are ;

- a) compression in both the upper and the lower surfaces.
- b) compression in the upper surface , tension in the lower surface.
- c) tension in the upper surface , compression in the lower surface.
- d) tension in both the upper and the lower surfaces.

Answer ;

- b) compression in the upper surface , tension in the lower surface.

---

0427. In the airflow through a single-spool axial flow turbo-jet engine , the axial velocity of the air is greatest ;

- a) within the combustion chamber.
- b) on exit from the propelling nozzle.
- c) as it leaves the turbine.
- d) as it leaves the compressor.

Answer ;

- b) on exit from the propelling nozzle.

---

0428. In the axial flow compressor of a turbo-jet engine , the flow duct is tapered. Its shape is calculated so as to ;

- a) maintain a constant axial speed in cruising flight.
- b) reduce the axial speed in cruising flight.
- c) maintain a constant axial speed whatever the engine rating.
- d) reduce the axial speed , whatever the engine rating.

Answer ;

- a) maintain a constant axial speed in cruising flight.

---

0429. In the cabin , when the oxygen mask is pulled downwards , the passenger breathes ;

- a) cabin air under pressure.
- b) pure oxygen at the ambient pressure.
- c) pure oxygen under pressure.
- d) a mixture of oxygen and cabin air.

Answer ;

- d) a mixture of oxygen and cabin air.

---

0430. In the cockpit of a transport airplane , at least one manual fire-extinguisher must be conveniently located containing ;

- a) halon.
- b) powder.
- c) special fluids.
- d) water.

Answer ;

- a) halon.



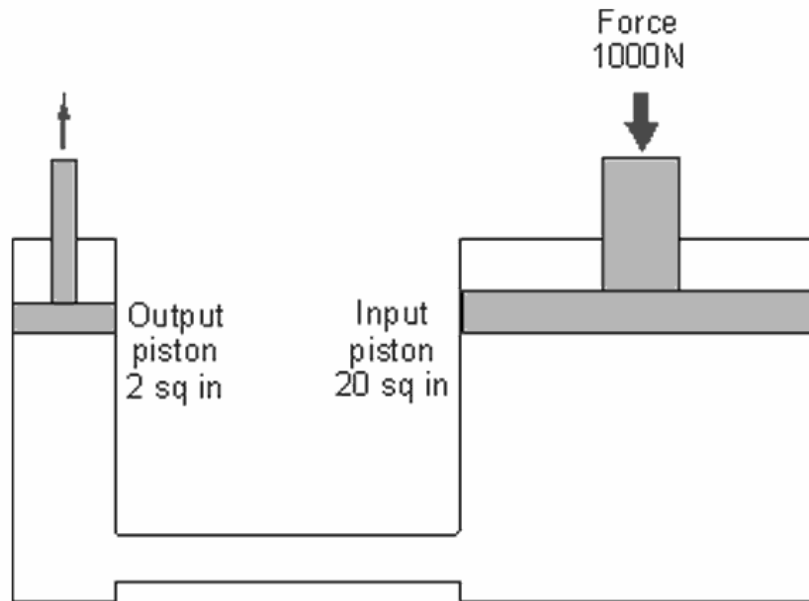
0431. In the event of an AC generator fault , on a twin engine aeroplane , the protection device will open ;

- a) the exciter control relay and the generator breaker.
- b) the bus tie breaker.
- c) the generator breaker and the bus tie breaker.
- d) the exciter control relay , the generator breaker and the bus tie breaker.

Answer ;

- a) the exciter control relay and the generator breaker.
- 

0432. ( For this question refer to Image I )



**Image I**

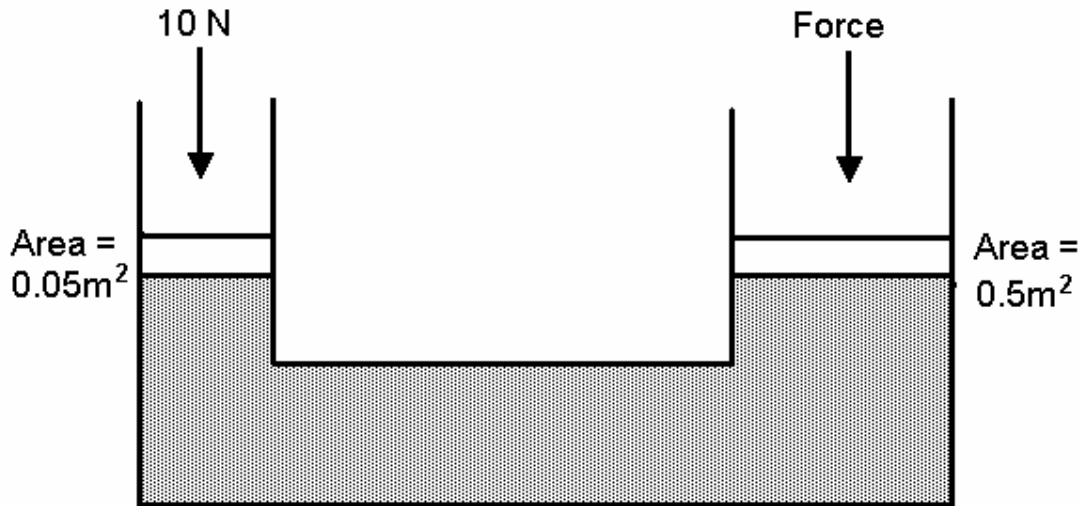
In the hydraulic press schematically shown , what balancing Force would be acting on the left hand side ? ( The diagram is not to scale )

- a) 20 N.
- b) 1 N.
- c) 100 N.
- d) 1000 N.

Answer ;

- c) 100 N.
-

0433. ( For this question use annex 021-6716A )



**Annex 021-6716A**

In the hydraulic press schematically shown , what balancing Force would be acting on the right hand side ? ( The diagram is not to scale )

- a) 20 N.
- b) 100 N.
- c) 1 N.
- d) 1000 N.

Answer ;

**b) 100 N.**

---

0434. In the propagation of MF waves , the phenomenon of FADING is particularly found ;

- a) at night and when raining.
- b) at night , due to the combination of the sky and ground waves.
- c) by day , due to the combination of sky and ground waves.
- d) by day and when raining.

Answer ;

**b) at night , due to the combination of the sky and ground waves.**

---

0435. In the response curve of an amplifier , the bandwidth is ;

- a) the frequency band corresponding to maximum gain less 3 decibels.
- b) the frequency band corresponding to maximum gain.
- c) the frequency band corresponding to maximum gain , increased by 10 kHz at each end.
- d) the frequency band corresponding to maximum gain less 20 decibels.

Answer ;

**a) the frequency band corresponding to maximum gain less 3 decibels.**

---

0436. In the stator of a turbine , the gas velocity  $V$  and static pressure  $P_s$  vary as follows ;

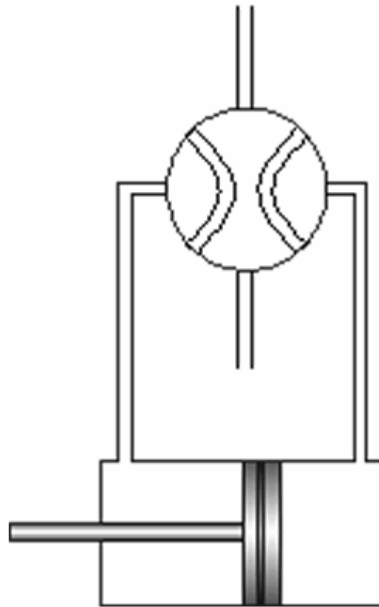
- a)  $V$  increases ,  $P_s$  increases.
- b)  $V$  increases,  $P_s$  decreases.
- c)  $V$  decreases ,  $P_s$  increases.
- d)  $V$  decreases ,  $P_s$  decreases.

Answer ;

**b)  $V$  increases,  $P_s$  decreases.**

---

0437. ( For this question refer to Image II )



**Image II**

In the typical hydraulic system represented in the graphic , assuming hydraulic pressure throughout and no internal leakage ;

- a) the piston moves to the right due to equal pressure acting on differential areas.
- b) a condition of hydraulic lock exists and no movement of the piston will take place.
- c) the piston moves to the left due to pressure acting on differential areas.
- d) the piston is free to move in response to external forces since pressures are equal.

Answer ;

- b)** a condition of hydraulic lock exists and no movement of the piston will take place.

---

0438. In twin-engine aeroplanes with right turning propellers ;

- a) the left engine is the critical motor.
- b) the left engine produces a higher yaw moment if the right engine fails than vice versa.
- c) the “minimum control speed” is determined by the failure of the right engine.
- d) the right engine is the critical motor.

Answer ;

- a)** the left engine is the critical motor.

---

0439. In very cold weather , a slightly higher than normal engine oil pressure during start up ;

- a) is unacceptable and requires the engine to be shut down.
- b) is acceptable , if it decreases after startup.
- c) is unacceptable but does not require the engine to be shut down.
- d) requires an oil change.

Answer ;

- b)** is acceptable , if it decreases after startup.

---

0440. In very cold weather the pilot notices during startup , a slightly higher than normal engine oil pressure. This higher pressure ;

- a) is normal , if it decreases after startup.
- b) is abnormal but does not require the engine to be shut down.
- c) is abnormal and requires the engine to be shut down.
- d) requires an oil change.

Answer ;

- a)** is normal , if it decreases after startup.

---

0441. In what state is the breathing oxygen for the cockpit of jet transport aeroplanes stored ;

- a) gaseous.
- b) a chemical compound.
- c) a gaseous or chemical compound.
- d) liquid.

Answer ;

- a)** gaseous.

0442. In which part of a gas turbine engine are high temperature requirements most critical for the materials used ?

- a) The combustion chamber and front part of the HP turbine.
- b) The turbine and exhaust nozzle.
- c) The combustion chamber and exhaust nozzle.
- d) The last part of the HP compressor and front part of the HP turbine.

Answer ;

a) The combustion chamber and front part of the HP turbine.

---

0443. In which sections of the carburettor would icing most likely occur ?

- a) float chamber and fuel inlet filter.
- b) main air bleed and main discharge nozzle.
- c) accelerator pump and main metering jet.
- d) venturi and the throttle valve.

Answer ;

d) venturi and the throttle valve.

---

0444. Increasing the frequency of variations in thrust setting during normal operation is detrimental to the life of HP turbine blades because ;

- a) creep will be more severe.
- b) the EGT limit would be exceeded more often.
- c) low cycle fatigue damage is more severe.
- d) fatigue due to vibration is more severe.

Answer ;

c) low cycle fatigue damage is more severe.

---

0445. Increasing the power setting of a gas turbine engine with a constant speed propeller and free power turbine causes ;

- a) HP spool speed to increase and EGT to remain constant.
- b) both HP spool speed and EGT to increase.
- c) HP spool speed to remain constant and EGT to increase.
- d) HP spool speed to decrease and EGT to increase.

Answer ;

b) both HP spool speed and EGT to increase.

---

0446. Internal leakage in a hydraulic system will cause ;

- a) fluid loss.
- b) an increased fluid temperature.
- c) a decreased fluid temperature.
- d) an increased fluid pressure.

Answer ;

b) an increased fluid temperature.

---

0447. It may be determined that an aircraft is not properly bonded if ;

- a) static noises can be heard on the radio.
- b) a circuit breaker pops out.
- c) there is heavy corrosion on the fuselage skin mountings.
- d) there is interference on the VOR receiver.

Answer ;

a) static noises can be heard on the radio.

---

0448. Jet engine total efficiency indicates the efficiency at which ;

- a) chemical power in the fuel is transformed into an increase of jet kinetic energy per second.
- b) potential energy in the jet is transformed into propulsive energy.
- c) the increase of jet kinetic energy is transformed into propulsive energy.
- d) chemical power in the fuel is transformed into propulsive power (  $T \times V$  ).

Answer ;

d) chemical power in the fuel is transformed into propulsive power (  $T \times V$  ).

---

0449. Labyrinth seals in the lubrication system of a gas turbine engine are designed to provide a seal that is ;

- a) completely tight , allowing some axial movement between rotating and static parts.
- b) completely tight , allowing no movement between rotating and static parts.
- c) not completely tight , allowing only some radial movement between rotating and static parts.
- d) not completely tight , allowing some movement between rotating and static parts.

Answer ;

d) not completely tight , allowing some movement between rotating and static parts.

---

0450. Landing gear torque links are used to ;
- a) maintain the compass heading throughout taxiing and take-off.
  - b) take up the lateral stresses to which the gear is subjected.
  - c) prevent the extension of the landing gear oleo strut rod.
  - d) prevent rotation of the landing gear piston in the oleo strut.

Answer ;

d) prevent rotation of the landing gear piston in the oleo strut.

---

0451. Large transport aeroplane hydraulic systems usually operate with a system pressure of approximately ;

- a) 2000 psi.
- b) 3000 psi.
- c) 4000 psi.
- d) 1000 psi.

Answer ;

b) 3000 psi.

---

0452. Loads on the cylindrical part of the fuselage during pressurisation are carried by the ;

- a) spars.
- b) skin.
- c) stringers.
- d) ribs.

Answer ;

b) skin.

---

0453. Main cabin temperature is ;

- a) only controllable at maximum cabin differential pressure.
- b) controlled by individual passenger.
- c) not controllable at the maximum cabin differential pressure.
- d) controlled automatically , or by flight crew selection.

Answer ;

d) controlled automatically , or by flight crew selection.

---

0454. Maximum exhaust Gas Temperature ( EGT ) is theoretically associated with ;

- a) full rich setting.
- b) mixture ratio very close to idle cut-out.
- c) cruising mixture setting.
- d) mass ratio of 1/15.

Answer ;

d) mass ratio of 1/15.

---

0455. Maximum exhaust Gas Temperature ( EGT ) of a piston engine is theoretically associated with a ;

- a) fuel to air ratio of 1:15.
- b) full rich setting.
- c) cruising mixture setting.
- d) mixture ratio very close to idle cut-out.

Answer ;

a) fuel to air ratio of 1:15.

---

0456. Maximum power output and low mass of aeroplane hydraulic systems can be achieved with ;

- a) medium system pressure and low volume flow.
- b) medium system pressure and high volume flow.
- c) high system pressure and low volume flow.
- d) low system pressure and high volume flow.

Answer ;

c) high system pressure and low volume flow.

---

0457. Modern pressurized transport airplanes are equipped with ;

- a) only portable oxygen bottles.
- b) only one oxygen system supplying the whole aircraft.
- c) two independent oxygen systems , one supplying the cockpit , the other the cabin.
- d) two oxygen systems both supplying the cockpit and the cabin.

Answer ;

c) two independent oxygen systems , one supplying the cockpit , the other the cabin.

---

0458. Most high-bypass ( unmixed ) turbofan engines are equipped with a cold exhaust reverser ( fan reverser ) only in order to ;
- a) reduce the risk of exhaust gas re-ingestion.
  - b) save costs and mass at the expense of a reduction in available reverse thrust.
  - c) prevent disturbance of cold exhaust reversed airflow by the hot exhaust reversed gas flow.
  - d) prevent extra fuel consumption due to operation of the hot exhaust reverse ( turbine reverser ).

Answer ;

- b) save costs and mass at the expense of a reduction in available reverse thrust.
- 

0459. Most large conventional aeroplanes are not provided with aileron and rudder trim tabs. Is it still possible to trim these control surfaces ?

- a) Yes , but trimming is only possible when before the flight , the respective auxiliary surfaces are correctly adjusted for cruising conditions by the maintenance department.
- b) Yes , but trimming is only possible when the autopilot is engaged.
- c) No , because without trim tabs trimming is not possible.
- d) Yes , trimming is possible by adjusting the neutral point of the artificial feel mechanism by means of a trim switch.

Answer ;

- d) Yes , trimming is possible by adjusting the neutral point of the artificial feel mechanism by means of a trim switch.
- 

0460. Most transport aeroplanes are provided with protection against control jamming. This means that ;

- a) the flight control system has provisions to disconnect the part of the control system that becomes blocked.
- b) the aerolane is protected against the adverse effects of strong electromagnetic radiation.
- c) in case of seizure , engine control is taken over automatically by an alternate electric circuit.
- d) seized brakes can be released from the cockpit.

Answer ;

- a) the flight control system has provisions to disconnect the part of the control system that becomes blocked.
- 

0461. “Nose wheel shimmy” may be described as ;

- a) the amount of free movement of the nose wheel before steering takes effect.
- b) the oscillatory movement of the nose wheel when extended prior to landing.
- c) a possibly damaging vibration of the nose wheel when moving on the ground.
- d) aircraft vibration caused by the nose wheel upon extension of the gear.

Answer ;

- c) a possibly damaging vibration of the nose wheel when moving on the ground.
- 

0462. On a 13653 a fire detection system includes ;

- a) a warning light for each engine and a single alarm bell common to all engines.
- b) a single warning light but a separate alarm bell for each engine.
- c) both a warning light and an alarm bell unique to each engine.
- d) a single warning light and a single alarm bell.

Answer ;

- a) a warning light for each engine and a single alarm bell common to all engines.
- 

0463. On a FADEC gas turbine engine during normal flight , the power for the FADEC’s sensors comes from ;

- a) the FADEC power source(s) on the engine.
- b) the aircraft generator.
- c) the APU generator.
- d) the aircraft battery.

Answer ;

- a) the FADEC power source(s) on the engine.
- 

0464. On a FADEC jet engine , the meaning of F.A.D.E.C. on a gas turbine engine is ;

- a) FAILURE ANALYSIS / DISPATCH EVALUATION COMPUTER.
- b) FAILSAFE ADAPTER DIRECT EFFECT COMPUTATION.
- c) FULL AUTHORITY DIGITAL ENGINE CONTROL.
- d) FOOL ADAPTER DUAL ELECTRONIC COMPUTER.

Answer ;

- c) FULL AUTHORITY DIGITAL ENGINE CONTROL.
-

**0465.** On a high by-pass ratio ( fan ) engine , due to the distribution of the thrust , it is normal for reverse thrust to be achieved by reversing ;

- a) only the direction of the fan airflow.
- b) the direction of rotation of the fan.
- c) the direction of both the exhaust gas flow ( hot stream ) and the fan airflow ( cold stream ).
- d) only the direction of the exhaust gas flow.

**Answer ;**

- a)** only the direction of the fan airflow.
- 

**0466.** On a high by-pass ratio ( fan ) engine , reverse thrust is achieved by reversing ;

- a) only the direction of the fan airflow.
- b) the direction of rotation of the fan.
- c) only the direction of the exhaust gas flow.
- d) the direction of both the exhaust gas flow ( hot stream ) and the fan airflow ( cold stream ).

**Answer ;**

- a)** only the direction of the fan airflow.
- 

**0467.** On a jet aircraft fuel heaters are ;

- a) installed in each tank.
- b) not necessary at all.
- c) installed only in the centre tank.
- d) located on the engines.

**Answer ;**

- d)** located on the engines.
- 

**0468.** On a large transport aeroplane , the auto-slat system ;

- a) assist the ailerons.
- b) ensures that the ( part of ) slats are always extended when the ground/flight system is in the “ground” position.
- c) extends ( part of ) the slats automatically when a certain value of angle of attack is exceeded.
- d) provides for automatic slat retraction after take-off.

**Answer ;**

- c)** extends ( part of ) the slats automatically when a certain value of angle of attack is exceeded.
- 

**0469.** On a modern aeroplane , to avoid the risk of tyre burst from overheating , due for example to prolonged braking during an aborted take-off , there is ;

- a) a hollow bolt screwed into the wheel which melts at a given temperature ( thermal fuse ) and deflates the tyre.
- b) a pressure relief valve situated in the filler valve.
- c) water injection triggered at a fixed temperature in order to lower tyre temperature.
- d) the “Emergency Burst” function of the anti-skid system which adapts braking to the tyre temperature.

**Answer ;**

- a)** a hollow bolt screwed into the wheel which melts at a given temperature ( thermal fuse ) and deflates the tyre.
- 

**0470.** On a modern large pressurized transport aircraft , the maximum cabin differential pressure is approximately;

- a) 3 - 5 psi.
- b) 7 - 9 psi.
- c) 22 psi.
- d) 13 - 15 psi.

**Answer ;**

- b)** 7 - 9 psi.
- 

**0471.** On a multi-engined aircraft a fire detection system includes ;

- a) a single warning light and a single alarm bell.
- b) both a warning light and an alarm bell unique to each engine.
- c) a single warning light but a separate alarm bell for each engine.
- d) a warning light for each engine and a single alarm bell common to all engines.

**Answer ;**

- d)** a warning light for each engine and a single alarm bell common to all engines.
-

0472. On a non-stressed skin type wing , the wing structure elements which take up the vertical bending moments  $M_x$  are ;
- a) the webs.
  - b) the spars.
  - c) the skin.
  - d) the ribs.

Answer ;

b) the spars.

---

0473. On a normally aspirated aero-engine fitted with a fixed pitch propeller ;
- a) the propeller setting is constant at all indicated airspeeds.
  - b) manifold pressure decreases as the aircraft climbs at a fixed throttle setting.
  - c) in a descent at a fixed throttle setting manifold pressure will always remain constant.
  - d) in level flight , manifold pressure will remain constant when the rpm is increased by opening the throttle.

Answer ;

b) manifold pressure decreases as the aircraft climbs at a fixed throttle setting.

---

0474. On a normally aspirated engine ( non turbo-charged ) , the manifold pressure gauge always indicates ;
- a) a lower value than atmospheric pressure when the engine is running.
  - b) zero on the ground when the engine is stopped.
  - c) a greater value than atmospheric pressure when the engine is running.
  - d) a value equal to the atmospheric pressure when the engine is at full power on the ground.

Answer ;

a) a lower value than atmospheric pressure when the engine is running.

---

0475. On a transport type aircraft the fuel tank system is vented through ;
- a) ram air scoops on the underside of the wing.
  - b) the return lines of the fuel pumps.
  - c) bleed air from the engines.
  - d) a pressure regulator in the wing tip.

Answer ;

a) ram air scoops on the underside of the wing.

---

0476. On an aeroplane using AC as primary power supply , the batteries are charged in flight from ;
- a) a DC transformer and rectifier.
  - b) a static inverter.
  - c) a Transformer Rectifier Unit.
  - d) the AC bus via current limiters.

Answer ;

c) a Transformer Rectifier Unit.

---

0477. On an aircraft landing gear , an under-inflated tyre ;
- a) will have an increased critical hydroplaning speed.
  - b) will be more subject to viscosity aquaplaning on dry runway.
  - c) it's tread will deteriorate faster.
  - d) will wear at the shoulders.

Answer ;

d) will wear at the shoulders.

---

0478. On an aircraft provided with resistance and capacitance variation type fire detection loops , a fire alarm is initiated by a temperature increase detected ;

- a) only at an isolated point of the loops.
- b) at any isolated point of the loops or else generally on all the loops.
- c) on at least one loop.
- d) only in a uniform way along the loops.

Answer ;

b) at any isolated point of the loops or else generally on all the loops.

---

0479. On an aircraft , the Krueger flap is a ;
- a) leading edge flap close to the wing root.
  - b) leading edge flap close to the wing tip.
  - c) trailing edge flap.
  - d) leading edge flap.

Answer ;

d) leading edge flap.

---



- 0480.** On an airplane , the Krueger flaps are ;
- a) trailing edge flaps close to the wing tip.
  - b) trailing edge flaps close to the wing root.
  - c) leading edge flaps close to the wing root.
  - d) leading edge flaps close to the wing tip.

**Answer ;**

**c) leading edge flaps close to the wing root.**

---

- 0481.** On-board electrical systems are protected against faults of the following type ;

- 1. AC generator over-voltage.
- 2. AC generator under-voltage.
- 3. over-current.
- 4. over-speed.
- 5. under-frequency.
- 6. undue vibration of AC generators.

The combination of correct statements is ;

- a) 1 , 2 , 3 , 4 , 5.
- b) 1 , 2 , 4 , 6.
- c) 2 , 3 , 4 , 5 , 6.
- d) 1 , 3 , 5 , 6.

**Answer ;**

**a) 1 , 2 , 3 , 4 , 5.**

---

- 0482.** On board present aircraft , the batteries used are mainly Cadmium-Nickel. Their advantages are ;

- 1. low risk of thermal runaway.
- 2. high internal resistance , hence higher power.
- 3. good charging and discharging capability at high rating.
- 4. wider permissible temperature range.
- 5. good storage capability.
- 6. sturdiness owing to its metal casing.
- 7. the electrolyte density remains unchanged during charging.

The combination of correct statement is ;

- a) 3 , 4 , 5 , 6.
- b) 1 , 2 , 5 , 6 , 7.
- c) 2 , 3 , 4 , 5 , 6.
- d) 3 , 4 , 6 , 7.

**Answer ;**

**a) 3 , 4 , 5 , 6.**

---

- 0483.** On design purpose the relationship between the fuel octane rating and the maximum compression ratio of a piston engine is ;

- a) the maximum compression ratio is independent of the octane rating.
- b) the higher the octane rating is , the higher the maximum compression ratio is.
- c) the higher the octane rating is , the lower the maximum compression ratio is.
- d) the lower the octane rating is , the higher the maximum compression ratio is.

**Answer ;**

**b) the higher the octane rating is , the higher the maximum compression ratio is.**

---

- 0484.** On detection of a persistent overvoltage fault on an AC generator connected to the aircraft AC busbars , the on-board protection device opens ;

- a) the generator breaker and tie breaker.
- b) the exciter breaker , the generator breaker and tie breaker.
- c) the generator breaker.
- d) the exciter breaker and the generator breaker.

**Answer ;**

**d) the exciter breaker and the generator breaker.**

---

- 0485.** On detection of a persistent phase imbalance between an AC generator connected to the main bus bars and other AC generators , the protection device that opens is/are the ;

- a) generator breaker and the tie breaker(s).
- b) exciter control relay and the generator breaker.
- c) tie breaker(s).
- d) generator breaker.

**Answer ;**

**c) tie breaker(s).**

---

**0486.** On detection of a persistent phase unbalance on an AC generator connected to the aircraft mains , with another AC generator , the protection device opens ;

- a) the generator breaker.
- b) the generator breaker and the tie breaker.
- c) the exciter control relay and the generator breaker.
- d) the tie breaker.

**Answer ;**

**d) the tie breaker.**

---

**0487.** On four-stroke piston engines , the theoretical valve and ignition settings are readjusted in order to increase the ;

- a) piston displacement.
- b) overall efficiency.
- c) compression ratio.
- d) engine RPM.

**Answer ;**

**b) overall efficiency.**

---

**0488.** On landing and prior to shut down , a gas turbine engine should be run at “GROUND IDLE” for a short period. This allows ;

- a) the turbine blade temperature to reduce only.
- b) the engine to be compressor washed.
- c) cool oil flushing of bearings , particularly those in the turbines.
- d) the core temperature to diminish.

**Answer ;**

**d) the core temperature to diminish.**

---

**0489.** On large aeroplanes equipped with power brakes , the main source of power is derived from ;

- a) pressure to the rudder pedals.
- b) the master cylinders.
- c) the aeroplane’s hydraulic system.
- d) the brake actuators.

**Answer ;**

**c) the aeroplane’s hydraulic system.**

---

**0490.** On modern transport aircraft , cockpit windows are protected against icing by ;

- a) vinyl coating.
- b) rain repellent system.
- c) electric heating.
- d) anti-icing fluid.

**Answer ;**

**c) electric heating.**

---

**0491.** On most modern airliners the cabin pressure is controlled by regulating the ;

- a) RPM of the engine.
- b) bleed air valve.
- c) airflow leaving the cabin.
- d) airflow entering the cabin.

**Answer ;**

**c) airflow leaving the cabin.**

---

**0492.** On most transport aircraft , the low pressure pumps of the fuel system are ;

- a) diaphragm pumps.
- b) centrifugal pumps.
- c) piston pumps.
- d) gear type pumps.

**Answer ;**

**b) centrifugal pumps.**

---

**0493.** On most transport aircraft , the low pressure pumps of the fuel system are ;

- a) centrifugal pumps , driven by an electric motor.
- b) removable only after the associated tank has been emptied.
- c) electro-mechanical wobble pumps, with self-regulated pressure.
- d) mechanically driven by the engine's accessory gearbox.

**Answer ;**

**a) centrifugal pumps , driven by an electric motor.**

---

0494. On most transport jet aircraft , the low pressure pumps of the fuel system are supplied with electric power of the following type ;

- a) 115 V DC.
- b) 115 V AC.
- c) 28 V AC.
- d) 28 V DC.

Answer ;

b) 115 V AC.

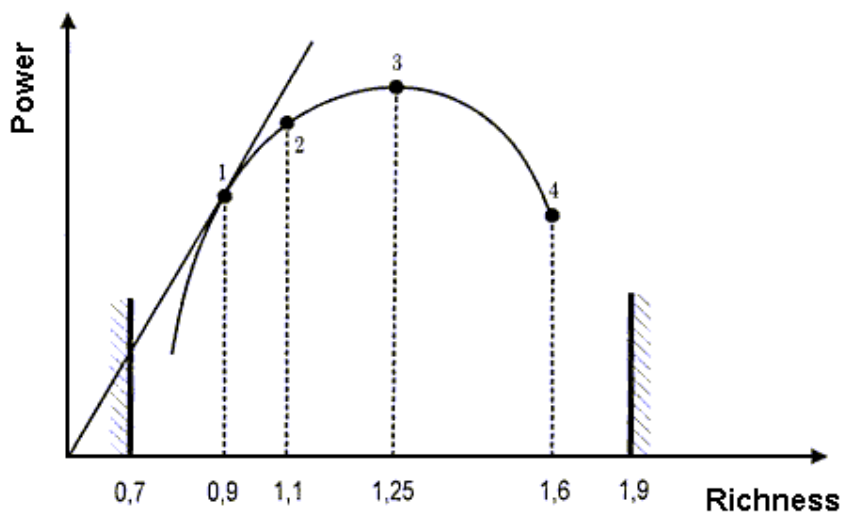
0495. On starting , in a brushless AC generator with no commutator rings , the generator is activated by ;

- a) the main field winding.
- b) the auxiliary winding.
- c) a set of permanent magnets.
- d) the stabilizer winding jointly with the voltage regulator.

Answer ;

c) a set of permanent magnets.

0496. ( For this question use annex 021-11507A )



Effect of richness on power at a given RPM and manifold pressure

Annex 021-11507A

On the attached diagram showing the power output of a piston engine as a function of mixture richness , best economy is at the point marked ;

- a) 3.
- b) 4.
- c) 1.
- d) 2.

Answer ;

c) 1.

0497. On the flight deck , an oil operated CSD unit is normally provided with means of monitoring the ;

- a) oil temperature and synchronous speed.
- b) oil over-temperature and low oil pressure.
- c) output speed and oil pressure.
- d) low oil temperature and low oil quantity.

Answer ;

b) oil over-temperature and low oil pressure.

**0498.** On the ground , the Auxiliary Power Unit ( APU ) can provide ;

1. electricity.
2. air for starting system.
3. hydraulic pressure.
4. air for air conditioning.

The combination which regroups all of the correct statements is ;

- a) 1.
- b) 1 , 2 , 4.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3 , 4.

**Answer ;**

**b) 1 , 2 , 4.**

---

**0499.** Once the engine has started , ignition systems of piston engines are ;

- a) independent of the electrical system of the aircraft.
- b) dependent on the DC-Generator.
- c) dependent on the battery.
- d) dependent on the AC-Generator.

**Answer ;**

**a) independent of the electrical system of the aircraft.**

---

**0500.** One advantage of having three instead of two spools in some high-bypass turbofan engine is that ;

- a) fewer variable stator vanes and variable inlet guide vanes are required to prevent compressor stall at high rotor speeds.
- b) there is a lower fuel consumption due to a higher overall pressure ratio.
- c) there is a lower fuel consumption and more thrust due to higher overall pressure ratio.
- d) fewer variable stator vanes and variable inlet guide vanes are required to prevent compressor stall at low rotor speeds.

**Answer ;**

**d) fewer variable stator vanes and variable inlet guide vanes are required to prevent compressor stall at low rotor speeds.**

---

**0501.** One design method to avoid control surface flutter is ;

- a) through the correct use of balance tabs.
- b) ensuring correct mass distribution within the control surface.
- c) providing the wing structure with sufficient flexibility.
- d) through the correct use of trim tabs.

**Answer ;**

**b) ensuring correct mass distribution within the control surface.**

---

**0502.** One function of a rib is ;

- a) to maintain the aerodynamic shape of the wing.
- b) to carry tensile load of the fuselage pressure hull.
- c) to be the primary structural member to carry wings loads.
- d) to stabilise the fuselage skin against buckling.

**Answer ;**

**a) to maintain the aerodynamic shape of the wing.**

---

**0503.** One indication of inadequate bonding of aircraft components may be ;

- a) a circuit breaker pops out.
- b) there is heavy corrosion on the fuselage skin mountings.
- c) there is interference on the VOR receiver.
- d) static noises can be heard on the radio.

**Answer ;**

**d) static noises can be heard on the radio.**

---

**0504.** One of the advantages of a turbocharger is ;

- a) to make the power available less affected by altitude.
- b) that there is no torsion at the crankshaft.
- c) an increased propulsive efficiency.
- d) that there is no danger of detonation.

**Answer ;**

**a) to make the power available less affected by altitude.**

---

0505. One of the advantages of a turbocharger is ;
- a) that there is no danger of detonation.
  - b) to make the power available less dependent on altitude.
  - c) that there is no torsion at the crankshaft.
  - d) that it has a better propulsive efficiency.

Answer ;

- b) to make the power available less dependent on altitude.
- 

0506. One of the advantages of an annular combustion chamber is ;
- a) a reduced thermal stress on the turbine.
  - b) a higher structural strength.
  - c) a simpler assembly.
  - d) that fewer igniters are required.

Answer ;

- a) a reduced thermal stress on the turbine.
- 

0507. One of the advantages of an annular combustion chamber over a multiple can is that ;
- a) the thermal load distribution on the HP turbine is more favourable.
  - b) only one igniter is required.
  - c) assembly and disassembly is easier.
  - d) engine length is shorter because no diffuser is required.

Answer ;

- a) the thermal load distribution on the HP turbine is more favourable.
- 

0508. One of the advantages of multiple-spool jet engine designs is that ;
- a) a smaller air starter driving only a single spool can be used.
  - b) if one spool seizes , the remaining spool(s) will continue to operate normally.
  - c) engine length can be reduced.
  - d) a compressor stall cannot occur under any condition.

Answer ;

- a) a smaller air starter driving only a single spool can be used.
- 

0509. One of the causes of high oil temperature in a gas turbine engine at constant power setting is ;
- a) unusually high scavenge pump oil flow.
  - b) high outside air temperature.
  - c) malfunction of the heat exchanger.
  - d) high oil pressure.

Answer ;

- c) malfunction of the heat exchanger.
- 

0510. One of the functions of a gas turbine engine subsonic intake is to ;
- a) decrease static temperature.
  - b) increase total pressure.
  - c) decrease airflow velocity.
  - d) increase airflow velocity.

Answer ;

- c) decrease airflow velocity.
- 

0511. One of the functions of an accumulator in a hydraulic system is ;
- a) to damp pressure surges in the system.
  - b) to store fluid.
  - c) to store pressure.
  - d) to act as a pressure relief valve.

Answer ;

- a) to damp pressure surges in the system.
- 

0512. One of the purposes of compressor bleed valves is to ;
- a) decrease the airflow across the rear compressor stages , whilst that across the early stages is increased.
  - b) decrease the load on the compressor when the engine is approaching maximum rpm.
  - c) limit the acceleration period of the engine.
  - d) increase the airflow across the rear compressor stages , whilst that across the early stages is decreased.

Answer ;

- a) decrease the airflow across the rear compressor stages , whilst that across the early stages is increased.
-

0513. One of the purposes of the bleed valves fitted to axial flow compressors is to ;
- a) spill compressor air should the engine overspeed thus controlling the speed.
  - b) enable an external air supply to spin up the compressor for engine starting.
  - c) reduce the tendency to compressor stall.
  - d) control the acceleration time of the engine.

Answer ;

c) reduce the tendency to compressor stall.

---

0514. One of the purposes of the fuel system booster pumps being submerged in fuel is ;
- a) to shorten the fuel lines , so minimising the pressure losses.
  - b) to facilitate the priming of the pumps.
  - c) to improve their efficiency.
  - d) to cool the pumps.

Answer ;

d) to cool the pumps.

---

0515. One reason to fit a tip shroud to a turbine blade is to ;
- a) reduce turbine temperatures.
  - b) provide a seal for the cooling airflow.
  - c) increase turbine efficiency.
  - d) prevent blade creep.

Answer ;

c) increase turbine efficiency.

---

0516. One stage of an axial compressor comprises ;
- a) a row of rotor blades followed by a row of stator vanes.
  - b) a row of rotor blades followed by a row of inlet guide vanes.
  - c) inlet guide vanes followed by a row of rotor blades and a row of stator vanes.
  - d) a row of inlet guide vanes followed by a row of rotor blades.

Answer ;

a) a row of rotor blades followed by a row of stator vanes.

---

0517. Overcentre mechanisms in landing gear systems are used to ;
- a) ensure the nose-wheel is positioned fore and aft prior to retraction.
  - b) lock the landing gear in the up position only.
  - c) ensure that the nose-wheel does not exceed the maximum steering arc.
  - d) lock the landing gear in the up and/or down position.

Answer ;

d) lock the landing gear in the up and/or down position.

---

0518. Overheating of a piston engine is likely to result from an excessively ;
- a) low barometric pressure.
  - b) weak mixture.
  - c) rich mixture.
  - d) high barometric pressure.

Answer ;

b) weak mixture.

---

0519. Oxygen systems on pressurized airplanes are used to provide oxygen ;
1. In an emergency in the case of depressurization.
  2. In an emergency in the case of the indisposition of a passenger.
  3. During a normal flight.
  4. In an emergency in case of smoke or toxic gases in the cockpit.

The combination regrouping all the correct statements is ;

- a) 2 , 3.
- b) 1 , 2 , 4.
- c) 1 , 4.
- d) 3.

Answer ;

c) 1 , 4.

---

0520. Parameters to monitor a hydraulic system in the cockpit can be ;

- a) pressure and RPM of the hydraulic pump.
- b) pressure and hydraulic pump output.
- c) pressure , fluid temperature and quantity.
- d) pressure and fluid viscosity.

Answer ;

c) pressure , fluid temperature and quantity.

---

0521. Pneumatic mechanical devices that provide ice protection ;

- a) can only be used as anti-icing devices.
- b) are usually used as de-icing devices.
- c) require large quantities of bleed air.
- d) are usually used on aeroplanes equipped with turbo-fan engines.

Answer ;

b) are usually used as de-icing devices.

---

0522. Pneumatic mechanical ice protection system are mainly used for ;

- a) propellers.
- b) wings.
- c) pitot tubes.
- d) windscreens.

Answer ;

b) wings.

---

0523. Powerplant fire extinguishers are operated by ;

- a) an electrically fired cartridge rupturing a seal in the head of the bottle.
- b) an electrically heated capsule rupturing a seal in the pipeline at the engine.
- c) manual remote control opening a discharge valve.
- d) a mechanical link withdrawing a safety pin and allowing a spring operated plunger to rupture the bottle seal.

Answer ;

a) an electrically fired cartridge rupturing a seal in the head of the bottle.

---

0524. Pre-ignition refers to the condition that may arise when ;

- a) the mixture is ignited before the piston has reached top dead centre.
- b) the sparking plug ignites the mixture too early.
- c) a rich mixture is ignited by the sparking plugs.
- d) the mixture is ignited by abnormal conditions within the cylinder before the spark occurs at the plug.

Answer ;

d) the mixture is ignited by abnormal conditions within the cylinder before the spark occurs at the plug.

---

0525. Primary flight controls are ;

- a) ailerons , elevators , rudder and flaps.
- b) control wheel or stick , rudder pedals , flap lever and throttle.
- c) ailerons , elevators and rudder.
- d) control wheel stick , rudder pedals and speed brake.

Answer ;

c) ailerons , elevators and rudder.

---

0526. Prolonged running at low rpm can have an adverse effect on the functioning of the ;

- a) oil pump.
- b) carburettor.
- c) fuel filter.
- d) spark plugs.

Answer ;

d) spark plugs.

---

0527. Pulling the fire shutoff handle causes a number of devices to disconnect. In respect of the AC generator it can be said that the ;

- a) exciter control relay opens.
- b) exciter control relay and the generator breaker open.
- c) exciter control relay , the generator breaker and the tie breaker open.
- d) generator breaker opens.

Answer ;

b) exciter control relay and the generator breaker open.

---

0528. Purposes of an accumulator in an hydraulic system are ;
1. to damp pressure fluctuations.
  2. to cool the hydraulic fluid.
  3. to serve as a limited alternate source of pressure.
  4. to serve as a main pressure source for normal operation.

The combination that regroupes all of the correct statements is ;

- a) 1 , 3.
- b) 1 , 4.
- c) 2 , 3.
- d) 2 , 4.

Answer ;

- a) 1 , 3.
- 

0529. Reflection from ionospheric layers is used in the following radio frequencies ;

- a) UHF
- b) VHF
- c) VLF
- d) HF

Answer ;

- d) HF
- 

0530. Regarding (1) a fuse and (2) a circuit breaker ;

- a) (1) is suitable for high currents , (2) is not suitable for high currents.
- b) (1) and (2) are not resettable.
- c) (1) is not suitable for high currents , (2) is suitable for high currents.
- d) (1) is not resettable , (2) is resettable.

Answer ;

- d) (1) is not resettable , (2) is resettable.
- 

0531. Regarding a jet engine ;

- I. The maximum thrust decreases as the pressure altitude increases.
  - II. The specific fuel consumption decreases slightly as the pressure altitude increases at constant TAS.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is incorrect.

Answer ;

- a) I is correct , II is correct.
- 

0532. Regarding a jet engine ;

- I. The maximum thrust decreases as the pressure altitude increases.
  - II. The specific fuel consumption is inversely proportional to pressure altitude , at constant TAS.
- a) I is incorrect , II is correct.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- d) I is correct , II is incorrect.
- 

0533. Regarding a jet engine:

- I. The maximum thrust decreases slightly as the pressure altitude decreases.
  - II. The specific fuel consumption increases slightly as the pressure altitude decreases at constant TAS.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

- a) I is incorrect , II is correct.
-



0534. Regarding a jet engine ;

- I. The maximum thrust increases as the pressure altitude decreases.
  - II. The specific fuel consumption increases slightly as the pressure altitude decreases at constant TAS.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

c) I is correct , II is correct.

---

0535. Regarding a jet engine ;

- I. The maximum thrust increases as the pressure altitude decreases.
  - II. The specific fuel consumption is proportional to pressure altitude , at constant TAS.
- a) I is correct , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

a) I is correct , II is incorrect.

---

0536. Regarding a jet engine ;

- I. The maximum thrust increases slightly as the pressure altitude increases.
  - II. The specific fuel consumption decreases slightly as the pressure altitude increases at constant TAS.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

c) I is incorrect , II is correct.

---

0537. Regarding a jet engine ;

- I. The maximum thrust remains constant as the pressure altitude decreases.
  - II. The specific fuel consumption decreases slightly as the pressure altitude decreases at constant TAS.
- a) I is incorrect , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

a) I is incorrect , II is incorrect.

---

0538. Regarding a jet engine ;

- I. The maximum thrust remains constant as the pressure altitude decreases.
  - II. The specific fuel consumption remains constant as the pressure altitude decreases at constant TAS.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

c) I is incorrect , II is incorrect.

---

0539. Regarding a jet engine ;

- I. The maximum thrust remains constant as the pressure altitude increases.
  - II. The specific fuel consumption increases slightly as the pressure altitude increases at constant TAS.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

a) I is incorrect , II is incorrect.

---

0540. Regarding a jet engine;

- I. The maximum thrust remains constant as the pressure altitude increases.
- II. The specific fuel consumption remains constant as the pressure altitude increases at constant TAS.
- a) I is incorrect , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is correct , II is correct.
- d) I is correct , II is incorrect.

Answer ;

- a) I is incorrect , II is incorrect.
- 

0541. Regarding a thermal wing anti-icing system , the correct statement is ;

- a) Aerodynamic performances of the wings are not maintained and there is a reduction of maximum engine thrust.
- b) Aerodynamic performances of the wings are maintained and there is no reduction of maximum engine thrust.
- c) Aerodynamic performances of the wings are not maintained and there is no reduction of maximum engine thrust.
- d) Aerodynamic performances of the wings are maintained and there is a reduction of maximum engine thrust.

Answer ;

- d) Aerodynamic performances of the wings are maintained and there is a reduction of maximum engine thrust.
- 

0542. Regarding fire detection , ion detectors are used to detect ;

- a) smoke.
- b) overtemperature.
- c) overtemperature and fire
- d) fire

Answer ;

- a) smoke.
- 

0543. Regarding Ohm's law ;

- a) The current in a circuit is directly proportional to voltage.
- b) The current in a circuit is inversely proportional to voltage.
- c) The current in a circuit is directly proportional to the resistance of the circuit.
- d) The power in the circuit is inversely proportional to the square of the current.

Answer ;

- a) The current in a circuit is directly proportional to voltage.
- 

0544. Regarding the chemical oxygen generator , to enable the oxygen to flow , the passenger must ;

- a) firmly pull the mask towards his face.
- b) operate the relevant switch in his armrest.
- c) turn the oxygen valve to open.
- d) firmly pull the door compartment of the oxygen mask storage.

Answer ;

- a) firmly pull the mask towards his face.
- 

0545. Regarding the oxygen systems of a JAR 25 aeroplane ;

- a) the seals must be carefully greased to avoid sparks.
- b) the passenger source of supply never uses chemically generated oxygen.
- c) the same source of supply is used by the crew and passengers.
- d) with the setting on "NORMAL" , the crew breathes a mixture of oxygen and cabin air.

Answer ;

- d) With the setting on "NORMAL" , the crew breathes a mixture of oxygen and cabin air.
- 

0546. Rotor brake discs are manufactured from ;

- a) steel or titanium.
- b) aluminium or steel.
- c) steel or carbon fibre.
- d) boron aluminide or titanium.

Answer ;

- c) steel or carbon fibre.
-

0547. Rudder trim adjustment in an aeroplane with irreversible flight controls is ;
- a) unnecessary because this aeroplane does not need rudder trim.
  - b) an adjustment of the rudder ratio changer.
  - c) an adjustment of the rudder trim tab.
  - d) an adjustment of the zero force rudder position.

Answer ;

d) an adjustment of the zero force rudder position.

---

0548. Severe compressor stall can be indicated amongst others by ;
- a) a decrease in TGT and a change in vibration level.
  - b) a rise in TGT and an increase in fuel flow.
  - c) a rise in turbine gas temperature ( TGT ) and a change in vibration level.
  - d) an increase in engine pressure ratio ( EPR ) and a decrease in TGT and fuel flow.

Answer ;

c) a rise in turbine gas temperature ( TGT ) and a change in vibration level.

---

0549. Severe compressor stall can be observed amongst others by ;
- a) a decrease in TGT and an increase in vibration level.
  - b) a rise in turbine gas temperature ( TGT ) and an increase in vibration level.
  - c) a rise in TGT and an increase in fuel flow.
  - d) an increase in engine pressure ratio ( EPR ) and a decrease in TGT and fuel flow.

Answer ;

b) a rise in turbine gas temperature ( TGT ) and an increase in vibration level.

---

0550. Shimmy occurs on the nose wheel landing gear during taxiing when ;
- 1. the wheels tend to describe a sinusoidal motion on the ground.
  - 2. the wheels no longer respond to the pilot's actions.
- This effect is overcome by means of ;
- 3. the torque link.
  - 4. an accumulator associated with the steering cylinder.

The combination of correct statements is:

- a) 1 , 4.
- b) 2 , 3.
- c) 2 , 4.
- d) 1 , 3.

Answer ;

a) 1 , 4.

---

0551. Should a hydraulic pump seize during operation ;
- a) the freewheel unit will disengage the pump from the gearbox.
  - b) the inner barrel of the pump will rotate thus offloading and protecting the gearbox.
  - c) the quill drive will shear to offload and protect the gearbox.
  - d) a red light will illuminate in the cockpit to warn the pilot of the failure.

Answer ;

c) the quill drive will shear to offload and protect the gearbox.

---

0552. Shuttle valves will automatically ;
- a) switch hydraulically operated units to the most appropriate pressure supply.
  - b) reduce pump loads.
  - c) shut down systems which are overloaded.
  - d) guard systems against overpressure.

Answer ;

a) switch hydraulically operated units to the most appropriate pressure supply.

---

0553. Significant torsion effects in a wing during flight can be caused by ;
- a) wing dihedral.
  - b) aileron deflection.
  - c) propwash.
  - d) wing tip vortices.

Answer ;

b) aileron deflection.

---

0554. Skip distance is the ;
- a) thickness of the ionosphere.
  - b) range from the transmitter to the first sky wave.
  - c) wavelength distance of a certain frequency.
  - d) highest critical frequency distance.

Answer ;

b) range from the transmitter to the first sky wave.

---

0555. Smoke detectors fitted on transport aircraft are of the following type ;

- a) electrical.
- b) magnetic.
- c) optical or ionization.
- d) chemical.

Answer ;

c) optical or ionization.

---

0556. Smoke detector systems are installed in the ;

- a) fuel tanks.
- b) upper cargo compartments ( class E ).
- c) engine nacelles
- d) wheel wells

Answer ;

b) upper cargo compartments ( class E ).

---

0557. Some gas turbine engines have multiple stages in the HP and LP turbines ;

- a) to facilitate balancing of the turbine assembly.
- b) because one stage does not generate sufficient shaft power.
- c) to help stabilise the pressure between the compressor and the turbine.
- d) to reduce noise in the propelling nozzle.

Answer ;

b) because one stage does not generate sufficient shaft power.

---

0558. Spark plug fouling is more likely to happen if ;

- a) the engine runs at the authorized maximum continuous power for too long.
- b) the aircraft climbs without mixture adjustment.
- c) the aircraft descends without a mixture adjustment.
- d) power is increased too abruptly.

Answer ;

b) the aircraft climbs without mixture adjustment.

---

0559. Spark plug fouling is more likely to happen if ;

- a) the aircraft descends continuously at idle power.
- b) the aircraft cruises at low speed with the correct mixture.
- c) the engine runs at maximum continuous power for too long.
- d) power is increased too abruptly.

Answer ;

a) the aircraft descends continuously at idle power.

---

0560. Spark timing is related to engine speed in the way that the ;

- a) slower the engine functions , the more the spark is advanced.
- b) faster the engine functions , the more retarded the spark is.
- c) faster the engine functions , the more the spark is advanced.
- d) faster the engine functions , the further past TDC the spark occurs.

Answer ;

c) faster the engine functions , the more the spark is advanced.

---

0561. Specific fuel consumption for a turbo-prop engine can be expressed in kg per ;

- a) hour per unit of shaft power.
- b) hour per nm.
- c) per unit of shaft power.
- d) hour per km.

Answer ;

a) hour per unit of shaft power.

---

0562. Specific fuel consumption is defined as the ;
- a) maximum fuel consumption of the aircraft.
  - b) mass of fuel required to produce unit power for unit time.
  - c) designed fuel consumption for a given rpm.
  - d) quantity of fuel required to run the engine for one minute at maximum operating conditions.

Answer ;

- b) mass of fuel required to produce unit power for unit time.
- 

0563. Static dischargers ;
- 1. are used to set all the parts of the airframe to the same electrical potential.
  - 2. are placed on wing and tail tips to facilitate electrical discharge.
  - 3. are used to reset the electrostatic potential of the aircraft to a value approximating 0 volts.
  - 4. are located on wing and tail tips to reduce interference with the on-board radiocommunication systems to a minimum.
  - 5. limit the risks of transfer of electrical charges between the aircraft and the electrified clouds.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 5.
- b) 1 , 3 , 4.
- c) 2 , 4 , 5.
- d) 3 , 4 , 5.

Answer ;

- c) 2 , 4 , 5.
- 

0564. Static gas temperature decrease when flowing at a subsonic speed through a jet engine exhaust nozzle because the ;
- a) ambient air cools the exhaust casing.
  - b) exhaust nozzle is convergent.
  - c) velocity of the gas stream decreases.
  - d) static pressure increases when passing the nozzle.

Answer ;

- b) exhaust nozzle is convergent.
- 

0565. The ABS ( Auto Brake System ) is being disconnected after landing ;
- a) by pilot action.
  - b) the system is always armed.
  - c) at a certain low speed.
  - d) automatically.

Answer ;

- a) by pilot action.
- 

0566. The accessory units driven by the accessory gearbox of a turbo-jet engine are the ;
- 1. thrust reverser pneumatic motors.
  - 2. AC generator through its Constant Speed Drive ( CSD ).
  - 3. oil pumps.
  - 4. hydraulic pumps.
  - 5. high pressure fuel pumps.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 2 , 3 , 4 , 5.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3 , 4 , 5.

Answer ;

- b) 2 , 3 , 4 , 5.
- 

0567. The accumulator in a hydraulic system works as ;
- a) a fluid storage.
  - b) a pressure storage.
  - c) a volume storage.
  - d) an energy storage.

Answer ;

- d) an energy storage.
-

**0568.** The advantages of a chemical oxygen source for the passenger cabin are ;

1. reduced weight and volume.
2. easy storage and maintenance.
3. greater autonomy.
4. no risk of explosion.
5. reversible functioning.
6. no maintenance.

The combination regrouping all the correct statements is ;

- a) 1, 3, 4, 5.
- b) 2, 3, 5.
- c) 1, 2, 4, 6.
- d) 1, 2, 3, 4, 5, 6.

**Answer ;**

- c) 1, 2, 4, 6.**
- 

**0569.** The advantages of a gaseous oxygen source compared with a chemical source for the passenger cabin are ;

1. no risk of fire and/or explosion.
2. possibility to regulate flow.
3. more capacity.

- a) 2, 3.
- b) 1.
- c) 2.
- d) 1, 2.

**Answer ;**

- a) 2, 3.**
- 

**0570.** The advantages of a gaseous oxygen source for the passenger cabin are ;

1. a greater autonomy.
2. no risk of explosion.
3. reversible functioning.
4. easy storage and maintenance.
5. possibility to regulate flow.

The combination regrouping all the correct statements is ;

- a) 2, 4, 5.
- b) 1, 3, 5.
- c) 2, 3, 4.
- d) 1, 2, 3, 4, 5.

**Answer ;**

- b) 1, 3, 5.**
- 

**0571.** The advantages of alternating current on board an aircraft are ;

1. simple connection.
2. high starting torque.
3. flexibility in use.
4. lighter weight of equipment.
5. easy to convert into direct current.
6. easy maintenance of machines.

The combination of correct statements is ;

- a) 3, 4, 5, 6.
- b) 1, 4, 6.
- c) 1, 2, 3, 5, 6.
- d) 1, 2, 3, 4, 5, 6.

**Answer ;**

- a) 3, 4, 5, 6.**
-

0572. The advantages of fly-by-wire control are ;

1. reduction of the electric and hydraulic power required to operate the control surfaces.
2. lesser sensitivity to lightning strike.
3. direct and indirect weight saving through simplification of systems.
4. immunity to different interfering signals.
5. improvement of piloting quality throughout the flight envelope.

The combination regrouping all the correct statements is ;

- a) 1 and 2.
- b) 3 and 5.
- c) 3 , 4 and 5.
- d) 1 and 5.

Answer ;

**b) 3 and 5.**

---

0573. The advantages of grounding the negative pole of the aircraft structure are ;

1. Weight saving.
2. Easy fault detection.
3. Increase of short-circuit risk.
4. Reduction of short-circuit risk.
5. Circuits are not single-wired lines.

The combination regrouping all the correct statements is ;

- a) 1 , 3 , 5.
- b) 1 , 2 , 4.
- c) 2 , 3 , 5.
- d) 1 , 2 , 3.

Answer ;

**b) 1 , 2 , 4.**

---

0574. The advantages of Nickel-Cadmium compared with lead-acid batteries are ;

1. lower risk of thermal runaway.
2. higher internal resistance , hence higher power.
3. reduced charging time.
4. constant output voltage.

The combination which regroups all the correct statements is ;

- a) 1 , 3.
- b) 2 , 4.
- c) 1 , 2.
- d) 3 , 4.

Answer ;

**d) 3 , 4.**

---

0575. The air flow passages between adjacent axial compressor rotor blades are divergent , but the absolute exit velocity is higher than the absolute entry velocity of the rotor. This is because ;

- a) the compressor rotational velocity slows down the air flow.
- b) the gas expands between the blades.
- c) part of the compressor work delivered to the air is converted into kinetic energy.
- d) it is a constant temperature process.

Answer ;

**c) part of the compressor work delivered to the air is converted into kinetic energy.**

---

0576. The air in a piston engine supercharger enters the centrifugal compressor at ;

- a) the diffuser and is fed to the impeller at the optimum angle of attack.
- b) a tangent to the rotor and leaves via the stator.
- c) the eye of the impeller and leaves it almost at a tangent to the periphery.
- d) the periphery and leaves via the eye of the impeller.

Answer ;

**c) the eye of the impeller and leaves it almost at a tangent to the periphery.**

---

0577. The air in a piston engine turbo-supercharger centrifugal compressor ;

- a) enters via the diffuser and is fed to the impeller at the optimum angle of attack.
- b) enters the eye of the impeller and leaves at a tangent to the periphery.
- c) enters at the periphery and leaves via the eye of the impeller.
- d) enters at a tangent to the rotor and leaves via the stator.

Answer ;

**b) enters the eye of the impeller and leaves at a tangent to the periphery.**

---

0578. The aircraft hydraulic system is designed to produce ;

- a) high pressure and small flow.
- b) small pressure and small flow.
- c) high pressure and large flow.
- d) small pressure and large flow.

Answer ;

c) high pressure and large flow.

---

0579. The alternators , when connected , are usually connected ;

- a) in parallel mode.
- b) dependant on the type of engine.
- c) dependant on the type of generator.
- d) in series mode.

Answer ;

a) in parallel mode.

---

0580. The anti-icing method for the wings of large jet transport aeroplanes most commonly used in flight is ;

- a) electrical ( electrical resistance ).
- b) thermal ( use of hot air ).
- c) mechanical ( pneumatic boots ).
- d) chemical ( glycol-based liquid ).

Answer ;

b) thermal ( use of hot air ).

---

0581. The application of carburettor heat reduces ;

- a) the density of air entering the carburettor , thus enriching the fuel/air mixture.
- b) it reduces the density of air entering the carburettor , thus leaning the fuel/air mixture.
- c) the volume of air entering the carburettor , thus enriching the fuel/air mixture.
- d) the volume of air entering the carburettor , thus leaning the fuel/air mixture.

Answer ;

a) the density of air entering the carburettor , thus enriching the fuel/air mixture.

---

0582. The application of carburettor heating reduces the ;

- a) density of air entering the carburettor , thus leaning the mixture.
- b) volume of air entering the carburettor , thus enriching the mixture.
- c) volume of air entering the carburettor , thus leaning the mixture.
- d) density of air entering the carburettor , thus enriching the mixture.

Answer ;

d) density of air entering the carburettor , thus enriching the mixture.

---

0583. The auto-brake system is disconnected during landing ;

- a) below a specific speed.
- b) after a preset elapsed time.
- c) when selecting the reverse thrust.
- d) by pilot action.

Answer ;

d) by pilot action.

---

0584. The Automatic Direction Finder uses the following wavelengths ;

- a) decimetric.
- b) hectometric or kilometric.
- c) metric.
- d) centimetric.

Answer ;

b) hectometric or kilometric.

---

0585. The automatic fuelling shut off valve ;

- a) cuts off the fuel in case of engine fire.
- b) stops fuelling as soon as a certain pressure is reached.
- c) stops fuelling as soon as a certain fuel level is reached inside the tank.
- d) stops fuelling as soon as the fuel spills into the vent line.

Answer ;

c) stops fuelling as soon as a certain fuel level is reached inside the tank.

---



0586. The automatic ground spoiler extension system is normally activated during landing by ;
- a) brake pressure application.
  - b) ground spoiler handle.
  - c) main wheel spin up.
  - d) idle thrust selection.

Answer ;

c) main wheel spin up.

---

0587. The Auxiliary Power Unit ( APU ) has its own AC generator that ;
- a) must have the same characteristics as the main AC generator so that it can be easily coupled with the latter.
  - b) is excited by its Generator Control Unit ( GCU ) as soon as the APU starts up.
  - c) is driven at constant speed through a Constant Speed Drive ( CSD ) , in the same way as the main AC generator.
  - d) supplies the aircraft with three-phase 115-200 V , 400 Hz AC.

Answer ;

d) supplies the aircraft with three-phase 115-200 V , 400 Hz AC.

---

0588. The axial compressor of a gas turbine engine usually has more stages than its driving turbine because ;
- a) the compressor consumes much more power than the turbine.
  - b) turbine speed is much higher than compressor speed.
  - c) bleed air is taken from the compressor.
  - d) the power output of a turbine stage is higher than the power consumption of a compressor stage.

Answer ;

d) the power output of a turbine stage is higher than the power consumption of a compressor stage.

---

0589. The bending loads on a cantilever wing due to lift are carried by the ;
- 1. upper skin surface.
  - 2. lower skin surface.
  - 3. wing root fairing.
  - 4. spar or spars.

The combination that regroups all of the correct statements is ;

- a) 3 , 4.
- b) 1 , 2 , 4.
- c) 2 , 3.
- d) 1 , 3 , 4.

Answer ;

b) 1 , 2 , 4.

---

0590. The bimetal strip detectors of an engine fire detection system ;
- 1. are arranged in series.
  - 2. are arranged in parallel.
  - 3. open during a fire.
  - 4. close during a fire.

The combination containing all of the correct statements is ;

- a) 1 and 3 only.
- b) 2 and 3 only.
- c) 2 and 4 only.
- d) 1 and 4 only.

Answer ;

c) 2 and 4 only.

---

0591. The blade angle of a constant-speed propeller ;
- a) is independent of the true air speed.
  - b) increases with increasing true air speed.
  - c) only varies with engine RPM.
  - d) decreases with increasing true air speed.

Answer ;

b) increases with increasing true air speed.

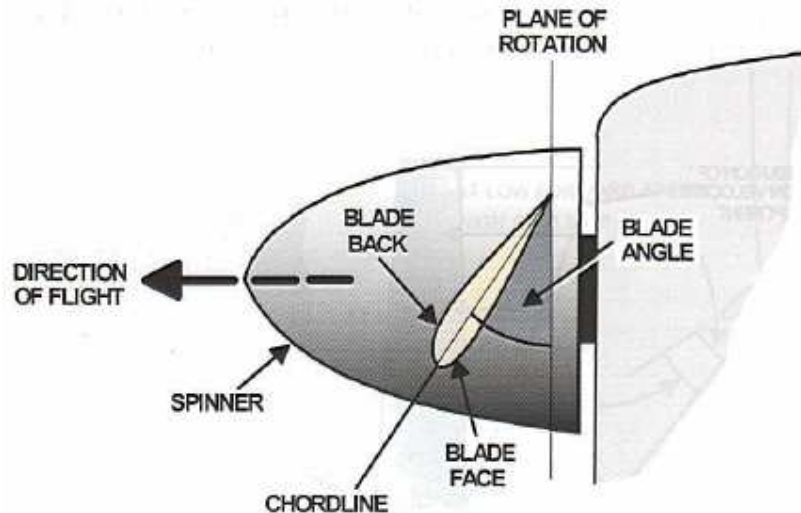
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0592. The blade angle of a propeller is the angle between the ;
- a) reference chord line and the propeller plane of rotation.
  - b) propeller plane of rotation and the relative airflow.
  - c) propeller reference chord line and the extremity of the propeller.
  - d) propeller reference chord line and the relative airflow.

Answer ;

a) reference chord line and the propeller plane of rotation.

Açıklama ;



0593. The blade angle of a propeller is the angle between the ;
- a) reference chord line and the relative airflow.
  - b) reference chord line and the propeller axis of rotation.
  - c) plane of rotation and the relative airflow.
  - d) reference chord line and the propeller plane of rotation.

Answer ;

d) reference chord line and the propeller plane of rotation.

Açıklama ;

Soru 0601'in açıklamasına bakınız.

0594. The blades in an axial flow compressor decrease in size from the ;
- a) low pressure to the high pressure stages in an attempt to sustain axial velocity.
  - b) root to the tip in order to maintain a constant angle of attack.
  - c) high pressure to the low pressure stages in order to allow for expansion at increased velocity.
  - d) root to the tip in order to increase tip clearance.

Answer ;

a) low pressure to the high pressure stages in an attempt to sustain axial velocity.

0595. The built-in passenger oxygen system be activated by ;

- a) opening the oxygen-bottle valves.
- b) switching the diluter demand regulator ON.
- c) switching the diluter demand regulator and the passenger oxygen ON.
- d) switching the passenger oxygen ON.

Answer ;

d) switching the passenger oxygen ON.

0596. The by-pass air in a by-pass type gas turbine engine promotes lower specific fuel consumption by causing an overall ;
- a) decrease in exhaust gas flow average velocity and higher propulsive efficiency.
  - b) increase in exhaust gas flow average velocity thus increasing engine pressure ratio.
  - c) decrease in combustion chamber temperature and higher thermal efficiency.
  - d) increase in the mean jet temperature and increase in thermal efficiency.

Answer ;

a) decrease in exhaust gas flow average velocity and higher propulsive efficiency.

0597. The by-pass air in a turbo fan engine has the effect of a lower specific fuel consumption by causing an overall ;
- a) decrease in combustion chamber temperature and higher thermal efficiency.
  - b) increase in average exhaust gas flow velocity thus increasing engine pressure ratio.
  - c) decrease in average exhaust gas flow velocity and a higher mass flow.
  - d) increase in the mean jet temperature and increase in thermal efficiency.

Answer ;

- c) decrease in average exhaust gas flow velocity and a higher mass flow.
- 

0598. The bypass ratio ;

- I. is the ratio of bypass air mass flow to HP compressor mass flow.
  - II. can be determined from the inlet air mass flow and the HP compressor mass flow.
- a) I is incorrect , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

- c) I is correct , II is correct.
- 

0599. The bypass ratio ;

- I. is the ratio of bypass air mass flow to HP compressor mass flow.
  - II. can be determined from the inlet air volume flow and the HP compressor volume flow.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

- b) I is correct , II is incorrect.
- 

0600. The bypass ratio ;

- I. is the ratio of inlet air mass flow to exhaust air mass flow.
  - II. can be determined from the bypass air mass flow and the HP compressor mass flow.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- d) I is incorrect , II is correct.
- 

0601. The bypass ratio ;

- I. is the ratio of inlet air mass flow to HP compressor mass flow.
  - II. can be determined from the inlet air volume flow and the HP compressor volume flow.
- a) I is incorrect , II is correct.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- c) I is incorrect , II is incorrect.
- 

0602. The bypass ratio ;

- I. is the ratio of inlet airflow to HP compressor flow.
  - II. can be determined from the inlet air mass flow and the HP compressor mass flow.
- a) I is correct , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

- c) I is incorrect , II is correct.
-

0603. The bypass ratio ;
- I. is the ratio of the volume of air passing through the inlet to that through the HP compressor.
  - II. can be determined from the inlet air mass flow and the bypass mass flow.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- d) I is incorrect , II is correct.
- 

0604. The “cabin differential pressure” is ;
- a) the pressure differential between the air entering and leaving the cabin.
  - b) approximately 5 psi at maximum.
  - c) cabin pressure minus ambient pressure.
  - d) approximately 15 psi at maximum.

Answer ;

- c) cabin pressure minus ambient pressure.
- 

0605. The cabin heating supply in a heavy jet transport aircraft is obtained from ;
- a) an electrical heater system.
  - b) a fuel heater system.
  - c) hot air coming from the engine’s turbines.
  - d) hot air coming from the engine’s compressors.

Answer ;

- d) hot air coming from the engine’s compressors.
- 

0606. The cabin pressure is regulated by the ;
- a) Air conditioning pack.
  - b) Air cycle machine.
  - c) Outflow valve.
  - d) Cabin inlet airflow valve.

Answer ;

- c) Outflow valve.
- 

0607. The cabin rate of descent ;
- a) is always the same as the airplane's rate of descent.
  - b) results in a cabin pressure decrease.
  - c) is not possible at constant airplane altitudes.
  - d) results in a cabin pressure increase.

Answer ;

- d) results in a cabin pressure increase.
- 

0608. The capacitance type fuel gauging system indicates the fuel quantity by measuring the ;
- a) electrical resistance change.
  - b) density variation of the fuel.
  - c) resistivity variation of the fuel.
  - d) dielectric change between fuel and air.

Answer ;

- d) dielectric change between fuel and air.
- 

0609. The capacity of a battery is given in ;
- a) Ohms.
  - b) Watts.
  - c) Ampere hours.
  - d) Amperes/volts.

Answer ;

- c) Ampere hours.
- 

0610. The capacity of a battery is the ;
- a) no-load voltage of the battery multiplied by its rated output current.
  - b) number of cycles ( charging and discharging ) that a battery can withstand without deterioration of its cells.
  - c) intensity withstood by the battery during charging.
  - d) amount of Ampere hours that a fully charged battery can supply.

Answer ;

- d) amount of Ampere hours that a fully charged battery can supply.
-

- 0611.** The capacity of an accumulator is ;
- the quantity of electricity that the battery can supply during discharge.
  - the number of cycles ( charging and discharging ) that a battery can withstand without deterioration of its cells.
  - the no-load voltage of the battery multiplied by its rated output current.
  - the intensity withstood by the battery during charging.

**Answer ;**

- a)** the quantity of electricity that the battery can supply during discharge.
- 

- 0612.** The Central Processing Unit ( CPU ) of a computer essentially consists of ;
- an Arithmetic and Logic Unit ( ALU ).
  - a control and timing unit.
  - registers.

The combination which regroups all the correct statements is ;

- 1 , 2.
- 2 , 3.
- 1 , 3.
- 1 , 2 , 3.

**Answer ;**

- d)** 1 , 2 , 3.
- 

- 0613.** The characteristics of a centrifugal compressor are ;
- a high pressure ratio by stage.
  - a large diameter.
  - a low pressure ratio by stage.
  - a small diameter.

The combination containing all of the correct statements is ;

- 2 and 3 only.
- 3 and 4 only.
- 1 and 4 only.
- 1 and 2 only.

**Answer ;**

- d)** 1 and 2 only.
- 

- 0614.** The characteristics of an axial compressor are ;
- a low pressure ratio by stage.
  - a high pressure ratio by stage.
  - the possibility of compressing a large mass airflow.
  - the inability of compressing a large mass airflow.

The combination that regroups all the correct statements is ;

- 2 and 4 only.
- 2 and 3 only.
- 1 and 4 only.
- 1 and 3 only.

**Answer ;**

- d)** 1 and 3 only.
- 

- 0615.** The chemical oxygen generator supplies oxygen for about ;
- 30 minutes.
  - 2 hours.
  - 15 minutes.
  - 5 minutes.

**Answer ;**

- c)** 15 minutes.
- 

- 0616.** The colour of a fresh synthetic hydraulic fluids is ;
- red.
  - blue.
  - purple.
  - pink.

**Answer ;**

- c)** purple.
-

0617. The component that converts hydraulic pressure into linear motion is called ;

- a) an actuator or jack.
- b) a hydraulic pump.
- c) an accumulator.
- d) a pressure regulator.

Answer ;

a) an actuator or jack.

---

0618. The compression ratio of a piston engine is the ratio of the ;

- a) diameter of the bore to the piston stroke.
- b) area of the piston to the cylinder volume.
- c) weight of the air induced to its weight after compression.
- d) volume of the cylinder with the piston at bottom dead centre to that with the piston at top dead centre.

Answer ;

d) volume of the cylinder with the piston at bottom dead centre to that with the piston at top dead centre.

---

0619. The compressor blades in a rotating gas turbine engine are held in position by ;

- a) a centrifugal latch.
- b) abradable seals and aerodynamic forces.
- c) the mountings and thermal expansion only.
- d) the mountings and the resultant of aerodynamic and centrifugal forces.

Answer ;

d) the mountings and the resultant of aerodynamic and centrifugal forces.

---

0620. The compressor pressure ratio of an axial flow compressor is primarily determined by the ;

- a) number of stages.
- b) compressor inlet pressure design value.
- c) compressor inlet temperature design value.
- d) rotor diameter.

Answer ;

a) number of stages.

---

0621. The compressor surge effect during acceleration is prevented by the ;

- a) inlet guide vanes.
- b) surge bleed valves.
- c) Fuel Control Unit ( FCU ).
- d) variable setting type nozzle guide vanes.

Answer ;

c) Fuel Control Unit ( FCU ).

---

0622. The conditions most likely to produce the highest engine power are ;

- a) warm air at low pressure with high relative humidity.
- b) cold and dry air at high pressure.
- c) cold air at high pressure with high relative humidity.
- d) warm and dry air at high pressure .

Answer ;

b) cold and dry air at high pressure.

---

0623. The conditions to be met to activate a shunt generator are ;

1. presence of a permanent field.
2. closed electrical circuit.
3. generator terminals short-circuited.
4. minimum rotation speed.

The combination of correct statements is ;

- a) 1 , 4.
- b) 1 , 3.
- c) 2 , 3.
- d) 1 , 2.

Answer ;

a) 1 , 4.

---

0624. The conditions under which you obtain the highest engine power are ;

- a) cold and dry air at high pressure.
- b) warm and humid air at low pressure.
- c) cold and humid air at high pressure.
- d) warm and dry air at high pressure.

Answer ;

a) cold and dry air at high pressure.

---

0625. The conditions which can cause detonation are ;

- a) low manifold pressure and high fuel flow.
- b) high manifold pressure and low revolutions per minute.
- c) high manifold pressure and high revolutions per minute.
- d) low manifold pressure and high revolutions per minute.

Answer ;

b) high manifold pressure and low revolutions per minute.

---

0626. The connection in parallel of two 12 volt/ 40 amp hours batteries , will create a unit with the following characteristics ;

- a) 24 volt / 80 amp hours.
- b) 24 volt / 40 amp hours.
- c) 12 volt / 80 amp hours.
- d) 12 volt / 40 amp hours.

Answer ;

c) 12 volt / 80 amp hours.

---

0627. The contents of an oxygen bottle can be ensure by the ;

- 1. weight.
- 2. pressure.
- 3. level.
- 4. volume.

The combination containing all of the correct statements is ;

- a) 1 and 2 only.
- b) 2 and 3 only.
- c) 1 and 4 only.
- d) 1 and 3 only.

Answer ;

a) 1 and 2 only.

---

0628. The control of free turbine engines on turboprops , is accomplished by ;

a propeller control lever used to select ;

- 1. propeller RPM.
- 2. turbine temperature.
- 3. turbine RPM.

a fuel control lever used to select ;

- 4. propeller RPM.
- 5. torque.
- 6. turbine temperature.

The combination which regroups all of the correct statements is ,

- a) 1 , 5 , 6.
- b) 1 , 3 , 5.
- c) 3 , 4 , 6.
- d) 2 , 4 , 5.

Answer ;

a) 1 , 5 , 6.

---

0629. The correct formula to calculate the total displacement of a multi-cylinder piston engine is ;

- a) cylinder length \* cylinder diameter.
- b) cylinder volume \* number of cylinders.
- c) piston area \* piston stroke.
- d) piston area \* piston stroke \* number of cylinders.

Answer ;

d) piston area \* piston stroke \* number of cylinders.

---

0630. The correct order of decreasing freezing points of the three mentioned fuels is ;

- a) Jet A-1 , Jet A , Jet B.
- b) Jet A , Jet A-1 , Jet B.
- c) Jet B , Jet A , Jet A-1.
- d) Jet B , Jet A-1 , Jet A.

Answer ;

- b) Jet A , Jet A-1 , Jet B.
- 

0631. The correct statement about rain protection for cockpit windshields is that ;

- a) wipers are sufficient under heavy rain conditions to provide adequate view through the cockpit windows.
- b) rain repellent should never be sprayed onto the windshield unless the rainfall is very heavy.
- c) the alcohol de-icing system for cockpit windows is also suitable for rain protection.
- d) the electric de-icing system for cockpit windows is also suitable for rain protection.

Answer ;

- b) rain repellent should never be sprayed onto the windshield unless the rainfall is very heavy.
- 

0632. The crank assembly consists of ;

- a) crankshaft , connecting rods and pistons.
- b) crankshaft , camshaft , valves , valve springs and push rods.
- c) propeller , crankshaft , pistons and connecting rods.
- d) Crankcase , crankshaft , connecting rods and pistons.

Answer ;

- a) crankshaft , connecting rods and pistons.
- 

0633. The crank assembly of a piston engine comprises the ;

- a) propeller , crankshaft , pistons and connecting rods.
- b) crankshaft , connecting rods and pistons.
- c) crankcase , crankshaft , connecting rods and pistons.
- d) crankshaft , camshaft , valves , valve springs and push rods.

Answer ;

- b) crankshaft , connecting rods and pistons.
- 

0634. The crash/fire axe is part of the safety equipment fitted to passenger aircraft. Its function is to ;

- a) activate a radio survival beacon by cutting off the red coloured top.
- b) settle an escalating conflict with unreasonable passengers , who threaten flight safety.
- c) free exits in case of evacuation via the sides.
- d) obtain forced access to a fire behind a panel and a general purpose tool during evacuation.

Answer ;

- d) obtain forced access to a fire behind a panel and a general purpose tool during evacuation.
- 

0635. The cross-feed fuel system can be used to ;

- 1. feed any engine from any fuel tank.
  - 2. dump the unusable fuel.
  - 3. adjust fuel distribution.
  - 4. to transfer fuel from one tank to another located in the same wing.
- a) 1 , 4.
  - b) 2 , 3.
  - c) 1 , 3.
  - d) 2 , 4.

Answer ;

- c) 1 , 3.
- 

0636. The cross-feed fuel system enables ;

- a) the supply of the outboard jet engines from any outboard fuel tank.
- b) the supply of any jet engine from any fuel tank.
- c) the supply of the jet engines mounted on a wing from any fuel tank within that wing.
- d) only the transfer of fuel from the centre tank to the wing tanks.

Answer ;

- b) the supply of any jet engine from any fuel tank.
-



0637. The cross-feed fuel system is used to ;
- a) allow the unusable fuel elimination.
  - b) allow the fuel to be quickly thrown away in case of emergency
  - c) automatically fill every tank up to the desired level.
  - d) feed every engine from any fuel tank.

Answer ;  
d) feed every engine from any fuel tank.

---

0638. The cylinder head and oil temperature gauges are to exceed the normal operating ranges if a pilot ;
- a) uses fuel that has a rating higher-than-specified for the engine.
  - b) uses fuel that has a rating lower-than-specified for the engine.
  - c) operating with higher-than-normal oil pressure.
  - d) operates with the mixture control set too rich.

Answer ;  
b) uses fuel that has a rating lower-than-specified for the engine.

---

0639. The cylinder head and oil temperatures may exceed their normal operating ranges if ;
- a) the engine is operated at a higher than normal oil pressure.
  - b) a higher octane rating than specified for the engine is used.
  - c) a lower octane rating than specified for the engine is used.
  - d) the engine is operated at too rich mixture.

Answer ;  
c) a lower octane rating than specified for the engine is used.

---

0640. The damping element in a landing gear shock absorber used on large aircrafts is ;
- a) Oil.
  - b) Nitrogen.
  - c) Oxygen.
  - d) Springs.

Answer ;  
a) Oil.

---

0641. The de-icing system which is mostly used for the wings of modern turboprop aeroplanes is:
- a) Fluid de-icing.
  - b) Electrical heating.
  - c) Pneumatic boots.
  - d) Thermal anti-icing.

Answer ;  
c) Pneumatic boots.

---

0642. The demand valve of a diluter-demand type oxygen regulator in normal mode , operates when the ;
- a) pressure in the oxygen reservoir is more than 500 psi.
  - b) user requires 100 percent oxygen.
  - c) user breathes in.
  - d) diluter control is in normal position.

Answer ;  
c) user breathes in.

---



**0646.** The disadvantages of axial flow compressors compared to centrifugal flow compressors are ;

1. more expensive to manufacture.
2. lower airflow.
3. greater vulnerability to foreign object damage.
4. lower pressure ratio.

The combination which regroups all of the correct answers is ;

- a) 1 , 3.
- b) 1 , 2.
- c) 2 , 4.
- d) 2 , 3.

**Answer ;**

**a) 1 , 3.**

---

**0647.** The Distance Measuring Equipment ( DME ) uses the following wavelengths ;

- a) centimetric.
- b) decimetric.
- c) hectometric.
- d) metric.

**Answer ;**

**b) decimetric.**

---

**0648.** The effect of ram air on the operation of a gas turbine engine is ;

- a) an increase in the engine rpm for take-off.
- b) a lower efficiency of the engine with an increase in aeroplane speed.
- c) an increase in the efficiency of the engine.
- d) an increase in the indicated air speed ( IAS ) at which the aeroplane should be flown for optimum range.

**Answer ;**

**c) an increase in the efficiency of the engine.**

---

**0649.** The electronic engine control of a FADEC gas turbine engine checks ;

- a) input data only.
- b) ACTIVE CHANNEL input and output data only.
- c) output data only.
- d) all input and output data.

**Answer ;**

**d) all input and output data.**

---

**0650.** The elements specifically protected against icing on transport aircraft are ;

1. engine air intake and pod.
2. front glass shield.
3. radome.
4. pitot tubes and waste water exhaust masts.
5. leading edge of wing.
6. cabin windows.
7. trailing edge of wings.
8. electronic equipment compartment.

The combination regrouping all the correct statements is ;

- a) 1 , 2 , 3 , 8.
- b) 1 , 2 , 5 , 6.
- c) 1 , 2 , 4 , 5.
- d) 1 , 4 , 5 , 7.

**Answer ;**

**c) 1 , 2 , 4 , 5.**

---

**0651.** The Engine Pressure Ratio ( EPR ) is the ratio of ;

- a) the total turbine inlet pressure to the total compressor outlet pressure.
- b) the total turbine outlet pressure to the total compressor inlet pressure.
- c) the total turbine inlet pressure to the total compressor inlet pressure.
- d) the total turbine outlet pressure to the total compressor outlet pressure.

**Answer ;**

**b) the total turbine outlet pressure to the total compressor inlet pressure.**

---

0652. The equipment of an oxygen supply installation must be kept absolutely free of oil or grease traces as ;
- a) these substances catch fire spontaneously in the presence of oxygen under pressure.
  - b) these substances mixed with oxygen could catch fire in the presence of a spark.
  - c) the oxygen system would be contaminated.
  - d) these substances could plug the oxygen masks filters.

Answer ;

- a) these substances catch fire spontaneously in the presence of oxygen under pressure.
- 

0653. The essential difference between aircraft AC alternators and DC generators ( dynamos ) is that the ;
- a) induced ( output ) windings of the alternators are fixed ( stator ) , and the dynamos have a fixed inductor ( field ) coil.
  - b) alternators supply all of the output current through the commutators and brush assemblies.
  - c) induced windings of the alternators are rotating ( rotor ) , and the dynamos have a rotary inductor coil.
  - d) The alternators generate much less power than DC generators.

Answer ;

- a) induced ( output ) windings of the alternators are fixed ( stator ) , and the dynamos have a fixed inductor ( field ) coil.
- 

0654. The expression “primary flight control” applies to the ;

- 1. stabiliser.
- 2. rudder.
- 3. speed brake.
- 4. aileron.

The combination that regroups all of the correct statements is ;

- a) 1 , 4.
- b) 2 , 3.
- c) 2 , 4.
- d) 1 , 2 , 3 , 4.

Answer ;

- c) 2 , 4.
- 

0655. The expression “primary flight control” applies to the ;

- 1. elevator.
- 2. speed brake.
- 3. lift-augmentation devices.
- 4. roll spoiler.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 2 , 4.
- c) 1 , 4.
- d) 1 , 2 , 3 , 4.

Answer ;

- c) 1 , 4.
- 

0656. The expression “secondary flight control” applies to the ;

- 1. elevator.
- 2. speed brake.
- 3. lift-augmentation devices.
- 4. roll spoiler.

The combination that regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1 , 4.
- c) 2 , 4.
- d) 2 , 3.

Answer ;

- d) 2 , 3.
-

0657. The expression “secondary flight control” applies to the ;

- 1. stabiliser.
- 2. rudder.
- 3. speed brake.
- 4. aileron.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 2 , 3 , 4.
- c) 1 , 3.
- d) 2 , 4.

Answer ;

c) 1 , 3.

---

0658. The FADEC of a gas turbine engine consists of ;

- a) an electronic engine control unit and its peripherals ( fuel metering , actuators , electrical wiring , sensors ).
- b) all electronic engine control units installed on the aircraft.
- c) an electronic engine control unit and the engine display units in the cockpit.
- d) an electronic engine control unit only.

Answer ;

a) an electronic engine control unit and its peripherals ( fuel metering , actuators , electrical wiring , sensors ).

---

0659. The FADEC of a gas turbine engine normally includes ;

- a) the engine fire loops.
- b) a overheat protection for the electronic engine control unit.
- c) an excessive fuel flow detection function.
- d) an engine overspeed and/or an EGT protection function.

Answer ;

d) an engine overspeed and/or an EGT protection function.

---

0660. The fan in a high by-pass ratio turbo-jet engine produces ;

- a) none of the thrust.
- b) the lesser part of the thrust.
- c) the greater part of the thrust.
- d) half the thrust.

Answer ;

c) the greater part of the thrust.

---

0661. The fan of a turbo-fan engine is driven by ;

- a) airflow drawn across it by the HP compressor.
- b) the LP turbine.
- c) the HP compressor through reduction gearing.
- d) the HP turbine.

Answer ;

b) the LP turbine.

---

0662. The first indication of carburettor icing in aeroplanes equipped with constant speed propellers would most likely be a ;

- a) decrease in rpm.
- b) rough running engine followed by an increase in manifold pressure.
- c) rough running engine followed by loss in rpm.
- d) decrease in manifold pressure.

Answer ;

d) decrease in manifold pressure.

---

0663. The following problems may occur when starting a twin-spool turbofan engine ;

- a) no N2 rotation , although N1 may accelerate normally.
- b) a hung ( abortive ) start with no light-up indication after “fuel-on” selection.
- c) a hung ( abortive ) start with fuel flow indication zero after “fuel-on” selection.
- d) no N1 rotation , although N2 may accelerate normally.

Answer ;

d) no N1 rotation , although N2 may accelerate normally.

---

0664. The frequency of an AC generator is dependent on the ;
- a) number of individual poles and the field strength.
  - b) number of pairs of poles and the speed of the moving part.
  - c) number of individual poles only.
  - d) field strength and the speed of the moving part.

Answer ;

- b) number of pairs of poles and the speed of the moving part.
- 

0665. The frequency of the current provided by an alternator depends on ;
- a) the strength of the excitation current.
  - b) its rotation speed.
  - c) its phase balance.
  - d) its load.

Answer ;

- b) its rotation speed.
- 

0666. The frequency which corresponds to a wavelength of 12 cm is ;
- a) 3600 MHz.
  - b) 2500 kHz.
  - c) 360 MHz.
  - d) 2500 MHz.

Answer ;

- d) 2500 MHz.

Çözüm ;

$f \text{ (frekans)} = C \text{ (Işık hızı)} : \lambda \text{ (Dalga boyu)}$   
Işık hızı olarak alınan  $(300 \times 10^6)$  değeri m/sec cinsinden olduğu için ;  
 $12 \text{ cm} = 0,12 \text{ m}$  , ve de ;  
 $f = (300 \times 10^6) : (0,12 \times 10^6) \rightarrow f = 300 : 0,12 = 2500 \text{ MHz}$  olarak bulunur.

---

0667. The fuel cross-feed system ;
- a) is only used to feed an engine from the tank of the opposite wing.
  - b) is only used on the ground for fuel transfer from one tank to another.
  - c) is only used in flight for fuel transfer from one tank to another.
  - d) allows feeding of any engine from any fuel tank.

Answer ;

- d) allows feeding of any engine from any fuel tank.
- 

0668. The fuel cross-feed system enables ;
- a) the supply of the outboard engines from any outboard fuel tank.
  - b) the transfer of fuel only from the centre tank to the wing tanks.
  - c) the supply of any engine from any fuel tank.
  - d) the supply of the engines mounted on a wing from any fuel tank within that wing.

Answer ;

- c) the supply of any engine from any fuel tank.
- 

0669. The fuel-oil heat exchanger ;
- a) cools both the oil and the fuel.
  - b) cools the oil and heats the fuel.
  - c) heats the fuel only.
  - d) cools the oil only.

Answer ;

- b) cools the oil and heats the fuel.
- 

0670. The fuel supply system on a jet engine includes a fuel heating device , upstream of the main fuel filter so as to ;
- a) ease low pressure pumps work by increasing fuel fluidity.
  - b) maintain and improve fuel heating power.
  - c) prevent fuel from freezing in fuel pipes due to low temperatures at high altitude.
  - d) prevent , at low fuel temperature, the risk of ice formation from water contained in the fuel.

Answer ;

- d) prevent , at low fuel temperature, the risk of ice formation from water contained in the fuel.
-

0671. The fuel system boost pumps are used to ;
- a) feed the lines with fuel for directing it to the engine at a positive pressure.
  - b) feed the fuel control units , which inject the pressurized fuel into the engine.
  - c) avoid the bubbles accumulation.
  - d) avoid the bubbles accumulation and feed the lines with fuel for directing it to the engine at a positive pressure.

Answer ;

- d) avoid the bubbles accumulation and feed the lines with fuel for directing it to the engine at a positive pressure.
- 

0672. The fuel tanks of aircrafts must be checked for water ;
- a) during refuelling.
  - b) before the first flight of the day or after a long turnaround time.
  - c) immediately after every refuelling.
  - d) before each flight.

Answer ;

- b) before the first flight of the day or after a long turnaround time.
- 

0673. The fuel temperature , at which , under standard conditions , the vapour ignites in contact with a flame and extinguishes immediately , is the ;

- a) flash point.
- b) self ignition point.
- c) fire point.
- d) combustion point.

Answer ;

- a) flash point.
- 

0674. The function of a constant speed drive ( CSD ) in an AC generating system is to ;
- a) vary generator rpm in order to compensate for various AC loads.
  - b) drive the generator at a constant speed.
  - c) directly maintain a constant proportion between the rpm of an engine and a generator.
  - d) vary the engine rpm ( within limits ) to compensate for various AC loads.

Answer ;

- b) drive the generator at a constant speed.
- 

0675. The function of a constant speed drive ( CSD ) is to ensure ;
- a) an equal AC voltage output from all generators.
  - b) that the CSD output remains at a constant RPM irrespective of generator RPM.
  - c) that the generator produces a constant frequency.
  - d) that the starter-motor maintains a constant RPM irrespective of engine acceleration/deceleration.

Answer ;

- c) that the generator produces a constant frequency.
- 

0676. The function of a CSD in an AC generating system is to ;
- a) drive the generator at a constant speed.
  - b) vary the engine rpm ( within limits ) to compensate for various AC loads.
  - c) vary generator rpm in order to compensate for various AC loads.
  - d) directly maintain a constant proportion between the rpm of engine and generator.

Answer ;

- a) drive the generator at a constant speed.
- 

0677. The function of a feed box in the fuel tank is to ;
- a) distribute the fuel to the various tanks during refuelling.
  - b) increase the fuel level at the boost pump location.
  - c) trap fuel sediments or sludge in the lower part of the tank.
  - d) ventilate the tank during refuelling under high pressure.

Answer ;

- b) increase the fuel level at the boost pump location.
-

0678. The function of a fusible plug is to ;

- a) function as a special circuit breaker in the electric system.
- b) protect the brake against brake disk fusion due to excessive temperature.
- c) protect against excessive pressure in the pneumatic system.
- d) protect the tyre against explosion due to excessive temperature.

Answer ;

- d) protect the tyre against explosion due to excessive temperature.
- 

0679. The function of a hydraulic selector valve is to ;

- a) discharge some hydraulic fluid if the system pressure is too high.
- b) select the system to which the hydraulic pump should supply pressure.
- c) automatically activate the hydraulic system.
- d) direct system pressure to either side of the piston of an actuator.

Answer ;

- d) direct system pressure to either side of the piston of an actuator.
- 

0680. The function of a NOT logic gate within a circuit is to ;

- a) ensure the input signal is DC only.
- b) invert the input signal such that the output is always of the opposite state.
- c) ensure the input signal is AC only.
- d) ensure the output signal is of the same state as the input signal.

Answer ;

- b) invert the input signal such that the output is always of the opposite state.
- 

0681. The function of a scissor ( torsion link ) in a landing gear is to ;

- a) transfer the rudder pedal deflection into nose wheel steering commands.
- b) prevent any rotation of the oleo strut in the landing gear shock absorber.
- c) create the wheel pitch on bogie gears.
- d) make the body gears pivot when the nosewheel is turned through more than 20°.

Answer ;

- b) prevent any rotation of the oleo strut in the landing gear shock absorber.
- 

0682. The function of a vibrator in some gas turbine ignition systems is to provide ;

- a) a pulsating AC voltage in order to generate a more powerful spark.
- b) a DC current to the transformer to facilitate the generation of a higher voltage to the ignition.
- c) an AC voltage in order to facilitate transformation to a higher voltage
- d) DC current if the ignition system is fed by AC voltage.

Answer ;

- b) a DC current to the transformer to facilitate the generation of a higher voltage to the ignition.
- 

0683. The function of an accumulator in a hydraulic brake system is ;

- a) to store the hydraulic energy recovered by the anti skid system to prevent wheel blocking.
- b) to damp pressure fluctuations of the auto brake system.
- c) to supply a limited amount of brake energy in case the hydraulic system normally powering the brakes does not function anymore.
- d) to function as a buffer to assist the hydraulic system during heavy braking.

Answer ;

- c) to supply a limited amount of brake energy in case the hydraulic system normally powering the brakes does not function anymore.
- 

0684. The function of an air cycle machine is to ;

- a) decrease the pressure of the bleed air.
- b) pump the conditioned air into the cabin.
- c) cool the bleed air.
- d) remove the water from the bleed air.

Answer ;

- c) cool the bleed air.
- 

0685. The function of ribs in a wing is to ;

- a) withstand all the structural loads.
- b) give the wing the desired aerodynamic shape.
- c) allow installation of fuel cells in the wing.
- d) withstand the torsional loads.

Answer ;

- b) give the wing the desired aerodynamic shape.
-



0686. The function of the Generator Breaker is to close when the voltage of the ;
- a) generator is greater than battery voltage and to open when the opposite is true.
  - b) battery is greater than the generator voltage and to open when the opposite is true.
  - c) battery is greater than the alternator voltage and to open when the opposite is true.
  - d) alternator is greater than the battery voltage and to open when the opposite is true.

Answer ;

- a) generator is greater than battery voltage and to open when the opposite is true.
- 

0687. The function of the primer pump in a reciprocating engine is to ;
- a) inject additional fuel during engine acceleration
  - b) to serve as main supply pump in a fuel injection system.
  - c) serve as an alternate pump in case of an engine driven pump failure.
  - d) provide additional fuel for an engine start.

Answer ;

- d) provide additional fuel for an engine start.
- 

0688. The function of the rudder limiter system is ;
- a) to reduce pilot's workload during engine failure.
  - b) to restrict rudder deflection during flight at high IAS.
  - c) to limit pedal movement in heavy turbulence.
  - d) to restrict rudder deflection during flight with at Mach numbers.

Answer ;

- b) to restrict rudder deflection during flight at high IAS.
- 

0689. The function of the selector valve is to ;
- a) automatically activate the hydraulic system.
  - b) discharge some hydraulic fluid if the system pressure is too high.
  - c) communicate system pressure to either side of an actuator.
  - d) select the system to which the hydraulic pump should supply pressure.

Answer ;

- c) communicate system pressure to either side of an actuator.
- 

0690. The function of the swirl vanes round the fuel nozzles is to ;
1. increase air pressure at the fuel nozzles.
  2. reduce the average axial flow speed in order to stabilise the flame front.
  3. generate a swirl of incoming air to enhance mixing of the fuel with air.

The combination that regroups all of the correct statements is ;

- a) 1 , 2.
- b) 1 , 2 , 3.
- c) 2 , 3.
- d) 1 , 3.

Answer ;

- c) 2 , 3.
- 

0691. The functions of a LP booster pump in a gas turbine fuel system are to ;
- a) pressurise the fuel dump system and increase the pressure during refuelling.
  - b) avoid vapour locking and increase the pressure during refuelling.
  - c) avoid vapour locking and prevent cavitation of the HP fuel pump.
  - d) increase the pressure during refuelling and prevent cavitation of the HP fuel pump.

Answer ;

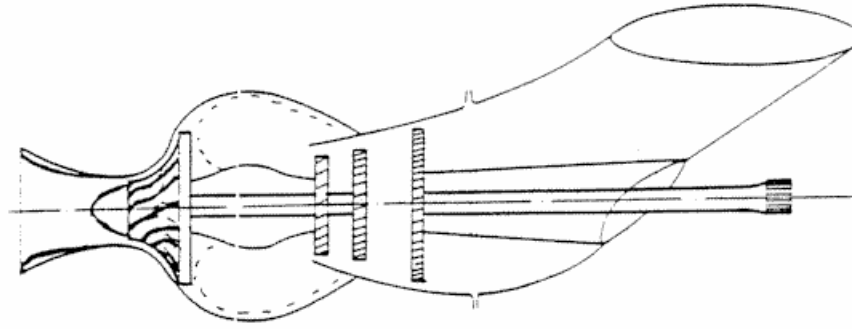
- c) avoid vapour locking and prevent cavitation of the HP fuel pump.
- 

0692. The fuselage structure of a pressurised transport aeroplane is an example of a ;
- a) pure monocoque structure.
  - b) semi-monocoque structure.
  - c) truss type structure.
  - d) sandwich structure.

Answer ;

- b) semi-monocoque structure.
-

0693. ( For this question use annex 021-11474A )



**Annex 021-11474A**

The gas turbine illustrated is of the following type ;

- a) free turbine and centrifugal compressor.
- b) free turbine and axial compressor.
- c) single shaft turbine and centrifugal compressor.
- d) single shaft turbine and axial compressor.

Answer ;

**a) free turbine and centrifugal compressor.**

0694. The greatest risk created by a free turbine overspeed is ;

- a) creep of the turbine blades.
- b) a surge of the gas generator.
- c) bursting of the free turbine disk.
- d) an engine overtemperature.

Answer ;

**c) bursting of the free turbine disk.**

0695. The heat for the anti-icing system of a turbofan engine intake is provided by ;

- a) bleed air from the HP compressor.
- b) a dedicated generator on the gearbox of that engine.
- c) bleed air from the LP compressor.
- d) the electrical system of the aircraft.

Answer ;

**a) bleed air from the HP compressor.**

0696. The heating facility for the windshield of an aircraft is ;

- a) harmful to the integrity of the windows in the event of a bird strike.
- b) used only at low altitudes where there is a risk of ice formation.
- c) used on a continual basis as it reduces the thermal gradients which adversely affect the useful life of the components.
- d) only used when hot-air demisting is insufficient.

Answer ;

**c) used on a continual basis as it reduces the thermal gradients which adversely affect the useful life of the components.**

0697. The high Altitude Radio Altimeter uses the following wavelengths ;

- a) decimetric.
- b) myriametric.
- c) hectometric.
- d) metric.

Answer ;

**a) decimetric.**

0698. The highest load on the torsion link in a bogie gear is ;

- a) when braking with an inoperative anti-skid system.
- b) whilst turning on the ground with a small radius.
- c) when gear is selected down.
- d) on touchdown with a strong crosswind.

Answer ;

**b) whilst turning on the ground with a small radius.**

0699. The hydraulic device similar to an electronic diode is a ;

- a) shutoff valve.
- b) flow control valve.
- c) distribution valve.
- d) check valve.

Answer ;

d) check valve.

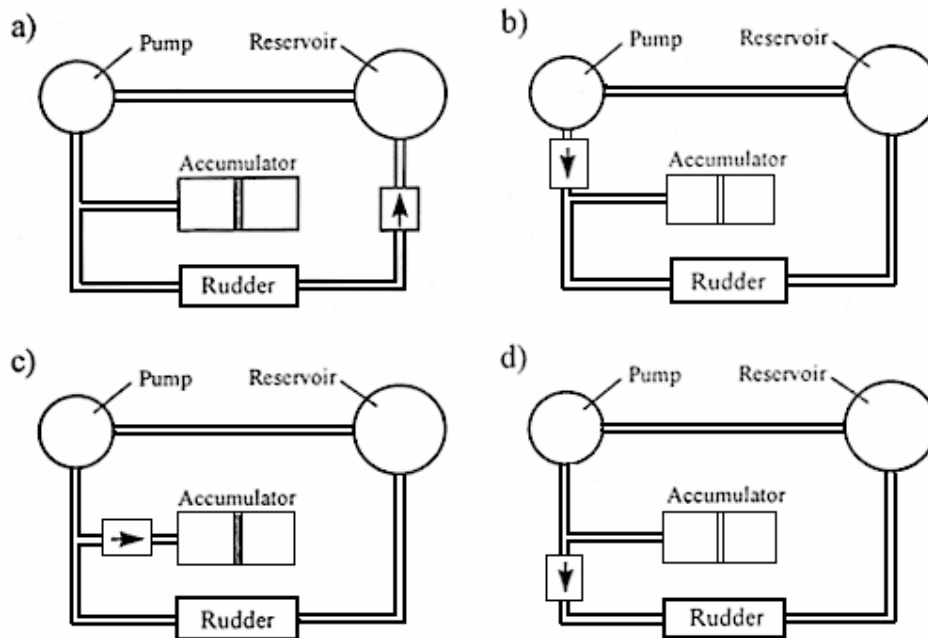
0700. The hydraulic device , which functionality is comparable to an electronic diode , is a ;

- a) shutoff valve.
- b) check valve.
- c) flow control valve.
- d) distribution valve.

Answer ;

b) check valve.

0701. ( For this question use annex 021-5248A )



Annex 021-5248A

The hydraulic systems which works correctly is shown in the figure ;

- a) c.
- b) a.
- c) b.
- d) d.

Answer ;

c) b.

0702. The ice protection for propellers of modern turboprop aeroplanes works ;

- a) electrically.
- b) with hot air.
- c) with anti-icing fluid.
- d) pneumatically.

Answer ;

a) electrically.

0703. The ice protection system currently used for the most modern jet aeroplanes is the ;

- a) liquid de-icing system.
- b) electrical de-icing system.
- c) hot air system.
- d) pneumatic system with expandable boots.

Answer ;

c) hot air system.

0704. The ignition system generally used for small aircrafts is a ;

- a) high tension system.
- b) low tension system.
- c) high intensity system.
- d) battery ignition system.

Answer ;

a) high tension system.

---

0705. The ignition system of a running piston engine receives electrical energy from ;

- a) generators.
- b) rotating permanent magnets.
- c) capacitors.
- d) batteries.

Answer ;

b) rotating permanent magnets.

---

0706. The illumination of the green landing gear light indicates that the landing gear is ;

- a) locked-down and its door is locked.
- b) not in the required position.
- c) locked-down.
- d) in the required position.

Answer ;

c) locked-down.

---

0707. The indication of the fire detection systems is performed by a ;

- a) gear warning.
- b) warning light and a warning bell ( or aural alert ).
- c) warning bell.
- d) warning light.

Answer ;

b) warning light and a warning bell ( or aural alert ).

---

0708. The inner surface of a heated windscreen is of ;

- a) glass.
- b) triplex.
- c) soft polycarbonate.
- d) hard perspex.

Answer ;

c) soft polycarbonate.

---

0709. The input and output of a static inverter are respectively ;

- a) AC and AC.
- b) AC and DC.
- c) DC and AC.
- d) DC and DC.

Answer ;

c) DC and AC.

---

0710. The installation and use of on-board oxygen generators is such that ;

1. the smoking ban is imperative when used.
2. in case of accidental drop of the "continuous flow" passenger masks , no crew action is required.
3. no trace of grease must be found in the system assembly.
4. the system's filling adaptors must be greased with non-freezing or graphite grease.

The combination which regroups all of the correct statements is ;

- a) 1 , 3.
- b) 2 , 4.
- c) 1 , 4.
- d) 2 , 3.

Answer ;

a) 1 , 3.

---

0711. The Instrument Landing System ( ILS ) uses the following wavelengths ;

- a) hectometric.
- b) centimetric.
- c) decimetric.
- d) metric.

Answer ;

d) metric.

0712. The internal geometry of a turbofan intake for a subsonic commercial aeroplane is ;

- a) convergent in order to increase airflow velocity and reduce static pressure in front of the fan.
- b) divergent in order to increase both airflow velocity and static pressure in front of the fan.
- c) divergent in order to reduce airflow velocity and increase static pressure in front of the fan.
- d) convergent in order to reduce airflow velocity in front of the fan.

Answer ;

c) divergent in order to reduce airflow velocity and increase static pressure in front of the fan.

0713. The kind of compressor normally used as a supercharger is ;

- a) a piston compressor.
- b) a hybrid compressor.
- c) a radial compressor.
- d) an axial compressor.

Answer ;

c) a radial compressor.

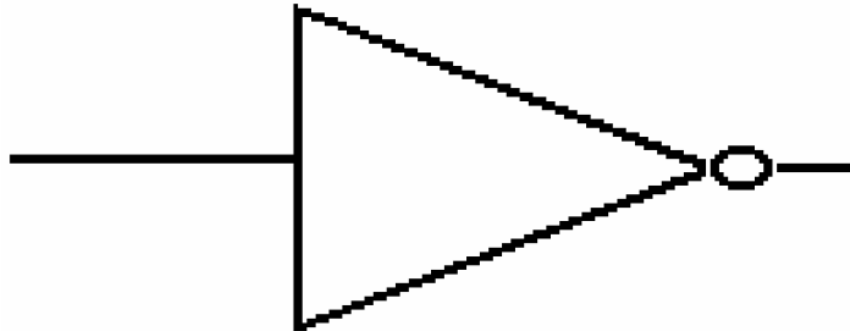
0714. The level in a hydraulic reservoir will ;

- a) initially increase with system pressurisation.
- b) always remain the same.
- c) increase as ambient temperature decreases.
- d) fluctuate with accumulator pressure.

Answer ;

d) fluctuate with accumulator pressure.

0715. ( For this question use annex 021- 6660A )



*Annex 021-6660A*

The logic symbol shown represents ( assuming positive logic ) ;

- a) an INVERT or NOT gate.
- b) a NOR gate.
- c) a NAND gate.
- d) an EXCLUSIVE gate.

Answer ;

a) an INVERT or NOT gate.

0716. The low pressure switch of a hydraulic circuit sets off an alarm if ;

- a) the reservoir level is at the normal operation limit.
- b) there is a leak in the reservoir return line.
- c) the pump power accumulator is deflated.
- d) the pump output pressure is insufficient.

Answer ;

d) the pump output pressure is insufficient.

0717. The lubricating system of an air cooled piston engine is used to ;
- a) keep the engine warm.
  - b) reduce internal friction and provide cooling.
  - c) to operate the fuel control unit.
  - d) operate constant speed propellers.

Answer ;

b) reduce internal friction and provide cooling.

---

0718. The lubricating system of an aircraft engine is used to ;
- a) operate ground adjustable propellers.
  - b) prevent inter-crystalline corrosion.
  - c) aid in dissipation of heat.
  - d) keep the engine warm.

Answer ;

c) aid in dissipation of heat.

---

0719. The main advantage of a constant speed propeller as compared to a fixed pitch propeller is a ;
- a) high efficiency in a much wider speed range of the aeroplane.
  - b) constant efficiency in the whole speed range of the aeroplane.
  - c) lower propeller blade stress.
  - d) higher efficiency in cruising range.

Answer ;

a) high efficiency in a much wider speed range of the aeroplane.

---

0720. The main feature of BCF fire extinguishers is that they ;
- a) are electrical conductors.
  - b) act as flame inhibitors by absorbing the air's oxygen.
  - c) are highly corrosive particularly for aluminium alloys.
  - d) use the cooling effect created by the venturi during discharge.

Answer ;

b) act as flame inhibitors by absorbing the air's oxygen.

---

0721. The main function of an air cycle machine is to ;
- a) remove the water from the bleed air.
  - b) cool the bleed air.
  - c) decrease the pressure of the bleed air.
  - d) pump the conditioned air into the cabin.

Answer ;

b) cool the bleed air.

---

0722. The main purpose of a Constant Speed Drive unit is to ;
- a) mechanically protect the alternator drive shaft during coupling.
  - b) maintain a constant frequency.
  - c) take part in the voltage regulation.
  - d) take part in the balancing of reactive loads.

Answer ;

b) maintain a constant frequency.

---

0723. The main purpose of a turbocharger is to ;
- a) maintain power with increasing altitude.
  - b) provide a leaner mixture at sea level.
  - c) provide a richer mixture at high altitudes.
  - d) reduce the fuel flow.

Answer ;

a) maintain power with increasing altitude.

---

0724. The main purpose of the mixture control is to ;
- a) increase the oxygen supplied to the engine.
  - b) decrease the air supplied to the engine.
  - c) decrease the oxygen supplied to the engine.
  - d) adjust the fuel flow to obtain the correct fuel/air ratio.

Answer ;

d) adjust the fuel flow to obtain the correct fuel/air ratio.

---

0725. The main reason for opening the cowl flaps is to control the ;

- a) CHT ( cylinder head temperature ).
- b) oil temperature.
- c) cabin temperature.
- d) EGT ( exhaust gas temperature ).

Answer ;

a) CHT ( cylinder head temperature ).

---

0726. The maximum cabin differential pressure of a pressurised aeroplane operating at FL370 is approximately ;

- a) 9.0 psi.
- b) 15.5 psi.
- c) 3.5 psi.
- d) 13.5 psi.

Answer ;

a) 9.0 psi.

---

0727. The measured output power components of a constant frequency AC system are ;

- a) kVA and kVAR.
- b) volts and amperes.
- c) volts and kilowatts.
- d) amperes and kilowatts.

Answer ;

a) kVA and kVAR.

---

0728. The mechanism to change the propeller blade angle of a small piston engine aeroplane in flight is operated ;

- a) by aerodynamic forces.
- b) hydraulically by engine oil.
- c) manually by the pilot.
- d) hydraulically by hydraulic fluid.

Answer ;

b) hydraulically by engine oil.

---

0729. The mechanism to change the propeller blade pitch of modern small piston engine aeroplanes is operated ;

- a) hydraulically by engine oil.
- b) hydraulically by hydraulic fluid.
- c) by an electrical actuator.
- d) by a mechanical linkage.

Answer ;

a) hydraulically by engine oil.

---

0730. The minimum airborne equipment required for operation of the VHF direction finder is a ;

- a) cathode-ray tube.
- b) VHF receiver operating in the 118 MHz to 136 MHz range.
- c) VHF compass operating in the 200 kHz to 1750 kHz range.
- d) VHF transmitter-receiver operating in the 118 MHz to 136 MHz range.

Answer ;

d) VHF transmitter-receiver operating in the 118 MHz to 136 MHz range.

---

0731. The mixture control for a carburettor achieves its control by ;

- a) altering the depression on the main discharge tube.
- b) varying the air supply to the main discharge tube.
- c) varying the fuel supply to the main discharge tube.
- d) moving the butterfly valve through a separate linkage to the main throttle control.

Answer ;

c) varying the fuel supply to the main discharge tube.

---

0732. The mixture controller of a carburettor ;

- a) moves the butterfly valve through a separate linkage to the main throttle control.
- b) alters the pressure drop at the main discharge nozzle.
- c) varies the air supply at the main discharge nozzle.
- d) varies the fuel supply to the main discharge nozzle.

Answer ;

d) varies the fuel supply to the main discharge nozzle.

---

0733. The modern anti-skid processes are based on the use of a computer whose input data is ;

1. idle wheel speed ( measured ).
2. braked wheel speed ( measured ).
3. brake temperature ( measured ).
4. desired idle wheel train slipping rate.
5. tire pressure.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 1 , 3.
- c) 1 , 2 , 4.
- d) 1 , 2 , 3 , 4 , 5.

Answer ;

c) 1 , 2 , 4.

---

0734. The most common extinguishing agent used in gas turbine engine fire protection system is ;

- a) Powder.
- b) Water.
- c) Halon.
- d) CO<sub>2</sub>.

Answer ;

c) Halon.

---

0735. The most common material used in the construction of gearbox casing is ;

- a) Titanium alloy.
- b) Magnesium alloy.
- c) Aluminium alloy.
- d) Steel alloy.

Answer ;

b) Magnesium alloy.

---

0736. The most common type of starter used on a helicopter engine is ;

- a) a DC starter motor.
- b) a DC starter/generator.
- c) an AC starter/generator.
- d) an AC starter.

Answer ;

b) a DC starter/generator.

---

0737. The most important oil system instrument indication during gas turbine engine start is ;

- a) oil pressure.
- b) oil temperature.
- c) oil filter clogged warning signal.
- d) oil quantity.

Answer ;

a) oil pressure.

---

0738. The most unfavourable conditions for adequate inlet air flow of a turbofan engine are ;

- a) maximum thrust during climb out with strong crosswind.
- b) maximum take-off thrust , zero forward speed and strong crosswind conditions.
- c) idle thrust , zero forward speed with no wind.
- d) maximum thrust at rotation speed on the ground during take-off.

Answer ;

b) maximum take-off thrust , zero forward speed and strong crosswind conditions.

---

0739. The most widely used electrical frequency in aircraft is ;

- a) 200 Hz.
- b) 400 Hz.
- c) 60 Hz.
- d) 50 Hz.

Answer ;

b) 400 Hz.

---



0740. The moving part in an AC generator is usually referred to as the ;

- a) rotor.
- b) stator.
- c) oscillator.
- d) slip ring.

Answer ;

a) rotor.

---

0741. The number of hand fire extinguishers which have to be installed in the passenger cabin according to JAR-OPS depends on the number of ;

- a) passengers in the cabin.
- b) emergency exists in the cabin
- c) seat rows in the cabin.
- d) seats in the cabin.

Answer ;

d) seats in the cabin.

---

0742. The octane rating of a fuel characterises the ;

- a) fuel electrical conductivity.
- b) fuel volatility.
- c) the anti-knock capability.
- d) quantity of heat generated by its combustion.

Answer ;

c) the anti-knock capability.

---

0743. The octane rating of a fuel characterises the ;

- a) quantity of heat generated by its combustion.
- b) resistance to detonation.
- c) fuel electrical conductivity.
- d) fuel volatility.

Answer ;

b) resistance to detonation.

---

0744. The oil supply to gas turbine engine bearings is maintained in the event of the main oil filter becoming clogged by ;

- a) differential pressure opening the by-pass valve of the main oil filter.
- b) the scavenge pumps.
- c) the HP fuel pump.
- d) a back-up oil system.

Answer ;

a) differential pressure opening the by-pass valve of the main oil filter.

---

0745. The oil system for a piston engine incorporates an oil cooler that is fitted ;

- a) in the return line to the oil tank after the oil has passed through the scavenge pump.
- b) after the pressure pump but before the oil passes through the engine.
- c) between the oil tank and the pressure pump.
- d) after the oil has passed through the engine and before it enters the sump.

Answer ;

a) in the return line to the oil tank after the oil has passed through the scavenge pump.

---

0746. The oil system of a gas turbine engine may be fitted with magnetic plugs , or chip detectors , that ;

- a) provide warning of impending failure without having to remove the filters for inspection.
- b) are fitted in the pressure line upstream of the oil filter.
- c) dispense with the requirement to fit an oil filter.
- d) monitor oil pressure and oil temperature.

Answer ;

a) provide warning of impending failure without having to remove the filters for inspection.

---

0747. The opening of the doors giving access to the oxygen masks for the passengers is ;

1. pneumatic for the gaseous oxygen system.
2. electrical for the chemical oxygen system.
3. pneumatic for the chemical oxygen system.
4. electrical for the gaseous oxygen system.

The combination regrouping all the correct statements is ;

- a) 1 , 2.
- b) 2 , 3.
- c) 1 , 3.
- d) 2 , 4.

Answer ;

- a) 1 , 2.
- 

0748. The operating principle of float-type carburettors is based on the ;

- a) difference in air pressure at the venturi throat and the air inlet.
- b) increase in air velocity in the throat of a venturi causing an increase in air pressure.
- c) measurement of the fuel flow into the induction system.
- d) automatic metering of air at the venturi as the aircraft gains altitude.

Answer ;

- a) difference in air pressure at the venturi throat and the air inlet.
- 

0749. The operating principle of magnetos in a piston engine ignition system consists of ;

- a) obtaining a high amperage , low voltage current in order to generate the spark.
- b) creating a brief high intensity magnetic field that will be sent through the distributor at the appropriate time.
- c) accumulating in a capacitor a low voltage current from the battery and inducing it as a high voltage current at the moment the spark is generated.
- d) breaking the primary circuit in order to induce a low amperage , high voltage current , which is distributed to spark plugs.

Answer ;

- d) breaking the primary circuit in order to induce a low amperage , high voltage current , which is distributed to spark plugs.
- 

0750. The output of a generator is controlled by ;

- a) varying the field strength.
- b) the reverse current relay circuit breaker.
- c) varying the speed of the engine.
- d) varying the length of wire in the armature windings.

Answer ;

- a) varying the field strength.
- 

0751. The pack cooling fan provides ;

- a) cooling air to the primary and secondary heat exchanger during cruise.
- b) cooling air to the primary and secondary heat exchanger during slow flight and ground operation.
- c) air to the eyeball outlets at the Passenger Service Unit ( PSU ).
- d) cooling air to the pre-cooler.

Answer ;

- b) cooling air to the primary and secondary heat exchanger during slow flight and ground operation.
- 

0752. The part of a piston engine that transforms reciprocating movement into rotary motion is termed the ;

- a) camshaft.
- b) piston.
- c) reduction gear.
- d) crankshaft.

Answer ;

- d) crankshaft.
- 

0753. The part of the flight that will cause the highest loads on the torsion link in a bogie gear is ;

- a) taxiing with a small turning radius.
- b) touch down with crosswind.
- c) gear down selection.
- d) braking with an inoperative anti skid system.

Answer ;

- a) taxiing with a small turning radius.
-

0754. The passenger oxygen mask will supply ;
- a) a mixture of cabin air and oxygen.
  - b) a mixture of oxygen and freon gas.
  - c) 100 % oxygen.
  - d) a mixture of compressed air and oxygen or 100 % oxygen.

Answer ;

a) a mixture of cabin air and oxygen.

---

0755. The phase relationship between two unparallelled AC generators ;

- a) is unimportant.
- b) must be synchronous.
- c) must be 90° out of synchronisation.
- d) must be in opposition.

Answer ;

a) is unimportant.

---

0756. The power of a piston engine decreases during a climb with a constant power lever setting because of the decreasing ;

- a) air density.
- b) engine temperature.
- c) temperature.
- d) humidity.

Answer ;

a) air density.

---

0757. The power of a piston engine which will be measured by using a friction brake is ;

- a) Brake horse power.
- b) Indicated horse power.
- c) Friction horse power.
- d) Heat loss power.

Answer ;

a) Brake horse power.

---

0758. The power output of a diesel engine without a turbo charger is regulated by ;

- a) fuel flow only.
- b) mixture.
- c) airflow only.
- d) fuel flow and airflow.

Answer ;

a) fuel flow only.

---

0759. The power output of a generator is controlled by ;

- a) varying the speed of the engine.
- b) varying the length of wire in the armature windings.
- c) the reverse current relay circuit breaker.
- d) varying the field strength.

Answer ;

d) varying the field strength.

---

0760. The power output of a piston engine can be calculated by ;

- a) Torque times RPM.
- b) Force times distance.
- c) Pressure times arm.
- d) Work times velocity.

Answer ;

a) Torque times RPM.

---

0761. The power output of a piston engine without supercharging increases with increasing altitude in standard atmosphere at constant Manifold Air Pressure ( MAP ) and RPM because of the ;

- a) lower losses during the gas change.
- b) increase of the air density behind the throttle valve.
- c) lower friction losses.
- d) leaner mixture at higher altitudes.

Answer ;

b) increase of the air density behind the throttle valve.

---

0762. The power required for field excitation of the main rotor in modern constant-frequency alternators is directly controlled by ;
- a) the voltage regulator.
  - b) a battery.
  - c) an excitation alternator with a permanent magnet generator.
  - d) an AC generator.

Answer ;

a) the voltage regulator.

---

0763. The pressure usually produced by the Boost Pumps ( BP ) of the fuel supply system of a large jet transport airplane is within the following range ;

- a) 3000 to 5000 psi.
- b) 20 to 50 psi.
- c) 300 to 500 psi.
- d) 5 to 10 psi.

Answer ;

b) 20 to 50 psi.

---

0764. The pressurisation load on the skin of a fuselage is ;

- a) tension.
- b) compression.
- c) torsion.
- d) bending.

Answer ;

a) tension.

---

0765. The pressurising function of the pressurising and dump valve in a gas turbine engine fuel system is ;

- a) to prevent overpressure at the fuel nozzles.
- b) to admit fuel to the nozzles only if the fuel pressure is high enough for sufficient atomisation.
- c) to prevent fuel leakage in the combustion chamber after engine shutdown.
- d) to admit oil to the fuel-oil heat exchanger preventing ice crystals blocking the fuel lines.

Answer ;

b) to admit fuel to the nozzles only if the fuel pressure is high enough for sufficient atomisation.

---

0766. The pressurization of fuel tanks is maintained by ;

- a) the fuel vent system.
- b) the fuel top off unit.
- c) the fuel tank drains.
- d) the fuel dump system.

Answer ;

a) the fuel vent system.

---

0767. The pressurization system of an aeroplane ;

- a) only pressurises the flight deck area.
- b) will maintain a zero cabin differential pressure at all altitudes.
- c) has the capability to maintain a cabin pressure higher than ambient pressure.
- d) will maintain a sea level cabin altitude at all altitudes.

Answer ;

c) has the capability to maintain a cabin pressure higher than ambient pressure.

---

0768. The primary function of the diffuser , located between the compressor and the combustion chamber , of a gas turbine engine is to ;

- a) convert kinetic energy into mechanical energy.
- b) increase the temperature of the compressed air.
- c) decrease the flow velocity.
- d) increase total pressure.

Answer ;

c) decrease the flow velocity.

---

0769. The primary input signals for a hydro-mechanical gas turbine engine fuel control system include ;

- a) N1 , N2 , bypass ratio , fuel shut-off , thrust lever angle.
- b) N2 , bypass ratio , EPR , fuel shut-off , thrust lever angle.
- c) N2 , compressor discharge pressure , compressor inlet temperature , fuel shut-off , thrust lever angle.
- d) N2 , turbine inlet temperature , fuel shut-of , generator , thrust lever angle.

Answer ;

c) N2 , compressor discharge pressure , compressor inlet temperature , fuel shut-off , thrust lever angle.

---

0770. The primary performance indicator ( cockpit instrument ) for the thrust of a turbofan engine is ;

- a) either EPR or N1.
- b) either torque or RPM.
- c) either N1 or N2.
- d) either N1 or fuel flow.

Answer ;

- a) either EPR or N1.
- 

0771. The primary purpose of a supercharger is to ;

- a) maintain power at altitude.
- b) provide leaner mixtures at altitudes below 5000 ft.
- c) provide a richer mixture at high altitudes.
- d) increase quantity of fuel at metering jet.

Answer ;

- a) maintain power at altitude.
- 

0772. The primary reason for a limitation being imposed on the temperature of gas flow in a gas turbine is to ;

- a) ensure that the maximum acceptable temperature within the combustion chamber is not exceeded.
- b) prevent damage to the jet pipe from overheating.
- c) prevent overheating and subsequent creep of the nozzle guide vanes.
- d) ensure that the maximum acceptable temperature at the turbine blades is not exceeded.

Answer ;

- d) ensure that the maximum acceptable temperature at the turbine blades is not exceeded.
- 

0773. The primary reason for venting an engine oil tank is to ;

- a) prevent excessive pressure in the tank.
- b) allow for expansion of hot oil.
- c) cool the oil.
- d) prevent overflow.

Answer ;

- a) prevent excessive pressure in the tank.
- 

0774. The principle of aeroplane propulsion is to generate a propelling force by ;

- a) accelerating air or gas in order to obtain a reaction force.
- b) pressurising air or gas in order to obtain a reaction force.
- c) heating up air in order to obtain a reaction force.
- d) generating a high velocity jet pushing against the outside air.

Answer ;

- a) accelerating air or gas in order to obtain a reaction force.
- 

0775. The principle of damage tolerance in structural design of an aircraft is based on the ;

- a) capability to withstand a certain amount of weakening of the structure without catastrophic failure.
- b) replacement of parts after a given number of cycles or hours of use.
- c) fact that there is no need to inspect the structure.
- d) monitoring of critical parameters and the replacement of parts if a limit value is exceeded.

Answer ;

- a) capability to withstand a certain amount of weakening of the structure without catastrophic failure.
- 

0776. The principle of fail safe design of an aircraft is based on the ;

- a) replacement of parts after a given number of cycles or hours of use.
- b) capability to withstand a certain amount of weakening of the structure without catastrophic failure.
- c) redundancy of the structure or equipment.
- d) monitoring of critical parameters and the replacement of parts if a limit value is exceeded.

Answer ;

- c) redundancy of the structure or equipment.
- 

0777. The principle of on condition maintenance is based on the ;

- a) monitoring of critical parameters and the replacement of parts if a limit value is exceeded.
- b) replacement of parts after a given number of cycles or hours of use.
- c) capability to withstand a certain amount of weakening of the structure without catastrophic failure.
- d) redundancy of the structure or equipment.

Answer ;

- a) monitoring of critical parameters and the replacement of parts if a limit value is exceeded.
-

0778. The principle of the safe life design of an aircraft is based on the ;
- a) monitoring of critical parameters and the replacement of parts if a limit value is exceeded.
  - b) replacement of parts after a given number of cycles or hours of use.
  - c) capability to withstand a certain amount of weakening of the structure without catastrophic failure.
  - d) redundancy of the structure or equipment.

Answer ;

- b) replacement of parts after a given number of cycles or hours of use.
- 

0779. The purpose of a battery control unit is generally to isolate the battery ;
- 1. from the bus when the battery charge has been completed.
  - 2. when there is a battery overheat condition.
  - 3. in case of an internal short circuit.
  - 4. in case of a fault on the ground power unit.

The combination which regroups all of the correct statements is ;

- a) 1 , 2 , 3 , 4.
- b) 1.
- c) 1 , 2.
- d) 1 , 2 , 3.

Answer ;

- d) 1 , 2 , 3.
- 

0780. The purpose of a chip detector in the oil system of an engine/gearbox is to indicate that ;

- a) the piston rings are worn.
- b) the seals are worn.
- c) the oil temperature is too high.
- d) there are metal particles in the oil.

Answer ;

- d) there are metal particles in the oil.
- 

0781. The purpose of a compressor bleed valve is to prevent surging ;

- a) at low compressor rpm.
- b) generated by foreign object ingestion.
- c) with altitude.
- d) of the first compressor stages.

Answer ;

- a) at low compressor rpm.
- 

0782. The purpose of a diluter demand regulator in an oxygen system is to ;

- a) deliver oxygen flow when inhaling.
- b) be used only when smoke is present in the cockpit.
- c) mix air and oxygen in a passenger oxygen mask.
- d) deliver oxygen flow only above FL 100.

Answer ;

- a) deliver oxygen flow when inhaling.
- 

0783. The purpose of a distributor in an ignition system is to distribute ;

- a) primary current to the condenser.
- b) secondary current to the condenser.
- c) primary current to the sparking plugs.
- d) secondary current to the sparking plugs.

Answer ;

- d) secondary current to the sparking plugs.
- 

0784. The purpose of a trim tab ( device ) is to ;

- a) reduce or to cancel control forces.
- b) trim the aeroplane during normal flight.
- c) lower manoeuvring control forces.
- d) trim the aeroplane at low airspeed.

Answer ;

- a) reduce or to cancel control forces.
-

0785. The purpose of a variable bypass valve , fitted between the LP and HP compressor of an axial flow compressor , is to ;
- a) prevent stall in the LP compressor.
  - b) enable to start the engine with an external air supply.
  - c) prevent the engine from an overspeed.
  - d) control the acceleration of the engine.

Answer ;

- a) prevent stall in the LP compressor.
- 

0786. The purpose of a voltage regulator is to control the output voltage of the ;
- a) generator at varying loads and speeds.
  - b) batteries at varying loads.
  - c) TRU.
  - d) generators at varying speeds and the batteries at varying loads.

Answer ;

- a) generator at varying loads and speeds.
- 

0787. The purpose of an accumulator in a hydraulic system is ;
- a) to enable the starting of hydraulic devices.
  - b) to bypass the pumps in the hydraulic system.
  - c) to damp the fluid pressure variations.
  - d) to eliminate the fluid flow variations.

Answer ;

- c) to damp the fluid pressure variations.
- 

0788. The purpose of an accumulator in a hydraulic system is to ;
- a) store energy.
  - b) store pressure.
  - c) store fluid.
  - d) cool the fluid.

Answer ;

- a) store energy.
- 

0789. The purpose of an ignition switch is to ;
- a) connect the battery to the magneto.
  - b) connect the secondary coil to the distributor.
  - c) connect the contact breaker and condenser in series with the primary coil.
  - d) control the primary circuit of the magneto.

Answer ;

- d) control the primary circuit of the magneto.
- 

0790. The purpose of baffle check valves fitted in aircraft fuel tanks is to ;
- a) close the vent lines in case of turbulence.
  - b) prevent fuel movement to the wing tip.
  - c) damp out movement of the fuel in the tank.
  - d) prevent positive pressure build up inside the tank.

Answer ;

- b) prevent fuel movement to the wing tip.
- 

0791. The purpose of baffles in an aeroplane's integral fuel tank is to ;
- a) prevent mixture of the fuel and hydraulic fluid.
  - b) prevent overpressure in the tank.
  - c) prevent the fuel from flowing in the vent lines.
  - d) restrict the fuel movements in the tank.

Answer ;

- d) restrict the fuel movements in the tank.
- 

0792. The purpose of bonding the metallic parts of an aircraft is to ;
- a) prevent electrolytic corrosion between mating surfaces of similar metals.
  - b) isolate all components electrically.
  - c) provide a single earth for electrical devices.
  - d) provide safe distribution of electrical charges and currents.

Answer ;

- d) provide safe distribution of electrical charges and currents.
-

0793. The purpose of cabin air flow control valves in a pressurization system is to ;
- a) discharge cabin air to atmosphere if cabin pressure rises above the selected altitude.
  - b) maintain a constant and sufficient mass air flow to ventilate the cabin and minimise cabin pressure surges.
  - c) regulate cabin pressure at the maximum cabin pressure differential.
  - d) regulate cabin pressure to the selected altitude.

Answer ;

- b) maintain a constant and sufficient mass air flow to ventilate the cabin and minimise cabin pressure surges.
- 

0794. The purpose of gas turbine engine trending is ;
- a) to obtain a timely indication when an engine requires to be shut down due to problems.
  - b) to enhance maintenance planning and detect anomalies.
  - c) to collect metal particles from the oil system.
  - d) to assess engine condition using non-destructive inspection.

Answer ;

- b) to enhance maintenance planning and detect anomalies.
- 

0795. The purpose of static wick dischargers is to ;
- a) dissipate static charge of the aircraft in flight thus avoiding radio interference as a result of static electricity.
  - b) provide a path to ground for static charges when refuelling.
  - c) be able to fly higher because of less electrical friction.
  - d) dissipate static charge from the aircraft skin after landing.

Answer ;

- a) dissipate static charge of the aircraft in flight thus avoiding radio interference as a result of static electricity.
- 

0796. The purpose of stringers , used in fuselage construction , is to ;
- a) withstand the shear stresses.
  - b) assist the skin in absorbing the longitudinal traction-compression stresses.
  - c) integrate the strains due to pressurization to which the skin is subjected and convert them into a tensile stress.
  - d) provide sound and thermal isolation.

Answer ;

- b) assist the skin in absorbing the longitudinal traction-compression stresses.
- 

0797. The purpose of stringers , used in fuselage construction , is to ;
- a) withstand the shear stresses.
  - b) provide sound and thermal isolation.
  - c) assist the skin withstand longitudinal compressive loads.
  - d) carry the loads due to pressurisation and convert them into tensile stress.

Answer ;

- c) assist the skin withstand longitudinal compressive loads.
- 

0798. The purpose of the barometric correction in a fuel controller is to ;
- a) maintain the correct fuel to air mass ratio when the altitude increases.
  - b) reduce the fuel to air mass ratio when the altitude increases.
  - c) increase the fuel to air mass ratio when the altitude increases.
  - d) maintain a constant fuel flow whatever the altitude is.

Answer ;

- a) maintain the correct fuel to air mass ratio when the altitude increases.
- 

0799. The purpose of the blow-in-doors at the air inlets is to ;
- a) serve to increase the relative velocity at the first compressor stage.
  - b) provide the engine with additional air at high power settings and low air speeds.
  - c) feed cooling air to the engine cowling.
  - d) provide the engine with additional air at high power settings at cruising speed.

Answer ;

- b) provide the engine with additional air at high power settings and low air speeds.
-



- 0800.** The purpose of the bonding the metallic parts of an aircraft is to ;
1. prevent electrolytic corrosion between mating surface of similar metals.
  2. ensure zero voltage difference between aircraft components.
  3. isolate all coponents electrically.
  4. provide a single earth for electrical devices.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 4.
- c) 1 , 3.
- d) 2 , 4.

**Answer ;**

**d) 2 , 4.**

---

- 0801.** The purpose of the cabin pressure controller , in the automatic mode , is to perform the following functions ;

1. control of cabin altitude.
2. control of cabin altitude rate-of-change.
3. limitation of differential pressure.
4. balancing aircraft altitude with cabin altitude.
5. cabin ventilation.
6. keeping a constant differential pressure throughout all the flight phases.

The combination regrouping all the correct statements is ;

- a) 5 , 6 , 1.
- b) 4 , 5 , 3.
- c) 1 , 2 , 3.
- d) 2 , 6 , 4.

**Answer ;**

**c) 1 , 2 , 3.**

---

- 0802.** The purpose of the contact breaker is to ;

- a) connect the secondary coil to the distributor.
- b) connect the contact breaker and condenser in series with the primary coil.
- c) control the primary circuit of the magneto.
- d) connect the battery to the magneto.

**Answer ;**

**c) control the primary circuit of the magneto.**

---

- 0803.** The purpose of the floor proximity emergency escape path marking system is to ;

- a) replace the overhead emergency lighting in event of failure.
- b) give additional guidance during an emergency evacuation at night only.
- c) give additional guidance during an emergency evacuation with a reduced cabin visibility.
- d) mark only the threshold exits.

**Answer ;**

**c) give additional guidance during an emergency evacuation with a reduced cabin visibility.**

---

- 0804.** The purpose of the pack cooling fans in the air conditioning system is to ;

- a) supply the Passenger Service Unit ( PSU ) with fresh air.
- b) supply the heat exchangers with cooling air during slow flights and ground operation.
- c) cool the APU compartment.
- d) supply the heat exchangers with cooling air during cruise flight.

**Answer ;**

**b) supply the heat exchangers with cooling air during slow flights and ground operation.**

---

- 0805.** The purpose of the “Pressure Relief Valve” in a high pressure oxygen system is to ;

- a) act as a manual shut-off valve.
- b) maximize the charging pressure of the system.
- c) relieve overpressure if the pressure reducing valve malfunctions.
- d) reduce pressure in the oxygen reservoir to a suitable manifold pressure for the regulator.

**Answer ;**

**c) relieve overpressure if the pressure reducing valve malfunctions.**

---

0806. The purpose of the proximity of the emergency evacuation path marking system is to ;
- replace the overhead emergency lighting in case of failure.
  - guide the emergency evacuation during night only.
  - mark only the exists at the floor level.
  - give additional guidance during an emergency evacuation with a thick smoke.

Answer ;

- d) give additional guidance during an emergency evacuation with a thick smoke.
- 

0807. The purpose of the regulator of a flight deck oxygen system is to ;
1. decrease oxygen pressure from 1800 PSI ( in the bottles ) down to about 50-75 PSI ( low pressure system ).
  2. supply 100% oxygen.
  3. supply diluted oxygen.
  4. supply oxygen on demand.
  5. supply oxygen at emergency/positive pressure.
  6. trigger the continuous cabin altitude warning at 10000 ft cabin altitude.

The combination regrouping all the correct statements is ;

- 2 , 3 , 4 , 5.
- 1 , 2 , 3 , 4.
- 1 , 3 , 4 , 6.
- 3 , 4 , 5 , 6.

Answer ;

- a) 2 , 3 , 4 , 5.
- 

0808. The purpose of the venturi in a carburettor is to ;
- create a rise in pressure at the throat before the mixture enters the induction system.
  - create the suction necessary to cause fuel to flow through the carburettor main jets.
  - prevent enrichment of the mixture due to high air velocity through the carburettor.
  - ensure complete atomisation of the fuel before entering the injection system.

Answer ;

- b) create the suction necessary to cause fuel to flow through the carburettor main jets.
- 

0809. The purpose of variable by-pass valves in a high bypass turbofan engine between the LP compressor and the HP compressor ( HPC ) is to ;
- prevent HPC stall during engine start.
  - provide sufficient air to the HPC during engine start.
  - prevent fan stall and LP compressor stall during engine start and low rotor speeds.
  - prevent fan surge at high rotor speeds ( high rpm ).

Answer ;

- c) prevent fan stall and LP compressor stall during engine start and low rotor speeds.
- 

0810. The purposes of the oil and the nitrogen in an oleo-pneumatic strut are ;
- the oil supplies the damping and lubrication function , the nitrogen supplies the heatdissipation function.
  - the oil supplies the spring function and the nitrogen supplies the damping function.
  - the oil supplies the sealing and lubrication function , the nitrogen supplies the damping function.
  - the oil supplies the damping function and the nitrogen supplies the spring function.

Answer ;

- d) the oil supplies the damping function and the nitrogen supplies the spring function.
- 

0811. The Ram Air Turbine ( RAT ) provides emergency hydraulic power for ;
- flap extension only.
  - undercarriage selection and automatic brake system.
  - nose wheel steering after the aeroplane has landed.
  - flight controls in the event of loss of engine driven hydraulic power.

Answer ;

- d) flight controls in the event of loss of engine driven hydraulic power.
- 

0812. The rating of electrical fuses is expressed in ;
- volts.
  - watts.
  - ohms.
  - amperes.

Answer ;

- d) amperes.
-

- 0813.** The reading on the oil pressure gauge is the ;
- a) pressure of the oil on the outlet side of the pressure pump.
  - b) difference between the pressure pump pressure and the scavenge pump pressure.
  - c) pressure of the oil on the inlet side of the pressure pump.
  - d) pressure in the oil tank reservoir.

**Answer ;**

- a) pressure of the oil on the outlet side of the pressure pump.
- 

- 0814.** The reason for a double switch on the elevator trim is ;
- a) to reduce the probability of a trim runaway.
  - b) to prevent both pilots from performing opposite trim inputs.
  - c) because there are two trim motors.
  - d) to be able to use two different trim speeds , slow trim rate at high speed and high trim rate at low speed.

**Answer ;**

- a) to reduce the probability of a trim runaway.
- 

- 0815.** The reason for disconnecting a constant speed drive ( CSD ) of an AC generator in flight are ;
- 1. low oil pressure in the CSD.
  - 2. slight variation about the normal operating frequency.
  - 3. high oil temperature in the CSD.
  - 4. excessive variation of voltage and kVAR.

The combination that regroups all of the correct statements is ;

- a) 2 , 4.
- b) 1 , 3.
- c) 2 , 3.
- d) 1 , 4.

**Answer ;**

- b) 1 , 3.
- 

- 0816.** The reason for fitting thermal plugs to aircraft wheels is that they ;
- a) prevent the brakes from overheating.
  - b) release air from the tyre in the event of overpressure.
  - c) release air from the tyre in the event of overheating.
  - d) prevent heat transfer from the brake disks to the tyres.

**Answer ;**

- c) release air from the tyre in the event of overheating.
- 

- 0817.** The reason for having a low pressure fuel-cooled oil cooler in a recirculatory type oil system is to ;
- a) cool the oil and heat the fuel.
  - b) cool both the oil and the fuel.
  - c) heat the fuel only.
  - d) cool the oil only.

**Answer ;**

- a) cool the oil and heat the fuel.
- 

- 0818.** The reason for the fact that an aeroplane designed for long distances cannot simply be used for short haul flights at higher frequencies is that ;
- a) the procedures and checklists for this kind of aeroplanes will take too much time.
  - b) in that case some fuel tanks remain empty during the whole flight , which stresses the aeroplane's structure in an unacceptable way.
  - c) these aeroplanes often consume too much fuel on short haul flights.
  - d) the lifetime of the fatigue sensitive parts has been based on a determined load spectrum.

**Answer ;**

- d) the lifetime of the fatigue sensitive parts has been based on a determined load spectrum.
- 

- 0819.** The reason for the trim switch on a control column to consist of two separate switches is ;
- a) to be able to use two different trim speeds, slow trim rate at high speed and high trim rate at low speed.
  - b) to prevent that both pilots perform opposite trim inputs.
  - c) because there are two trim motors.
  - d) to reduce the probability of a trim-runaway.

**Answer ;**

- d) to reduce the probability of a trim-runaway.
-

0820. The reason for using inverters in an electrical system is ;

- a) to change AC into DC.
- b) to avoid a short circuit.
- c) to change DC into AC.
- d) to change the DC voltage.

Answer ;

c) to change DC into AC.

---

0821. The relationship between compression ratio of a piston engine and the required fuel octane rating is that the ;

- a) higher the compression ratio , the lower the octane rating required.
- b) higher the compression ratio , the higher the octane rating required.
- c) lower the compression ratio , the higher the octane rating required.
- d) required octane rating is indepenent of the compression ratio.

Answer ;

b) higher the compression ratio , the higher the octane rating required.

---

0822. The replenishment of an onboard oxygen bottle must be done ;

- a) with a ground power unit connected , electrical power supplied.
- b) with the engine(s) on.
- c) with the engine(s) off.
- d) with the engine(s) a idle.

Answer ;

c) with the engine(s) off.

---

0823. The reservoirs of a hydraulic system can be pressurized ;

- a) by an auxiliary system.
- b) by bleed air from the engine.
- c) in flight only.
- d) by the air conditioning system.

Answer ;

b) by bleed air from the engine.

---

0824. The resistors R1 and R2 are connected in parallel. The value of the equivalent resistance ( Req ) so obtained is given by the following formula ;

- a)  $Req = R1 \times R2$
- b)  $1/Req = 1/(R1 + R2)$
- c)  $Req = R1 + R2$
- d)  $1/Req = 1/R1 + 1/R2$

Answer ;

d)  $1/Req = 1/R1 + 1/R2$

---

0825. The richness of a fuel/air mixture ratio is the ;

- a) volume of fuel relative to the mass of the volume of air.
- b) mass of fuel relative to the volume of air.
- c) actual mixture ratio relative to the chemically correct ratio.
- d) volume of fuel relative to the volume of air.

Answer ;

c) actual mixture ratio relative to the chemically correct ratio.

---

0826. The richness of a fuel/air mixture ratio is the ;

- a) volume of fuel relative to the volume of air.
- b) mass of fuel relative to the volume of air.
- c) volume of fuel relative to the mass of the volume of air.
- d) real mixture ratio relative to the theoretical ratio.

Answer ;

d) real mixture ratio relative to the theoretical ratio.

---

0827. The rotor blades of an axial compressor of a gas turbine engine are often mounted on a disk or drum using a “loose fit” because this ;

- a) limits FOD damage.
- b) limits damage due to creep.
- c) limits damage to the blades due to vibration.
- d) facilitates easy assembly and removal.

Answer ;

c) limits damage to the blades due to vibration.

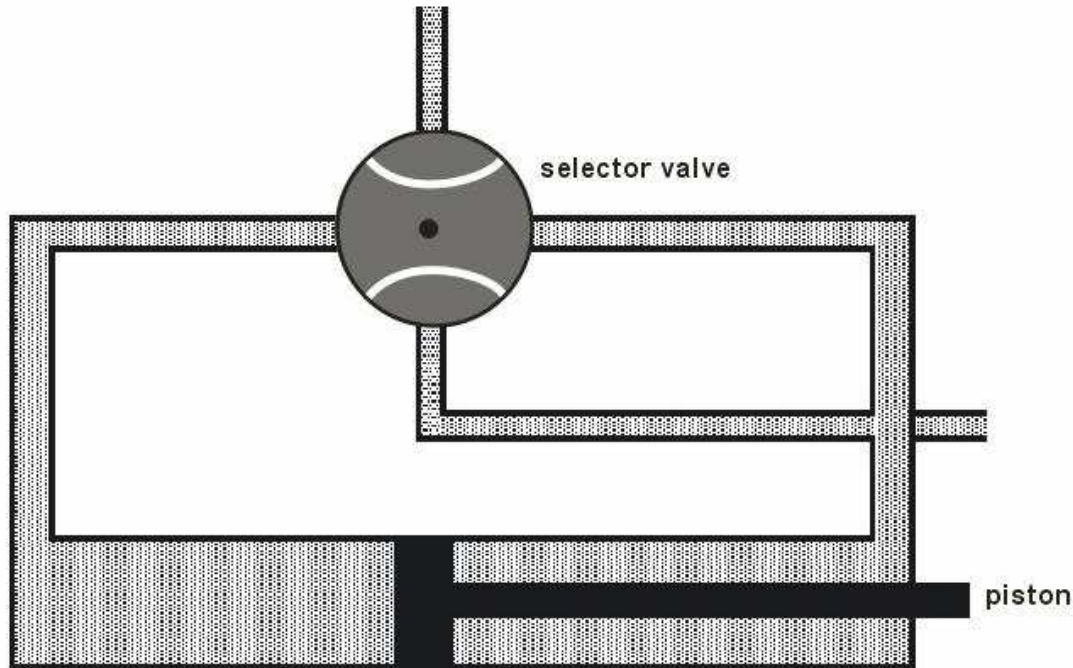
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0828. The sand filter located at the air intake of a gas turbine engine ;
- a) has no effect on the efficiency of the air intake.
  - b) reduces the efficiency of the air intake.
  - c) increases the efficiency of the air intake.
  - a) has an effect on the efficiency of the compressor only.

Answer ;

b) reduces the efficiency of the air intake.

0829. ( For this question use annex 021-6736A )



**Annex 021-6736A**

The schematic diagram annexed illustrates an actuator and a selector valve in a typical hydraulic system. Assuming hydraulic pressure throughout and no internal leakage ;

- a) the piston moves to the left due to pressure acting on differential areas.
- b) since pressures are equal , the piston is free to move in response to external forces.
- c) the piston moves to the right due to equal pressure acting on differential areas.
- d) a condition of hydraulic lock exists and no movement of the piston will take place.

Answer ;

d) a condition of hydraulic lock exists and no movement of the piston will take place.

0830. The selection of bleed air to “ON” will cause ;

- a) a reduction of torque and gas generator RPM with an increase of exhaust temperature.
- b) an increase in exhaust temperature , gas generator speed and power.
- c) a decrease in power and an increase in exhaust gas temperature.
- d) a decrease in gas generator RPM with no change in power or exhaust temperatures.

Answer ;

c) a decrease in power and an increase in exhaust gas temperature.

0831. The sequential pneumatic impulses used in certain leading edge de-icing devices ;

1. prevent ice formation.
2. can be triggered from the flight deck after icing has become visible.
3. will inflate each pneumatic boot for a few seconds.
4. will repeat more than ten times per second.

The combination that regroups all of the correct statements is ;

- a) 2 , 3.
- b) 1 , 3.
- c) 2 , 4.
- d) 1 , 4.

Answer ;

a) 2 , 3.

0832. The skip zone of HF-transmission will increase when the following change in circumstance occurs ;
- a) Higher frequency and lower position of the reflecting ionospheric layer.
  - b) Lower frequency and lower position of the reflecting ionospheric layer.
  - c) Lower frequency and higher position of the reflecting ionospheric layer.
  - d) Higher frequency and higher position of the reflecting ionospheric layer.

Answer ;

d) Higher frequency and higher position of the reflecting ionospheric layer.

---

0833. The “slipstream effect” of a propeller is most prominent at ;

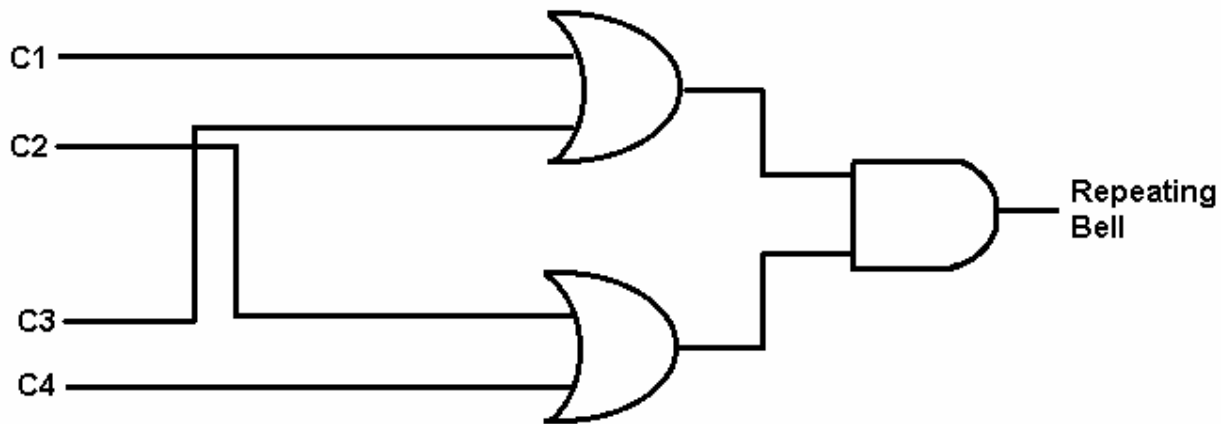
- a) low airspeeds with low power setting.
- b) low airspeeds with high power setting.
- c) high airspeeds with high power setting.
- d) high airspeeds with low power setting.

Answer ;

b) low airspeeds with high power setting.

---

0834. ( For this question use annex 021-10164A )



**Annex 021-10164A**

The smoke detection in the aircraft cargo compartments is carried out by four sensors ; C1 , C2 , C3 and C4. They are associated with the logic circuit as described in the annex. The repeating bell is activated when:

- a) only one sensor detects smoke.
- b) the C1 and C3 sensors detect smoke.
- c) the C2 and C4 sensors detect smoke.
- d) the C1 and C2 sensors detect smoke.

Answer ;

d) the C1 and C2 sensors detect smoke.

---

0835. The so-called “hot buses” or “direct buses” are ;

- a) automatically connected to the battery if generators have failed.
- b) directly connected to the battery.
- c) kept in operating conditions by an electrical resistance in the case of energy failure.
- d) providing an alternative current.

Answer ;

b) directly connected to the battery.

---

0836. The speed of an asynchronous four-pole motor fed at a frequency of 400 Hertz is ;

- a) 800 revolutions per minute.
- b) 1600 revolutions per minute.
- c) 6000 revolutions per minute.
- d) 12000 revolutions per minute.

Answer ;

d) 12000 revolutions per minute.

---

0837. The static thrust of a turbo-jet , at the selection of full power ;
1. equals the product of the exhaust gas mass flow and the exhaust gas velocity.
  2. is obtained by pressure of the exhaust gas on the ambient air.
  3. is equivalent to zero mechanical power since the aeroplane is not moving.
  4. is independent of the outside air temperature.

The combination which regroups all of the correct statements is ;

- a) 1 , 3.
- b) 1 , 2.
- c) 4.
- d) 2 , 3.

Answer ;

- a) 1 , 3.
- 

0838. The systems used for emergency extension of landing systems may comprise ;

1. Compressed CO<sub>2</sub>.
2. Compressed nitrogen.
3. Compressed oxygen.
4. Auxiliary hydraulic system.
5. Freefall.

The combination containing all of the correct statements is ;

- a) 1 , 2 , 5.
- b) 2 , 3 , 4.
- c) 2 , 4 , 5.
- d) 1 , 3 , 4.

Answer ;

- c) 2 , 4 , 5.
- 

0839. The take off of a jet aeroplane may be performed with “reduced thrust” in order to ;

- a) reduce the risk of engine failure during take-off.
- b) save fuel.
- c) reduce take-off noise.
- d) reduce maintenance costs.

Answer ;

- d) reduce maintenance costs.
- 

0840. The task of the primer pump is to ;

- a) inject additional fuel during engine acceleration.
- b) serve as an alternate pump in case of an engine driven pump failure.
- c) provide additional fuel for an engine start.
- d) to serve as main supply pump in a fuel injection system.

Answer ;

- c) provide additional fuel for an engine start.
- 

0841. The term “bootstrap” , when used to identify a cabin air conditioning and pressurisation system , refers to the ;

- a) source of the charge air.
- b) cold air unit ( air cycle machine ) arrangement.
- c) means by which pressurisation is controlled.
- d) charge air across the inter-cooler heat exchanger.

Answer ;

- b) cold air unit ( air cycle machine ) arrangement.
- 

0842. The term “pressure cabin” applies when an aeroplane ;

- a) has the ability to maintain a constant cabin differential pressure at all flight altitudes.
- b) is only pressurised in the area of the control cabin.
- c) has the ability to maintain a constant cabin altitude at all flight altitudes.
- d) has the means to maintain cabin pressure higher than ambient pressure.

Answer ;

- d) has the means to maintain cabin pressure higher than ambient pressure.
-

0843. The term used to describe a permanent and cumulative deformation of the turbine blades in a gas turbine engine is ;

- a) creep.
- b) twist.
- c) stretch.
- d) distortion.

Answer ;

a) creep.

---

0844. The thermal efficiency of a diesel engine is higher than that of a petrol engine because ;

- a) the compression ratio is much higher.
- b) the EGT is higher.
- c) the air mass flow through the engine is higher.
- d) the calorific value of the fuel is higher.

Answer ;

a) the compression ratio is much higher.

---

0845. The thermal efficiency of a gas turbine engine will increase with a ;

- a) increase in humidity.
- b) decrease in ambient air pressure.
- c) increase in ambient air temperature.
- d) decrease in ambient air temperature.

Answer ;

d) decrease in ambient air temperature.

---

0846. The thermal efficiency of a piston engine is about ;

- a) 80%.
- b) 30%.
- c) 70%.
- d) 50%.

Answer ;

b) 30%.

---

0847. The Throttle Lever Angle ( TLA ) information of a FADEC controlled gas turbine engine is provided ;

- a) to a single dedicated channel of the FADEC.
- b) to the hydro mechanical main engine control.
- c) via cables and pulleys.
- d) to the electronic engine control unit by a direct electrical wiring connection.

Answer ;

d) to the electronic engine control unit by a direct electrical wiring connection.

---

0848. The thrust reversers of jet engines may use clamshell doors in order to ;

- a) reduce the pressure of the exhaust gas stream by diffusing it to atmosphere.
- b) stop the flow of the exhaust gas stream.
- c) change the direction of the exhaust gas stream.
- d) absorb any surge in engine thrust.

Answer ;

c) change the direction of the exhaust gas stream.

---

0849. The torque of an aeroplane engine can be measured at the ;

- a) propeller blades.
- b) accessory gear box.
- c) gear box which is located between the engine and the propeller.
- d) camshaft.

Answer ;

c) gear box which is located between the engine and the propeller.

---

0850. The trim tab ;

- a) increases hinge moment and control surface efficiency.
- b) reduces hinge moment and control surface efficiency.
- c) reduces hinge moment and increases control surface efficiency.
- d) increases hinge moment and reduces control surface efficiency.

Answer ;

b) reduces hinge moment and control surface efficiency.

---



0851. The turbine in a gas turbine engine generates shaft power using ;
- a) expansion of hot gas followed by conversion of kinetic energy ( velocity ) into ( mechanical ) work.
  - b) compression of hot gas followed by conversion of potential energy ( pressure ) into kinetic energy ( velocity ).
  - c) compression of hot gas followed by conversion of kinetic energy into ( mechanical ) work.
  - d) expansion of hot gas followed by conversion of potential energy ( pressure ) into ( mechanical ) work.

Answer ;

- a) expansion of hot gas followed by conversion of kinetic energy ( velocity ) into ( mechanical ) work.
- 

0852. The two deformation modes that cause wing flutter are ;

- a) shearing and elongation.
- b) torsion and shearing.
- c) bending and elongation.
- d) torsion and bending.

Answer ;

- d) torsion and bending.
- 

0853. The type of an aircraft oxygen system intended for use by passengers , is mostly ;

- a) a continuous flow system.
- b) a pressure demand system.
- c) portable equipment only.
- d) an air recycle system.

Answer ;

- a) a continuous flow system.
- 

0854. The type of brake unit found on most transport aeroplanes is a ;

- a) Drum type brake.
- b) Multiple disk brake.
- c) Belt brake.
- d) Single disk brake.

Answer ;

- b) Multiple disk brake.
- 

0855. The type of gate which requires all conditions with logic 1 for engagement is ;

- a) AND.
- b) OR.
- c) NOR.
- d) NAND.

Answer ;

- a) AND.
- 

0856. The type of hydraulic fluid which has the greatest resistance to cavitation is ;

- a) synthetic fluid.
- b) water and glycol based fluid.
- c) vegetable oil based fluid ( castor oil ).
- d) mineral oil based fluid.

Answer ;

- a) synthetic fluid.
- 

0857. The type of hydraulic oil used in modern hydraulic systems is ;

- a) mixture of mineral oil and alcohol.
- b) synthetic oil.
- c) mineral oil.
- d) vegetable oil.

Answer ;

- b) synthetic oil.
- 

0858. The type of windings commonly used in DC starter motors are ;

- a) compound wound.
- b) shunt wound.
- c) series wound.
- d) series shunt wound.

Answer ;

- c) series wound.
-

0859. The types of fuel tanks used on most modern transport aircraft are ;

- a) Cell tanks.
- b) Fixed built-in tanks.
- c) Integral tanks.
- d) Combined fuel tanks.

Answer ;

c) Integral tanks.

---

0860. The use of a hot air wing anti-icing system ;

- a) reduces aerodynamic performance of the wing and causes no reduction in maximum thrust.
- b) does not affect aerodynamic performance of the wing and causes a reduction in maximum thrust.
- c) does not affect aerodynamic performance of the wing and causes no reduction in maximum thrust.
- d) reduces aerodynamic performance of the wing and causes a reduction in maximum thrust.

Answer ;

b) does not affect aerodynamic performance of the wing and causes a reduction in maximum thrust.

---

0861. The use of igniters in a gas turbine engine is required ;

- 1. throughout the operating range of the engine.
- 2. during engine acceleration.
- 3. for in-flight relights.

The combination that regroups all of the correct statements is ;

- a) 3.
- b) 1 , 3.
- c) 2.
- d) 1 , 2.

Answer ;

a) 3.

---

0862. The use of igniters is necessary or recommended on a turbo-jet ;

- 1. throughout the operating range of the engine.
- 2. for accelerations.
- 3. for ground starts.
- 4. for in-flight relights.
- 5. during turbulence in flight.
- 6. under heavy precipitation or in icing conditions.

The combination which regroups all of the correct statements is ;

- a) 2 , 3 , 4.
- b) 3.
- c) 3 , 4 , 5 , 6.
- d) 1.

Answer ;

c) 3 , 4 , 5 , 6.

---

0863. The use of too low an octane fuel may cause ;

- a) a cooling effect on cylinders.
- b) detonation.
- c) a prompt pre-ignition reaction.
- d) higher manifold pressure.

Answer ;

b) detonation.

---

0864. The useful work area in an ideal Otto engine indicator diagram is enclosed by the following gas state change lines ;

- a) 2 adiabatic and 1 isothermic lines.
- b) 2 adiabatic and 2 isobaric lines.
- c) 2 adiabatic , 1 isochoric and 1 isobaric lines.
- d) 2 adiabatic and 2 isochoric lines.

Answer ;

d) 2 adiabatic and 2 isochoric lines.

---

0865. The vapour lock is ;
- a) the exhaust gases obstructions caused by an engine overheating.
  - b) a stoppage in a fuel feeding line caused by a fuel vapour bubble.
  - c) the effect of the water vapour bubbles in the induction manifold caused by the condensation.
  - d) the abnormal mixture enrichment caused by a greater gasoline vaporisation in the carburettor.

Answer ;

- b) a stoppage in a fuel feeding line caused by a fuel vapour bubble.
- 

0866. The ventilation system in a fuel tank ;
- a) prevents vapour lock in the fuel lines.
  - b) it prevents a surge in the tank of an iced aeroplane.
  - c) prevents low pressure or excessive overpressure in the tank.
  - d) can be used to drain the tanks , for daily checks.

Answer ;

- c) prevents low pressure or excessive overpressure in the tank.
- 

0867. The very rapid magnetic field changes ( flux changes ) in the primary coil of a magneto are accomplished by the ;
- a) distributor arm aligning with one of the high tension segments.
  - b) contact breaker points opening.
  - c) contact breaker points closing.
  - d) rotor turning past the position of maximum flux in the armature.

Answer ;

- b) contact breaker points opening.
- 

0868. The VHF direction finder uses the following wavelengths ;
- a) hectometric.
  - b) metric.
  - c) centimetric.
  - d) decimetric.

Answer ;

- b) metric.
- 

0869. The VHF Omnidirectional Range ( VOR ) uses the following wavelengths ;
- a) hectometric.
  - b) decimetric.
  - c) metric.
  - d) centimetric.

Answer ;

- c) metric.
- 

0870. The viscosity of a hydraulic fluid should be ;
- a) the highest to minimize power consumption and resistance to flow.
  - b) the lowest to minimize power consumption and resistance to flow.
  - c) the highest to provide excellent lubrication properties.
  - d) the lowest to provide excellent lubrication properties.

Answer ;

- b) the lowest to minimize power consumption and resistance to flow.
- 

0871. The voltage of a fully charged lead-acid battery cell is ;
- a) 1.2 volts.
  - b) 2.2 volts.
  - c) 1.8 volts.
  - d) 1.4 volts.

Answer ;

- b) 2.2 volts.
- 

0872. The voltage regulator of a DC generator is connected in ;
- a) parallel with the armature.
  - b) parallel with the shunt field coil.
  - c) series with the shunt field coil.
  - d) series with the armature.

Answer ;

- c) series with the shunt field coil.
-

0873. The wavelength of a non-directional beacon ( NDB ) at a frequency of 300 kHz is ;

- a) 1 metre.
- b) 100 metres.
- c) 1000 metres.
- d) 10 metres.

Answer ;

c) 1000 metres.

Çözüm ;

$f$  ( frekans ) =  $C$  ( Işık hızı ) :  $\lambda$  ( Dalga boyu )

Değerimiz kHz , yani dalga boyu kilometrik değerde olacağı için ışık hızını (  $300 \times 10^3$  ) km/sec olarak alıyoruz ;  
 $300 = ( 300 \times 10^3 ) : \lambda \rightarrow \lambda = 300\,000 : 300 = 1000 \text{ metre}$  olarak bulunur.

---

0874. The wavelength of a radio signal transmitted at the frequency of 75 MHz is ;

- a) 40 m.
- b) 7.5 m.
- c) 75 m.
- d) 4 m.

Answer ;

d) 4 m.

Çözüm ;

$f$  ( frekans ) =  $C$  ( Işık hızı ) :  $\lambda$  ( Dalga boyu )

Değerimiz MHz , yani dalga boyu metrik değerde olacağı için ışık hızını (  $300 \times 10^6$  ) m/sec olarak alıyoruz ;

Burada frekans değeri Hz cinsinden olmalıdır ;  $75 \text{ MHz} = 75\,000\,000 \text{ Hz}$  ;

$75 = ( 300 \times 10^6 ) : \lambda \rightarrow \lambda = 300\,000\,000 : 75\,000\,000 = 4 \text{ m}$  olarak bulunur.

---

0875. The wavelength of a radio signal transmitted at the frequency 118.7 MHz is ;

- a) 2.53 m.
- b) 25.3 m.
- c) 2.53 cm.
- d) 25.3 cm.

Answer ;

a) 2.53 m.

Çözüm ;

$f$  ( frekans ) =  $C$  ( Işık hızı ) :  $\lambda$  ( Dalga boyu )

Değerimiz MHz , yani dalga boyu metrik değerde olacağı için ışık hızını (  $300 \times 10^6$  ) m/sec olarak alıyoruz ;

Burada frekans değeri Hz cinsinden olmalıdır ;  $118.7 \text{ MHz} = 118\,700\,000 \text{ Hz}$  ;

$118\,700\,000 = ( 300 \times 10^6 ) : \lambda \rightarrow \lambda = 300\,000\,000 : 118\,700\,000 = 2.5273 \approx 2.53 \text{ m}$  olarak bulunur.

---

0876. The wavelength of a radio transmitted on frequency 121.95 MHz is ;

- a) 2.46 m
- b) 24.60 cm
- c) 24.60 m
- d) 2.46 cm

Answer ;

a) 2.46 m

Çözüm ;

$f$  ( frekans ) =  $C$  ( Işık hızı ) :  $\lambda$  ( Dalga boyu )

Değerimiz MHz , yani dalga boyu metrik değerde olacağı için ışık hızını (  $300 \times 10^6$  ) m/sec olarak alıyoruz ;

Burada frekans değeri Hz cinsinden olmalıdır ;  $121.95 \text{ MHz} = 121\,950\,000 \text{ Hz}$  ;

$121\,950\,000 = ( 300 \times 10^6 ) : \lambda \rightarrow \lambda = 300\,000\,000 : 121\,950\,000 = 2.46002 \approx 2.46 \text{ m}$  olarak bulunur.

---

0877. The wing anti-ice system has to protect ;

- a) the whole upper wing surface and the flaps.
- b) the leading edge or the slats , either partially or completely.
- c) the whole leading edge except the slats because they cannot be de-iced when extended.
- d) the whole leading edge and the whole under wing surface.

Answer ;

b) the leading edge or the slats , either partially or completely.

---

0878. The wing ice protection system currently used for most large jet transport aeroplanes is a(n) ;

- a) liquid de-icing system.
- b) electrical de-icing system.
- c) hot air system.
- d) pneumatic system with inflatable boots.

Answer ;

c) hot air system.

---

0879. The wing ice protection system currently used for most large turboprop transport aeroplanes is a(n) ;

- a) hot air system.
- b) electrical de-icing system.
- c) pneumatic system with inflatable boots.
- d) liquid de-icing system.

Answer ;

c) pneumatic system with inflatable boots.

---

0880. The wing of an aircraft in flight , powered by engines located under the wing , is subjected to a bending moment due to thrust and drag. The loading on the front spar of the torsion box from the wing root to the wing tip is ;

- a) compression.
- b) tension , and then compression.
- c) tension.
- d) compression , and then tension.

Answer ;

d) compression , and then tension.

---

0881. The working cycle of a four-stroke engine is ;

- a) induction , power , compression , exhaust.
- b) compression , induction , power , exhaust.
- c) induction , compression , power , exhaust.
- d) induction , compression , expansion , power.

Answer ;

c) induction , compression , power , exhaust.

---

0882. The working principle of the anti-skid system is ;

- a) reduction of the brake pressure at the slower turning wheels.
- b) increase of the brake pressure at the slower turning wheels.
- c) increase of the brake pressure at the faster turning wheels.
- d) reduction of the brake pressure at the faster turning wheels.

Answer ;

a) reduction of the brake pressure at the slower turning wheels.

---

0883. Thermal plugs are installed in ;

- a) cabin windows.
- b) fire warning systems.
- c) cargo compartments.
- d) wheel rims.

Answer ;

d) wheel rims.

---

0884. To achieve control flutter damping the balance mass must be located ;

- a) directly below the control surface hinge.
- b) in front of the control surface hinge.
- c) behind the control surface hinge.
- d) directly above the control surface hinge.

Answer ;

b) in front of the control surface hinge.

---

0885. To achieve reverse thrust on a “big fan” turbine engine , it is necessary to ;

- a) reduce the flow of hot stream air.
- b) stop the fan.
- c) deflect the fan thrust.
- d) reverse the direction of rotation of the fan.

Answer ;

c) deflect the fan thrust.

---

**0886.** To achieve reverse thrust on a turbine engine with a high by-pass ratio , generally ;

- a) the fan flow is deflected.
- b) the fan is stopped.
- c) the flow of hot stream air is reversed.
- d) the direction of rotation of the fan is reversed.

**Answer ;**

**a) the fan flow is deflected.**

---

**0887.** To allow for failure of the normal method of system pressure limiting control , a hydraulic system often incorporates ;

- a) auxiliary hydraulic motors.
- b) a high pressure relief valve.
- c) an accumulator.
- d) a stand-by hydraulic pump.

**Answer ;**

**b) a high pressure relief valve.**

---

**0888.** To avoid the risk of tyre burst from overheating there is ;

- a) water injection triggered at a fixed temperature in order to lower tyre temperature.
- b) a thermal plug that deflates the tyre at a specific temperature.
- c) a pressure relief valve situated in the filler valve.
- d) the “Emergency Burst” function of the anti-skid system that adapts braking to the tyre temperature.

**Answer ;**

**b) a thermal plug that deflates the tyre at a specific temperature.**

---

**0889.** To ensure correct load sharing between AC generators operating in parallel ;

- a) only real loads need to be matched.
- b) both real and reactive loads must be matched.
- c) only reactive loads need to be matched.
- d) the matching of loads is unimportant.

**Answer ;**

**b) both real and reactive loads must be matched.**

---

**0890.** To ensure that the fuel flow is kept directly proportional to the mass of air flowing through the choke , thus preventing the main jet supplying excessive fuel as engine speed is increased , a carburettor is fitted with ;

- a) a power jet.
- b) a diffuser.
- c) a mixture control.
- d) an accelerator pump.

**Answer ;**

**b) a diffuser.**

---

**0891.** To maintain a constant mixture at low and high power settings in a piston engine , a carburettor is fitted with ;

- a) a diffuser ( compensating jet ).
- b) a mixture control.
- c) a power jet.
- d) an accelerator pump.

**Answer ;**

**a) a diffuser ( compensating jet ).**

---

**0892.** To maintain the correct air/fuel ratio of a piston engine , independent of power setting , a carburettor is fitted with ;

- a) a power jet.
- b) an accelerator pump.
- c) a diffuser ( compensating jet ).
- d) a mixture control.

**Answer ;**

**c) a diffuser ( compensating jet ).**

---

**0893.** To protect against excessive system pressure , a hydraulic system usually incorporates ;

- a) a stand by hydraulic pump.
- b) auxiliary hydraulic motors.
- c) a high pressure relief valve.
- d) an accumulator.

**Answer ;**

**c) a high pressure relief valve.**

---

0894. To reverse the direction of rotation of shunt-type ( parallel field ) DC electric motor , it is necessary to ;
- a) connect a phase-shift capacitor to the field circuit.
  - b) reverse the polarity of either the stator and the rotor.
  - c) change the connections from shunt to series.
  - d) reverse the polarity of the motor connections.

Answer ;

- b) reverse the polarity of either the stator and the rotor.
- 

0895. Torsion in a wing can be caused by ;
- a) wing tip vortex.
  - b) dihedral.
  - c) positive sweep.
  - d) propwash.

Answer ;

- c) positive sweep.
- 

0896. Trimming of aileron and rudder in an irreversible flight control system ;
- a) is not possible.
  - b) is achieved by adjusting the neutral point of the flight control actuator.
  - c) is achieved by adjusting the “zero force point” of the feel system.
  - d) is not necessary.

Answer ;

- c) is achieved by adjusting the “zero force point” of the feel system.
- 

0897. Turbine blade stages may be classed as either “impulse” or “reaction”. In an impulse blade section the pressure (i)..... across the nozzle guide vanes and (ii)..... across the rotor blades ;
- a) (i) remains constant , (ii) drops
  - b) (i) rises , (ii) remains constant
  - c) (i) remains constant , (ii) rises
  - d) (i) drops , (ii) remains constant

Answer ;

- d) (i) drops , (ii) remains constant
- 

0898. Turbine blade stages may be classed as either “impulse” or “reaction”. In an impulse turbine stage ;
- a) the pressure remains constant across stator blades and rises across the rotor blades.
  - b) the pressure rises across the stator blades and remains constant across the rotor blades.
  - c) the pressure drops across the stator blades and remains constant across the rotor blades.
  - d) the pressure remains constant across the stator blades and drops across the rotor blades.

Answer ;

- c) the pressure drops across the stator blades and remains constant across the rotor blades.
- 

0899. Tyre “creep” may be described as the ;
- a) gradual circumferential increase of tyre wear.
  - b) the increase in inflation pressure due to decrease in ambient temperature.
  - c) the decrease in inflation pressure due to increase in ambient temperature.
  - d) circumferential movement of the tyre in relation to the wheel flange.

Answer ;

- d) circumferential movement of the tyre in relation to the wheel flange.
- 

0900. Under normal conditions ( JAR 25 ) the cabin pressure altitude is not allowed to exceed ;
- a) 6000 ft.
  - b) 8000 ft.
  - c) 4000 ft.
  - d) 10000 ft.

Answer ;

- b) 8000 ft.
- 

0901. Under normal flight conditions , cabin pressure is controlled by ;
- a) regulating the discharge of air through the outflow valve(s).
  - b) pressurisation duct relief valve(s).
  - c) engine rpm.
  - d) inward relief valve(s).

Answer ;

- a) regulating the discharge of air through the outflow valve(s).
-

0902. Under normal running conditions a magneto draws primary current ;

- a) directly from the aircraft batteries.
- b) from the booster coil.
- c) from a self-contained electro-magnetic induction system.
- d) from the aircraft batteries via an inverter.

Answer ;

c) from a self-contained electro-magnetic induction system.

---

0903. Using compressor bleed air to power systems ;

- a) is limited to the phases of take-off and landing.
- b) decreases aircraft performance.
- c) increases aircraft performance.
- d) has no influence on aircraft performance.

Answer ;

b) decreases aircraft performance.

---

0904. Vapour lock is ;

- a) the inability of a fuel to vaporize in the carburettor.
- b) the formation of water vapour in a fuel system.
- c) vaporizing of fuel prior to reaching the carburettor.
- d) vaporizing of fuel in the carburettor.

Answer ;

c) vaporizing of fuel prior to reaching the carburettor.

---

0905. "Vapour lock" is the phenomenon by which ;

- a) water vapour plugs are formed in the intake fuel line following the condensation of water in fuel tanks which have not been drained for sometime.
- b) burnt gas plugs forming and remaining in the exhaust manifold following an overheat and thereby disturbing the exhaust.
- c) abrupt and abnormal enrichment of the fuel/air mixture following an inappropriate use of carburettor heat.
- d) heat produces vapour plugs in the fuel line.

Answer ;

d) heat produces vapour plugs in the fuel line.

---

0906. Vapour locking is caused by ;

- a) vaporizing of fuel prior to the engine.
- b) vaporizing of fuel in the nozzles.
- c) the inability of fuel to vaporize in the nozzles.
- d) the formation of water vapour in a fuel system.

Answer ;

a) vaporizing of fuel prior to the engine.

---

0907. Variations in mixture ratios for carburettors are achieved by the adjustment of ;

- a) fuel flow and air flow.
- b) air flow.
- c) fuel flow , air flow and temperature.
- d) fuel flow.

Answer ;

d) fuel flow.

---

0908. Vents in oil tanks are primarily to ;

- a) eliminate foaming.
- b) allow for expansion of hot oil.
- c) prevent excessive pressure from building up in tank.
- d) prevent overthrow.

Answer ;

c) prevent excessive pressure from building up in tank.

---

0909. Vibration sensors are used in a gas turbine engine to indicate ;

- a) imbalance of one or both of the spools.
- b) bearing problems only.
- c) that there is too much free movement of the engine strut ( pylon ).
- d) that tip clearance of turbine blades has decreased below a minimum value.

Answer ;

a) imbalance of one or both of the spools.

---



0910. Viscosity is ;

- a) the flow velocity inside the oil lines.
- b) the temperature dependence of an oil.
- c) the tendency of a liquid or gas to resist flow.
- d) the pressure resistance of an oil.

Answer ;

c) the tendency of a liquid or gas to resist flow.

---

0911.  $V_{LE}$  is the maximum ;

- a) speed with flaps extended in a given position.
- b) speed at which the landing gear can be operated with full safety.
- c) flight speed with landing gear down.
- d) speed authorized in flight.

Answer ;

c) flight speed with landing gear down.

---

0912.  $V_{LO}$  is the maximum ;

- a) cruising speed not to be exceeded except in still air with caution.
- b) speed at which the landing gear can be operated with full safety.
- c) speed with flaps extended in a given position.
- d) flight speed with landing gear down.

Answer ;

b) speed at which the landing gear can be operated with full safety.

---

0913. What are the three elements of the fuselage structure of a large transport aeroplane ?

- a) Skin , ribs , formers.
- b) Skin , frames , stringers.
- c) Skin , spars , ribs.
- d) Skin , girders , webs.

Answer ;

b) Skin , frames , stringers.

---

0914. What is breathed in when using a passenger oxygen mask ?

- a) Cabin air and oxygen.
- b) A mixture of oxygen and freon gas.
- c) Cabin air and oxygen or 100% oxygen.
- d) 100% oxygen.

Answer ;

a) Cabin air and oxygen.

---

0915. What is the load in the upper respectively lower girder of a spar of a cantilever wing during straight and level flight ?

- a) Tension in lower - and upper girder.
- b) Torsion in the lower - and upper girder.
- c) Tension in the upper girder and compression in the lower girder.
- d) Compression in the upper girder and tension in the lower girder.

Answer ;

d) Compression in the upper girder and tension in the lower girder.

---

0916. What may happen during a continuous climb with a mixture setting fully rich ?

- a) Increase of the power available.
- b) The engine will operate smoother even through fuel consumption is increased.
- c) Fouling of spark plugs.
- d) The engine will overheat.

Answer ;

c) Fouling of spark plugs.

---

0917. What will eventually happen during a continuous climb with a mixture setting full rich ?

- a) The engine will operate smoother even though fuel consumption is increased.
- b) Fouling of spark plugs.
- c) The engine will overheat.
- d) Increase of the power available.

Answer ;

b) Fouling of spark plugs.

---

**0918.** What will happen to the geometrical pitch angle of a “constant speed propeller” if the manifold pressure is increased ?

- a) It will decrease so that the engine can increase.
- b) It will remain the same.
- c) It will increase.
- d) It will increase and after a short time it will be the same again.

**Answer ;**

**c) It will increase.**

---

**0919.** When a battery is almost fully discharged there is a tendency for the ;

- a) electrolyte to “boil”.
- b) current produced to increase due to the reduced voltage.
- c) voltage to increase due to the current available.
- d) voltage to decrease under load.

**Answer ;**

**d) voltage to decrease under load.**

---

**0920.** When a bimetallic strip is used as a switch in a fire detection loop , a fire alarm is triggered after a delay. The purpose of this delay is to ;

- a) wait for the triggering of the second fire detection loop in order to confirm the fire.
- b) avoid false alarms in case of vibrations.
- c) delay the triggering of the fire extinguishers and increase their efficiency.
- d) allow temperatures to equalise.

**Answer ;**

**b) avoid false alarms in case of vibrations.**

---

**0921.** When a cabin oxygen mask is pulled downwards , the passenger breathes ;

- a) cabin air under pressure.
- b) a mixture of oxygen and cabin air.
- c) pure oxygen at the ambient pressure.
- d) pure oxygen under pressure.

**Answer ;**

**b) a mixture of oxygen and cabin air.**

---

**0922.** When a conductor cuts the flux of a magnetic field ;

- a) the field will collapse.
- b) there will be no effect on the conductor.
- c) current will flow in accordance with Flemings left hand rule.
- d) an electromotive force ( EMF ) is induced in the conductor.

**Answer ;**

**d) an electromotive force ( EMF ) is induced in the conductor.**

---

**0923.** When a continuous loop wire type fire detection system is tested ;

- a) a part of the wire is heated.
- b) the whole wire is heated.
- c) the wiring and the warning are tested.
- d) only the warning function is tested.

**Answer ;**

**c) the wiring and the warning are tested.**

---

**0924.** When a gas turbine engine accelerates from idle to take-off thrust and the compressor rpm is too high in comparison with the air inlet velocity , this may result in ;

- a) supersonic airflow at the compressor blade trailing edges.
- b) a compressor stall.
- c) stagnation.
- d) a compressor blade airflow angle of attack that is too small.

**Answer ;**

**b) a compressor stall.**

---

**0925.** When a gas turbine engine is fitted with can-type combustion chambers ;

- a) each chamber is fitted with two igniters.
- b) a total of only two igniters are usually fitted since the chambers are inter-connected.
- c) each chamber is fitted with one igniter and one glow plug.
- d) each chamber has its own igniter.

**Answer ;**

**b) a total of only two igniters are usually fitted since the chambers are inter-connected.**

---

0926. When a landing gear wheel is hydroplaning , its friction factor is equal to ;

- a) 1
- b) 0.1
- c) 0.5
- d) 0

Answer ;

d) 0

---

0927. When a part of a continuous loop fire detection system is heated ;

- 1. its resistance decreases.
- 2. its resistance increases.
- 3. the leakage current increases.
- 4. the leakage current decreases.

The combination regrouping all the correct statements is ;

- a) 2 , 4.
- b) 2 , 3.
- c) 1 , 3.
- d) 1 , 4.

Answer ;

c) 1 , 3.

---

0928. When a persistent top excitation limit fault on an AC generator connected to the mains with another AC generator , the overexcitation protection device opens ;

- a) the exciter breaker and the generator breaker.
- b) the exciter breaker , the generator breaker and the tie breaker.
- c) the generator breaker.
- d) the tie breaker.

Answer ;

b) the exciter breaker , the generator breaker and the tie breaker.

---

0929. When a persistent underexcitation fault is detected on an AC generator connected to the aircraft mains with another AC generator , the protection device opens ;

- a) the exciter control relay , the generator breaker and the tie breaker.
- b) the generator breaker and the tie breaker.
- c) the tie breaker.
- d) the generator control relay and the generator breaker.

Answer ;

a) the exciter control relay , the generator breaker and the tie breaker.

---

0930. When a turbo-prop aeroplane decelerates during the initial part of the landing roll , a constant speed propeller within the beta range ;

- a) is feathered.
- b) produces positive thrust.
- c) produces negative thrust.
- d) produces zero thrust.

Answer ;

c) produces negative thrust.

---

0931. When a wing bends downwards , aileron flutter might occur if the aileron deflects ;

- a) downwards , because the location of the aileron centre of gravity lies in front of the hinge line.
- b) downwards , because the location of the aileron centre of gravity lies behind the hinge line.
- c) upwards , because the location of the aileron centre of gravity lies in front of the hinge line.
- d) upwards , because the location of the aileron centre of gravity lies behind the hinge line.

Answer ;

d) upwards , because the location of the aileron centre of gravity lies behind the hinge line.

---

0932. When a wing bends upwards , aileron flutter might occur if the aileron deflects ;

- a) upwards , because the location of the aileron centre of gravity lies in front of the hinge line.
- b) upwards , because the location of the aileron centre of gravity lies behind the hinge line.
- c) downwards , because the location of the aileron centre of gravity lies in front of the hinge line.
- d) downwards , because the location of the aileron centre of gravity lies behind the hinge line.

Answer ;

d) downwards , because the location of the aileron centre of gravity lies behind the hinge line.

---

0933. When AC generators are operated in parallel , they must be of the same ;

- a) amperage and kVAR.
- b) frequency and amperage.
- c) voltage and frequency.
- d) voltage and amperage.

Answer ;

c) voltage and frequency.

---

0934. When air is compressed for pressurization purposes , the percentage oxygen content is ;

- a) dependent on the degree of pressurisation.
- b) decreased.
- c) increased.
- d) unaffected.

Answer ;

d) unaffected.

---

0935. When altitude increases without adjustment of the mixture ratio , the piston engine performance is affected because of ;

- a) an increase of air density with a fuel flow which becomes too high.
- b) an increase of air density with a fuel flow which becomes too low.
- c) a decrease of air density with a fuel flow which becomes too low.
- d) a decrease of air density with a fuel flow which becomes too high.

Answer ;

d) a decrease of air density with a fuel flow which becomes too high.

---

0936. When an “open circuit” occurs in an electrical supply system , the ;

- a) load as indicated by the ammeter will increase.
- b) loss of continuity will prevent its working components from functioning.
- c) fuse or CB should isolate the circuit due to excess current drawn.
- d) component will operate normally , but will not switch off.

Answer ;

b) loss of continuity will prevent its working components from functioning.

---

0937. When an underspeed fault is detected on an AC generator connected to the aircraft AC busbar , the protection device opens the ;

- a) exciter breaker.
- b) generator breaker.
- c) exciter breaker and generator breaker.
- d) exciter breaker , generator breaker and tie breaker.

Answer ;

b) generator breaker.

---

0938. When applying carburettor heating ;

- a) a decrease in RPM results from the lean mixture.
- b) no change occurs in the mixture ratio.
- c) the mixture becomes richer.
- d) the mixture becomes leaner.

Answer ;

c) the mixture becomes richer.

---

0939. When carrying out battery condition check using the aircraft’s voltmeter ;

- a) the load condition is unimportant.
- b) no load should be applied to the battery because it would depress the voltage.
- c) a load should be applied to the battery in order to give a better indication of condition.
- d) the battery should be isolated.

Answer ;

c) a load should be applied to the battery in order to give a better indication of condition.

---

0940. When changing power on engines equipped with constant-speed propeller , engine overload is avoided by ;

- a) reducing the RPM before reducing the manifold pressure.
- b) adjusting Fuel Flow before the manifold pressure.
- c) increasing the manifold pressure before increasing the RPM.
- d) increasing the RPM before increasing the manifold pressure.

Answer ;

d) increasing the RPM before increasing the manifold pressure.

---

0941. When detonation is recognised in a piston engine , the correct procedure is to ;

- a) increase manifold pressure and fully enrich the mixture.
- b) reduce manifold pressure and lean the mixture.
- c) reduce manifold pressure and fully enrich the mixture.
- d) increase manifold pressure and lean the mixture.

Answer ;

c) reduce manifold pressure and fully enrich the mixture.

---

0942. When excessively leaning the mixture for a better fuel economy , but still on the rich side of the peak EGT , the following engine parameter(s) may exceed their normal operating ranges ;

- a) cylinder head and exhaust gas temperature.
- b) oil temperature.
- c) manifold pressure.
- d) engine RPM.

Answer ;

a) cylinder head and exhaust gas temperature.

---

0943. When in flight , a piston engine is stopped and the propeller blade angle is near 90° , the propeller is said to be ;

- a) at zero drag.
- b) feathered.
- c) windmilling.
- d) transparent.

Answer ;

b) feathered.

---

0944. When increasing true airspeed with a constant engine RPM , the angle of attack of a fixed pitch propeller ;

- a) stays constant.
- b) increases.
- c) reduces.
- d) stays constant because it only varies with engine RPM.

Answer ;

c) reduces.

---

0945. When leaning the mixture for the most economic cruise fuel flow , excessive leaning will cause ;

- a) high manifold pressure.
- b) high cylinder head.
- c) high engine RPM.
- d) low cylinder head.

Answer ;

b) high cylinder head.

---

0946. When operating two AC generators unparallelled , the phase relationship of each generator ;

- a) must be synchronous.
- b) must be 90° out of synchronisation.
- c) must be in opposition.
- d) is unimportant.

Answer ;

d) is unimportant.

---

0947. When powering up a hydraulic system , the level in the reservoir will ;

- a) always remain the same.
- b) increase as ambient temperature decreases.
- c) initially increase with system pressurisation.
- d) decrease slightly.

Answer ;

d) decrease slightly.

---

0948. When quick donning masks are in use , the pilot is ;

- a) able to radiotelephone.
- b) only able to transmit.
- c) only able to receive.
- d) not able to do any radio communication.

Answer ;

a) able to radiotelephone.

---

0949. When selected to normal , the oxygen/air ratio of the mixture supplied by the cockpit oxygen system regulator ;
- a) is 100%.
  - b) increases when the altitude increases.
  - c) decreases when the altitude increases.
  - d) is constant whatever the altitude.

Answer ;

b) increases when the altitude increases.

---

0950. When starting an engine , or when the engine runs at an idle rpm on ground , the mixture is ;
- a) lean , to prevent that the engine consumes too much fuel.
  - b) rich , to make starting possible and to cool the engine sufficiently when idling.
  - c) rich , because carburettor heat is switched on.
  - d) rich , because the choke valve is closed.

Answer ;

b) rich , to make starting possible and to cool the engine sufficiently when idling.

---

0951. When starting the engine , or when the engine running at idle rpm on the ground , the mixture is ;
- a) weak , to prevent the engine consuming too much fuel.
  - b) rich , because carburettor heat is switched on.
  - c) rich , because the choke valve is closed.
  - d) rich , to make starting possible and to cool the engine sufficiently when idling.

Answer ;

d) rich , to make starting possible and to cool the engine sufficiently when idling.

---

0952. When TAS increases , the blade angle of a constant speed propeller will ( RPM and throttle levers are not moved ) ;
- a) increase.
  - b) remain constant.
  - c) first decrease and after a short time increase to its previous value.
  - d) decrease.

Answer ;

a) increase.

---

0953. When the AC generators are connected in parallel , the reactive loads are balanced by means of the ;
- a) torque of the Constant Speed Drive ( CSD ).
  - b) frequency or load controller.
  - c) voltage controller.
  - d) excitation current.

Answer ;

d) excitation current.

---

0954. When the AC generators are connected in parallel , the reactive loads are balanced by means of the ;
- a) energizing current.
  - b) voltage.
  - c) frequency.
  - d) torque of the Constant Speed Drive ( CSD ).

Answer ;

a) energizing current.

---

0955. When the AC voltage across a capacitor is kept constant and the frequency is increased , the current through the capacitor will ;

- a) remain same.
- b) increase.
- c) decrease.
- d) be zero.

Answer ;

b) increase.

---

0956. When the combustion gases pass through a turbine the ;
- a) axial velocity of the flow decreases.
  - b) pressure drops.
  - c) temperature increases.
  - d) pressure rises.

Answer ;

b) pressure drops.

---

0957. When the door of a transport airplane equipped with evacuation slides in the armed position is opened from the outside , the slide ;

- a) unfolds and becomes inflated.
- b) is disarmed automatically.
- c) unfolds but does not become inflated.
- d) becomes inflated in its container thus preventing its unfolding.

Answer ;

b) is disarmed automatically.

---

0958. When the oxygen system of a large aeroplane is used ;

- a) with the setting on “NORMAL” , the cockpit crew is supplied with a mixture of oxygen and cabin air.
- b) the same source of supply is used by the cockpit crew and passengers.
- c) the same source of supply is used by the cockpit crew and cabin crew.
- d) with the setting on “NORMAL” , the passengers are supplied with a mixture of oxygen and cabin air.

Answer ;

a) with the setting on “NORMAL” , the cockpit crew is supplied with a mixture of oxygen and cabin air.

---

0959. When the pilot moves the mixture lever of a piston engine towards a lean position the ;

- a) volume of air entering the carburettor is reduced.
- b) amount of fuel entering the combustion chamber is increased.
- c) amount of fuel entering the combustion chamber is reduced.
- d) volume of air entering the carburettor is increased.

Answer ;

c) amount of fuel entering the combustion chamber is reduced.

---

0960. When the supply frequency in a circuit with a capacitor is increased , the current in this circuit will ;

- a) remain the same.
- b) decrease.
- c) increase.
- d) be zero.

Answer ;

c) increase.

---

0961. When the wing skin is not able to carry loads , the structural elements of the wing , which carry the bending moment , are ;

- a) the ribs.
- b) the webs.
- c) the spars.
- d) the rivets.

Answer ;

c) the spars.

---

0962. When two DC generators are operating in parallel , control of load sharing is achieved by ;

- a) the synchronous bus-bar.
- b) an equalising circuit which , in conjunction with the voltage regulators , varies the field excitation current of the generators.
- c) carrying out systematic load-shedding procedures.
- d) an equalising circuit which , in turn , controls the speed of the generators.

Answer ;

b) an equalising circuit which , in conjunction with the voltage regulators , varies the field excitation current of the generators.

---

0963. When using an emergency oxygen mask , a pilot is able to ;

- a) both transmit and receive.
- b) receive only.
- c) neither transmit or receive.
- d) transmit only.

Answer ;

a) both transmit and receive.

---

0964. Which of the following parameters remains constant in a divergent ( subsonic ) gas turbine engine intake ?

- a) Static pressure.
- b) Total temperature.
- c) Static temperature.
- d) Dynamic pressure.

Answer ;

b) Total temperature.

---

0965. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?

- I. The dynamic pressure decreases in flow direction.
  - II. The static temperature decreases in flow direction.
- a) I is correct , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

b) I is correct , II is incorrect.

---

0966. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?

- I. The dynamic pressure decreases in flow direction.
  - II. The static temperature increases in flow direction.
- a) I is incorrect , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

c) I is correct , II is correct.

---

0967. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?

- I. The dynamic pressure increases in flow direction.
  - II. The static temperature decreases in flow direction.
- a) I is incorrect , II is correct.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

c) I is incorrect , II is incorrect.

---

0968. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?

- I. The dynamic pressure increases in flow direction.
  - II. The static temperature increases in flow direction.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

c) I is incorrect , II is correct.

---

0969. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?

- I. The dynamic pressure remains constant in flow direction.
  - II. The static temperature remains constant in flow direction.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

d) I is incorrect , II is incorrect.

---



0970. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?
- I. The speed decreases in flow direction.
  - II. The total pressure decreases in flow direction.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is correct.

Answer ;

b) I is correct , II is incorrect.

---

0971. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?
- I. The speed decreases in flow direction.
  - II. The total pressure remains constant in flow direction.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

d) I is correct , II is correct.

---

0972. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?
- I. The speed remains constant in flow direction.
  - II. The total pressure decreases in flow direction.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

a) I is incorrect , II is incorrect.

---

0973. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?
- I. The speed remains constant in flow direction.
  - II. The total pressure remains constant in flow direction.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

d) I is incorrect , II is correct.

---

0974. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?
- I. The static pressure decreases in flow direction.
  - II. The total temperature increases in flow direction.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

c) I is incorrect , II is incorrect.

---

0975. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?
- I. The static pressure increases in flow direction.
  - II. The total temperature increases in flow direction.
- a) I is correct , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

a) I is correct , II is incorrect.

---

0976. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?

- I. The static pressure increases in flow direction.
- II. The total temperature remains constant in flow direction.
- a) I is incorrect , II is incorrect.
- b) I is correct , II is incorrect.
- c) I is correct , II is correct.
- d) I is incorrect , II is correct.

Answer ;

c) I is correct , II is correct.

---

0977. Which of the following statements is correct about the flow in a divergent ( subsonic ) gas turbine engine intake ?

- I. The static pressure remains constant in flow direction.
- II. The total temperature decreases in flow direction.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is correct , II is correct.
- d) I is incorrect , II is incorrect.

Answer ;

d) I is incorrect , II is incorrect.

---

0978. Which of the following statements regarding variable inlet guide vanes ( VIGVs ) and variable stator vanes ( VSVs ) on a high-bypass engines is correct ?

- a) If VIGVs move towards the open position , VSVs move towards the closed position.
- b) VIGVs prevent LP compressor stall and VSVs prevent HP compressor stall.
- c) VIGVs are located at the front of the HP compressor and , together with the VSVs , control compressor airflow.
- d) VIGVs are located in the front of the fan and control both fan and compressor airflow. VSVs are used in the compressor only and are used to increase pressure ratio.

Answer ;

c) VIGVs are located at the front of the HP compressor and , together with the VSVs , control compressor airflow.

---

0979. Which of these statements about a gust lock system are correct or incorrect ?

- I. A gust lock can be used in flight to reduce the effects of turbulence.
- II. There is no need for a gust lock on reversible flight controls.
- a) I is correct , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is incorrect , II is correct.
- d) I is correct , II is incorrect.

Answer ;

b) I is incorrect , II is incorrect.

---

0980. Which of these statements about a gust lock system are correct or incorrect ?

- I. Irreversible flight controls should have a gust lock.
- II. There is no need for a gust lock on manual flight controls.
- a) I is correct , II is incorrect.
- b) I is correct , II is correct.
- c) I is incorrect , II is correct.
- d) I is incorrect , II is incorrect.

Answer ;

d) I is incorrect , II is incorrect.

---

0981. Which of these statements about a gust lock system are correct or incorrect ?

- I. There is no need for a gust lock on irreversible flight controls.
- II. Manual flight controls should have a gust lock.
- a) I is correct , II is correct.
- b) I is correct , II is incorrect.
- c) I is incorrect , II is correct.
- d) I is incorrect , II is incorrect.

Answer ;

a) I is correct , II is correct.

---

0982. Which of these statements about a gust lock system are correct or incorrect ?
- I. There is no need for a gust lock on irreversible flight controls.
  - II. There is no need for a gust lock on manual flight controls.
- a) I is incorrect , II is correct.
  - b) I is correct , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is incorrect.

Answer ;

c) I is correct , II is incorrect.

---

0983. Which of these statements about a gust lock system are correct or incorrect ?
- I. There should be suitable design precautions to prevent flight with the gust lock engaged.
  - II. Reversible flight controls should have a gust lock.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

d) I is correct , II is correct.

---

0984. Which of these statements about a gust lock system are correct or incorrect ?
- I. There should be suitable design precautions to prevent flight with the gust lock engaged.
  - II. There is no need for a gust lock on reversible flight controls.
- a) I is correct , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

b) I is correct , II is incorrect.

---

0985. Which of these statements about an aeroplane's hydraulic system is correct ?
- a) The filters , the pressure relief valve(s) , the by-pass valve(s) , and the fire shut-off valve are safety features installed in the system.
  - b) A hydraulic leak in a return line does not affect the functioning of the hydraulic system.
  - c) The hydraulic reservoir contains membrane and is pressurised by nitrogen on one side of this membrane.
  - d) The pumps are always electrically driven because they need to deliver a high pressure of 3000 psi

Answer ;

a) The filters , the pressure relief valve(s) , the by-pass valve(s) , and the fire shut-off valve are safety features installed in the system.

---

0986. Which of these statements about an impulse turbine are correct or incorrect ?
- I. The pressure drops across the nozzle guide vanes.
  - II. The pressure drops across the rotor blades.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

b) I is correct , II is incorrect.

---

0987. Which of these statements about an impulse turbine are correct or incorrect ?
- I. The pressure drops across the nozzle guide vanes.
  - II. The pressure remains constant across the rotor blades.
- a) I is correct , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

d) I is correct , II is correct.

---

**0988.** Which of these statements about an impulse turbine are correct or incorrect ?

- I. The pressure drops across the nozzle guide vanes.
- II. The pressure rises across the rotor blades.
- a) I is correct , II is correct.
- b) I is correct , II is incorrect.
- c) I is incorrect , II is correct.
- d) I is incorrect , II is incorrect.

**Answer ;**

**b) I is correct , II is incorrect.**

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**0989.** Which of these statements about an impulse turbine are correct or incorrect ?

- I. The pressure remains constant across the nozzle guide vanes.
- II. The pressure drops across the rotor blades.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is correct.
- d) I is incorrect , II is correct.

**Answer ;**

**b) I is incorrect , II is incorrect.**

---

**0990.** Which of these statements about an impulse turbine are correct or incorrect ?

- I. The pressure remains constant across the nozzle guide vanes.
- II. The pressure remains constant across the rotor blades.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is correct.

**Answer ;**

**b) I is incorrect , II is correct.**

---

**0991.** Which of these statements about an impulse turbine are correct or incorrect ?

- I. The pressure remains constant across the nozzle guide vanes.
- II. The pressure rises across the rotor blades.
- a) I is incorrect , II is correct.
- b) I is correct , II is incorrect.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is correct.

**Answer ;**

**c) I is incorrect , II is incorrect.**

---

**0992.** Which of these statements about an impulse turbine are correct or incorrect ?

- I. The pressure rises across the nozzle guide vanes.
- II. The pressure drops across the rotor blades.
- a) I is incorrect , II is incorrect.
- b) I is correct , II is incorrect.
- c) I is correct , II is correct.
- d) I is incorrect , II is correct.

**Answer ;**

**a) I is incorrect , II is incorrect.**

---

**0993.** Which of these statements about an impulse turbine are correct or incorrect ?

- I. The pressure rises across the nozzle guide vanes.
- II. The pressure remains constant across the rotor blades.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is correct.

**Answer ;**

**b) I is incorrect , II is correct.**

---

**0994.** Which of these statements about an impulse turbine are correct or incorrect ?

- I. The pressure rises across the nozzle guide vanes.
- II. The pressure rises across the rotor blades.
- a) I is incorrect , II is incorrect.
- b) I is correct , II is incorrect.
- c) I is incorrect , II is correct.
- d) I is correct , II is correct.

**Answer ;**

**a) I is incorrect , II is incorrect.**

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**0995.** Which of these statements about an reaction turbine are correct or incorrect ?

- I. The pressure drops across the nozzle guide vanes.
- II. The pressure drops across the rotor blades.
- a) I is correct , II is incorrect.
- b) I is correct , II is correct.
- c) I is incorrect , II is correct.
- d) I is incorrect , II is incorrect.

**Answer ;**

**b) I is correct , II is correct.**

---

**0996.** Which of these statements about an reaction turbine are correct or incorrect ?

- I. The pressure drops across the nozzle guide vanes.
- II. The pressure remains constant across the rotor blades.
- a) I is incorrect , II is incorrect.
- b) I is correct , II is correct.
- c) I is correct , II is incorrect.
- d) I is incorrect , II is correct.

**Answer ;**

**c) I is correct , II is incorrect.**

---

**0997.** Which of these statements about an reaction turbine are correct or incorrect ?

- I. The pressure drops across the nozzle guide vanes.
- II. The pressure rises across the rotor blades.
- a) I is correct , II is correct.
- b) I is incorrect , II is correct.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is incorrect.

**Answer ;**

**d) I is correct , II is incorrect.**

---

**0998.** Which of these statements about an reaction turbine are correct or incorrect ?

- I. The pressure remains constant across the nozzle guide vanes.
- II. The pressure drops across the rotor blades.
- a) I is correct , II is correct.
- b) I is incorrect , II is correct.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is incorrect.

**Answer ;**

**b) I is incorrect , II is correct.**

---

**0999.** Which of these statements about an reaction turbine are correct or incorrect ?

- I. The pressure remains constant across the nozzle guide vanes.
- II. The pressure remains constant across the rotor blades.
- a) I is correct , II is correct.
- b) I is incorrect , II is correct.
- c) I is incorrect , II is incorrect.
- d) I is correct , II is incorrect.

**Answer ;**

**c) I is incorrect , II is incorrect.**

---

1000. Which of these statements about an reaction turbine are correct or incorrect ?
- I. The pressure remains constant across the nozzle guide vanes.
  - II. The pressure rises across the rotor blades.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

b) I is incorrect , II is incorrect.

---

1001. Which of these statements about an reaction turbine are correct or incorrect ?
- I. The pressure rises across the nozzle guide vanes.
  - II. The pressure drops across the rotor blades.
- a) I is incorrect , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

b) I is incorrect , II is correct.

---

1002. Which of these statements about an reaction turbine are correct or incorrect ?
- I. The pressure rises across the nozzle guide vanes.
  - II. The pressure remains constant across the rotor blades.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is incorrect.

Answer ;

d) I is incorrect , II is incorrect.

---

1003. Which of these statements about an reaction turbine are correct or incorrect ?
- I. The pressure rises across the nozzle guide vanes.
  - II. The pressure rises across the rotor blades.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

b) I is incorrect , II is incorrect.

---

1004. Which of these statements about chemical oxygen generators are correct or incorrect ?
- I. Chemical oxygen generators are often used to supply passenger oxygen.
  - II. The oxygen supply of chemical oxygen generators , once activated , can be switched off.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

c) I is correct , II is incorrect.

---

1005. Which of these statements about chemical oxygen generators are correct or incorrect ?
- I. Chemical oxygen generators are often used to supply passenger oxygen.
  - II. The oxygen supply of chemical oxygen generators , once activated , cannot be switched off.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

a) I is correct , II is correct.

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1006. Which of these statements about chemical oxygen generators are correct or incorrect ?

- I. Chemical oxygen generators are used to supply the cockpit oxygen system.
  - II. The oxygen supply of chemical oxygen generators , once activated , can be switched off.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

- b) I is incorrect , II is incorrect.
- 

1007. Which of these statements about chemical oxygen generators are correct or incorrect ?

- I. Chemical oxygen generators are used to supply the cockpit oxygen system.
  - II. The oxygen supply of chemical oxygen generators , once activated , cannot be switched off.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- b) I is incorrect , II is correct.
- 

1008. Which of these statements about composite and metal structures are correct or incorrect ?

- I. For a structural component with given dimensions composite materials enable a structural component's strength to be tailored to the direction of the load.
  - II. Composite materials enable structures with higher strength/weight ratio than metal structures.
- a) I is correct , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- a) I is correct , II is correct.
- 

1009. Which of these statements about composite and metal structures are correct or incorrect ?

- I. For a structural component with given dimensions composite materials enable a structural component's strength to be tailored to the direction of the load.
  - II. Composite materials enable structures with lower strength/weight ratio than metal structures.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- c) I is correct , II is incorrect.
- 

1010. Which of these statements about composite and metal structures are correct or incorrect ?

- I. In a structural component with given dimensions constructed of composite materials , the strength is the same in all directions.
  - II. Composite materials enable structures with higher strength/weight ratio than metal structures.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- b) I is incorrect , II is correct.
- 

1011. Which of these statements about composite and metal structures are correct or incorrect ?

- I. In a structural component with given dimensions constructed of composite materials , the strength is the same in all directions.
  - II. Composite materials enable structures with lower strength/weight ratio than metal structures.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- c) I is incorrect , II is incorrect.
-

- 1012.** Which of these statements about compressor stall are correct or incorrect ?
- I. A compressor stall is most likely to occur in a compressor designed for a high pressure ratio operating at high RPM.
  - II. A compressor stall is most likely to occur when the air flow stagnates in the front stages of the compressor.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

**Answer ;**

**b)** I is incorrect , II is incorrect.

---

- 1013.** Which of these statements about compressor stall are correct or incorrect ?
- I. A compressor stall is most likely to occur in a compressor designed for a high pressure ratio operating at high RPM.
  - II. A compressor stall is most likely to occur when the air flow stagnates in the rear stages of the compressor.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

**Answer ;**

**b)** I is incorrect , II is correct.

---

- 1014.** Which of these statements about compressor stall are correct or incorrect ?
- I. A compressor stall is most likely to occur in a compressor designed for a high pressure ratio operating at low RPM.
  - II. A compressor stall is most likely to occur when the air flow stagnates in the front stages of the compressor.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is correct.

**Answer ;**

**c)** I is correct , II is incorrect.

---

- 1015.** Which of these statements about compressor stall are correct or incorrect ?
- I. A compressor stall is most likely to occur in a compressor designed for a high pressure ratio operating at low RPM.
  - II. A compressor stall is most likely to occur when the air flow stagnates in the rear stages of the compressor.
- a) I is incorrect , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

**Answer ;**

**d)** I is correct , II is correct.

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- 1016.** Which of these statements about rudder limiting are correct or incorrect ?
- I. A rudder ratio changer system limits both rudder and rudder pedal deflection as the IAS increases.
  - II. A variable stop system reduces the rudder deflection for a given rudder pedal deflection as the IAS increases.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

**Answer ;**

**b)** I is incorrect , II is incorrect.

---



1017. Which of these statements about rudder limiting are correct or incorrect ?

- I. A rudder ratio changer system reduces the rudder deflection for a given rudder pedal deflection as the IAS increases.
  - II. A variable stop system limits both rudder and rudder pedal deflection as the IAS decreases.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- d) I is correct , II is incorrect.
- 

1018. Which of these statements about rudder limiting are correct or incorrect ?

- I. A rudder ratio changer system reduces the rudder deflection for a given rudder pedal deflection as the IAS increases.
  - II. A variable stop system limits both rudder and rudder pedal deflection as the IAS increases.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

- a) I is correct , II is correct.
- 

1019. Which of these statements about sandwich structural parts are correct or incorrect ?

- I. A sandwich structural part consists of fibres and a resin.
  - II. A sandwich structural part is not suitable for absorbing concentrated loads.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- b) I is incorrect , II is correct.
- 

1020. Which of these statements about sandwich structural parts are correct or incorrect ?

- I. A sandwich structural part consists of fibres and a resin.
  - II. A sandwich structural part is suitable for absorbing concentrated loads.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

- d) I is incorrect , II is incorrect.
- 

1021. Which of these statements about sandwich structural parts are correct or incorrect ?

- I. A sandwich structural part consists of two thin sheets enclosing a light core material.
  - II. A sandwich structural part is not suitable for absorbing concentrated loads.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

- b) I is correct , II is correct.
- 

1022. Which of these statements about sandwich structural parts are correct or incorrect ?

- I. A sandwich structural part consists of two thin sheets enclosing a light core material.
  - II. A sandwich structural part is suitable for absorbing concentrated loads.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- c) I is correct , II is incorrect.
-

1023. Which of these statements about structural design principles are correct or incorrect ?
- I. In structural design , fail safe implies parallel structural parts.
  - II. In structural design , safe life implies the structure will never fail during a declared time period or number of cycles.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

d) I is correct , II is incorrect.

---

1024. Which of these statements about structural design principles are correct or incorrect ?
- I. In structural design , fail safe implies redundant load paths.
  - II. A safe life structure is based on a declared number of cycles or time period.
- a) I is incorrect , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

d) I is correct , II is correct.

---

1025. Which of these statements about structural design principles are correct or incorrect ?
- I. In structural design , fail safe implies the structure will never fail.
  - II. A safe life structure is based on a declared time period or number of cycles.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

b) I is incorrect , II is correct.

---

1026. Which of these statements about structural design principles are correct or incorrect ?
- I. In structural design , fail safe implies the structure will never fail.
  - II. In structural design , safe life implies the structure will never fail during a declared time period or number of cycles.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

d) I is incorrect , II is incorrect.

---

1027. Which of these statements about structural design principles are correct or incorrect ?
- I. The damage tolerance principle assumes cracks in the structure will never occur.
  - II. The safe life principle is based on the replacement of parts after a given number of cycles or time period.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

b) I is incorrect , II is correct.

---

1028. Which of these statements about structural design principles are correct or incorrect ?
- I. The damage tolerance principle takes cracking of the structure into account.
  - II. The safe life principle is based on the replacement of parts after a given number of cycles or flight hours.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

d) I is correct , II is correct.

---

1029. Which of these statements about the wing structure is correct ?

- a) A torsion box is formed by wing spars , ribs and wing skin reinforced by stringers.
- b) A semi-monocoque structure consists of the skin and frames.
- c) The slats are a part of the torsion box.
- d) A wing main spar consists of a web with stringers.

Answer ;

- a) A torsion box is formed by wing spars , ribs and wing skin reinforced by stringers.
- 

1030. Which of these statements about trimming in a irreversible flight control system of a conventional aeroplane are correct or incorrect ?

- I. The zero force position of the control column changes when using the elevator trim.
  - II. The zero force position of the control wheel changes when using the aileron trim.
- a) I is correct , II is correct.
  - b) I is incorrect , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is incorrect.

Answer ;

- b) I is incorrect , II is correct.
- 

1031. Which of these statements about trimming in a irreversible flight control system of a conventional aeroplane are correct or incorrect ?

- I. The zero force position of the control column changes when using the elevator trim.
  - II. The zero force position of the control wheel does not change when using the aileron trim.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

- b) I is incorrect , II is incorrect.
- 

1032. Which of these statements about trimming in a irreversible flight control system of a conventional aeroplane are correct or incorrect ?

- I. The zero force position of the control column does not change when using the elevator trim.
  - II. The zero force position of the control wheel changes when using the aileron trim.
- a) I is correct , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

- a) I is correct , II is correct.
- 

1033. Which of these statements about trimming in a irreversible flight control system of a conventional aeroplane are correct or incorrect ?

- I. The zero force position of the control column does not change when using the elevator trim.
  - II. The zero force position of the control wheel does not change when using the aileron trim.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- a) I is correct , II is incorrect.
- 

1034. Which of these statements concerning a sandwich structural part are correct or incorrect ?

- I. The main function of the core material is sound insulation.
  - II. A sandwich structural part is unsuitable for absorbing concentrated loads.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

- a) I is incorrect , II is correct.
-

1035. Which of these statements concerning a sandwich structural part are correct or incorrect ?
- I. The main function of the core material is sound insulation.
  - II. A sandwich structural part is well suited for absorbing concentrated loads.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

- a) I is incorrect , II is incorrect.
- 

1036. Which of these statements concerning a sandwich structural part are correct or incorrect ?
- I. The main function of the core material is to stabilise the covering sheets.
  - II. A sandwich structural part is unsuitable for absorbing concentrated loads.
- a) I is correct , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- a) I is correct , II is correct.
- 

1037. Which of these statements concerning a sandwich structural part are correct or incorrect ?
- I. The main function of the core material is to stabilise the covering sheets.
  - II. A sandwich structural part is well suited for absorbing concentrated loads.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

- a) I is correct , II is incorrect.
- 

1038. Which of these statements regarding cockpit windows are correct or incorrect ?
- I. Cockpit windows never have an additional speed restriction , related to bird impact , when window heating is inoperative.
  - II. Cockpit side windows are always provided with a de-icing system.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

- c) I is incorrect , II is incorrect.
- 

1039. Which of these statements regarding cockpit windows are correct or incorrect ?
- I. Cockpit windows never have an additional speed restriction , related to bird impact , when window heating is inoperative.
  - II. Cockpit side windows are usually defogged only.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is incorrect.
  - d) I is correct , II is correct.

Answer ;

- a) I is incorrect , II is correct.
- 

1040. Which of these statements regarding cockpit windows are correct or incorrect ?
- I. On some aeroplanes the cockpit windows have an additional speed restriction , related to bird impact , when window heating is inoperative.
  - II. Cockpit side windows are always provided with a de-icing system.
- a) I is incorrect , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

- b) I is correct , II is incorrect.
-

1041. Which of these statements regarding cockpit windows are correct or incorrect ?

- I. On some aeroplanes the cockpit windows have an additional speed restriction , related to bird impact , when window heating is inoperative.
  - II. Cockpit side windows are usually defogged only.
- a) I is incorrect , II is incorrect.
  - b) I is correct , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is correct.

Answer ;

c) I is correct , II is correct.

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1042. Which of these statements regarding most gust lock systems is correct ?

- a) On reversible flight controls , there is no need for a gust lock.
- b) When the gust lock is ON there is protection to prevent take-off.
- c) A gust lock is only fitted on the elevator and the rudder.
- d) A gust lock can be used in flight to reduce the effect of turbulence.

Answer ;

b) When the gust lock is ON there is protection to prevent take-off.

---

1043. Which one of the following factors would be most likely to increase the possibility of detonation occurring within a piston engine ?

- a) the use of a fuel with a high octane rating as compared to the use of one with a low octane rating.
- b) using an engine with a low compression ratio.
- c) slightly retarding the ignition timing.
- d) high cylinder head temperature.

Answer ;

d) high cylinder head temperature.

---

1044. Which one of the following factors would be most likely to increase the possibility of detonation occurring within a piston engine ?

- a) using an engine with a low compression ratio.
- b) using too lean a fuel/air mixture.
- c) slightly retarding the ignition timing.
- d) the use of a fuel with a high octane rating as compared to the use of one with a low octane rating.

Answer ;

b) using too lean a fuel/air mixture.

---

1045. Which statement is correct ?

- I. The freezing point for Jet A is at a higher temperature than that for Jet B.
  - II. The flash point for Jet A is at a higher temperature than that for Jet B.
- a) I is correct , II is correct.
  - b) I is correct , II is incorrect.
  - c) I is incorrect , II is incorrect.
  - d) I is incorrect , II is correct.

Answer ;

a) I is correct , II is correct.

---

1046. Which statement is correct ?

- I. The freezing point for Jet A is at a higher temperature than that for Jet B.
  - II. The flash point for Jet A is at a lower temperature than that for Jet B.
- a) I is incorrect , II is incorrect.
  - b) I is incorrect , II is correct.
  - c) I is correct , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

d) I is correct , II is incorrect.

---

1047. Which statement is correct ?

- I. The freezing point for Jet A is at a lower temperature than that for Jet B.
- II. The flash point for Jet A is at a higher temperature than that for Jet B.
- a) I is incorrect , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is incorrect.
- d) I is correct , II is correct.

Answer ;

- a) I is incorrect , II is correct.
- 

1048. Which statement is correct ?

- I. The freezing point for Jet A is at a lower temperature than that for Jet B.
- II. The flash point for Jet A is at a lower temperature than that for Jet B.
- a) I is correct , II is incorrect.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is correct.
- d) I is incorrect , II is correct.

Answer ;

- b) I is incorrect , II is incorrect.
- 

1049. Which statement is correct ?

- I. The freezing point for Jet A is at a lower temperature than that for Jet B.
- II. The flash point for Jet A is at about the same temperature as that for Jet B.
- a) I is incorrect , II is correct.
- b) I is correct , II is incorrect.
- c) I is correct , II is correct.
- d) I is incorrect , II is incorrect.

Answer ;

- d) I is incorrect , II is incorrect.
- 

1050. Which statement is correct ?

- I. The freezing point for Jet A is at about the same temperature as that for Jet B.
- II. The flash point for Jet A is at a higher temperature than that for Jet B.
- a) I is correct , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is incorrect.
- d) I is incorrect , II is correct.

Answer ;

- d) I is incorrect , II is correct.
- 

1051. Which statement is correct about the flow in a divergent jet engine inlet ?

- I. The speed increases in flow direction.
- II. The total pressure increases in flow direction.
- a) I is incorrect , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is correct , II is correct.
- d) I is correct , II is incorrect.

Answer ;

- a) I is incorrect , II is incorrect.
- 

1052. Which statement is correct about the flow in a divergent jet engine inlet ?

- I. The static pressure remains constant in flow direction.
- II. The total temperature remains constant in flow direction.
- a) I is incorrect , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is correct , II is correct.
- d) I is correct , II is incorrect.

Answer ;

- b) I is incorrect , II is correct.
-

1053. Which statement is correct concerning the effect of the application of carburettor heat ?
- a) The volume of air entering the carburettor is reduced , thus enriching the fuel/air mixture.
  - b) The volume of air entering the carburettor is reduced , thus leaning the fuel/air mixture.
  - c) The density of the air entering the carburettor is reduced , thus leaning the fuel/air mixture.
  - d) The density of the air entering the carburettor is reduced , thus enriching the fuel/air mixture.

Answer ;

- d) The density of the air entering the carburettor is reduced , thus enriching the fuel/air mixture.
- 

1054. Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is decreased ?

- I. The HP spool speed decreases.
  - II. The EGT decreases.
- a) I is correct , II is incorrect.
  - b) I is correct , II is correct.
  - c) I is incorrect , II is correct.
  - d) I is incorrect , II is incorrect.

Answer ;

- b) I is correct , II is correct.
- 

1055. Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed decreases.
  - II. The EGT decreases.
- a) I is correct , II is incorrect.
  - b) I is incorrect , II is incorrect.
  - c) I is incorrect , II is correct.
  - d) I is correct , II is correct.

Answer ;

- b) I is incorrect , II is incorrect.
- 

1056. Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed decreases.
  - II. The EGT increases.
- a) I is correct , II is incorrect.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is incorrect , II is correct.

Answer ;

- d) I is incorrect , II is correct.
- 

1057. Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed decreases.
  - II. The EGT remains constant.
- a) I is incorrect , II is correct.
  - b) I is incorrect , II is incorrect.
  - c) I is correct , II is correct.
  - d) I is correct , II is incorrect.

Answer ;

- b) I is incorrect , II is incorrect.
- 

1058. Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed increases.
  - II. The EGT decreases.
- a) I is correct , II is incorrect
  - b) I is incorrect , II is incorrect
  - c) I is incorrect , II is correct
  - d) I is correct , II is correct

Answer ;

- a) I is correct , II is incorrect
-

**1059.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed increases.
- II. The EGT increases.
- a) I is correct , II is incorrect.
- b) I is correct , II is correct.
- c) I is incorrect , II is correct.
- d) I is incorrect , II is incorrect.

**Answer ;**

- b)** I is correct , II is correct.
- 

**1060.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed increases.
- II. The EGT remains constant.
- a) I is correct , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is incorrect.
- d) I is incorrect , II is correct.

**Answer ;**

- c)** I is correct , II is incorrect.
- 

**1061.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed remains constant.
- II. The EGT decreases.
- a) I is incorrect , II is incorrect.
- b) I is correct , II is incorrect.
- c) I is correct , II is correct.
- d) I is incorrect , II is correct.

**Answer ;**

- a)** I is incorrect , II is incorrect.
- 

**1062.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The gas generator speed remains constant.
- II. The EGT increases.
- a) I is incorrect , II is incorrect.
- b) I is incorrect , II is correct.
- c) I is correct , II is incorrect.
- d) I is correct , II is correct.

**Answer ;**

- b)** I is incorrect , II is correct.
- 

**1063.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The HP spool speed decreases.
- II. The EGT increases.
- a) I is incorrect , II is incorrect.
- b) I is correct , II is incorrect.
- c) I is correct , II is correct.
- d) I is incorrect , II is correct.

**Answer ;**

- d)** I is incorrect , II is correct.
- 

**1064.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The HP spool speed increases.
- II. The EGT increases.
- a) I is correct , II is correct.
- b) I is correct , II is incorrect.
- c) I is incorrect , II is incorrect.
- d) I is incorrect , II is correct.

**Answer ;**

- a)** I is correct , II is correct.
-



**1065.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The HP spool speed increases.
- II. The EGT remains constant.
- a) I is correct , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is incorrect.
- d) I is incorrect , II is correct.

**Answer ;**

- c) I is correct , II is incorrect.**
- 

**1066.** Which statement is correct for a gas turbine engine with a constant speed propeller and free power turbine , when the power setting is increased ?

- I. The HP spool speed remains constant.
- II. The EGT increases.
- a) I is incorrect , II is correct.
- b) I is incorrect , II is incorrect.
- c) I is correct , II is incorrect.
- d) I is correct , II is correct.

**Answer ;**

- a) I is incorrect , II is correct.**
- 

**1067.** Whilst in level cruising flight , an aeroplane with a pressurised cabin experiences a malfunction of the pressure controller. If the cabin rate of climb indicator reads -200 ft/min ;

- a) a descent must be initiated to prevent the oxygen masks dropping when the cabin altitude reaches 14000 ft.
- b) the aircraft has to climb to a higher flight level in order to reduce the cabin altitude to its initial value.
- c) the crew has to intermittently cut off the incoming air flow in order to maintain a zero cabin altitude.
- d) the differential pressure will rise to its maximum value , thus causing the safety relief valves to open.

**Answer ;**

- d) the differential pressure will rise to its maximum value , thus causing the safety relief valves to open.**
- 

**1068.** Whilst stationary on the ground in a hangar the most important loads on a cantilever wing are ;

- a) compression in both the upper and the lower surfaces.
- b) compression in the upper surface , tension in the lower surface.
- c) tension in the upper surface , compression in the lower surface.
- d) tension in both the upper and the lower surfaces.

**Answer ;**

- c) tension in the upper surface , compression in the lower surface.**
- 

**1069.** Windshield heating of a transport aeroplane is ;

- a) not affecting the strength of a cockpit windows.
- b) essential to improve the strength of the cockpit windows.
- c) used only at low altitudes where there is a risk of ice formation.
- d) only used when hot-air demisting is insufficient.

**Answer ;**

- b) essential to improve the strength of the cockpit windows.**
- 

**1070.** With a piston engine , when detonation is recognised , you ;

- a) increase manifold pressure and lean the mixture.
- b) reduce manifold pressure and enrich the mixture.
- c) increase manifold pressure and enrich the mixture.
- d) reduce manifold pressure and lean the mixture.

**Answer ;**

- b) reduce manifold pressure and enrich the mixture.**
- 

**1071.** With an aircraft fitted with a fixed pitch propeller , during flight at normal cruising speed , one magneto fails completely. This will first cause ;

- a) engine to overheat.
- b) excessive vibration.
- c) additional load on the other magneto.
- d) loss of approximately 100 rpm.

**Answer ;**

- d) loss of approximately 100 rpm.**
-

1072. With regard to a twin spool turbofan , an abortive start ( hung start ) will result in ;
- a) N2 speed stabilising at a higher than normal rpm and a subsequent decrease in EGT.
  - b) N2 speed stabilising at a lower than normal rpm and a subsequent decrease in EGT.
  - c) N2 speed stabilising at a higher than normal rpm and a subsequent increase in EGT.
  - d) N2 speed stabilising at a lower than normal rpm and a subsequent increase in EGT.

Answer ;

- d) N2 speed stabilising at a lower than normal rpm and a subsequent increase in EGT.**
- 

1073. With regard to an aircraft structure , “fail safe” is one ;
- a) that is only used for a limited time.
  - b) that is easily manufactured.
  - c) in which the load is carried by other components if a part of the structure fails.
  - d) used for small aircraft only.

Answer ;

- c) in which the load is carried by other components if a part of the structure fails.**
- 

1074. With regard to pneumatic mechanical devices that afford ice protection the only correct statement is ;
- a) They can be used as both de-icing and anti-icing devices.
  - b) They can only be used as de-icing devices.
  - c) They can only be used as anti-icing devices.
  - d) They are used extensively on modern aircraft as they are inexpensive and easy to maintain.

Answer ;

- b) They can only be used as de-icing devices.**
- 

1075. With respect to a piston engine aircraft , ice in the carburettor ;
- a) will only form at OAT's below +10°C.
  - b) may form at OAT's higher than +10°C.
  - c) will only form at outside air temperatures ( OAT's ) below the freezing point of water.
  - d) will only form at OAT's below the freezing point of fuel.

Answer ;

- b) may form at OAT's higher than +10°C.**
- 

1076. With which instrument(s) do you monitor the power output of an aeroplane fitted with a fixed pitch propeller ?
- a) RPM indicator.
  - b) Fuel Flow indicator.
  - c) Cylinder head temperature indicator.
  - d) RPM and MAP indicator.

Answer ;

- a) RPM indicator.**
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